

REPUBLIC OF KENYA



MINISTRY OF LANDS, PUBLIC WORKS, HOUSING AND URBAN DEVELOPMENT

STATE DEPARTMENT OF HOUSING AND URBAN DEVELOPMENT

BILL OF QUANTITIES

FOR:

**PROPOSED CONSTRUCTION OF EMURUA DIKIRR AFFORDABLE HOUSING PROJECT IN
NAROK COUNTY**

Ministry of Lands, Public Works, Housing and Urban Development

State Department for Housing and Urban Development,

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PREAMBLES

BILL NO. 1: PREAMBLES

EXCAVATION AND EARTHWORK

Nature of Excavation

- A The Contractor must ascertain for himself the nature of the materials to be excavated and price the work accordingly as no allowance will be made beyond the Contract Sum for any alleged ignorance in this respect.

Site Clearance

- B. The Contractor shall clear the construction areas within the site of all bushes, roots, brush, boulders, natural obstructions, rubbish and any other natural or artificial obstructions which would interfere with construction of buildings, roads, paths and drains.
- C. Clear away all anti/termite hills and nests over the area of the site, excavate for, locate and destroy queens.
- D. Treat the cavity formed by the removal of the nest as described hereinafter under "Soil Sterilization" and backfill with approved material well rammed and consolidated in layers not exceeding 300 mm thick.
- E. All areas of the site must be thoroughly proofed against rodents and special care must be taken to ensure that no unconsolidated areas are left near banks and ditches.

Commencing Levels

- F. Unless specifically stated otherwise the commencing levels for excavation shall be deemed to be existing ground level or underside of reduced level excavation.
- G. All measurements are based upon reduced level excavation being executed first and no adjustment will be made should a differing sequence of operations be adopted, unless specifically ordered by the Architect in writing.

Excavations

- H. Excavations shall be to the widths and depths indicated the drawings or to such lesser or greater depths as the Architect may deem necessary and so instruct the Contractor in order to obtain satisfactory foundations.
- J. Any difference in the quantity of works actually executed under such instructions and that provided in the Bills of Quantities shall be measured and valued by the Quantity Surveyor as a variation under the relevant Conditions of Contract.

- A. If, however, the Contractor excavates to any greater depth or widths than are shown on the drawings or directed by the Architect, then the Contractor shall at his own expense fill in such extra depths and widths with concrete similar to that described for foundations to the satisfaction of the Architect.

Bottoms to Excavation

- B. The Contractor shall report to the Architect as and when a secure bottom to the excavations has been obtained and the same is ready to receive concrete. Any excess depth unnecessarily excavated below the formation level shall be backfilled with and compacted as directed by the Architect and no payment shall be made for excess excavation or for the fillings & compaction
- C. Any concrete or other work put in before excavations have been inspected and approved shall, if so directed, be removed and new work substituted after excavations have been approved all at the Contractor's expense.
- D. If so directed, the Contractor shall water and well ram the bottoms of excavations to the satisfaction of the architect.

Measurement of Excavation Work

- E. Excavation work is measured net as before digging and the Contractor must allow for increase in bulk after digging.

Trenches for pipes, cables kerbs, etc., other than drain pipes

- F. Prices for excavation of trenches for pipes, cables, kerbs, etc., shall include for grading and ramming bottoms to the levels required, all necessary planking and strutting, carefully returning, filling and ramming selected excavated materials and for carting away any surplus materials.

Rock

- G. Any rock or other hard materials encountered in excavating to the required depth which, in the opinion of the Architect, can only be removed by wedges or compressor plant shall be paid for as an extra and the price shall include for trimming and levelling. No blasting will be allowed. Hard compacted murrum which can be removed by pick will not be classed as rock notwithstanding that the Contractor may decide to remove it by wedges or compressor plant.
- H. The Contractor must give notification to the Architect or his representative when such material is encountered and its extent must be agreed with Architect or Quantity Surveyor or their authorised representative before the work is carried out. No allowance will be made for rock excavation unless the foregoing procedure has been followed.

Rates for Excavation

- J. The rates for excavation shall include for excavating by hand or machine in all types of materials except rock, as previously specified.

- A. Excavations for plain concrete foundations have been measured to the **net sizes** required by concrete dimensions.
- B. An allowance for working space and formwork has been measured to reinforced concrete foundation, but if the Architect's approval is given to pouring concrete against the face of the excavations these items will be measured and adjusted in the Final Account.
- C. The rates for excavation must include for such excavating in all types of ground encountered including sand, murram, hard murram, tree roots and loose boulders.

Levelling

- D. No item is measured for levelling and consolidating ground and rates for excavations must include for levelling and preparing the ground for concrete or other works including ramming or rolling.

Disposal of Water

- E. The Contractor shall keep the excavations free from standing water and silt (or excavated materials softened by water) and he shall include for the cost of pumping, construction of temporary drains, soakaway pits, etc., as deemed necessary to achieve this. An item has been included for this in the Bills of Quantities in each relevant section. The cost of pumping to dispose of any spring or running water has been covered by Provisional Sum. If spring or running water is encountered, the cost of any pumping ordered by the Architect will be paid for in accordance with the Dayworks Schedule.

Planking and Strutting

- F. Sides of all excavations must be supported in order to prevent falls from or collapse of the earth face. The term "planking and strutting" is deemed to include any method or methods which the Contractor elects to adopt, uphold, protect and maintain the sides of excavations. The Contractor will be responsible for any consequences of his failure in this respect including clearing away fallen materials and any extra concrete or other works including formwork ordered by the Architect due to such failure. An item has been included in these Bills of Quantities in each relevant section.

Return, Fill in and Ram

- H. Material returned around foundations externally shall be selected hard, dry excavated materials arising from the excavations free from vegetable soil, roots and rubbish carefully filled in, spread, watered and compacted in layers not exceeding 200 mm thick. Backfilling internally shall be hardcore, or selected hard dry granular materials as above to approval.
- J. No excavations or foundation work shall be filled in or covered up until all measurements necessary for the adjustment of variations have been made. Walling shall not be built upon the foundations until four days after deposition of concrete.

Cart Away

- A. All surplus excavated material, where so directed, and all rubbish is to be removed from the site and the Contractor is to find his own dump and pay all charges.

Approval Before Filling

- B. No fill materials shall be placed before approval has been given by the Architect for filling to begin.

Measurement of Filling Generally

- C. Filling is measured net as after consolidation.

Earth Filling

- D. Levels specified to be made up with surplus soil, etc., are to be filled in with selected soil free from vegetable growth to the approval of the Architect and is to be laid in layers not exceeding 200 mm thick, each layer to be levelled, well rammed and consolidated and watered if necessary.

Hardcore Filling

- E. Hardcore shall consist of clean hard broken stone or rubble graded to pass in all directions a 100 mm ring with sufficient sand added to fill the interstices. The hardcore shall be well packed, rammed and where possible, rolled with a heavy roller. Where rolling is impossible compaction shall be by hand or by mechanical tampers. Before any concrete is laid on hardcore, the hardcore shall be levelled and blinded with sand, rolled and well watered through a sprinkler rose.

Borrow Pits

- F. No borrow pits will be allowed to be opened on the site

Soil Sterilization

- G. Anti-termite treatment is to be carried out using one of the chemicals below and the Contractor will be required, upon completion of the soil sterilization, to furnish a written guarantee certifying the following:-

- (a) That the chemicals applied comply with the requirements specified herein for chemical concentration and rates of application.
- (b) That the treatment will remain effective against termite infestation for a period of five years.
- (c) Application shall not be done whilst its raining or to surface of filling which are wet, and strictly in accordance to manufacturer's instructions

Soil Sterilization

- A. The chemicals used shall be one of the following:-
- 5 Termicide A; 1 part to 45 parts water
 - 7 Pentachlorophenol; 5% in oil solution
 - 8 Trichlorobenzene; 1 part to 3 parts oil
- B. Some of the chemicals listed above are toxic to animals and plant life and must, therefore, be applied only with caution by an experienced person. Where individual water supply systems are proposed, precautions must be taken to prevent infiltrating and endangering the water supply. Treatment shall not be made when soils or fill is excessively wet or immediately after heavy rains.
- C. Precautions must also be taken to prevent disturbance of the treatment by animals or human contact with the treated soil. The treated areas are to be covered as quickly as possible after treatment.
- D. The rate of applications is to be 7 litres per square meter and the areas measured include those under floor and around wall and column foundations.

CONCRETE WORK

Code of Practice for Reinforced Concrete Work

- A. All workmanship, materials and tests in connection with reinforced concrete work are to be conformity with B.S. Code of Practice B S 8110 : 1985 - The Structural Use of Concrete.

Generally

- B. A competent person shall be employed whose first duty it will be to supervise all stages in the preparation and placing of concrete. All cubes shall be made and site tests carried out under his direct supervision.
- C. All materials which have been damaged, contaminated or have deteriorated. or which do not comply in any way with the requirements of the specification, shall be rejected and shall be immediately removed from the site.
- D. No materials shall be stored or stacked on suspended floors without the Engineer's prior approval.

Samples

- E. Samples of all materials are to be submitted for approval of the Engineer at least one week before it is desired to commence deliveries. All condemned materials are to be removed from the site within 24 hours.

Cement

- F. Cement used shall be ordinary Portland cement and shall be obtained only from manufacturers approved by the Engineer, and shall comply in every respect with B.S. 197-1. The Contractor at his own expense may use rapid hardening Portland Cement (to B.S. 197-1) in order to speed up progress of the Works. If rapid hardening Portland Cement is used, the prior approval of the Engineer shall be obtained in writing.
- G. Each consignment of cement shall be accompanied by the manufacturer's certificate showing that a representative sample of the consignment has been tested and complies with the appropriate specification. From time to time as requested by the Engineer, copies of the cement manufacturer's test certificates shall be delivered to the Engineer or his representative on the site promptly, but such documents shall not preclude the Engineer from rejecting any cement which does not in every way comply with the specification.

Cement Storage

- H. The cement must be delivered in the manufacturer's sealed and branded bags and stored separately in dry, water-tight stores with their floors raised above ground level and shall be at all times carefully protected from moisture.
- I. The cement shall be stored in such a way that each consignment may be identified and used in the order of its delivery. Cement may be delivered in bulk containers provided additional suitable arrangements are made for bulk storage on site to the approval of the Engineer.

Inferior Cement

- A. Any cement which has failed to pass the tests or has been damaged by water or contaminated in any way on site shall immediately be put into bags and removed from the site.

Aggregate

- B. Aggregates shall be granite or other equal and approved obtained from an approved source and shall comply with B.S. 1260. They must be chemically inert, strong, hard, durable, free from adhering coating, salts, organic or other impurities and shall be washed or screened as directed.

Fine Aggregate and Sand

- C. Fine aggregate and sand shall be clean, sharp, coarse, hard siliceous materials and equal at all times to the samples which shall be deposited with and approved by the Architect or Engineer. It shall comply with the requirements of B.S. 1260, Table 2, Zones 1,2 or 3. The caustic soda tests for organic impurities shall show a colour not deeper than that of the standard solution. The settling test for natural sand shall be made and after being allowed to settle for three hours the layer of silt deposit on the coarse materials shall not exceed 10% for crushed stone and 3% for natural sand or crushed gravel.
- D. The Contractor shall supply all necessary equipment for testing of fine aggregate and sand for use of the Clerk of Works.

Coarse Aggregates

- E. Coarse aggregates shall be granite from approved quarries, clean, free from earth and extraneous matter, and shall conform to B.S. 1260. The amount of fine particles occurring in a free state or as loose adherent shall not exceed 1% when determined by the laboratory sedimentation test.
- F. After twenty-four hours in water, a previously dried sample shall not gain more than 10% in weight for crushed stone or 3% for natural sand or crushed gravel
- G. The four nominal aggregate sizes shall be 40 mm (1.5"): 20 mm (.75"): 10 mm (3/8"): 6 mm (1/4") : and the grading when analyzed as described in B.S. 812 shall be within the limits given in B.S. 1260.

Aggregate Storage

- A. Each grade of aggregate shall be stored in the works in separate heaps so that there shall be no possibility of any inter-mixing. Any materials which have become inter-mixed shall be removed from the site forthwith by the Contractor.
- B. The materials shall be stored on a timber or concrete floor and the piles shall be as large as possible, flat topped and drained.

Water

- C. All water used on the Works shall be clean, free from earthy vegetable and organic matter and from acidic and alkaline substances in suspension or solution. It shall preferably be obtained from the water mains of the Ministry of Water and Energy Department or Water Authority and shall be stored in proper water storage tanks to the approval of the Architect or Engineer. Any approved water shall be tested in accordance with B.S. EN 1008.

Admixtures

- D. Admixtures of any kind for accelerating the setting of cement, plasticiser, hardeners, water proof etc., shall be used only if approved or specified by the Architect or Engineer.

Proportion of Concrete Mix

- E. The quantity of cement shall be measured by weight and each batch of concrete is to use one or more whole bags. The quantity of fine aggregate and coarse aggregate shall be measured separately by weight in an approved weight batching plant. Volume mixing will not be permitted. The weight of damp aggregates must be adjusted to take into account the weight of water in the aggregates, and must be adjusted to take into account the weight of water in the aggregates, and this in turn will affect the amount of water to be added into the mix.
- F. Throughout the carrying out of the Contract "Work Tests" are to be made from concrete drawn from newly laid concrete or concrete about to be placed in position, such cubes being made when directed by the Clerk of Works and in his presence. Such cubes shall be made in 150 mm or six inch cube steel or cast from mould and shall be marked and cured strictly in accordance with Appendices of the Code of Practice, and shall be forwarded carriage paid in time for testing at the required age to a testing laboratory to be nominated by the Architect or Engineer.
- G. Six cubes shall be made on each occasion, and cured in compliance with B.S. 1881 Part 3, 1983 concrete for each cube being from a different batch. Three cubes shall be forwarded in time for testing at the age of seven days from casting and three cubes in time to testing in twenty-eight days. Each cube shall be marked with the date of casting and a distinctive reference number in accordance with a system agreed by the Engineer. A record shall be kept of the position from which the concrete for each set of cube was drawn, or to which it was about to be placed.

Concrete Work Cont'd

- A. At least three sets of six cubes shall be cast during each week concrete is being cast including sets of cubes for each quality of concrete used during the period.
- B. Concrete is required to have the properties and give the strength in Newtons per square millimetre as set out in the table below which is to be considered as the minimum standard that will be accepted in the finished Works.
- C. The workability of the fresh concrete should be such that concrete is suitable for handling, placing and compaction so that it surrounds the reinforcement, tendons and ducts and completely fills the formwork.

Grade	Quality	Maximum size of coarse aggregate	Maximum Water Cement Ratio by weight of Aggregate	Minimum Crushing Strength of Works Test Cubes	
				7 days	28 days
30	1:1:2	20	0.45	30	36
25	1:1.5:3	20	0.55	21	26
25	1:1.5:3	10	0.55	21	26
20	1:2:4	20	0.60	14	21
20	1:2:4	10	0.60	14	21
15	1:3:6	10	0.60	-	12
10	1:3:6	10	0.60	8	10
7	1:4:8	40	0.60	-	7
-	1:10	All in Aggregate		-	-

- D. If the strengths required in the table are not attained and maintained throughout the carrying out of the Contract, the Contractor will be required to increase the proportion of cement or substitute better aggregate at his own cost so as to give concrete which does comply with the requirements of this Clause. The Contractor may be required to remove and replace at his own cost any concrete which fails to attain the required strength as ascertained by the Works Cube Tests.

Unsatisfactory Concrete Work

- E. Should in the opinion of the Engineer any of the results of the specified tests of concrete or materials be unsatisfactory, the Engineer may order the work to be stopped pending his further instructions. Executed work for which test cubes are unsatisfactory shall be liable to rejection and, if so directed by the Engineer, the work represented by the tests shall be cut out and re-executed at the Contractor's expense.

- A. In the case of seven day Works Cube Tests proving unsatisfactory, the work may be stopped, but shall not be liable to rejection until the result of the twenty-eight day test is known.
- B. In the event of the results of the twenty-eight day Works Cube Tests proving unsatisfactory, the work represented shall be immediately liable to rejection. The Contractor may, however, be given the option of cutting three specimens from the completed work subject to the direction of the Engineer, and preparing therefrom test cubes or cores Tests in accordance with the requirements of Part 4 of B.S. 1881 Part 3, 1983, which shall be sent to the Testing Laboratory for testing as for Works Cube
- C. Should the average strength of these specimens attain the specified minimum twenty-eight day strength, the work will, subject to the Engineer's discretion be accepted. Alternatively, the Engineer may instruct the Contractor to make a loading test as described hereinafter. The cost of all cutting, preparation of specimens, testing and making good the portions of the structure affected, shall be borne by the Contractor. The cost of all delays on site due to concrete not attaining the desired strength, or caused by investigation of defects, cutting away and making good, shall be entirely the Contractor's responsibility.

Structural Test

- D. If, in the Engineer's opinion, there is a doubt as to the strength of a structure, solely or in part, for the reason that the site-made concrete cubes fail to attain the specified strength, the Contractor shall be reimbursed for the cost of the test. If the result of the test is not satisfactory, the Contractor shall bear the cost of the test and the cost of correcting any defects in accordance with the instructions of the Engineer, strength, or because of one or more circumstances attributable to alleged negligence on the part of the Contractor to make a loading test on the Works or any part thereof. The nature of the test and the loading shall be in accordance with Clause 605 of C.P. 114. If the result of the test is satisfactory, except where the test has been made because test cubes

Formwork

- E. The formwork shall be so constructed as to remain sufficiently rigid during the placing compaction of the concrete and shall be sufficiently tight to prevent loss of liquid from the concrete. Vertical strutting shall be carried down to such construction as is sufficiently strong to afford the required support without injury. All rubbish, chippings, shavings and sawdust shall be removed from the interior of the forms before the concrete is placed, and suitable washout holes shall be provided to facilitate this, and the formwork in contact with the concrete shall be clean and thoroughly wetted and treated with the approved mould oil. Care shall be taken that such oil is kept out of contact with the reinforcement and shall be used as sparingly as possible. In no circumstances shall forms be struck until the concrete reaches a cube strength of at least twice the stress to which the concrete may be subjected at the time of striking, and in any case the minimum permissible times shall be as follows:-

Vertical sides of wall and columns	2 days
Sides of beams and lintels	2 days
Soffits of slabs (Subject to retention of props until 21 days	14 days
Soffits of beams and lintels (Subject to retention of props until 21 days)	14 days

- A. No formwork is to be removed if, in the opinion of the Engineer, the concrete has not hardened sufficiently. Approval of the Engineer shall not relieve the Contactor of his liability to make good any concrete which may be damaged by premature removal or collapse of forms. Notwithstanding any other clauses in this specification the responsibility for the safe removal of the formwork rests with the Contactor.
- B. All formwork shall be removed without such shock or vibration as would damage the reinforced concrete.
- C. Forms shall be true to lines and levels and braced and strutted to prevent deformation.
- D. Before placing of the concrete, bolts and fixings shall be in position and cores and other devices used for forming openings, holes pockets, recesses, ducts or other cavities shall be fixed to the shuttering.
- E. Concrete shall not be poured in horizontal layers to a depth exceeding 1500 mm in formwork, except where prior approval of the Engineer has been obtained.
- F. Formwork is measured to the actual net surface of the concrete to be supported and the Contractor shall allow in his prices for any waste, fixing at the various levels, straight cuttings, splayed edges, notchings, fillets to form chamfered arises, extra materials, joints, overleaves for angles, extra labour for narrow widths and small quantities, props, stays, struts, hangers, brackets, edges, wiring, bolts, and everything necessary to keep all quite firm and rigid, and any other labour and materials necessary to fix, ease, adjust and remove the formwork as described.

Normal Finish to Faces of Structural Concrete

- G. After removal of shuttering, unless instructed to the contrary, the face of exposed concrete is to be rubbed down immediately to remove fins or other irregularities. In the event of parts of the concrete being honeycombed, such portions are to be cut to a depth and shape required by the Engineer and made up with fine concrete of equal quality in such a manner as shall be directed. The face of concrete for which shuttering is not provided, other than slab, is to be smoothed with a wooden float to give a finish equal to that of the rubbed-down surface where shuttering is provided. The top face of a slab which is to not intended to cover with other materials is to be levelled and floated before setting to a smooth finish at the level or falls shown on the drawings or elsewhere. The floating must be carried out in such a way as will prevent an excess of mortar being brought to the surface of the concrete. The top face of a slab intended to be surfaced with mortar, granolithic, or similar materials is to be brushed with a stiff broom while still green to remove any laitence © and to provide a roughened surface.

Fairfaced Concrete

- A. Where so described or measured, faced of concrete shall be finished fair by means of formwork lined with approved waterproof plywood so as to produce a perfectly true surface and shall have all imperfections in the concrete face cut out, made good in cement mortar and rubbed down with carborundum stone and finally bag rubbed with cement slurry to finish to a high standard without trace of shuttering marks, joints or other disfigurements.

Wrought Boarded Face Formwork to give a Board Mark Finish

- B. Where so described or measured, faces of concrete shall be finished fair by means of 100 mm or 150 mm (nominal) width tongued and grooved boarding of 25 mm (minimum) thickness. The edges of all boards shall be nominal 2 mm chamfer to form controlled fins.
- C. Such formwork to column faces shall be of continuous length boards between construction joints.
- D. End joints will be permitted to beams faces, etc., and shall be tongued, staggered and well distributed.
- E. All imperfections shall be cut out and made good in concrete of equal quality.
- F. The resulting concrete shall show grain and individual board marks, be free from honeycombing and excessive air holes, of uniform colour and to the entire satisfaction of the Engineer.

Wall Ties

- G. Where blockwalls abut columns or solid concrete walls two 6 mm diameter steel reinforcing bar ties are to be cast into the concrete at vertical intervals of 400 mm. Ties to be 300 mm long and project 150 mm into blockwork.

Holes, Pipes Etc.

- H. The Contractor shall be responsible for the co-ordination with sub-contactors for incorporating any electrical conduits pipes, fixing blocks, chases, holes, etc., in the concrete members as required. The Contractor shall submit full details of these items to the Engineer for approval before the work is put in hand. Concrete fixing blocks may be embedded in the concrete provided that the strength or effective cover of any part of the structure is not adversely affected nor the finished work damaged by any movement of the blocks. All fixing blocks, chases, holes etc., to be left in concrete shall be accurately set out and cast with the concrete. No openings, chases, holes or other voids shall be cut or formed in concrete without the approval of the Engineer.

Blinding Concrete

- A. No casting of any concrete on the ground shall take place until the ground has been passed as satisfactory by the Engineer. All ground to carry reinforced concrete shall be covered with a 50 mm minimum blinding layer of concrete 1:4:8. The cover for concrete under reinforcement shall be entirely above the blinding layer.

Mixing

- B. Concrete is to be mixed in a batch mixer of approved type having a drum rotating about a horizontal or inclined axis. The speed of the drum is to be not more than twenty and not less than fourteen revolutions per minute. Each mixer is to be fitted with a water measuring device capable of accurate measurement to one gallon for one cubic yard mixers and pro rate for smaller sizes and so arranged that the accuracy is not affected by variations in the pressure of the water supply line.
- C. The fine and coarse aggregate and the cement are to be mixed for at least four turns of the drum, after which the required amount of water is to be added gradually while the drum is in motion and the concrete then mixed for at least one and a half minutes and until a mix of uniform colour and consistency is attained.
- D. The volume of concrete mixed in any one batch is not to exceed the rated capacity of the mixer.
- E. The whole of the mixed batch is to be removed before materials for a fresh batch enter the drum.
- F. On cessation of work, including all stoppages exceeding twenty minutes, the mixers and all handling plant are to be washed out with clean water.
- G. Concrete mixed as above is not to be modified by the addition of water or otherwise in order to facilitate handling, or for any other purpose.
- H. At least one slump test shall be made each day concreting is in progress under the supervision of the Clerk of Works. The slump shall not exceed 75 mm but at 25 mm slump may be allowed by the Engineer in certain structural members.

Transporting

- J. Concrete is to be handled from the place of mixing to the place of final deposit as rapidly as practicable by the methods which will prevent segregation or loss of ingredients and maintain the required workability. It should be deposited as nearly as practicable in its final position to avoid rehandling.
- K. Concrete shall be placed into the forms from as small a height as possible and shall in no case be dropped from a height of more than 1500 mm except with the approval of the Engineer.

- A. When chuting is used, the inclination of the chute must be such as to allow the concrete to flow without the use of excessive water and without segregation or loss of the ingredients. Details of any proposed chuting plant must be approved by the Engineer before the plant is delivered to the site.
- B. If the Contractor wishes to distribute concrete by means of pumps, full details of the system must be made available to the Engineer for approval.

Placing and Consolidation

- C. The concrete shall be placed before setting has commenced and in any case within thirty minutes from the time the water is added, and must not be subsequently disturbed. Concrete shall be thoroughly compacted during the operation of placing, and thoroughly worked around the reinforcement, around embedded fixtures, and into corners of the formwork. Mechanical vibration with an approved type insertion vibrator shall be used.
- D. The use of mechanical vibration will not relieve the Contractor of his responsibility for making good work which may be damaged by excessive or ill-applied vibration.
- E. All methods of placing and consolidation of the concrete are to be such as not to cause any disturbance or movement to the formwork or reinforcement. After being placed in position, the concrete is to be left absolutely undisturbed by any movements or thrusts while setting.
- F. An accurate record is to be kept by the Contractor showing dates and times when various portions of the work were concreted. The concreting foreman must not vary the approved mix or water content without the permission of the representative of the Engineer. it may occasionally be found that in constructed structural members or where the proportion of reinforcement to concrete is high, the workability of the concrete must be increased locally in order to effect full compaction. Such increase in workability shall be achieved by an increase in the cement content of not more than 10% of the concrete by weight in any single batch and must be made only with the approval of the representative of the Engineer.
- G. The workability of the concrete must never be altered by the use of additional water or sand alone.

Construction Joint

- H. The form and location of all construction joints shall be approved by the Engineer before commencement of work.

- A The Centering to form the stop shall be fitted with splay fillets on the concrete face and will be firmly fixed and scribed around the reinforcing steel. If any concrete shall flow past the stop, it shall be hacked off as soon as the concrete has set. Before any new concrete is placed up against the stopped face, the concrete previously placed shall be hacked and scoured with a wire brush to remove the scum. The joint shall then be soaked with water and covered with a sand cement mortar of proportions in the same ration in the concrete used. In all cases of application of mortar the punning must be adequate to incorporate the mortar in the body of the concrete. In no circumstances shall the concrete be allowed to finish at a break running down a rough slope. Such cases, if found, will be treated as contrary to the specification and the Contractor will be required to cut out the member and re-cast. In the case of horizontal joints, any excess water and laitence shall be removed from the surface after the concrete is deposited and before it has set.
- B Before casting slabs the haunchings or seatings for the slab shall be thoroughly hacked, scoured and washed and covered with at least 5 mm of mortar immediately before the slab is cast.
- C Slabs to be cast using alternate bay construction, maximum size of single panel 40 square meters.

Column Plinths

- D Column kicker plinths 75 mm high not cast monolithically with the beam or slab will be allowed only at the discretion of the Engineer and special precautions must be taken if permission is granted, especially in regard to the quality of the mix used and the curing of concrete.

Curing

- E The curing of the concrete must receive particularly careful attention. The concrete shall be covered with a layer of a sacking, canvas, hessian or suitable absorbent materials, and concrete, formwork and covering kept constantly wet for the first seven days after casting. Foundation concrete must be protected from falling earth and kept free from deleterious substances.

Dimensions of Finished Concrete

- F Except where specially noted, dimensions, levels, sizes, positions, and covers are to be exactly as dimensioned or specified with the following tolerances for concrete cast in situ.
- (a) For sizes of beams or columns, slab or wall thicknesses, not less than specified, nor more than 5 mm above. Dimensions between column faces not to have a greater tolerance than 10 mm.
- (b) For layout positions or dimensions horizontal or vertical 5 mm plus or minus.
- (c) Levels of floor, ceilings, beams, lintels, etc., (top and bottom), 5 mm plus or minus and no surface intended to be horizontal must slope more than 2 mm in 1 meter.
- (d) Errors in plumbing 5 mm plus or minus, and no line or surface intended to be vertical must slope more than 2 mm in 1 meter.
- (e) For cover of concrete around reinforcement 3 mm plus or minus.

Permissible tolerance shall not be cumulative.

Steel Reinforcement

- A. Mild steel rod reinforcement shall comply with B.S. 4449.
- B. High tensile steel rod reinforcement shall be hot rolled deformed steel complying with B.S. 4661 grade 460.
- C. Welded steel fabric reinforcement shall comply with B.S. 4483.
- D. The steel shall be stored so that it is kept clean and reasonably free from rust.
- E. All metal for reinforcement is to be free from loose mill scale, loose rust, oil and grease, or other harmful matter immediately before placing of the concrete.
- F. All reinforcement is to be placed and maintained in the positions shown on the drawings. Some definite method of ensuring the amount of cover required by the designer must be agreed between the Contractor and the Engineer.
- G. Reinforcement must be bent or straightened in a manner that will not injure the materials, and in accordance with B.S. 4466.
- H. All bars are to be bent cold.
- J. Starter bars are to be positioned accurately.
- K. All crossings of bars are to be securely wired.
- L. Bars at the top of slabs are to have substantial support.
- M. The prices of all rod reinforcement are to include for cutting to lengths and for all bending, hooked ends, etc., and for placing in position with distance pieces where necessary to ensure the rigidity of the bars and for tying together with approved wire in order to prevent displacement during concreting.
- N. The placing of all reinforcement shall be checked by the Engineer and in no circumstances is concrete to be deposited around any steel that has not been passed. At least forty eight (48) hours notice shall be given to the Engineer that reinforcement will be ready for inspection.
- O. Where bending schedules are provided, the measured weight of reinforcement for purposes of payment will be taken from the bending schedules and the Contractor must make due allowance in his rates for rolling margins and all the foregoing items and labour including cutting to waste from random lengths.

Cover to Reinforcement

- A. The thickness of the concrete cover to reinforcement shall conform in all respects to the B.S. Code of Practice B. S. 8110 : 1995 unless specifically shown on the drawings. Some approved method of ensuring the correct amount of cover shall be used.

Spacing Blocks and Chairs etc.

- B. Properly formed spacing blocks of concrete with wire ties or other approved means shall be securely wired or attached to the reinforcing bars to ensure the maintenance of the proper cover of concrete.
- C. These shall be dense concrete left with a wire brushed surface or dipped in grout before fixing. These blocks are particularly important where the surface of the concrete is exposed to the weather or dampness. The Contractor must ensure that the bars are securely fixed so as to maintain their indicated positions during the progress or pouring, tamping or vibration of concrete. Four chairs per drop are to be provided around columns to hold steel in positions and chairs are to be made up of 12 mm diameter mild steel bars. The cost of all such fixing steel must be allowed for the Contractor in his rates for reinforcement generally.

Precast Concrete

- D. Concrete shall all be cast in properly made strong mould to form shapes required. For work described as "finished fair" the mould shall be lined with sheet iron or other approved material.
- E. The coarse aggregate for precast concrete shall be 10 mm gauge where 1:1.5:3 mix concrete is specified.
- F. The concrete shall be of the mixes described and shall be thoroughly tamped in the mould and shall not be removed from them until seven days after placing the concrete, but the sides may be removed after three days providing the mould are such that the sides are easily removable without damaging the concrete.
- G. The precast work shall be cast under sheds and shall remain under same for seven days in the mould and a further seven days after removal from the moulds. During the whole of this period the concrete shall be shield by sacking or other approved materials and kept wet. It shall then be removed from the sheds and stacked in the open for at least seven days to season.
- H. All precast work shall be in lengths convenient for handling, unless otherwise described.
- J. Prices for precast concrete shall include for all moulds, hoisting and fixing to the levels required, bedding and pointing in cement mortar (1:3) and for finishing exposed faces fair and smooth where so described.

WALLING

Setting out Walling

- A. The Contractor shall provide proper setting out rods and set out all work on same for courses, openings, heights, etc., and shall build the walls and piers etc., to the widths, depths and heights indicated on the drawings and as directed and approved by the Architect.

Cement

- B. Cement shall be described in Concrete Work.

Fine Aggregate

- C. Fine aggregate for concrete blocks shall be as described for fine aggregate in Concrete Work.

Coarse Aggregate

- D. Coarse Aggregate for concrete blocks shall be good, hard, clean aggregates from approved quarries. It shall be free from all decomposed materials and shall be graded up to 10 mm all as described for coarse aggregate in Concrete Work.

Concrete Block

- E. Concrete blocks for walling shall be provided by the Contractor complying with B.S. 2028 Type A, and made in approved block making machines or a composition as follows:-

Portland Cement; 1 Cubic Meter

Fine Aggregate (graded up to 5 mm); 3 Cubic Meters

Coarse Aggregate (graded up to 10 mm); 6 Cubic Meters

- F. Blocks shall be solid or hollow two-hole type as specified and are to be made under sheds erected by the Contractor to the directions and approval of the Architect. In hollow blocks of the volume of the cavities shall be not less than 45% and not more than 50% of the gross

- G. The compressive strength Type A blocks shall be not less than:-

Average of 13 hollow blocks; 5.75 N/mm² gross area

Lowest individual hollow block; 4.0 N/mm² gross area

- A. The concrete is to be put into the machine's moulds in thin layers and all properly tamped therein. On removal from the machines the blocks are to be carefully deposited on wet the whole time, after which they shall be put out in the open on racks and protected with the approved matting, sacking or straw and kept wet for a further five days, then kept in the same position and under the same mat cover, but without wetting, for a further two days and then left in the open without matting or wetting for a further seven wet the whole time, after which they shall be put out in the open on racks and protected with the approved matting, sacking or straw and kept wet for a further five days, then kept in the same position and under the same mat cover, but without wetting, for a further two days and then left in the open without matting or wetting for a further seven days to season. All blocks must be left with good sharp edges. The blocks for use in the Works shall be 190 mm high and may vary in length from 300 mm to 400 mm and no variation above or below these lengths will be allowed except where required to form proper bonding at corners, round openings, sills, lintels, beams, etc., and the like positions and the Contractor must make or cut blocks to all the varying sizes required for these purposes and include this in his price. days to season. All blocks must be left with good sharp edges. The blocks for use in the Works shall be 190 mm high and may vary in length from 300 mm to 400 mm and no variation above or below these lengths will be allowed except where required to form proper bonding at corners, round openings, sills, lintels, beams, etc., and the like positions and the Contractor must make or cut blocks to all the varying sizes required for these purposes and include this in his price. racks under sheds erected by the Contractor to the direction and approval of the Architect and there left for three days and kept thoroughly

Bonding Walling

- B. The blocks shall be properly bonded together in such manner that no vertical joint in any one course shall be within 100 mm of a similar joint in the courses immediately above or below. Sufficient through bonders shall be provided as directed by the Architect. Alternate courses of walling at all angles and intersections shall be carried through the full thickness of the adjoining walls. All walling shall be built up entirely solid in blocks without void, allowance being made for joints 10 mm thick only. All perpend, reveals and other angles of the walling shall be built strictly true and square

Wall Reinforcement

- C. Where so specified hollow block walls shall be reinforced vertically with 10 mm diameter mild steel bars built into the cavities of the blocks at 400 mm centres, unless otherwise specified, all bars in walls to have a minimum lap of 350 mm.
- D. Prices for walling described as reinforced must include for all extra costs involved in slotting blocks over the vertical reinforcement.

Filling of Hollow Blockwork

- E. All cavities where specified and shown above ground and all cavities below ground level shall be filled in solid with concrete of the mix described and placed and consolidated in sections not exceeding 1190 mm in height.
- F. In reinforced walls the filling shall be carefully compacted around the reinforcement.

Blocks to be Wetted

- A All concrete blocks and stone walling shall be well wetted before being laid and the top of walling where left off shall be wetted before re-commencing building. Walls to be kept wet three days after building.

Mortar

- B Mortar to be used for all walling work shall be composed of 1 part of Portland Cement to 1 part lime to 6 parts of fine aggregate measured by volume in specially prepared dry on clean and watertight mixing platforms, with water added afterwards from a can with a fine rose until all parts are completely incorporated and brought to a proper consistency and then used within thirty minutes of mixing. gauge boxes and thoroughly mixed
- C No partially or wholly set mortar will be allowed to be used or re-mixed.

Fair Face Walling

- D Where walling is to be finished with a fair face, the concrete blocks are to be selected for freedom from defects and the joints raked out as the Works proceed and flush pointed with a neat joint in cement mortar.

Joints for Walling

- E The blocks shall be bedded and jointed in cement mortar as described with beds and joints 10 mm thick, full flushed up and grouted solid as the work proceeds. Joints shall be raked out where the surfaces or walling are to be plastered.
- F All walling shall be properly protected while mortar is setting as the Architect shall direct.

Building Walling

- G All walls throughout the Works shall be carried up evenly in 12 mm course, no part being allowed to be carried up more than 800 mm higher at one time than any other part and in such cases the jointing shall

Putlog Holes

- H Putlog holes shall be carefully, properly and completely filled up on completion of walling work.

Rough Cutting etc.

- H. The Contractor shall allow in his prices for the walling which is measured net herein, for all ordinary rough cutting, bonding, plumbing angles, forming reveals and fitting up to under side of concrete beams, slabs and lintels etc.

Stone Pitching

- A The ground to receive pitching shall be well compacted and the stones, which shall be flat bedded and not less than 230 mm either way along the bearing surface, shall be punned to the required falls and inclinations so that neither wedges nor spalls are required to keep the pitching rigidly in place. The joints shall be no more than 13 mm thick and shall be solidly filled with 1:3 cement mortar.
- B Stone for pitching shall be coral obtained from approved quarries. It shall be hard, sound, durable and clean.

Stone for Walling

- C Stone for walling shall be from an approved quarry, roughly square and built random and uncoursed in mortar as described. The stone shall be well bonded with a minimum of one good bond or through stone evenly spaced to each square meter. All cavities and joints in stonework are to be filled in and flushed up solid with mortar.
- D Jointing and pointing is as detailed or instructed.

Precast Screen and Louvre Block Walling

- E Precast concrete screen blocks shall be manufactured in concrete of 30.0 N/mm² strength using 10 mm aggregate, the blocks shall be 390 mm and 190 mm long x 190 mm high and 150 mm on bed in accordance with detailed drawings and finished fair on all surfaces and bedded, jointed and pointed in cement mortar with a neat flush joint.
- F Precast concrete louvre blocks shall be of similar concrete, similarly jointed and pointed and constructed to detail drawing.

Damp Proof Course

- G Damp proof courses shall be hessian based bituminous felt to B.S. 743 Type 5A laid on and including a levelling screed of cement and sand and lapped 230 mm at joints.

ROOFING - ASPHALT WORKS

APPROVED SUPPLIER

- A. All materials shall be supplied by a firm approved in writing by the Architect and the works executed by workmen approved by the supplier.

Guarantee

- B. The Contractor shall deposit with the Architect, a written guarantee and undertaking to the effect that during a period of not less than twelve calendar months from and after the certified date of completion of the whole of the works the contractor shall at his own expense make good to the satisfaction of the Architect all and any defects in the asphalt work which shall be attributed to improper materials or faulty workmanship and shall bear the cost of any consequential damage as shall be provided for in such guarantee.

Samples

- C. The Contractor shall when required by the Architect submit samples of any material for testing.

MATERIALS

Asphalt for roofing

- D. Asphalt for roofing shall comply with B.S. 1162 tropicalised mastic asphalt for roofing purposes.

Felt underlay

- E. The underlay shall be saturated "Cabro" sheathing felt complying with B.S. 1162 (or equivalent).

Insulating screeds

- F. Insulating screeds shall consist of lightweight concrete composed of one part Portland Cement and eight parts vermiculite aggregate and shall be covered with 10 mm cement and sand (1:4) screed wood floated to receive asphalt coverings.

WORKMANSHIP

Preparation of surfaces

- A. All surfaces to receive asphalt and other roof coverings are to be dry, wood floated and finished to suppliers specifications.

Laying

- B. Asphalt and other roof coverings shall be laid in bays generally not exceeding 2 m wide and succeeding coats shall be laid at breaking joint. Junctions between bays and fillets shall be properly married the whole being worked so that the joints are neatly made.
- C. Horizontal asphalt for roof coverings shall be 20 mm thick built up into two layers each 10 mm thick. The first layer shall be applied to sheathing felt and the final coat shall be left ready to receive roofing tiles.

Air pockets and stains

- D. Air pockets and stains on the asphalt and other roof coverings will not be permitted and the finished work shall not ring hollow over any parts of its surface.

Joints and fillets

- E. Joints in all asphalt work and other roof coverings shall be carefully made and complete fusion obtained to make them watertight. Fillets shall be run at all internal angles and in at least two operations. Perfectly watertight joints shall be made around pipes passing through walls and floors etc.

Felt underlay

- F. The felt underlay shall be fixed and laid loose or partially bonded in hot bitumen with but joints.

Testing for falls

- G. To ensure that asphalt and other roof coverings have been truly laid to falls, the contractor is to arrange for the roof areas and gutters to be flushed with water in the presence of the Architect. Any defects or depressions in the asphalt or other roof coverings are to be rectified and retested for approval.

CARPENTRY

Terminology

- A. All technical terms shall be as defined in the "Timber Act (amended 2012)".

Timber Generally

- B. The timber for carpentry and joinery shall be specified and obtained from an approved sawmill.
- C. The timber for carpentry shall be Second or Select Grade for strength.
- D. The timber shall be reasonably straight grained.
- E. All timber for the Works is to be purchased immediately the Contract is signed and is to be open-stacked for as long as possible before use or kiln drying.
- F. All timber as it arrives on the site shall be inspected by the Architect, and any timber brought on to the site and not approved must be removed forthwith.
- G. All timber and assembled woodwork shall be protected from the weather and stored in such a way as to prevent attack by termites, insects or fungi.

Species of Timber for Structural Work

- H. The following softwoods shall be used for structural work;

Standard Common Name	Botanical Name
Podo	Podocarpus
Cypress	Cuppressues Lusitanica

- J. Both to be second strength Grade P5 or equivalent. Whilst either timber is suitable, intermixing of species will not be accepted.
- K. The Contractor is permitted to propose substitute species but these shall not be used without the written approval of the Architect and no adjustment shall be made to the basic rates for softwood trusses in the event of a substitute species being accepted.

Insect Damage

- L. All timber shall be free from live borer beetle or other insect attack when brought upon the Site. The Contractor shall be responsible up to the end of the maintenance period for executing at his own cost all work necessary to eradicate insect attack of timber which becomes evident, including the replacement of timber attacked or suspected of being attacked, notwithstanding that the timber concerned may have already been inspected and passed as fit for use.

Seasoning of Timber

- A. All timber shall be seasoned to a moisture content of not more than 18% for carpentry and 15% for joinery. The Contractor's price must include for any kiln drying that may be necessary to achieve these figures.

Pressure Impregnation

- B. The softwood described as pressure impregnated shall be treated with the "Celcure A" "Tanalith C" full cell process. Timber must be seasoned to a moisture content not exceeding 25% before being treated. The treatment shall be to the minimum standard of:-

Solution concentration; 2%

Absorption of preservative; 520 Litres per cubic meter

Net dry salt retention; 10.4 Kg per cubic meter

- C. After treatment, the timber shall be seasoned to the specified moisture content.
- D. Cut ends and faces of timber sawn, drilled and cut after treatment are to be swabbed liberally with approved preservatives until saturated, allowed to dry and then treated with a second coat and rates for timber must include for this. Approved preservatives are: Atlas A; Brunophan Nr 2; Cuprinol Clear or Water Repellant Clear; Ensele Woodtreat 55.

Inspection and Testing

- E. The Architect shall be given facilities for inspection of all works in progress whether in workshops or on site. All timber as it arrives on the site must be inspected by the Architect and any timber brought onto the site and not approved by him must be removed forthwith, failing which he may arrange for the removal of the rejects and dispose of them as he may consider advisable at the Contractor's expense.
- F. Notwithstanding approval having been given above, any timber incorporated in the Works found to be in any way defective before the expiry of the maintenance period shall be removed and renewed at the Contractor's expense. The Contractor is to allow for testing of prototypes of special construction units and the Architect shall be at liberty to select any samples he may require for the purpose of testing, i.e. for moisture content, or identification of species, strength, etc.
- G. Where timbers need to be extended into a wall, they shall be thoroughly "brush treated" with Ensele in addition to preservative treatment as already described above, and as much clear air space maintained around the timber where it adjoins the wall as possible.

Clearing Up

- H. The Contractor is to clear out and destroy or remove all cut ends, shavings and other woodwaste from all parts of the building and the site generally, as the work progresses and at the conclusion of the Work.

Workmanship

- A. All carpentry shall be executed with workmanship of the best quality. Scantlings and boardings shall be accurately sawn and shall be of uniform width and thickness throughout. All carpenter's work shall be left with sawn surfaces except where particularly specified to be wrought.
- B. All carpentry shall be accurately set out in strict accordance with the drawings.
- C. All structural timbers shall be frame or jointed together as is most appropriate in the circumstances in accordance with the rules of good practice. Joints must be executed in strict conformity with the drawings.
- D. All joints shall be secured with a sufficient number of nails disposed as shown on the drawings and rates must include for the jointing of timbers. Surfaces must be in good contact over the whole area of the joint before securing. Holes for nails must be pre-drilled undersize; holes for bolts must be bored slightly over size from both sides of the timber and washers must be used under the nut which must be tightened sufficiently to permanently secure the joint but not to crush the timber.
- E. Actual dimensions of scantlings for carpentry shall not vary from the specified dimensions by more than 3 mm in deficiency or excess but must be uniform throughout. Boards 25 mm thick or less shall hold up to the specified size. All timbers shall be as long as possible and practicable, in order to eliminate joints.

Joints

- F. All nails, screws, bolts, connectors, etc., are to be as specified under "Metalwork" and as shown on the drawings.

General

- A. The provisions contained in the "Carpentry" section shall apply also to the Joinery Section where applicable.

Species of Timber

- B. The following timber of First or Prime Grade for appearance shall be used for Joinery Work in conjunction with the term "softwood" or "approved softwood":-

Standard Name; Botanical Name

Podo (for grounds, etc., only); Podocarpus spp.

African Mahogany; Khaya Nyasica

Mninga, Pterocarpus Angolensis

Iroko (Mvula); Chlorophora excelsa

- C. The following may also be used as "local hardwood" (referred to hereafter) with the Architect's approval:-

Adina; East African Afrormosia; East African Afzelia

Generally

- D. All joinery work shall be accurately set out on boards to full size for the information and guidance of the artisans before commencing the respective works, with all joints, iron work and other work connected therewith full delineated. Such setting out must be submitted to the Architect and approved before such respective works are commenced.
- E. **All joinery work shall be cut and framed together as soon after the commencement of the building as is practicable**, but not to be wedged up or glued until the building is ready for fixing same. Any portions that warp, wind or develop shakes or other defects within six months after completion of the Works shall be removed and new fixed in their place together with all other work which may be affected thereby, all at the Contractor's own expense.
- F. All work shall be properly morticed, tenoned, housed, shouldered, dovetailed, notched, wedged, pinned, bradded, etc., as directed and to the satisfaction of the Architect and all properly glued up with the best quality approved glue.

- A. Joints in joinery must be as specified or detailed, and so designed and secured so as to resist or compensate for any stresses to which they may be subjected. All nails, springs, etc., are to be punched and puttied. Loose joints are to be made where provision must be made for shrinkage; with glued joints where shrinkage need not be considered and where sealed joints are required. Glue for load-bearing joints or where conditions may be damp must be of the resin type. For non-load-bearing joints or where dry conditions may be guaranteed casein or organic glues may be used. All exposed surface of joinery work shall be wrought and all arises "eased-off" by planning and sand-papering to an approved finish suitable to the specified treatment.

Dimensions

- B 3 mm reduction off specified sizes will be allowed for each wrought face except where described as (f) i.e. **finished** size in which case joinery shall hold up to the full dimensions. Dimensions of 25 mm or less shall hold up to the specified sizes.

Fixing Joinery

- C All beads, fillets and small members shall be fixed with round or oval brads or nails well punched in and stopped. All large members shall be fixed with brass screws, the heads let in and pellated to march the grain where natural finish timber is specified.

Mastic

- D Mastic where specified for bedding, joinery, sills, water bars, etc., is to be approved non-hardening plastic, phylsulphide synthetic rubber or butyl composition filler or sealer.

Fiberboard

- E Fiberboard shall be "Celotex" or equal and approved.

Plywood

- F Plywood shall be from an approved source and comply with B.S. 1455, first or second grade, as required and unless otherwise stated shall be "interior" quality. Where veneered plywood is specified, samples must be submitted for prior approval. Where stated to be "exterior" quality, this shall be waterproof (Bonding W.B.P.).
- G Routine tests will be required from time to time to check the quality of manufacture. Plywood used in structural members shall be bonded with a suitable adhesive.

Chipboard

- H Chipboard shall be approved medium density resin bonded wood chipboard equivalent to B.S. 2604 with sanded finish or thickness stated. Where faced with plastic sheeting the chipboard shall be counterbalanced.

Blockboard

- J Blockboard shall be laminated board to B.S. 3444. Where faced with plastic sheeting the blockboard shall be counterbalanced.

Flush Doors

- A. Flush doors shall be from an approved source and manufacture, be hollow / semi-solid core constructed generally in accordance with B.S. 459-2 finished with 6 mm veneer plywood (to Architect's approval) and lipped all round with softwood 12 mm thick.
- B. The thickness stated is the overall finished thickness.

Plastic Sheeting

- C. Plastic sheeting shall be Formica or equal and approved laminated sheeting 1.5mm thick fixed with an approved adhesive. All colours are to be selected by the Architect.

Plugging Walls

- D. All work described as plugged shall be fixed with brass screws to plugs formed by drilling concrete, walls, etc., with a proper tool of suitable size at 500 mm spacings and filling the holes completely with an approved proprietary plugging compound used in accordance with the manufacturer's instructions.

Protect Joinery

- E. All fixed joinery which, in the opinion of the Architect, is liable to become bruised or damaged in any way shall be completely cased and protected by the Contractor until the completion of the Works.

Bottom Edges

- F. Bottom edges of doors shall be painted with one coat of approved primer before fixing .

Mosquito Screening

- G. Mosquito screening shall be "Alcad" or equal and approved aluminium fine wire mesh screening.

Bird Screening

- H. Bird screening shall be approved galvanized coffee tray wire.

Ironmongery

- J. All ironmongery shall be fixed with screws to match. Before the woodwork is painted, handles shall be removed, carefully stored and refixed after completion of painting, and locks oiled and left in perfect working order. Prices for fixing locks must include for organizing masterkeying systems if required and all keys shall be labelled with door references marked on approved labels before handing to the Architect on completion.

STRUCTURAL STEELWORK

Standard of Construction

- A. The whole of the structural steelwork and testing shall comply with the relevant clauses of B.S. 449, B.S 4360; 1980 and B.S. 5940 grade 43.

Fabrication by Specialist Firm.

- B. The steelwork shall be fabricated by a specialist firm or under proper factory conditions to be approved by the Architect.

Contractor to Submit Drawings

- C. The Contractor shall include for the preparation of all shop details from the drawing supplied by the Architect. All such details shall be approved in writing, by the Architect, before the work is put in hand. Every drawing shall show the number and sizes of all rivets and bolts, complete details of welds, type of electrodes, welding procedure, whether the welds are to be made in the shop or elsewhere and any other relevant information.

Accuracy of Drawings.

- D. The Contractor shall be responsible for the correctness of his shop details and for shop fittings and site connections.

Erection Scheme

- E. The Contractor shall submit to the Architect for approval, drawings showing the proposed erection scheme, together with all calculations for erection stresses, etc. The approval by the Architect will not absolve the Contractor in any way from his responsibility.

Dimensions to be Verified

- F. The Contractor shall take the dimensions from the site or buildings and he shall verify all dimensions given on the drawings before the work is put in hand.

Copies of Orders

- G. A copy of all orders for materials shall be supplied by the Contractor to the Architect at the time of ordering, for identification purposes.

Damage

- H. Any damage to materials on the site due to inadequate precautions being taken during the erection of the steelwork shall be made good to the satisfaction of the Architect at the Contractor's expense.

Materials

Quality of Steel

- A. (i) All structural and rivet mild steel shall comply with B.S 4360 Part 2
- (ii) Nil
- (iii) Nil
- (iv) All structural steel tubes shall comply with B.S. 1775 and B.S. 449
- (v) Mild steel and medium tensile steel electrodes for metal-arc welding shall comply with the requirements of B.S 2549.
- (vi) High tensile steel electrodes for metal-arc welding shall comply with the requirements of B.S. 2549.
- (vii) All mild steel bolts and nuts shall have a tensile strength of not less than 432 N per Square Millimetre (28 tons per square inch) and a minimum elongation of 17 percent as defined in Clause 2 of B.S. 916 or in B.S. 2708.
- (viii) All high tensile steel bolts, nuts and washers shall have a minimum tensile strength of 570 N per square millimetre (37 tones per square inch).
- (ix) High strength friction grip bolts and washers shall comply with B.S. 3139, Part 1.
- (x) All plan washers shall be of steel. Tapered or other specially shaped washers shall be made of steel or malleable case iron complying with B.S. 3410. shall be made of steel or malleable case iron complying with B.S. 3410.

Marking of Steel

- B. Each piece of steel shall be legibly marked with the maker's name or trade mark and with cast numbers or identification marks by which the steel can be traced to the cast from which it was made.
- C. For rivet bars and small pieces securely bundled, a metal tag marked with the cast number will be sufficient.

Standard Dimensions

- D. The dimensions and allied requirements of all structural rolled sections shall comply with B.S. 4. The dimensions, weight, tolerances etc., of all rivets, bolts, nuts, studs, etc., shall conform to the following standards. Rivets shall comply with the requirements of B.S. 275 for dimensions
- E. Black bolts, nuts, studs, lock nuts and washers shall comply with the requirements of B.S. 916 for dimensions and with B.S. 1580 for unified black bolts etc.

- A. Turned bolts shall have the shank turned to the specified diameter allowing only a minus tolerance up to 0.13mm (0.005 inch).

Weight of Steel

- B. For the purpose of measurement, the weight of mild steel shall be as given in B.S. 648 which will be the basis for measurement of variations. The weights per meter given on the drawings do not include the shelf angles riveted to webs, nor the plates riveted to the flanges of R.S. Js or other sections.

Conditions of Surfaces

- C. All surfaces of steel work shall be clean, free from loose millscale and loose rust.

Tests and Inspection

- D. Manufacturer's Mill Test Certificates for all structural steel shall be supplied to the Architect as and when required. Where and when directed by the Architect, the Contractor shall take and deliver samples of structural steel for testing to the Employer's Highways and Transportation Testing Station. Should the results of either test be unsatisfactory the whole consignment of steel which the sample represents shall be rejected and shall be replaced by other material of proper quality at the expense of the Contractor.
- E. The Architect or his representative shall at all reasonable times, be given free access to the Works.

Metallic Coatings

- F. Galvanized steelwork shall comply with B.S. 729 Part 1 entirely coated with zinc after fabrication by complete immersion in a zinc bath in one operation and excess carefully removed. The finished surfaces shall be clean and uniform.
- ii) Zinc sprayed steelwork shall comply with B.S. 2569 Part 1. The nominal thickness of zinc coating shall be not less than 0.102 mm (0.004 inches) and at no point less than 0.076mm (0.003 inches).

Generally

- G. The whole of the fabrication and erection of the steelwork shall be carried out in accordance with B.S. 449

Materials (Cont'd)

- A. The welding of steel to B.S. 1962 must conform to: B.S. 1856 - "General requirements for the metal-arc welding of mild steel" or B.S. 2642 - "Arc welding of mild steel". "General requirements for the arc welding of steel to B.S. 968 and similar steel"
- B. For welding any particular type of joint the Contractor shall provide evidence acceptable to the Architect that the welder has satisfactorily completed the appropriate tests as described in B.S. 5950 - 7
- C. Any welder's tests shall be made at the Contractor's expense and shall include the cost of any fees incurred by the Employer for witnessing of, or making such tests.
- D. The right is reserved to make non-destructive tests on the welding to determine if the welding conforms to the standards laid down in either B.S. 1856 or B.S. 2642 as applicable. This will normally consist of radiography on butt welds, ultrasonic examination of fillet welds or other tests as appropriate to the actual configuration of the weld in question.

Rejection

- E. Any portion of the work which, in the opinion of the Architect, is not in accordance with the drawings, or specification shall be rejected whether before or after delivery and must be removed from the site if delivered within 24 hours from receipt of such notice or rejection at the Contractor's expense. Any delay caused by such rejection will not in any way relieve the Contractor from his responsibility with regard to the provisions of the Contract. If any welding is found to be defective the cost of all remedial measures shall be borne by the Contractor, including the cost of re-testing the subsequent inspection of welds as referred to in the P.C. Sum hereafter.
- F. The Contractor is responsible for the good quality of all welding work and no exceptions will be made on the grounds that the Architect or his representative have inspected any part or parts of the work at some stage during production.

Fabrication

- G. As much of the work of fabrication of the steelwork as is reasonably practicable shall be completed in the manufacturer's works. Field connections shall be made in accordance with the approved drawings. The Contractor shall give four day's clear notice of steelwork ready for inspection at the manufacturer's works, to facilitate inspection before delivery.

Cast of Temporary Erection, etc.

- A. Trial erection of principal or other units may be called for at the discretion of the Architect or his representative.
- B. The cost of any necessary temporary erection, testing, packing, marking, carriage and delivery is deemed to be included by the Contractor in the Tender price.

Joints and Connections

- C. No variation of the number, type or position of the joints or connections shown on the drawings shall be made without the consent of the Architect. If such consent is desired the Contractor shall submit detailed drawings of the proposed joints for the approval of the Architect and no extra cost incurred by reason of such additions or alterations will be allowed to the Contractor.

Painting at Works

- D. Where described as primed at works, steelwork shall be freed of rust, millscale, welding slag and flux residue and shall be dry immediately prior to painting with primer as Clause Q 14 a.
- E. For joints with high strength friction grip bolts the contact surfaces shall be left unpainted but special care shall be taken after assembly to paint all edges and corners near the joints together with bolt head, nuts and washers to prevent the ingress of moisture.
- F. For joints made with other bolts and rivets the contact surfaces shall each be given a coat of priming paint and for shop connections the contact surfaces shall be brought together while the paint is still wet.
- G. For welded connections where the contact surfaces are not completely sealed the contact surfaces shall be painted to within 50mm of the edges that are to be welded.
- H. The primer shall be touched up with similar primer if damaged by subsequent handling.

METALWORK

Mild Steel

- A. Mild steel shall comply with B.S. 4360 Grade 1 and the sizes of all small sections shall be in accordance with B.S. 4 and 4A.

Galvanized Work

- B. Iron and steel, where galvanized, shall comply with B.S.1461 Part 1 entirely coated with zinc after fabrication by complete immersion in a zinc bath in one operation and all excess carefully removed. The finished surface shall be clean and uniform.

Aluminium

- C. Aluminium shall be of the alloys described in and shall comply with B.S. 485. Aluminium sheet for flashings shall be soft-temper, super purity (S1 or S1A) and not less than 20 s.w.g. (0.9mm) in thickness.

Smithying, Shearing and Cutting

- D. All smithying, welding, cutting and bending shall be soundly and neatly executed, care being taken not to overheat. All flame cut edges and welds shall be neatly ground off on completion.

Bolts

- E. Mild steel bolts, nuts and washers shall comply with B.S. 916 for black bolts with hexagonal heads and nuts. High tensile steel bolts and nuts shall be in accordance with B.S. 3139 Part 1.

Anchor Bolts

- F. Anchor bolts in concrete for steel works etc., are to be self drilling anchor bolts of one of the following types:-

Phillips redhead concrete anchors
Rawlplug super drillanchor
Spit self-drilling anchors

- G. Rates are to include for fixing complete with washer. Mortices in concrete have not been measured for this item.

Shop Inspection

- A. The Architect shall be granted full facilities and any necessary assistance for inspection of materials and assembled parts in the Contractor's (or his Sub-Contractor's) workshops. At least two weeks notice shall be given to the Architect in writing prior to the despatch of finished components to the site to enable the Architect to inspect and approve the materials and workmanship at the workshops. Approval of work at the workshop does not relieve the Contractor of this obligation to carry out the work complete at the site to the Architect's satisfaction in accordance with the Contract.

Marking

- B. All components delivered to the site are to be marked in paint with the Mark number in accordance with any shop and erection drawings.

Storage

- C. All components are to be stored at the site in proper racks provided for the purpose which provide full support to each member to obviate any deflection and distortion. Steelwork is to be stored at least 25cm clear of the ground and temporary protection is to be provided for protection against water and damage from any other source.

Erection

- D. Rates for all metalwork are to include for the complete erection including any temporary supports required and any necessary templates and wedges.

Painting

- E. All steel is to be thoroughly de-rusted and degreased prior to despatch to the site and is to be given one coat zinc chromate primer at the works. Further painting treatment will be carried out at the site. Painting is measured separately and the cost thereof is not to be included in the rates for metalwork.

PLUMBING AND ENGINEERING INSTALLATION

Execution of the Works

- A. The work shall be carried out strictly in accordance with:-
- (a) "British Standard Code of Practice" C.P. 310: 1965: Water Supply
 - (b) "British Standard Code of Practice" C.P. 404: 1968: Sanitary Pipework above ground
 - (c) All other relevant British Standard Specifications and Codes of Practice
 - (d) Bye-laws of the Local Authority
 - (e) The working drawings

Extent of Work

- B. The Contractor will be responsible for all below ground plumbing and drainage work and the installation of the Sanitary Fittings only, the remainder of the Plumbing and Engineering Installation will be executed by a Nominated Sub-Contractor.

Quality of Materials and Workmanship

- C. All materials, equipment and accessories are to be new and in accordance with the requirements of the current rules and regulations where such exist, or in their absence with the relevant British Standard Specification.
- D. Uniformity of type and manufacture of equipment or accessories is to be preserved as far as practicable throughout the whole work.
- E. The Contractor shall, if required by the Architect, submit samples of materials to the Architect for his approval before placing an order.
- F. If in these Preambles the practice is adopted of specifying a particular item as "similar" to that of a particular firm's product, it is to be clearly understood that this is to indicate the type and quality of the equipment required. No attempt is being made to give preference to the equipment supplied by the firm whose name or product is quoted.
- G. Where particular manufacturers are specified herein, no alternative makes will be considered and the Architect shall be allowed to reject any other makes.
- H. The Contractor will be entirely responsible for all materials, apparatus, equipment, etc., furnished by him in connection with his work, and shall take all special care to protect all parts of finished work from damage until handed over.
- J. The work shall be carried out by competent workmen under skilled supervision. The Architect shall have the authority to have any of the work taken down or changed which is executed in an unsatisfactory manner.

Galvanized Steel Tubes and Fittings

- A. Galvanized steel tubing shall comply with B.S. 1387 with plain galvanized malleable fittings complying with B.S. 143/1256.
- B. Tubes and fittings shall be jointed by means of screwed threads to B.S. 21, by means of P.T.F.E., tape or hemp and "Bosswhite". All joints shall be perfectly smooth inside without excrescences.
- C. Where sleeves are required for pipework passing through concrete, blockwork or below concrete slabs, they shall be galvanized steel tube or drain pipes of sufficient diameter to give at least 25mm clearance all round the water main.
- D. Galvanized water mains below ground level or below slabs shall be double wrapped in "Denso" tape.

Brasswork

- E. Stop valves shall comply with B.S. 1010 and shall be with crutch handles or loose keys where so described on the drawings. Draincocks shall comply with B.S. 2879.

Testing

- F. Upon completion the whole of the water main shall be tested to a pressure not less than twice times the working pressure for a period of thirty minutes.
- G. Notwithstanding the foregoing clauses, all water mains and fittings and installation thereof shall comply fully with the requirements of the Water Supply Authority.

Sanitary and Other Appliances

- H. The appliances shall be fixed in the positions shown on the drawings or as described by the Architect.
- J. The Contractor shall include in his rates for providing all necessary screws, bolts, etc., together with all jointing materials required and also for temporarily erecting and securing fittings in the required position or service and discharge pipes, taking down, storing and fixing after completion of wall finishings permanently fixing and connecting to service and discharge.
- K. Care shall be taken at all times and particularly after fixing, to protect appliances from damage.
- L. Upon completion of the work, all appliances shall be cleaned of plaster, paint, etc., and carefully examined for defects.

Fire Fighting Equipment

- A. The specified fire fighting equipment shall be supplied and installed by the Contractor in the positions shown on the drawings
- B. Portable fire extinguishers shall comply with the following British Standards:-
 - (a) Water type (soda acid); B.S. 138: 1948
 - (b) Foam type (chemicals); B.S. 740: Part 1: 1948
 - (c) Foam type (gas pressure); B.S. 740: Part 2: 1952
 - (d) Water type (gas pressure); B.S. 1382: 1948
 - (e) Carbon tetrachloride and chlorobromethane; B.S. 1721: 1960
 - (f) Carbon dioxide type; B.S. 3326: 1960
 - (g) Dry powder type; B.S. 3465: 1962
 - (h) Water type (store pressure); B.S. 3709: 1964
- C. Fire hose couplings and ancillary equipment shall comply with B.S. 336: 1965; rubber reel hose shall comply with B.S. 3169: 1959.
- D. Underground fire hydrants and surface box openings for same shall comply with B.S. 750: 1964.
- E. The installation of hydrants and fire extinguishers shall be in accordance with C.P. 402:101: 1952 and C.P. 402 part 3: 1964 respectively.
- F. If nothing else is specified, fire extinguishers and hose reels shall be supplied in the colour "fire red" and be similar to manufacture "ANGUS".

FLOOR WALL AND CEILING FINISHINGS

Sand

- A. Sand for backing, floor and wall finishes is to comply with B.S. 13139, Table 1.

Cement

- B. Cement is to be as described for "Concrete Work:.

Lime

- C. Lime is to be no-hydraulic hydrated lime to B.S. 459 Class "A" obtained from an approved source and run into putty at least 24 hours before use.

Workmanship

- D. All concrete beds or slabs shall be thoroughly brushed clean, hacked if necessary and well wetted and flushed over with a cement sand (1:1) grout immediately before screeds or pavings are laid.
- E. Screeds and cement pavings shall be laid in accordance with the relevant B.S. Code of Practice. Working joints between bays of the floor finish should be placed in accordance with the Architect's instructions and will be plain butt joints placed over joints in the concrete bed under. Pavings shall be damp cured with sand or sawdust and kept damp for at least 7 days after laying.
- F. All surfaces to be plastered or rendered must be brushed clean and well wetted before plaster is applied. Joints of walling shall be raked out and concrete hacked to form a key. Care shall be taken to see that paving and plastering do not dry out prematurely.
- G. Adequate time intervals must be left between successive coats in two-coat work in order that the drying shrinkage of the undercoat may be substantially complete. All internal and external angles shall be pencil rounded.

In-Situ Pavings Generally

- H. Before laying in-situ floor finishes, the concrete beds are to be thoroughly hacked for key, cleaned off, thoroughly wetted with clean water and coated with a stiff cement slurry and rates for screed, granolithic and terrazzo paving are to include for this. They are also to include for all necessary curing and protecting until the building is handed over.

Cement and Sand Paving

- J. The cement and sand paving shall be in proportions of 1:4 by volume and incorporating or treated with an approved hardener.

Polished Granolithic Paving

- A. The aggregate for granolithic paving shall be in accordance with B.S. 1201 and shall be mixed in the proportions of 1:1:1.50 cement, fine and coarse aggregate respectively. The mix shall incorporate an approved hardener suitable for incorporation and not for surface treatment. The water cement ratio shall be kept as low as possible and shall not in any case exceed 0.45. The paving is to be laid to the full thickness described and to be finished with a wood float and with no extra cement trowelled into the surface which is to be laid true and level. The paving is to be thoroughly cured after laying by covering with polythene sheeting and periodically watered to keep it moist for at least one week after laying. The surface is to be polished with approved rotary carborundum discs mechanically operated coarse and fine grain and with cement and sand slurry to produce a blemish-free surface.
- B. The granolithic shall be laid in bays not exceeding 3.50 square meters with ebonite dividing strips for the full depth of the paving and shall be executed by Specialist who have a thorough knowledge of the work.

Polished Terrazzo Paving

- C. The ins-Situ terrazzo shall consist of white or coloured cement and marble aggregate; the colours of the cement and aggregate shall be selected by the Architect. The mix shall comprise three parts of 6mm nominal aggregate to one part coloured cement by volume. The aggregate shall be clean and granular and shall not contain flaky particles or dust. The underbed shall be cement and sand 1:4 by volume.
- D. The terrazzo shall be laid in bays not exceeding 3.5 square meters with ebonite dividing strips for the full depth of the terrazzo and underbed, and shall be executed by Specialist who have a thorough knowledge of the work.
- E. The terrazzo topping shall be laid to a minimum of 12mm thickness in a plastic condition while the underbed is still green and this should be watered to minimise absorption from the topping. The terrazzo must be well tamped into position and rolled with a suitable hand roller. The topping should be allowed to take an initial set and then any surface voids must be grouted up with neat cement of the same colour used in the mix. The surface should be cured by keeping moist by covering with damp sacking for at least 72 hours. When dry and hard the surface shall be machine polished by grinding with carborundum or other stone discs of suitable grade and with rotary polishing pads.
- F. Rates must include for all necessary protection until the building is handed over to the Architect. The depths stated are for the full depth including topping and underbed.

P.V.C. Flooring and Skirting

- G. P.V.C. floor tiles shall comply with B.S. 10595. The tiles and accessories shall be supplied in the sizes and thickness specified in colours selected by the Architect and are to be fixed to the screed base with a suitable adhesive supplied (or recommended) by the Manufacturer and used in accordance with his instructions. Rates for floor tiles shall include for thoroughly washing and cleaning on completion and for the application of one coat of water based wax polish.

Brushed Terrazzo Rendering

- A. Brushed terrazzo rendering is to comprise two coats as described. The undercoat shall consist of cement and sand mixed in the proportion of (1:4) by volume and applied to a minimum thickness of 10mm finished with a wood float and scratched to provide key for top coat. The finishing coat shall consist of one part white cement to two parts marble chippings or approved size applied to a minimum thickness of 10mm and the final surface wet brushed to expose the aggregate.
- B. The Contractor will be required to produce a sample panel of rendering on site for the approval of the Architect.

Internal Plaster

- C. Internal plaster shall be applied in two coats and adequate time intervals must be allowed between successive coats in order that the drying shrinkage of the undercoat may be substantially complete. The first coat must be well scratched, keyed and wetted to receive the finishing coat. The finishing coat shall be finished smooth with a steel float but care must be taken not to overwork the surface in order to minimize the incidence of shrinkage cracks. All internal and external angles shall be pencil rounded.
- D. Internal plaster, unless otherwise described, shall be lime plaster of 12mm minimum overall finished thickness applied in two coats, the first coat consisting of cement, lime putty and sand mixed in the proportion of 1:2:9. The finishing coat shall be a skim coat comprising cement and lime putty in the proportion of 1:10.
- E. Cement plaster is to be employed where specified on the drawings and is to be applied in two coats of approximately equal thickness to a total of 12mm minimum overall finished thickness. The composition of both coats shall be the same and shall comprise cement and sand (1:3) but a small percentage addition (not more than 10%) lime putty may be permitted if the Architect considers that this will reduce the incidence of shrinkage cracks.
- F. The Contractor shall cut out and make good all cracks, blisters and other defects and leave the whole of the plastering and rendering perfect at completion. When making good defects the plaster shall be cut out to a rectangular shape with edges undercut to form dovetailed key, and all finished flush with the face of surrounding plaster.

Marmoran Finishings

- G. Prepare and prime surface, apply one coat 3 mm thick PVC Resin Bonded plaster with trowel, apply 3.2 mm thick stone chips with low pressure spray gun or by hand, roll flat by roller immediately after application. Colours and texture of the plaster and stone chips are subject to specifier's approval. Specifications must be strictly in accordance with manufacturer's instructions.

Wall Tiles

- A. Glazed wall tiles shall be from an approved manufacturer and shall conform with the requirements of B.S. 1281. Tiles shall be white with slightly rounded or "cushion" edges and unless otherwise specifically described shall be size 150 x 150 x 6mm thick. Tiles shall be laid with continuous straight joint and internal angles shall be butt jointed. Rounded on edge tiles shall be used at all external angles and at edges of panels. Tiles shall be bedded in approved tiles adhesive and pointed in white cement.
- B. Backing to tiles is to be cement and sand in the proportion of 1:4 rendering in one coat to a minimum thickness of 12mm trowelled smooth. Backings have been measured separately.

Carpet Tiles

- C. Carpet floor tiles shall be from an approved manufacturer and shall conform to specification as per M/s Protex of South Africa. Graveltex Protex Carpet tiles shall be of heavy duty grade, 100% stain proof miracle fibre with density of 920, g/sq.m (fibre) and 4500 g/sq.m (total) with fire resistance (S.A.B.S) of 3, lavender colour. The size shall be 500 x 500 x 9.50mm thick. Tiles shall be laid with continuous straight joint. Tiles shall be bedded in approved tiles adhesive.
- D. Beds to tiles are to be cement and sand in the proportion of 1:4 rendering in one coat to a minimum thickness of 30mm trowelled smooth. Backings have been measured separately.

Floor tiles

Porcelain tiles

- E. Porcelain floor tiles shall be from an approved manufacturer and shall be of black polished, cocowhite-polished or gardenia green matt. The size shall be size 400 x 400 x 10 x thick. Tiles shall be laid with continuous straight joint. Tiles shall be bedded in approved tiles adhesive.
- F. Beds and backings to tiles is to be cement and sand in the proportion of 1:4 rendering in one coat to a minimum thickness of 30mm trowelled smooth. Beds and backings have been measured separately.

Laying of Marble, Granite, Porcelain or Ceramic Floor Tiles

- G. Before laying the tiles, level the flooring area, ensure the surface is rough and clean.

Laying Floor tiles with Traditional Mortar

- H. The cement thickness needed to lay tiles should be around 40 mm. The mixture for indoor is 1 volume of Portland cement and 3 volumes of sand. The mixture must be made with appropriate quantity of water in order to dampen the materials. Clean and wet the flooring area, making sure to leave completely clean. Spread the mixture and level with a ruler, in order to reach the 40 mm of thickness. Spread dry cement over the mixture, until the water that remains over the surface has been completely absorbed. Lay the tiles, already mixed from different boxes, with a wide joint and in the desired way. Wet tiles, then cover to achieve a perfect level.

Laying Floor tiles with Adhesive

- A. The bed needed for this kind of laying, should be around 30 mm. The flooring area should be steel or wood trowelled and levelled. Spread the adhesive with a spatula with ridges. It is very important to lay a good quantity of adhesive so that there is no free space between the tiles.

Mixing the Colour Shades of Floor tiles

- B. Before laying the tiles at least 5 to 6 boxes must be laid over a dry surface in order to ensure that the different shades have a uniform look. The best result is obtained this way.

The Joints of Floor tiles

- C. The tiles have to be laid with a minimum of at least a joint separation between tiles of 3 to 10 mm.

Setting the Joints of Floor tiles

- D. The cord or wire system can be used in the 4 or 5 joints, ensuring they are all parallel with the reference joint. Plastic crosses used for this purpose, in different sizes, can be obtained in specialized shops, giving a much better finishing and final result.

Filling the Joints of Floor tiles

- E. Apply a mixture composed of 2 volumes of Portland cement and 1 of fine washed sand, with enough water in order to amplify the handing. There are suitable preparations for different uses and in different colours now available, so as to achieve the desired effects. Spread the substance by use of a rubber or plastic spatula. Clean the tiles before the mixture dries. After the joints are completely dry, wash with plenty of water several times.

Concrete Tiles

- F. Concrete tile for finishing the roofs shall be 25mm thick of natural colour with bevelled top arises on all sides and shall comply with B.S. 1197. The tiles shall be laid to regular pattern with open joints. Care should be taken to ensure that the surface level is even and follows accurately the levels of the roof finish. All cement stains shall be carefully removed.

Precast Concrete Paving Slabs and Kerbs

- A. Precast concrete paving slabs shall comply with B.S. 368. precast concrete kerbs shall comply with B.S. 340 figure 5 and shall be finished true and smooth on all exposed faces.
- B. Precast paving shall be bedded on a compacted sand bed with 6mm wide joints filled and pointed with cement mortar coloured to match the colour of the slabs. The pavings shall be finished true and even and to the falls shown with no surface irregularities.

GLAZING

Method of Glazing

- C. Notwithstanding reference in the descriptions of glazing method to glazing beads, or the like with associated fixings, and insulating strips, such components will be measured separately in accordance with the appropriate rules of the S.M.M.
- D. The provision of glazing compounds and putties and springs, clips and other sundry fixings shall be deemed to be included with all items of glazing.
- E. Distance pieces and setting blocks, in appropriate materials, shall be provided in accordance with good glazing practice and they shall be deemed to be included with all items of glazing.

MATERIALS

Glass generally

- F. All glass shall comply in all respects with the appropriate section of B.S. 952. Plain sheet clear glass shall be O.Q.; plate glass shall be GG. All glass shall comply in all respects with the latest British Standards including the British Codes of Safety.

Putting for glazing to wood

- G. Putty for glazing to wood shall comply with B.S. 544.

Samples

- H. Samples not less than 150 mm square, are to be submitted to the Architect for approval before any glass is cut.

WORKMANSHIP

Glass to be kept free from moisture

- J. All glass surfaces shall be kept dry during transit and storage. Glass becoming moist from condensation or other causes, shall be thoroughly dried and aired.

Rebates and beads

- A. All glazing beads in wood shall be primed, (as measured in Painting and Decorating), before glazing is commenced.

Edges of glass

- B. All glass shall have clean cut edges. The edges of louvres shall be rounded and polished.

Bead glazing

- C. Glazing fixed by beads shall have both glass and beads bedded and back puttied, and the putty trimmed off flush. Where sealing strip is used, it shall pass round both faces of the glass and be trimmed off flush on both sides. Metal surfaces to receive sealing strip shall be treated with mineral oil before glazing.

Method of measurement

- D. Beads and sealing strips have been measured separately. Prices for glazing with beads are to include for taking out and re-fixing beads as required, which shall be deemed to be bradded unless otherwise described.

PAINTING

Execution by a Specialist Firm

- E. All work under this section must be executed by a Specialist Firm, approved by the Architect.

Approved Paints

- G. All paints shall be obtained from the same manufacturer and shall be approved by the Architect.
- H. The Contractor must allow for providing the Architect with colour charts from the approved firm and for executing sample panels as required.

Generally

- I. All materials shall be delivered on site intact in the original drums or tins and shall be mixed and applied strictly in accordance with the manufacturer's instruction and to the approval of the Architect.
- J. The only addition which will be allowed to be made locally will be liquid thinners and driers supplied or recommended by the manufacturers and none shall be thinned more than approved by the Architect.

Preparation

- K. All surfaces to receive treatment are to be clean and dry before paint application and surface irregularities are to be removed by filling or the use of suitable abrasives.

External Rendered Surfaces

- A External cement slurry finished wall which are to be painted must be clean and must be thoroughly brushed and washed to remove any dust, loose flakes or other foreign matter and must be well wetted prior to the application of finish.

Plastered Surfaces

- B Internal plastered surfaces which are to be painted are to be allowed to dry out thoroughly prior to paint application. All cracks and surface imperfections are to be cut back and filled with a patent filler in accordance with the manufacturer's instructions and rubbed down to a true and even surface.

Woodwork Preparations

- C Large knots in woodwork are to be cut and replace with sound wood or scorched back and after priming the surface made good with stopping. All knots are to be treated with two thin coats of patent knotting free from resin. After priming, all nails holes and other imperfections shall be filled with stopping and the whole surface rubbed down to a smooth even finish. The stopping must be "Sadofill" or other approved make.

Woodwork - Fittings

- D Unless otherwise specified, fittings are to be treated with two coats of linseed oil.

Metalwork

- E All rust and loose scale on steel and iron work must be removed by wire brushing and rubbing with emery paper. Where patches of ingrained rust cannot be removed they are to be thoroughly rubbed down and treated with one coat of "Galvafroid" or other zinc rich paint in accordance with the manufacturer's instructions. One coat of zinc chromate primer will then be applied followed by two undercoat and one finishing coat of gloss paint as described for Woodwork above. The Contractor is to note that where mild steel burglar bars are housed into wood frames, the full length of the bar is to be treated before fixing.
- F Galvanized metalwork is to receive one coat of white spirit or mordant degreasing solution washed off prior to the application of calcium plumbate primer followed by two undercoats and one finishing coat of gloss as previously described.
- G Galvanized metal work is to be painted only where instructions are given by the Architect as in some cases galvanized metalwork is to be left untreated.

DRAINAGE

Generally

Preambles to Other Sections

- A The preambles contained in other sections of this document shall apply equally hereto where applicable, so far as is consistent with the clauses following.

Notices

- B The Contractor shall give all requisite noticed. Uncoloured plans will be supplied by the Architect at the Contractor's request.

Drainage Bye-Laws

- C All of the works shall comply with the requirements of the drainage bye-laws made by the Local Authority and shall be executed to the satisfaction of the Architect and Local Authority.

Inspections

- D The Contractor shall give written notice to the Architect for the purpose of inspections and measurements, whenever section of:-

- (a) excavations are completed
- (b) concrete beds are laid
- (c) drains are completed

and no further work shall be executed until each stage of the work has been inspected.

Levels of Existing Drains

- E The Contractor shall check the invert levels of existing drains, sewer and manholes before laying new drains, and shall notify the Architect immediately if the declared invert levels are found to be inaccurate

Pitch Impregnated Fibre Drain Pipes, Couplings and Fittings

- F Pitch impregnated fibre drain couplings and fittings shall comply with B.S. 2760.

UPVC Pipes and Fittings

- G UPVC pipe and fittings shall comply with B.S. 3506 Class O to be obtained from a manufacturing source approved by the Architect in writing.

Spun Cast Iron Drain Pipes and Cast Iron Fittings, Gullies etc.

- H Spun cast iron drain pipes shall be coated centrifugally cast (spun) iron pipes complying with B.S.1211 Class B.

- A Fittings, gullies, etc., shall be of coated cast iron and shall comply with B.S. 1130.

Concrete Pipes and Fittings

- B Concrete pipes and fittings shall comply with B.S. 556. They shall be reinforced, and of sulphate resisting cement if specified.

Manhole Covers and Road Gratings

- C Manhole covers and road gratings and frames shall comply with B.S. 497.

Step Irons

- D Step irons shall be galvanized malleable cast iron complying with B.S. 1247.

Mesh Reinforcement

- E Mesh reinforcement shall be steel fabric complying with B.S. 1221 Part A or B.S. 4483.

Setting Out

- F The Contractor shall set out all drains in accordance with the drawings, and provide all profiles, etc., necessary for the execution of the work.

Excavation

- G The bottoms of all excavations shall be trimmed and consolidated to the correct levels. Unauthorized excavations below the required levels shall be filled with concrete of the same composition as for drain beds, at the Contractor's expense.
- H Where the bottom is insufficiently firm, the Contractor shall excavate until, in the Architect's opinion, a firm bottom is obtained and the level shall be made up with concrete of the same composition as for drain beds. Particulars of such additional work shall be agreed with the Architect's representative before the work is covered up, otherwise no claim in this respect will be entertained.

Planking and Strutting

- I Care shall be taken not to undermine the foundations of the buildings and, if so directed by the Architect, planking and strutting shall be left in, or other means adopted to protect the foundations. Details of such additional items shall be agreed with the Architect's representative before the work is covered up, otherwise no claim in this respect will be entertained.

Backfilling

- J Trenches for pitch impregnated fibre of UPVC pipes shall first be filled with selected screened excavated materials carefully hand-tamped between the pipe and sides of the trench, followed by 150mm - 200mm of similar materials before the general filling is carried out.

- A Trenches for concrete or cast iron drains shall first be filled to a depth of 300mm with selected fine materials carefully hand-packed around the pipe. On no account shall materials be tipped into the trench until first 300mm has been completed.
- B Filling shall be continued in layers not exceeding 300mm thick, well rammed and, if necessary, watered.

Laying Drains

- C Drains shall be laid truly straight on line and gradient with sockets upstream and the full bore shall be unobstructed.

Pitch Impregnated Fibre Drains

- D All hard obstructions shall be removed from trench bottoms before laying pitch impregnated fibre pipes. The pipes shall be bedded in sand and laid and jointed in accordance with Appendix "C" to B.S. 2760.

UPVC Drains

- E UPVC drain pipes shall be laid and jointed with solvent welded joints entirely in accordance with the manufacturer's instructions.
- F Pipes shall be bedded in sand after all hard obstructions have been removed from trench bottoms.

Cast Iron Drains

- G Cast iron drains shall be laid on concrete beds where specified or shown on the drawings and shall be jointed with gasket of hemp, well caulked, to a depth of 30mm for 100mm pipes and 40mm for large pipes, and remainder of the socket shall be filled with molten lead or lead fibre solidly caulked.
- H Connection of iron to concrete drains shall be jointed as described for concrete drains.
- I Cast iron drains fixed to walls or beams shall be supported on brackets at 1,350mm centres.
- J Gullies, outlets, etc., on drains under concrete floors shall be set in position at correct levels before the floors are laid.

Concrete Drains

- K Concrete drains shall be jointed with one turn of tarred gaskin, well caulked and the remainder of the socket filled with cement and sand (1:3), finished with an angle fillet around the pipe. All surplus mortar shall be removed from the inside of the pipe with a badger. Where pipes are sulphate resisting, the jointing mortar shall contain sulphate resisting cement.

Concrete Beds, Haunches and Coverings

- A Where specified or shown on drawings, drains shall be laid on concrete, (105kg/sq.cm - 40mm aggregate), beds 100mm thick, 400mm wide for 100mm diameter drains and 450mm wide diameter drains. The concrete shall be haunched up both sides of the barrel to give lateral support.
- B Where drains, other than cast iron drains, are laid under buildings or pavings carrying vehicular traffic, they shall be completely surrounded in concrete, (105kg/sq.cm - 40mm aggregate), 150mm thick, (i.e. 400mm x 400mm overall for 100mm pipes and 450 x 450mm overall for 150mm pipes). Where directed, drain beds shall be reinforced.
- C Gullies shall be bedded and surrounded in concrete 105kg/sq.cm - 40mm aggregate minimum 150mm thick all round.

Sleeves

- D All drains passing through walls or foundations shall have sleeves of cast iron pipe of sufficient size to allow a clearance round the drain.

Benching

- E Benching in bottom of manholes shall be concrete (105kg/sq.cm - 40mm aggregate) to falls of not less than 10 degrees to channels finished with cement and sand (1:2), 25mm thick, trowelled hard and smooth with all angles rounded.

Bedding and Sealing Covers and Frames

- F Frames to manhole covers shall be bedded in cement mortar (1:3), and the covers in grease and sand.

Testing

- G All drains and manholes shall be tested for water tightness and straightness to the satisfaction, and in the present of, the Architects and the Local Authority. Drains shall be filled with water to a head of 1.50 meters and are to be tested in sections agreed with the Architect:-

- (i) after jointing
- (ii) after haunching and backfilling
- (iii) after completion of the works

- H The Contractor shall provide all necessary testing apparatus and shall carry out such other tests as are required by the Architect and the Local Authority.

Clean and Flush all Drains

- I All drains, gullies, manholes, etc., shall be cored, cleaned and flushed on completion.

Method of Measurement

- A Where not otherwise stated, the starting level for trench manhole excavation shall be:-
- (i) the formation level in areas where the site is excavated to reduce levels.
 - (ii) existing ground level in areas where no excavation is required, or where filling is required.
- B The depths of all the trenches in the following description lie within the same 1.5m stages as the average depths stated.
- C Prices for excavating pipes trenches shall be deemed to include keeping them free from general water (i.e. all water except spring or running water).
- D Notwithstanding the provisions of SMM Clause V.7 (a) to (c) the descriptions of excavating manholes, yard gullies, septic tanks and soakpits shall be deemed to include grading bottoms, planking and strutting, return filling and compacting, disposal of surplus soil and keeping excavation free from water.
- E Prices for building pipes into manholes shall include for building in on rake where necessary.
- F Prices for concrete beds, benchings and covering for pipes laid in trenches, shall be deemed to include for any necessary formwork. Formwork required for beds, etc., for pipes above ground, and for casing to vertical pipes, is referred to in the descriptions of such items.
- G Prices for all gullies shall be deemed to include for all necessary excavation, return filling, disposal of surplus excavated materials, planking and strutting, and trimming and ramming bottoms.

EXTERNAL PAVINGS

Generally

- A. The Preambles contained in other sections of the document shall apply equally to this sections so far as is consistent with the following clauses.

Materials

Soil for Planted Areas

- B. Soil for planted areas shall be vegetable soil free from roots and rubbish and treated with weed killer to prevent the growth of weeds.

Sand for Filling under Footpaths

- C. Sand for filing under footpaths shall be clean, dry, pit or river sand, free from vegetable soil, roots and rubbish.

Crusher Dust for Sub-Base Course of Macadam Paving

- D. Crusher dust shall be from an approved source and shall be free from clay or other deleterious matter.

Stone for Base Course to Macadam Paving.

- E. Stone for base course to macadam paving shall be 40mm gauge, clean and hard and free from clay or other deleterious matter.

Blinding For Stone Base Course

- F. Blinding for stone base course shall be 4mm gauge hard stone chippings, free from clay, dust or other deleterious matter.

Precast Paving Slabs

- G. Precast paving slabs shall comply with B.S. 368 except for sizes.

Kerbs

- H. Precast concrete kerbs shall comply with B.S. 340, and shall be finished true and smooth on all exposed faces.

Prime Coat for Macadam Paving

- J. The prime coat for macadam paving shall be bitumen grade M.C.I.

Bitumen for surfacing

- A. The bitumen for surfacing shall be made 500/700 grade bitumen.

Workmanship**Generally**

- B. The sub-grade, sub-base and base courses for roads and parking area shall be prepared and laid at a convenient time before completion of the contract, as shall be agreed between the Architect and the Contractor, together with their kerbs and foundations.
- C. The wiring course shall be applied at a later date, and prior to laying, the base course shall be made good in accordance with the requirements specified herein. The Contractor shall make good at his own expense any damage to kerbs.

Surveying

- D. The Contractor shall verify all dimensions and levels prior to the commencement of work.
- E. All surveying necessary for the accomplishment of the works shall be done by the Contractor at his own expense and he shall give notice of his intention to carry out such work in order that the arrangements can be made for supervision and checking. The Contractor shall also provide, without extra charge, all necessary instruments, appliances, labour and any other materials required for checking the survey work.
- F. The Contractor shall make all necessary surveys using given bench marks as reference points. These bench marks he shall carefully preserve.
- G. The Contractor shall draft, in accordance with these surveys, all plans and drawings which are necessary for the completion of the work, and shall submit these plans and drawings to the Architect for approval in writing.

Levels, Falls, Crossfalls and Cambers

- H. The works shall be executed to the levels, falls, crossfalls and cambers shown on the drawings

Accuracy

- J. The Contractor shall be responsible for ensuring that the works are carried out to the line, levels and dimensions shown on the drawings, and shall provide camber gauges and straight edges for checking to ensure that the surfaces are within the following tolerances:-

(a) **Sub-Grade**

The camber or crossfall shall not vary more than 20mm from that shown on the drawings. In the longitudinal direction the variations from a 3 meter straight edge placed parallel to the centre line of the road shall not exceed 12mm.

(b) **Base**

The camber or crossfall shall not vary more than 12mm from that shown on the drawings. The variation on the longitudinal section shall be as above for sub-grade

Sub-Grade

- A. The sub-grade shall be shaped to the required falls and cambers and any depressions filled with approved materials having a minimum C.B.R. of 8 percent. This value shall be obtained at optimum moisture content and compacted to 100 percent of the maximum dry density as determined by B.S x1377. The Contractor shall carry out standard compacting tests on the sub-grade in accordance with Test Nr 10 of B.S. 1377. Such tests shall be taken at 30metre intervals. The standard of compaction required shall be 98 percent of the maximum dry density as determined by Test No. 9 of B.S. 1377.
- B. The sub-grade shall be approved by the Architect before any materials to be used in construction of the carriageway are deposited or laid.

Sub-Base Course

- C. The sub-base shall consist of a layer of crusher dust finishing to the thickness specified after compaction. The bed shall be watered as necessary and rolled to produce a smooth and uniform surface with no irregularities.

Base Course

- D. The base course shall consist of a layer of stone in which the interstices shall be filled by application of crusher fines after the stone is in place, to finish to the thickness specified after compaction. The base course shall not be blinded with crusher fines, but with 4mm gauge stone chippings to provide a clean hard surface. If any irregularities develop, they should be corrected by loosening the material at these places and adding or removing material and recompaction until the surface is smooth and uniform with no irregularities.

Application of Bitumen

- E. The plant used by the Contractor for transporting, heating and spraying bitumen shall be in suitable rubber-tyred units and shall ensure adequate and uniform heating without the introduction of steam or moisture, and giving rise to the cooking or burning of the bitumen, and shall be fitted with a thermometer and heating control. Distributors shall be equipped to provide a constant rate of application per square meter of surface and there shall be visible speedometer indicating the speed of the vehicle in meters per minute.

- A. Spray bars shall be capable of spreading the bitumen evenly to the full width of the work. The bitumen shall be heated to the temperature specified below and sprayed on the clean surface of the base at the rates specified.
- B. Application temperatures shall be in accordance with those recommended by the manufacturer, or where this information is not available, they shall be as follows:-

Bitumen Grade	Sprayed Temperature (Degree Celcius)
N.C.I	54-80
500/700	124-149

Prime Coat

- C. Prior to the application of the prime coat, the surface of the base shall be swept clean of dust and foreign materials to the satisfaction of the Architect. Approximately 30 minutes before applying the bitumen the surface of the base shall be lightly sprayed with water.
- D. The prime coat shall be applied at the rate of 0.70 litres per square meter.

Wearing Course

- E. After the application of the priming coat, and where directed and approved by the Architect, the Contractor shall lay bitumen type 500/700 spread at the rate of 3 square meters per 5 litres immediately followed by spreading dry, clean approved 12mm chippings at the rate of 130 square meters per cubic meter, rolled six to eight passes of a six to eight tonne roller. A second and similar surfacing layer shall be laid at the end of the defects liability period.
- F. Alternatively, where specified, the wearing course shall consist of a premix macadam carpet of 500/700 grade bitumen and approved quality aggregate graded and mixed together prior to laying in the proportions and by the methods given in B.S. 1621 table 4, laid to finish to the thicknesses shown after compaction. The compaction shall be achieved with six to eight passes of a six to eight tonne roller.

Wet Weather

- G. No bitumen spraying shall be carried out when either the carriageway surface of the aggregate are wet, without the prior approval, in writing, of the Architect who may allow such work to proceed by the use of an approved adhesive agent at the Contractor's expense

Murram Roads

- H. Murram roads shall be laid in layers not exceeding 150mm compacted thickness, to finish compacted to the thicknesses shown on the drawings.
- J. Each layer shall be watered, rolled and compacted as previously described herein to produce a smooth dense surface free of all irregularities.

Laying Precast Paving Slabs

- A. Precast paving slabs shall be bedded on a sandbed compacted to the thickness specified with 6mm wide joints, filled and pointed with cement mortar coloured to match the colour of the slabs and recessed 5mm deep. The paving shall be finished true and even to the falls shown on the drawings with no surface irregularities.

Grassing

- B. Grassing shall be carried out by a Specialist using approved local grass. Prices for grass shall include for tending, watering, cutting and keeping weed free for a period of twelve months, to produce a dense and healthy weed free grass carpet.

Note:

The Contractor shall include here for any cost they may consider necessary and over and above costs which they believe they cannot recover in any other section of these Bills of Quantities.

GENERAL SPECIFICATIONS

SPECIFICATIONS

Reference is made to the General Specifications for Building Works (1976) by the Ministry of Works, Housing and Physical Planning.

A copy is available for perusing at the request of the procuring entity.

Contractors are required to adhere to the latest industry standards as outlined in the most recent version of KS (Kenyan Standards) / BS (British Standards) EN International standards. Failure to comply may result in project delays or financial penalties. It is the responsibility of the Contractor to stay informed about and apply the current industry standards throughout the construction process. Any disputes arising from non-compliance with updated standards will be subject to resolution through dispute resolution mechanism outlined in the contract.

PRELIMINARIES

ITEM	DESCRIPTION	AMOUNT
A	<u>BILL NO. 1</u>	
	<u>PARTICULAR PRELIMINARIES</u>	
	<u>PARTIES</u> The Employer is: Principal Secretary, Ministry of Lands, Public works,Housing and Urban Development, State Department of Housing and Urban Development P.O Box 30119 -00100 NAIROBI, KENYA The Engineer is: The term "PM" wherever used in these Bills of Quantities shall be deemed to imply the Engineer as defined in Condition 1 of the Conditions of Contract or such person or persons as may be duly authorised to represent him on behalf of the Government . The Architect is: Ministry of Lands, Public works,Housing and Urban Development, State Department of Housing and Urban Development P.O Box 30119 -00100 NAIROBI, KENYA The Quantity Surveyors is: Ministry of Lands, Public works,Housing and Urban Development, State Department of Housing and Urban Development P.O Box 30119 -00100 NAIROBI, KENYA The Structural/ Civil Engineers is: Ministry of Lands, Public works,Housing and Urban Development, State Department of Housing and Urban Development P.O Box 30119 -00100 NAIROBI, KENYA The Electrical / Mechanical Engineers is: Ministry of Lands, Public works,Housing and Urban Development, State Department of Housing and Urban Development P.O Box 30119 -00100 NAIROBI, KENYA	

ITEM	DESCRIPTION	AMOUNT
A	<u>LOCATION OF SITE</u> The site of the proposed works is located in Emurua Dikirr, Narok County The Contractor shall be deemed to have visited the site and satisfied himself as to:- a) The nature, position, topography and access of the site b) The amount of the rubbish or debris to be cleared away before commencement. c) The nature, current usage, proximity and size of adjoining property and buildings d) The availability of land for the erection and positioning of all temporary structures, plant and materials necessary for the execution of the works. The Contractor shall obtain approval from the relevant Local Authority in adherence to site access and erection of temporary structures and must ensure all matters relating to the requirements of these authorities. No claim will be allowed for travelling or other expenses which may be incurred by the Contractor in visiting the site or preparing the tender for the works.	
	<u>EXISTING SITE CONDITIONS</u> The Contractor is advised that the site is in a predominantly residential area and all measures should be taken to avoid nuisance to neighbours. All occupation health and safety requirements must be met as required by law. This includes prevention and or minimizing noise, dust, fumes e.t.c. Notices should be given prior to disruption of services	
C	<u>SCOPE OF CONTRACT</u> The Works under this contract comprises of the structures as detailed in the project data sheet	
	Carried to collection	

ITEM	DESCRIPTION	AMOUNT
A	<u>DESCRIPTION OF THE WORKS</u>	
	The construction comprises reinforced concrete foundations, masonry walling, reinforced concrete beams, column , staircases and suspended solid slabs, roof construction. The exterior facade consists of steel casement windows, steel and timber doors , render and paint finish, clay and stone facing finish to walls The interior works includes timber doors and finishes which are generally plaster and paint to walls, ceramic and non slip ceramic tiles to floors and walls. External works generally comprise of foul water drainage, storm water drainage, pathway, dryline area, septic tank, underground water tank. All mechanical / electrical services and other specialist works associated with the above works shall be executed by domestic/nominated sub contractors approved by the Engineer <u>CONTRACT PARTICULARS</u>	
B	<u>FORM OF CONTRACT</u>	
	The Contractor will be required to enter into a contract with the Employer under the Terms and Conditions of Contract as "Standard Tender Document for Procurement of Works (Building and Associated Civil Engineering Works) Issued by the Public Procurement Regulatory Authority in February 2021 (updated 2022) and in association with the latest applicable version of the Public Procurement and Asset Disposal Act. The Contractor's attention is called to the appendix of the conditions of Contract and additions and amendments thereto, which shall be read as incorporated herein and he shall allow any sums which he considers necessary for the observance of such conditions, together with sub clauses used in application. The priority of such documents shall be as stated in the conditions of agreement.	
	Carried to collection	

ITEM	DESCRIPTION	AMOUNT
A	<u>LIABILITY AGAINST INJURY TO PERSONS AND PROPERTY</u> Insurance against injury to persons and property NOTES In addition to the conditions of the contract and the requirement contained herein the contractor's all risk policy shall cover the full value of the following and allow for all costs thereof:- i) The works and temporary works erected in performance of this contract. ii) The materials on site, plant and tools iii) The cost and expense of removing debris of the property insured, destroyed or damaged by any peril insured. iv) Professional fees (to be allowed at 15% of the contract sum) v) Employer's liability (workman's compensation) ii) Third party (Public liability for an indemnity of not less than shs 15,000,000 for any accident or series of accidents arising from the same event (unlimited in aggregate) The contractor shall ensure that all sub-contractors effect and maintain such insurances as are necessary to cover their liabilities in respect of injury to persons and property and workman,s compensation. Should the contractor already hold annual insurances covering the whole of his activities, and the indemnity required under the existing policy/ies then further insurances shall be effected and maintained to cover such excess, the policies of insurances being suitably endorsed to cover this project	1,000,000.00
B	<u>Insurance of the works (contractors liability)</u> The Contractor shall insure as required in the Conditions of Contract. No payment on account of the work executed will be made to the Contractor until he has satisfied the Engineer either by production of an Insurance Policy or and Insurance Certificate that the provision of the foregoing Insurance Clauses have been complied with in all respects. Thereafter the Engineer shall from time to time ascertain that premiums are duly paid up by the Contractor who shall if called upon to do so, produce the receipted premium renewals for the Engineer's inspection.	5,000,000.00
	Carried to collection	6,000,000.00

ITEM	DESCRIPTION	AMOUNT
	<u>PERFORMANCE BOND</u>	
A	Performance bond for the works The Contractor shall submit on the Form of Tender an approved bank or approved (By PPRA) Insurance Company and who will be willing to be bound to the Employer in an amount(<i>as defined in the bidding documents</i>)..... for the due performances of the Contract up to the date of completion as certified by the Engineer and who will when and if called upon, sign a Bond to that effect on the relevant standard form as seen in the CONTRACT STANDARD FORMS (without the addition of any limitations) And should the surety fail to be approved, the Contractor shall furnish within seven days another Surety to the approval of the Employer. Note that no payments on account of works executed will be made to the Contractor until he has submitted the Performance bond, duly stamped signed and sealed by an approved bank or insurance company.	2,000,000.00
B	<u>POSSESSION AND COMMENCEMENT</u> The Contractor shall take possession of the site on the date indicated in the acceptance letter. The date of commencement of the works shall also be communicated to the Contractor and the contract period shall run from the commencement date. The Contractor is expected to utilize the period between possession and commencement to mobilise his resources to ensure smooth running of the works from the commencement date.	
	Carried to collection	2,000,000.00

ITEM	DESCRIPTION	AMOUNT
	<u>PROJECT SUPERVISION</u>	
A	The said works shall be executed under the direction and to the entire satisfaction of the Engineer and Clerk of works who shall have the Engineer's specifically delegated authority and shall at all times have access to the works, to the yards and workshops of the contractor or other places where goods are being prepared for the building.	
	<u>LABOUR CAMPS</u>	
B	The contractor will generally be permitted to house labour on site subject to approval by Architect	
	<u>DOWNTAKINGS</u>	
C	All materials arising from demolitions and downtakings are deemed to be the property of the employer. No claim will be entertained on account of employer excising this right to retain the materials All downtakings shall be carefully removed, taken down, dismantled and stored on site until instructed by the Engineer to remove from the site. Such materials shall only be incorporated in the new works if required by the Engineer in which case appropriate adjustments will be made in the final account for the cost of labour, screws etc for fixing such downtakings in the new works.	
	<u>DAMAGES</u>	
D	Damages for delay in completion shall be levied at the rate of Kshs(Refer to the special Conditions of Contract).....	
E	<u>BIOMETRIC INSTALLATIONS</u> Supply, Deliver to site, Install, test and commission a biometric and RFID card reader that supports RS485 and TCP/IP. supports both 125kHz & 13.56MHz, finger, card and pin authentication complete with Wiegand Interface, 12-digit Keypad, Power Supply unit c/w batteries, RFID Proximity ID Cards, Workstation and all necessary cabling and infrastructure required for a complete installation to Engineers approval. Allow for full graphic customization and programming of the installed system into the existing access control system	
F	<u>CCTV INSTALLATION</u> Supply, Deliver to site, Install, test and commission a CCTV installation during the construction period for monitoring and socio-economic data collection as per engineer's specifications. System should include but not limited to remote monitoring capabilities and atleast 30 days cloud storage, IP66, IK10 Bullet Network Camera 4MP resolution, H.265, H.264, MJPEG codec supported, Multiple streaming, Motion detection, Tampering, Defocus detection, micro SD/SDHC/SDXC memory slot (Max. 128GB), PoE / 12V DC, IR viewable length 20m (QNO-7010R), 25m (QNO-7020R) 30m (QNO-7030R), Warranty 3 years, 8 Channel Network Video Recorder, Maximum 12-megapixel camera supported complete with storage and all necessary active and passive devices, cabling, back up power and accessories required for a complete installation.	
G	<u>OTHER PRELIMINARIES</u> Allow for any other item necessary to execute the works and state them below; 	
	Carried to collection	-

ITEM	DESCRIPTION	AMOUNT
	<u>BILL NO. 2</u> <u>GENERAL PRELIMINARIES</u> <u>PRICING OF ITEMS OF PRELIMINARIES AND PREAMBLES</u> A Whenever in the Contractor's priced Bills of Quantities no price appears against an item of Preliminaries or Preambles or work items , the value of such item shall be deemed to be included in his prices for other items in the Bills of Quantities. <u>SUFFICIENCY OF TENDER</u> B The Contractor shall be deemed to have satisfied himself before tendering as to the correctness and sufficiency of his tender for the works and of the rates and prices stated in the priced Bills of Quantities. Rates and prices quoted shall cover all his obligations under the contract and all the matters and maintenance of the works <u>RECORDS</u> C The Contractor shall ensure proper records are kept and maintained for : Daily Reports on Personnel and Machinery; tracked programme; weather charts/reports; site instruction book and query book,a digital camera shall be provided for taking progress photos The contractor shall be required to provide equipment for taking ground and aerial photos or videos in relation to the progress of works when called upon to do so. <u>DEFINITIONS AND ABBREVIATIONS</u> D Throughout these Bills, units of measurements and terms are abbreviated and shall be interpreted as follows: mm shall mean millimeter lm shall mean linear meter sm shall mean square meter m² shall mean square meter cm shall mean cubic meter kg shall mean kilogramme N shall mean Newton KN shall mean KiloNewton Carried to collection	

ITEM	DESCRIPTION	AMOUNT
	<u>SITE LEVELS</u>	
A	Before commencing work the Contractor must arrange for and agree with the Architect, Engineer and Quantity Surveyor the existing site levels and similarly establish and agree on a bench mark. The Contractor shall provide a surveyor to ensure all levels are achieved as per the drawings and Architects/Structural Engineer's instructions	200,000.00
	<u>SETTING OUT</u>	
B	The contractor shall set out works in accordance with the dimensions and levels shown on the drawings and shall be responsible for the correctness of all dimensions and levels set out by him and he will be required to amend all errors arising from inaccurate setting out at his own cost and expenses. In the event of any error or discrepancy in the dimensions or levels marked on the drawings being discovered, such errors or discrepancies must be reported by the Contractor to the Engineer for his immediate attention. No work shall be commenced by the Contractor until he has received written instructions from the Engineer to adjust such discrepancies which may be proved, upon receipt of such instructions and no claim for extra expenses or relief from the provisions of Clause 5 of the Conditions of the Contract , any discrepancy or error in the dimensions or levels shown on the drawings may be made thereafter. The Contractor shall give the Engineer reasonable notice of the intention to set out or take levels for any part of the Works so that arrangements may be made for checking the work. The accuracy of setting out and leveling shall be within the tolerances specified in the Specifications or on the Drawings. The checking of setting out or leveling by the Engineer shall not relieve the Contractor of his duties or responsibilities under the Contract.	500,000.00
	<u>MEASUREMENTS</u>	
C	Measurements are based on Standard Methods of Measurement of Building Works and Associated Civil Works For Eastern Africa (SMM) Second Edition 2008. In the event of any discrepancies arising between the Bills of Quantities and the actual works, the site measurements shall generally take precedence.	
D	<u>GENERAL SPECIFICATIONS</u>	
	All works to be carried out in accordance with the Ministry of Roads, public Works and Housing General Specifications for Building Works issued in 1976 or as qualified and amended.	
	Carried to collection	700,000.00

ITEM	DESCRIPTION	AMOUNT
	<u>SAMPLES</u>	
A	The contractor shall furnish at the earliest possible opportunity before work commences and at his own cost any samples of materials or workmanship that may be called for by the Engineer for his approval or rejection until such samples are approved to be the acceptable standard for the work to which they apply. The samples shall be maintained and displayed on a designated section within the site for the duration of the project where practical and possible.	500,000.00
	<u>PROTECTION OF EXISTING PROPERTY</u>	
B	The contractor shall take every precaution to avoid damage to all existing property including boundary wall, carpark, roads, cables, drains, staircases, lift etc including other services and he will be held responsible for all damages hereto arising from the execution of his contract and he shall make good all such damages when directed at his own expense. Any damage or disturbances caused to any element shall be reported immediately to the Engineer and the relevant Authority and shall be made good to their satisfaction at the Contractor's expense.	
C	<u>PROTECTION / RELOCATION OF EXISTING SERVICES</u>	
	Prior to commencement of any work the Contractor is to ascertain from the relevant Authorities the exact position, depth and level of all existing electric cables, water pipes and all other services in the area and he shall make whatever provisions may be required by the authorities concerned for the support and protection and/or relocation of such services as will be necessitated. The contractor is also expected to generate a utility management plan to the approval of the Engineer . Any damage or disturbances caused to any service shall be reported immediately to the Engineer and the relevant Authority and shall be made good to their satisfaction at the Contractor's expense.	
	Carried to collection	500,000.00

ITEM	DESCRIPTION	AMOUNT
A	<u>MATERIALS, TOOLS, PLANT AND SCAFFOLDINGS</u> All materials and workmanship used in the execution of the works shall be of the best quality and description. Any materials for the works condemned by the Engineer shall immediately be removed from the site at the Contractor's expense. The Contractor shall be responsible for the provision of all materials, scaffolding, tools, plant, transport and workmen required for the works except in so far as may be stated otherwise herein and he shall allow for the provision of the foregoing except for such items specifically and only required for the use of Nominated Sub-contractors as described herein. No timber used for scaffolding, formwork or similar purpose shall be used afterwards in the permanent works. All such plant, tools and scaffolding shall comply with all regulations whether general or local in force including Environmental, Social, Health and Safety (ESHS) policies throughout the period of the contract and shall be required as may be necessary to comply with any amendments in or additions to such regulations The Contractor shall keep on the site and maintain in good condition one dumpy or quickset level, metric leveling staff and one 30 metre steel tape for the use of the Architect, Surveyor and Engineer. The contractor may be required to provide an appropriate tower crane as required during the project life. Where a crane is provided, it should meet all regulatory and technical standards, all licences in connection with erection, usage shall be at the Contractors expense. The contractor may be required to provide an appropriate tower crane as required during the project life. Where a crane is provided, it should meet all regulatory and technical standards, all licences in connection with erection, usage shall be at the contractors expense.	800,000.00
	Carried to collection	

ITEM	DESCRIPTION	AMOUNT
A	<u>LOCAL REGULATIONS AND BY-LAWS</u> The contractor is to comply with all local regulations and by-laws of the Local Authority including serving notices and paying of fees where applicable. These include, but not limited to: National Environmental Management Authority (NEMA), National Construction Authority (NCA), Water Resources Authority (WRA) The Contractor will be held responsible for serving on the Chief Inspector of Factories a written notice not later than seven days after the beginning of the building operations included in this contract stating the particulars required. <u>TRANSPORT TO AND FROM THE SITE</u>	
	B The Contractor shall include in his prices for the transport of materials, workmen etc to and from the site of the proposed works at such hours and by such routes as are permitted by the Authorities. All unit rates for local or imported goods are to include freight, insurance, handling and delivery costs to the project site together with import duties, sale tax, port charges etc and all other charges of whatever nature. Carried to collection	

ITEM	DESCRIPTION	AMOUNT
A	<u>FAIR WAGES</u>	
	The Contractor shall pay rates of wages and observe hours and conditions of labour not less favourable than the minimum conditions of employment applicable in the area in which the work is carried out. The relevant notice must be posted up and kept posted upon the site where it can be conveniently read by the employees concerned in languages they can understand.	
	The Contractor is to comply with the regulations of Wages and Conditions of Employment Act, Building and Construction Industry Wages Council and is to be responsible for compliance of the sub-contractors employed in the execution of the contract. If required he is to notify the Engineer of the names and addresses of all such Sub-contractors. Any Contractor or Sub-contractors not complying will not be permitted to tender for other work for such a period as the Engineer may determine	
	Should a claim be made to the Engineer alleging the Contractor's default in payment of fair wages to any workman employed on the contract and if proof thereof satisfactory to the Engineer, may failing payment by the Contractor, pay the claim out of any monies due or which may become due to the contractor under this contract.	
B	The Contractor is to furnish to the Engineer, if called upon to do so, such particulars of the rates of wages, hours and conditions of labour referred to above as the Engineer may direct	
	<u>SECURITY OF WORKS</u>	
	The Contractor shall be entirely responsible and shall pay security of all works, stores, materials, plant, personnel etc both his own and sub-contractors and shall also provide all necessary watching, lighting, and other precautions as necessary to ensure the security, the safety and protection of the public. He is to ensure that there is no informal business settlement near the establishment.	2,500,000.00
	Carried to collection	2,500,000.00

ITEM	DESCRIPTION	AMOUNT
A	<u>OCCUPATIONAL HEALTH AND SAFETY MEASURES</u> The Engineer expects the Contractor to adhere to strict safety measures. In this regard the Contractor should ensure that all his workers, the Consultants and his sub-Contractors workmen are wearing Personal Protective Equipment (PPE) before commencement of any work where applicable including overalls with the company name clearly printed on the back each with clearly marked Identification Numbers stitched or imprinted on. The Contractor shall allow for providing all watching, lighting, green netting, barriers, signs, covering open trenches and protection of the works, including Sub-Contract works, as may be necessary for the safety of the works and for the protection of the public and his own and Sub-Contractors' employees. He shall also ensure provision of a certified and qualified safety, health and environmental officer, access to ambulance services at all worksites and arrangement to access a local hospital/dispensary with qualified medical staff. The contractor shall take cognisance and shall fully adhere to the regulations of the Occupational Safety and Health Act of 2007 including all the associated revisions The Engineer shall expect full compliance to this regulation and no excuses will be entertained for non-compliance which may lead to suspension of works until the issue is addressed satisfactorily.	1,000,000.00
B	<u>PUBLIC, PRIVATE ROADS AND PAVEMENTS ETC</u> The contractor will be required to make good at his own expense any damages he may cause to the present approach and surrounding road surfaces during the period of the works	
C	<u>POLICE REGULATIONS</u> The contractor is to allow for complying with all Government Acts, orders or regulations in connection with employment of labour and other matters related to the execution of the works. The Contractor must acquit himself duly with current acts and regulations, including police regulations regarding the movement, housing, security and control of labour, labour camps, passes for transport, etc..	
	Carried to collection	1,000,000.00

ITEM	DESCRIPTION	AMOUNT
A	<u>AREA TO BE OCCUPIED BY CONTRACTOR</u> The area of the site which may be occupied by the Contractor for use as storage and for the purpose of erecting workshops etc shall be defined on the site by the Engineer	
B	<u>PROGRESS SCHEDULE</u> Immediately after signing the contract the Contractor is to prepare a Time Progress Chart showing the time and order in which he proposes to carry out the works within the total construction time stated in the contract. The chart will show in detail the construction time and order in which each section of the work is to be carried out and be sub-divided into trades and tasks. If the contractor proposes sectional completion of the project he must plan this in detail including access roads, and services and this shall be reflected on the chart Upon the letting of the Sub-Contractors work the Contractor is to incorporate times and details of each separate Sub-Contractor work which information is to be agreed by the Sub-Contractor and the chart will be so designed to accommodate this infantine. At the end of each week the Contractor is to mark on the chart in a different colour the actual time taken to complete the respective stages and sections of the work. The contractor shall obtain the Engineer's approval on the chart and then shall supply copies to the Engineer and Quantity Surveyor If at any time it should appear to the Engineer that the actual progress of the works does not conform to the approved programme progress schedule the Contractor shall produce at the request of the Engineer a revised programme showing the modifications and accelerations to the approved programme necessary to ensure completion of the works within the agreed contract period. The submission of and approval by the Engineer of such revisions and accelerations shall not entitle the Contractor to any extra payment or extension of time and shall not relieve the Contractor of any duties or obligations or responsibilities under the contract	250,000.00
	Carried to collection	250,000.00

ITEM	DESCRIPTION	AMOUNT
A	<u>OVERTIME</u> The Contractor shall be responsible for any extra costs for overtime working he considers will be necessary in order to complete the works within the contract period or time for completion apart from overtime working which may be authorised by the Engineer If overtime is worked out in accordance with a written instruction issued by the Engineer the contractor will be reimbursed in respect of such overtime to the unproductive time payable over and above the basic hourly rates as laid down by the Regulation of wages and Conditions of employment Act, Building and Construction Industry Wages Council and excluding any bonuses, profits and overheads.	
B	<u>WATER</u> The contractor shall provide at his own risk and cost all water for use in connection with the works including the work of sub-contractors make arrangements with the local authority for the installation of a separate meter where applicable and possible for all water used by him throughout the contract and pay all costs and fees in connection therewith. He shall also provide temporary storage tanks and tubing etc as he may consider necessary and clear away at completion. The contractor is to provide clean drinking water at the construction site for his workers at all times. All water shall be fresh, clean and pure, free from earthly vegetable or organic matter, acid or alkaline substance in solution or suspension.	1,000,000.00
C	<u>TELEPHONE</u> The contractor shall provide in the office, from the commencement to the completion of the works, a wireless or mobile phone and shall pay all charges or airtime necessary for its use	
D	<u>LIGHTING AND POWER</u> The contractor shall provide at his own risk and cost all temporary artificial lighting and power for use on the works including all sub-contractors and specialists requirements and including all temporary connections, wiring, fittings etc and clearing away on completion. The Contractor shall pay all fees and obtain all permits in connection therewith.	1,000,000.00
	Carried to Collection	2,000,000.00

ITEM	DESCRIPTION	AMOUNT
A	<u>TEMPORARY STRUCTURES</u> a) The Contractor shall allow for providing and clearing away on completion of the works such temporary hoarding , rubbish chutes, gates, planked walkways, guard rails etc. as may be necessary for the protection of the workers, the general public, and for the proper execution of the works. b) As such, temporary structures shall be constructed with the approval of the Engineer and to his full satisfaction and in such a manner as to cause minimum intrusiveness and disturbance to occupants of adjacent developments and users of the adjacent roads. c) All such temporary structures shall comply in all aspects with the national laws, rules, and regulations currently in force and applicable to such structures. d) All temporary structures shall be erected in a manner so that the unloading of materials causes minimum obstruction to the use of adjacent roads and other facilities e) All temporary structures shall be kept properly lighted throughout the periods of darkness and any corners or projections shall be painted white. g) Temporary structures shall not be used or permitted to be used for advertisement purposes except with the written consent of the Engineer h) All temporary structures shall be maintained at all times in good order and good condition to the satisfaction of the Engineer. i) All temporary structures shall be removed when so required by the Engineer or at the end of the period for which it is required. j) The Contractor shall indemnify and shall keep the employer idemnified against any expenses, loss, claim or suits arising out of or in connection with the temporary structures.	
	Carried to collection	

ITEM	DESCRIPTION	AMOUNT
A	<u>SITE OFFICE</u> The contractor shall supply, maintain, service, clean and light a fully furnished, suitable office having an approximate floor area of not less than 200sqm. The office shall have a sample room suitable dimensions with clean running water and electricity connected to the approval of the Engineer. The Contractor shall provide offices, messrooms and all other buildings required by the Contractor for his own use and the use of by Clerk of Works and Nominated SubContractors as required by the items or attendance The site office shall be equipped with a table and chairs of sufficient size and number for site meetings and plan chests for drawings shall also be provided by the contractor The Contractor shall allow for the cost of providing light refreshment for the consultants at site meetings.	1,000,000.00
	<u>TEMPORARY DISPOSAL OF RAIN WATER</u> The Contractor shall provide and maintain all necessary temporary gutters, downpipes, chutes, drains etc. for conveying rainwater from the buildings and storage tanks for rainwater harvesting. The Contractor shall allow for temporary drainage plumbing and piping for keeping the premises and site free from accumulation of water. He shall also allow for construction and maintaining any necessary storm water drainage structures as directed.	
	<u>CLEARING AWAY</u> The Contractor shall remove all temporary works, rubbish, debris and surplus materials from the site as they accumulate, on intervals as instructed by the Engineer and upon completion of the works, remove and clear away all plant, equipment, rubbish, unused materials and stains and leave in a clean and tidy state to the reasonable satisfaction of the Engineer. The whole of the works shall be delivered up clean, complete and in perfect condition in every respect to the satisfaction of the Engineer.	
	Carried to collection	1,000,000.00

ITEM	DESCRIPTION	AMOUNT
A	<u>SITE ACCOMODATION & STORAGE</u>	
	The Contractor shall provide sheds for storage accommodation for all goods and materials liable to suffer damage from exposure to sunlight or inclement weather.	800,000.00
	The Contractor shall provide offices, mess rooms and all the buildings required by the Contractor for his own use and the use of Nominated Sub-Contractors as required by the items of attendance only.	
	The Contractor shall provide at his own risk and cost where directed on the site weather proof lock-up sheds and make good damaged or disturbed surfaces upon completion to proof lock-up sheds and make good damaged or disturbed surfaces upon completion to the satisfaction of the Engineer	
	Upon completion all temporary buildings are to be removed and cleared away	
B	<u>SANITATION OF THE WORKS</u>	
	The sanitation of the works shall be provided, maintained and removed on completion by the Contractor to the satisfaction of the Engineers and local Authorities.	200,000.00
	The sanitary facilities shall be of generally acceptable standard regardless of the material being used to ensure ease of cleaning and maintain general well being of the users. Their location shall be agreed with the Engineers and the works shall not be commenced before the sanitary accommodation has been approved by the above mentioned authorities.	
	The Contractor will be required to pay all conservancy charges and shall ensure clean daily maintenance and disinfecting of the sanitary facilities, and not less than once per week, the whole area shall be sprayed with disinfectant and insecticides and any temporary drains shall be removed and all works and surfaces disturbed made good and then the whole area disinfected and left clean and free from pollution to the satisfaction of the Engineer and local authorities.	
C	<u>HOARDINGS</u>	
	The Contractor shall provide, erect and maintain throughout the course of the Contract and thereafter clear away and make good disturbed areas, temporary hoarding; approximate length of 1000 metres: 2400mm high above ground consisting of: 100mm diameter timber poles at 1200mm centres firmly founded and secured, 75x50mm horizontal timber rails at 900mm centres, painted GCI sheets, proper timber gates with suitable locks to Engineers approval.	1,500,000.00
	Carried to collection	2,500,000.00

ITEM	DESCRIPTION	AMOUNT
A	<u>DEMOLITIONS AND DOWNTAKINGS</u> The Contractor is to allow for all temporary protection required during the works including ordinary and special dust screens, hoardings, barriers, warning signs etc. as directed by the Engineer and as necessary for the adequate protection of adjacent property and finishes, workmen employed upon the site and the public. Any damage or loss incurred due to the insufficiency of such protection must be made good by the Contractor. All protective devices are to be removed on completion of the work and any necessary making good consequent upon this is to be executed to the satisfaction of the Engineer All materials arising from demolitions and downtakings are deemed to be the property of the employer. No claim will be entertained on account of employer excising this right to retain the materials unless otherwise stated. The Contractor shall allow in his rates the cost of handling and disposal of debris arising out of the demolition works All downtakings shall be carefully removed, taken down, dismantled and stored on site until instructed by the Engineer to remove from the site. Such materials shall only be incorporated in the new works if required by the Engineer in which case appropriate adjustments will be made in the final account for the cost of labour, screws etc for fixing such downtakings in the new works. The Contractor shall be entirely responsible for any breakage or damage which may occur to materials required for re-use, during their removal, unless it is certified by the Engineer that such damage or breakage was inevitable as a result of the condition of the item concerned.	
	<u>ACCESS TO SITE AND TEMPORARY ROADS</u> Means of access to the site shall be agreed with the Engineer prior to commencement of the works and the Contractor must allow for building and maintaining any temporary access roads for the transport of materials, plant and workmen as may be required for the complete execution of the works including the provision of temporary culverts, crossings, bridges or any other means of gaining access. Upon the completion the works the Contractor shall remove such temporary roads, temporary culverts bridges etc and make good and reinstate all works and services disturbed to the satisfaction of the Engineer.	
	Carried to collection	

ITEM	DESCRIPTION	AMOUNT
A	<u>SIGN BOARD</u> The Contractor shall provide and erect where directed and maintain during the whole period of the building operation and remove at completion, one approved sign board of approximately 3000x3000mm and approximately 5800mm overall height to the Architect's later design giving a brief description of the works, a 3D perspective image of the project, and showing the names of the employer and the consultants, with sufficient space to append the names of the sub-contractors and suppliers when known. The lettering concerning the Architect, Quantity Surveyor and Engineer is not to be more than 50mm high.	100,000.00
B	<u>PRIME COST SUMS</u> i) The words "Prime Cost" (or the initials "P.C") appearing in the contract documents shall mean net costs exclusive of any trade, cash or other discount whatsoever but inclusive of the costs of the packing, carriage and delivery. Such costs shall be the same due to the sub-contract or supplier after adjustments where applicable in respect of measurements of rates. ii) Any increase or decrease in the prime costs sums resulting from the adjustments and properly paid by the contractor shall be added or deducted from the contract sum in the final account. In substantiation the contractor will require to produce to the Quantity Surveyor all quotations, invoices and receipted accounts as shall be necessary to show the details of the sums actually paid. iii) Any sum added by the contractor in these Bills of Quantities in respect of profits upon any prime costs will be deducted at the final settlement of accounts and the sum will be added to the amount of which will bear the same proportion to the sum added as the net amount properly expended to the original P.C sum. The profit is a management fee for arranging and taking responsibility of the sub-contract works or arranging for and checking the supply of materials and goods from nominated suppliers.	
	Carried to collection	100,000.00

ITEM	DESCRIPTION	AMOUNT
A	<u>NOMINATED SUB-CONTRACTORS</u> The contractor shall accept responsibility for providing the following services for nominated sub-contractors. i) GENERAL ATTENDANCE: The following services are described as "allow for general attendance" . This shall mean: a) Use for the purpose of the sub-contract works of any scaffolding belonging to or provided by the contractor while it remains so erected upon site, provided that no warranty or other liability on the part of the contractor or of his other sub-contractors shall be created or implied in regard to the fitness, condition or suitability of the said scaffolding b) Provision of water, lighting, watching and attendance for the purpose of the sub-contract works. c) Use of sanitary accommodation, mess rooms and welfare facilities. d) Provision of space for erecting of offices or stores or space for storage of plant and materials. ii) SPECIAL ATTENDANCE: The following services are described as "allow for special attendance" . This shall mean: a) Taking delivery and including the provision of unskilled labour necessary to attend upon the sub-contractors workmen for the purpose of unloading plants/equipment and materials of significant weight and/or size, when received upon the site and placing in position within the sub-contractor's storage space or store. b) Special Scaffolding, scaffolding additional to the Contractors scaffolding or Reassembling of contractor's scaffolding. c) Facilitating special power requirements during the course of the works.	
	Carried to collection	

ITEM	DESCRIPTION	AMOUNT
	CLAIMS A It shall be a condition of this contract that upon it becoming reasonably apparent to the Contractor that he has incurred losses and / or expenses due to any of the contract conditions, or by any other reason whatsoever, he shall present such a claim or intent to claim notice to the Engineer within the contract period. No claim shall be entertained upon the expiry of the said contract period. PAYMENTS B The tenderer's attention is drawn to the fact that the payments shall be made in accordance with Clause 14 of the Conditions of Contract Agreement. In order to facilitate this, a list of the general component elements for the works is given at the summary page of these specifications and the tenderer is requested to break down his tender sum commensurate to the said elements. PREVENTION OF ACCIDENT, DAMAGE OR LOSS C The Contractor is thus instructed to take reasonable care in the execution of the works as to prevent accidents, damage or loss and disruption of activities being carried out. The Contractor shall allow in his rates any expense he deemed necessary by taking such care within the site.	
	Carried to collection	

ITEM	DESCRIPTION	AMOUNT
A	<u>NOMINATED SUPPLIERS</u> The contractor shall take delivery all materials or goods supplied by the Nominated suppliers and shall sign a receipt as having received them in good order and condition. He shall offload, transport to site, unload, hoist, provide safe storage and thereafter be responsible for any loss or damage or replacement of any such lost or damaged articles at his own expense and shall return case if so required. Provision is made herein following each appropriate P.C sums for the costs of the foregoing services against items reading "take delivery of and fix only" <u>Fix Only:-</u> "Fix Only" shall mean take delivery on site, unload, store, unpack, assemble as necessary, distribute to position, hoist and fix only.	
B	<u>DIRECT CONTRACTS</u> Notwithstanding the foregoing conditions, the Government reserves the right to place a "Direct Contract" for any goods or services required in the works which are covered by a P.C. Sum in the Bills of Quantities and to pay for the same direct. In any such instances, profit relative to the P.C. Sum in the priced Bills of Quantities will be adjusted as described for P.C. Sums is allowed.	
C	<u>PROTECTION OF THE WORK</u> The Contractor shall cover up and protect all finished work liable to damage including provision of temporary roof, gutters, drains etc until the completion of the works. In the event of any damages occurring to the works, materials, sewers, drains, gullies, paths or other works on site in temporary possession of the contractor for the purpose of this contract either from weather, want of proper protection, defects, or insufficiency of the works or any other causes or whatsoever during the progress of the works, the contractor shall be responsible and without extra charge, make good all damage and pay all costs which may be levied. BLASTING OPERATIONS	
D	Blasting will only be allowed with the express permission of the Engineer in writing. All blasting operations shall be carried out at the Contractor's sole risk and cost in accordance with any Government regulations in force for the time being, and any special regulations laid down by the Engineer governing the use and storage of explosives.	
	Carried to collection	

ITEM	DESCRIPTION	AMOUNT
A	<u>PREVENTION OF NUISANCE</u> The works and such sections of the site necessary thereafter shall be under the entire care and control of the contractor during the whole period of the contract and shall take all possible precautions to prevent any nuisance, inconvenience or injury to the holder or occupiers of the existing or surrounding properties and to the public generally, and shall at all times keep all paths and roads affected by the works in a safe and clear state, and shall use proper precautions to ensure the safety of all wheeled traffic and pedestrians. The contractor shall provide appropriate screens to seal off the working area.	
	<u>REMOVAL OF PLANT AND RUBBISH ETC</u> The Contractor shall upon completion of the works remove and clear away all temporary buildings, plant, rubbish and unused materials, and shall leave the whole of the site of the works in a clean and tidy state to the satisfaction of the Engineer. He shall also remove all rubbish and dirt from the site at intervals or as directed by the Engineer. Particular care shall be taken in leaving windows, floors and fittings clean and the removal of all paint and cement stains therefrom. The contractor is expected to have established a well planned method of solid disposal of debris/garbage on and off the camp site	
	<u>CONTRACTOR'S SUPERINTENDENCE/SITE AGENT</u> The Contractor shall constantly keep on the works a literate English speaking Agent or Representative, competent and experienced in the kind of work involved who shall give his whole experience in the kind of work involved and shall give his whole time to the superintendence of the works. Such Agent or Representative shall receive on behalf of the Contractor all directions and instructions from the Engineer and such directions shall be deemed to have been given to the Contractor in accordance with the Conditions of Contract.	
	Carried to collection	

ITEM	DESCRIPTION	AMOUNT
	<u>TRAINING LEVY</u> A The Contractor's attention is drawn to legal notice No. 237 of 2007 which requires payment by the Contractor for a training levy and the contractor shall allow in the preliminaries of this contract (basic rates column) for all costs arising or resulting therefrom. Proof of payment of this Levy should be provided at the request of the Engineer <u>STANDARDS LEVY</u> B The Contractor is required to make payments to the Kenya Bureau of Standards as Standard Levy inline with the current current and prevailing regulations. <u>VALUE ADDED TAX (V.A.T.)</u> C The Contractor's attention is drawn to V.A.T PUBLIC NOTICE NO. 6 of 5th August, 1993 regarding the Finance Bill 1993 which expanded the V.A.T base to cover construction services amongst other items. The Contractor's attention is also drawn to all other notices issued by the government in relation to taxation. The Contractor shall familiarise himself with the said notices and allow in all his Bills of Quantities rates (Excluding P.C and Provisional Sums) for the net tax. (i.e less input tax where applicable) as required by law. Please note that allowing a lump sum tax either in preliminaries or in summary page shall not be acceptable. Any additional information and assistance concerning the application of the said notice should be directed to the office of the Commissioner of Value Added Tax D <u>PUBLIC PROCUREMENT CAPACITY BUILDING LEVY</u> The Contractor's attention is drawn to Legal Notice No. 206 of 2023, which establishes a Capacity Building Levy on all public procurement contracts in Kenya to fund training and professional development in the public procurement sector. The contractor is required to allow for all costs arising from or resulting from payment of the Levy in the basic rates of the Bill of Quantities.	
	Carried to collection	

ITEM	DESCRIPTION	AMOUNT
	<u>BILL NO. 1</u>	
	<u>GENERAL PRELIMINARIES</u>	
	<u>COLLECTION</u>	
	Carried from page 1/9	-
	Carried from page 1/10	-
	Carried from page 1/11	700,000.00
	Carried from page 1/12	500,000.00
	Carried from page 1/13	-
	Carried from page 1/14	-
	Carried from page 1/15	2,500,000.00
	Carried from page 1/16	1,000,000.00
	Carried from page 1/17	250,000.00
	Carried from page 1/18	2,000,000.00
	Carried from page 1/19	1,000,000.00
	Carried from page 1/20	-
	Carried from page 1/21	1,000,000.00
	Carried from page 1/22	2,500,000.00
	Carried from page 1/23	-
	Carried from page 1/24	100,000.00
	Carried from page 1/25	-
	Carried from page 1/26	-
	Carried from page 1/27	-
	Carried from page 1/28	-
	Carried from page 1/29	-
	Total for General Preliminaries Carried to Summary of Bill No. 1	11,550,000.00

Bill No. 2

General Preliminaries

ITEM	DESCRIPTION	UNIT	AMOUNT
	<u>BILL NO. 3</u> <u>PROJECT EXPENSES</u> Disclaimer: The following items are provisional sums provided by the Engineer. They are to be expended at the written instruction of the Engineer. Note: Tenderers should not attach, revise or add any conditions or specifications to the project supervisor's expenses		
A	Provide and maintain equipment for the Engineer's site office for the duration of the project	Lump Sum	2,500,000.00
B	Provide for supervision as follows: 1 No. Clerks of works, (Building and services) for the duration of project and 1 no. Work Inspectors, and 1 No Surveyor to be engaged on need basis.	Lump Sum	3,600,000.00
C	Allow a provisional sum of Kshs. Eight Million Five Hundred Thousand (8,500,000) for Project Management Team and other stakeholders facilitation allowances during project implementation, as and whenever it is necessary.	Lump Sum	8,500,000.00
D	Allow a P.C. Sum of KShs. 10,000,000 for supply of 1 Nr. (one)4WD station wagon SUV zero mileage vehicles of 2500 cc, or approved equivalent, including road licenses, number plates, insurances, etc. The vehicles to revert to Employer after completion of Contract. Minimum specifications include but not limited to the following: • Engine - 2.5 litres Turbo Diesel • Rear Differential Gear Lock • Braking System to include ABS (Anti-Lock Brake System) • Power Steering with adjustable Steering Column • Electronic Fuel Injection System. • 5 Speed Semi-Auto Transmission • Power Windows • Immobilizer and Alarm System • Fuel tank capacity between 80 litres and 100 litres	Item	10,000,000.00
E	Provide for the driver, fuels, maintenance, lubricants and servicing of the vehicle for kilometrage over 1,500 km per vehicle month.	Item	1,800,000.00

BUILDERS WORK

BLOCK TYPE A

Item	Description	Unit	QTY	Rate	Block A G+9 Amount
	<u>PROPOSED SOCIAL + AFFORDABLE UNITS BLOCK (TYPE A G+9)</u>				
	BILL NO.4 -BLOCK A				
	ELEMENT NO 1 - SUBSTRUCTURES (ALL PROVISIONAL)				
	<i>Site Clearance</i>				
A	Clear site of all grass, hedges, shrubs, bushes grub up roots, load and remove from site and dispose at designated local authority areas.	SM	682	30	20,460.00
					-
B	Bulk excavation to reduce levels depth not exceeding 1.5m commencing from existing ground level	CM	1,022	300	306,600.00
					-
C	Ditto but exceeding 1.5 metres but not 3 metres deep	CM	682	400	272,800.00
					-
D	Extra over all type of excavation for excavating in soft rock	CM	85	1,000	85,000.00
					-
E	Ditto excavation in hard rock class I	CM	85	1,400	119,000.00
					-
	Disposal of water				-
F	Allow for keeping the whole of the excavation free from all spring and running water by pumping or any other such means as may be necessary	Item		50,000	50,000.00
					-
	Planking and strutting				-
					-
G	Allow for maintaining and upholding the sides of excavations and keeping excavations clear of all fallen materials, rubbish etc	Item		50,000	50,000.00
					-
					-
					-
					-
					-
	Carried to collection				903,860.00

Item	Description	Unit	QTY	Rate	Block A G+9 Amount
	<u>Disposal of excavated material</u>				
A	Load,wheel and cart away surplus excavated material to a Local Authority designated dumping site or fill soil heaps as away from site instructed by the Project Engineer.	CM	656	200	131,200.00
					-
B	Return, fill and ram selected excavated material around sides of foundations.	CM	1049	100	104,900.00
					-
	Fillings				-
					-
C	Make up levels using approved imported materials: compacted in layers not exceeding 300mm thick with a roller: to the satisfaction of the Structural Engineer.	CM	821	1,450	1,190,450.00
					-
D	300mm thick hardcore bed: hand packed : compacted in layers not exceeding 150mm thick: to the satisfaction of the Structural Engineer	SM	456	500	228,000.00
					-
E	50 mm Stone dust/ Murrum blinding to surfaces of hardcore	SM	456	120	54,720.00
					-
	Anti - termite to treatment				-
F	Approved anti-termite treatment, with ten-year guarantee, sprayed to surfaces of hardcore strictly in accordance with manufacturer's instructions.	SM	532	150	79,800.00
					-
	Damp-proof membrane				-
G	1000 gauge polythene or other equal and approved damp-proof membrane, laid over blinded hardcore (measured separately) with 300mm side and end laps (measured nett-allow for laps); 1 No. layer: bedded in and including cement and sand (1:3) mortar	SM	532	100	53,200.00
					-
	Concrete Blinding				-
					-
	In situ concrete Class 15MPa: vibrated:				-
					-
H	50 mm thick blinding under raft	SM	682	400	272,800.00
					-
I	50 mm thick blinding under ground beams	SM	76	400	30,400.00
					-
	In- situ vibrated reinforced concrete Class 25 MPa: in:				-
					-
J	Raft Foundation	CM	511	13,200	6,745,200.00
					-
K	Ground Beams	CM	15	13,200	198,000.00
					-
L	100mm thick surface bed	SM	532	1,320	702,240.00
					-
M	Steps	CM	1	13,200	13,200.00
					-
	In- situ vibrated reinforced concrete Class 30 MPa: in:				-
					-
N	Columns	CM	22	14,200	312,400.00
					-
O	200mm thick Lift pit shaft wall	SM	31	2,840	88,040.00
					-
	Carried to collection				10,204,550.00

Item	Description	Unit	QTY	Rate	Block A G+9 Amount
	Ribbed reinforcement steel bars to KS 573:2014 : Grade 500 high tensile strength including bends, hooks, tying wire and distance blocks; to S.E's detail (Provisional)				
A	Assorted reinforcement	KGS	24718	165	4,078,470.00
	Mesh fabric reinforcement to KS/EAS 412;2 (2019) BRC A142;200 x 200mm, weighing 2.22kg/m² (measured net - no allowance) for laps; in two layers - top & bottom; including bends, tying wire and spacer blocks)				-
B	In ground floor slab	SM	532	400	212,800.00
	<u>Modular steel frame with steel plates covering formwork and/or marine board formwork: to:</u>				-
C	Sides of raft foundation	SM	106	500	53,000.00
D	Sides of ground beams	SM	152	500	76,000.00
E	Vertical sides to columns	SM	261	500	130,500.00
F	Vertical sides to lift shaft walls	SM	61	500	30,500.00
G	Edge of slab not exceeding 150mm girth	LM	142	75	10,650.00
H	Edges of risers 75 - 150mm high	LM	12	75	900.00
	Foundation Walling				-
	Natural quarry stone walling with a minimum of 7.0 N/mm2 bedded and jointed in cement and sand (1:4) mortar, reinforced with and including 25 x 3 mm thick hoop iron strips at every alternate course as described in;				-
I	200mm thick walls in foundations	SM	724	1,450	1,049,800.00
	<u>Pavings</u>				-
J	Supply and lay 600 x 600 x 50mm reinforced concrete precast paving slabs around the building including laying, spreading and compacting 100mm thick approved sand bed blinding, on and including 150mm thick compacted hardcore to Engineer's approval.	SM	85	1,400	119,000.00
	<u>Plinth</u>				-
	<u>25mm Thick cement and sand (1:4) rendering on concrete or masonry ; wood float finished; to</u>				-
K	Plinths externally	SM	64	350	22,400.00
	<u>Two coats black bitumastic paint on:</u>				-
L	Rendered surfaces	SM	64	300	19,200.00
	<u>Cement/sand (1:3) screed with approved waterproofing admixture, steel trowelled hard and smooth to receive waterproofing (m/s)</u>				-
M	20mm thick water proof cement/sand (1:4) screed to lift pit floor prepared to receive water proofing	SM	7	350	2,450.00
N	12mm thick water proof cement/sand (1:4) render to wall prepared to receive water proofing	SM	32	350	11,200.00
	Carried to collection				5,816,870.00

Item	Description	Unit	QTY	Rate	Block A G+9 Amount
	<u>Waterproofing</u> <u>CRYSTALLINE WATERPROOFING</u> <u>All areas indicated shall be waterproofed by approved cementitious system , provide 10 year guarantee, all to manufacturer's specifications and instructions as described:</u> <u>Two coat slurry application: waterproofing powder: 1kg per m2 per coat, minimum 2 coats to seal all expansion joints, holes, repaired areas and angle fillet or as per manufacturers instructions</u> <u>Application of render coat: waterproofing powder: 1kg per m2 at 4.5mm thick on slabs. Rate shall allow for hacking and preparing all concrete surfaces or as per manufacturers instructions</u>				
A	Horizontal surfaces of lift base	SM	7	1,300	9,100.00
B	Vertical surfaces of Lift shaft walls	SM	32	1,300	41,600.00
Carried to collection					50,700.00
COLLECTION					
	Total brought forward from page no: 1				903,860.00
	Total brought forward from page no: 2				10,204,550.00
	Total brought forward from page no: 3				5,816,870.00
	Total brought forward from above				50,700.00
	<u>ELEMENT NO. 1</u> <u>SUBSTRUCTURES</u>	Carried to Main summary			16,975,980.00

Item	Description	Unit	QTY	Rate	Block A G+9 Amount
	<u>PROPOSED SOCIAL + AFFORDABLE UNITS BLOCK</u> <u>(TYPE A G+9)</u>				
	BILL NO.4 -BLOCK A				
	ELEMENT No 2 - R.C FRAME				
	<u>Insitu concrete class 30MPa: vibrated: reinforced</u>				
A	Columns	CM	301	14,200	4,274,200.00
B	200mm thick Lift shaft wall	SM	417	2,840	1,184,280.00
	<u>Insitu concrete class 25MPa: vibrated: reinforced</u>				
C	Beams	CM	279	13,200	3,682,800.00
D	Upper Roof Beams	CM	3	13,200	39,600.00
E	130mm thick suspended slabs	SM	3757	1,716	6,447,012.00
F	150mm thick suspended slabs	SM	1206	1,980	2,387,880.00
G	150mm thick Tank Slab over stairwell	SM	82	1,980	162,360.00
H	150 mm thick landing	SM	103	1,980	203,940.00
I	Staircases	CM	40	13,200	528,000.00
J	100 mm thick reinforced concrete benches reinforced with BRC mesh A142 complete with formwork and all necessary plasterworks/screed work	SM	24	2,250	54,000.00
	Ribbed reinforcement steel bars to KS 573:2014 : Grade 500 high tensile strength including bends, hooks, tying wire and distance blocks; to S.E's detail (Provisional)				
K	Assorted reinforcement	Kg	148150	165	24,444,750.00
	<u>Modular steel frame with steel plates covering formwork and/or marine board formwork: to</u>				
L	Sides of columns	SM	3597	500	1,798,500.00
M	Sides and soffits of beams	SM	2695	500	1,347,500.00
N	Soffits of suspended slabs	SM	5044	500	2,522,000.00
O	Edges of slab over 75mm but not exceeding 150mm girth	LM	3138	75	235,350.00
P	To sloping soffits of staircases	SM	148	500	74,000.00
Q	Soffits of landings	SM	103	500	51,500.00
R	Riser of steps over 150 mm but not exceeding 225 mm girth	LM	480	113	54,000.00
S	Staircase string 300mm extreme girth and cut to profile of steps	LM	247	150	37,050.00
T	Edges of landing over 150 but ot exceeding 225mm high	LM	80	113	9,000.00
U	Sides of lift walls	SM	834	500	417,000.00
V	Edges of lift shaft openings 150-225mm high	LM	114	113	12,825.00
	<u>ELEMENT NO. 2</u> Carried to				
	<u>R.C FRAME</u> Main summary				49,967,547.00

Item	Description	Unit	QTY	Rate	Block A G+9 Amount
<u>PROPOSED SOCIAL + AFFORDABLE UNITS BLOCK</u> <u>(TYPE A G+9)</u> BILL NO.4 -BLOCK A ELEMENT No 3-WALLING <u>WALLING</u> <u>External Walling</u> <i><u>Machine cut quarry stone walling with a minimum of 7.0 N/mm² average compressive strength ;bedded and jointed in cement and sand (1:4) mortar, reinforced with and including 25 x 3 mm thick hoop iron strips at every alternate course as described in;</u></i>					
A	200mm thick walling Externally	SM	2967	1,450	4,302,150.00
B	200mm thick parapet walling	SM	308	1,450	446,600.00
<u>Internal Walling</u> <i><u>Machine cut quarry stone walling with a minimum of 7.0 N/mm² average compressive strength ;bedded and jointed in cement and sand (1:4) mortar, reinforced with and including 25 x 3 mm thick hoop iron strips at every alternate course as described in;</u></i>					
C	200mm thick walling Internally	SM	2214	1,450	3,210,300.00
D	150mm thick walling Internally	SM	2781	1,300	3,615,300.00
E	100mm thick walling Internally	SM	238	1,300	309,400.00
F	Approved hessian based damp proof course to 200mm thick walling in cement/sand mortar	LM	266	100	26,600.00
G	Approved hessian based damp proof course to 150mm thick walling in cement/sand mortar	LM	150	75	11,250.00
<i><u>Precast Concrete Breeze Ventilation Blocks</u></i>					
H	150 x 150mm wide x 50mm thick Pre Cast Concrete flower 3D breeze ventilation blocks bedded and jointed in cement and sand (1:4) mortar	SM	233	1,800	419,400.00
<u>COPING</u>					
I	600 x 350 wide x 100mm thick insitu reinforced concrete class 20Mpa coping, throated and weathered and jointing to columns with cement sand 1:4 mortar	NO	79	700	55,300.00
J	Ditto 300 x 100mm insitu concrete coping to walls twice weathered and throated;	LM	257	600	154,200.00
<u>Lintols</u>					
K	200mm x 200mm Deep lintols in reinforced concrete class 20MPa with and including 4No T10 and T8 stirups at 200mm centres; complete with formwork	LM	592	1,200	710,400.00
<u>ELEMENT NO. 3</u> Carried to <u>WALLING</u> Main summary					
					13,260,900.00

Item	Description	Unit	QTY	Rate	Block A G+9 Amount
	<u>PROPOSED SOCIAL + AFFORDABLE UNITS BLOCK</u> <u>(TYPE A G+9)</u> BILL NO.4 -BLOCK A ELEMENT NO 4-WINDOWS <u>MILD STEEL WINDOWS</u> Supply, fabricate and fix the following purpose made small pane mild steel casement windows to be fabricated from approved mild steel sections (atleast 14g 2mm thick) comprising of frame and casement incorporating permanent hooded high level ventilation panels infilled with mosquito gauze : window supplied complete with and including 12mm solid square burglar proofing bars fixed at 200mm centres both ways and metal fixing lugs including building into wall and making good, and all necessary iron mongery viz hinges, fasteners, and hasp including shop priming window with red oxide primer before delivery to site:- <u>A prime cost sum of Kshs 6,000 per Sqm has been allowed for fabrication of the above specified Steel casement Windows by AHP juakali artisans as approved by the Project Manager/Architect.</u> <u>The contractor's unit rate shall include the cost of transport, storage, fixing and all associated accesories in addition to the PC Rate.</u>				
A	Window, overall size 1200 X 1500mm high to Architects Details (Lounge)	NO	111	12,600	1,398,600.00
B	Ditto Size 1200 x 1500mm high (bedroom)	NO	150	12,600	-
C	Ditto Size 1200 x 1200mm high (Kitchen)	NO	80	10,080	1,890,000.00
D	Ditto Size 600 x 900mm high (WC/SH)	NO	171	3,780	-
	<u>Glazing</u>				806,400.00
E	4mm Thick clear sheet glass panes over 0.1 but not exceeding 0.5 square meters; fixing with premium putty	SM	601	1,100	-
F	Ditto; obscure	SM	97	1,200	661,100.00
	<u>Painting and Decorations</u>				-
	<u>On Metal work</u>				-
	<u>Prepare and apply aerosol spray painting in two finishing coats of first grade paint as per the manufacturer's printed instructions to: -</u>				-
G	General window and grille surfaces; over 300mm girth internal	SM	1397	300	-
					419,100.00
	Carried to Collection				5,937,980.00

Item	Description	Unit	QTY	Rate	Block A G+9 Amount
A	<u>Precast concrete window cill finishing fair on all exposed surfaces and hoisting and placing in position, bedding, jointing and pointing in pigmented cement and sand (1:3) mortar</u> 150 x 25mm thick Precast concrete window sill <u>Curtain rods;</u> 1.5mm thick, 20mm thick diameter twin powdercoated mild steel rod complete accessories to approval	LM	634	550	348,700.00 - - -
B		LM	525	1,300	682,500.00
	Carried to collection				1,031,200.00
	COLLECTION				
	Total brought forward from page no: 7				5,937,980.00
	Total brought forward from above				1,031,200.00
	<u>ELEMENT NO. 4</u> <u>WINDOWS</u>	Carried to the Main summary			6,969,180.00

Item	Description	Unit	QTY	Rate	Block A G+9 Amount
	<u>PROPOSED SOCIAL + AFFORDABLE UNITS BLOCK (TYPE A G+9)</u> BILL NO.4 -BLOCK A ELEMENT NO 5-DOORS <u>External Doors</u> <u>Hardwood Panelled Doors</u> <u>50mm thick approved hardwood PANEL doors with 12.5mm thick mahogany</u> <u>hardwood lipping to Architect's details, specifications and approval</u> <u>A prime cost sum at the rate indicated below has been allowed for fabrication of the above specified Hardwood panelled door leaves by AHP juakali artisans as approved by the Project Manager/Architect.</u> <u>The contractor's unit rate shall include the cost of transport, storage, fixing and all associated accesories in addition to the PC Rate.</u>				
A	Double leaf door size 1500 x 2400mm high (PC Rate Kshs 41,000)	NO	3	43,200	129,600.00
B	Single leaf Size 900 x 2400mm high (- PC Rate Kshs 25,000)	NO	111	27,000	- 2,997,000.00
C	<u>Frames and frame finishes in mahogany timber:</u> 25 x 25mm quadrant beading (PC Rate Kshs 75)	LM	663	100	- 66,300.00
D	25 x 50mm architrave with two labours, plugged (PC Rate Kshs 150)	LM	663	200	- 132,600.00
E	150mm x 50mm frame with three labours; chamfered edges; plugged (PC Rate Kshs 1400)	LM	663	1,550	- 1,027,650.00
	Carried to collection				4,353,150.00

Item	Description	Unit	QTY	Rate	Block A G19 Amount
	<u>Internal Doors</u> <u>Flush timber doors</u> <u>50 mm thick Semi Solid cored flush doors with plywood facing to receive painting (m.s) all to Architects details, specifications and approval</u> <u>A prime cost sum at the rate indicated below for fabrication of the above specified Timber Flush door leaves by AHP juakali artisans as approved by the Project Manager/Architect.</u> <u>The contractor's unit rate shall include the cost of transport, storage, fixing and all associated accesories in addition to the PC Rate.</u>				
F	Door size 900mm x 2400mm High comprising of 1 No Opennable leaf size 800 x 2100mm high including fixed fanlight size 900 x 300mm high in 4mm clear glass (measured separately) (PC Rate Kshs 4,000)	NO	150	5,200	780,000.00
G	Ditto 800 x 2100mm high comprising of 1No. Opennable leaf size 700 x 2100mm high (PC Rate Kshs 3,500)	NO	171	4,700	803,700.00
	4mm Thick clear sheet glass fixing with matching timber glazing beads to timber frames				-
H	In panes exceeding 0.1 sqm but not exceeding 0.5 square metres.	SM	41	1,100	45,100.00
I	<u>Frames and frame finishes in cypress Timber</u> 25 x 25mm quadrant (PC Rate Kshs 75)	LM	1710	100	171,000.00
J	25 x 50mm architrave with two labours, plugged (PC Rate Kshs 150)	LM	1710	200	342,000.00
K	150mm x 50mm transome with three labours; chamfered edges; plugged (PC Rate Kshs 800)	LM	135	950	128,250.00
L	150mm x 50mm frame with three labours; chamfered edges; plugged (PC Rate Kshs 800)	LM	1710	950	1,624,500.00
	Carried to collection				3,894,550.00

Item	Description	Unit	QTY	Rate	Block A G+9 Amount
	<u>Painting and decorating</u>				
	<u>Priming back of frame with an aluminium or equivalent and approved wood primer</u>				
A	Surfaces not exceeding 100mm girth	LM	4745	30	142,350.00
B	Surfaces over 100mm but not exceeding 200mm girth	LM	2373	60	142,380.00
	<u>Prepare Knot, prime, stop and apply one undercoat and two finishing coats first grade timber quality paint to wood surfaces as per the manufacturer's printed instructions</u>				-
C	General timber surfaces	SM	1750	300	525,000.00
D	Surfaces over 200mm but not exceeding 300mm girth	LM	2373	90	213,570.00
E	Architraves: not exceeding 100 mm girth	LM	2373	30	71,190.00
F	Quadrant beading : not exceeding 100 mm girth	LM	2373	30	71,190.00
	Ironmongery				-
	Supply and fix the following ironmongery to timber complete with matching screws and keys to the approval of the Architect				-
G	100mm pressed steel Butt Hinges	PRS	657	350	229,950.00
H	Stainless steel 3 Lever Mortice Door Lock with handle furniture set;(keyhole escutcheons, cylinder and latch)	NO	114	3,500	399,000.00
I	Ditto: but 2 Lever Door Lock with handle	NO	321	2,800	898,800.00
J	Door fixing cramps	NO	435	120	52,200.00
K	200 x 75 x 3mm perspex door signage with door numbers as per Architect detail	NO	0	500	-
L	Block sign with block type, name and number in exterior quality paint as per Architect detail	NO	0	2,000	-
	Supply and fix the following purpose-made signage in printed vinyl panels in approved background and text colours in accordance to the architects later signage design fixed to finished surfaces in the approved adhesive and matching screws as applicable to;				-
M	200 x 50mm Main door signage	NO	130	400	52,000.00
N	150x450mm Floor Level Indicator Sign	NO	20	1,000	20,000.00
O	500x100 Floor Wayfinding/Directional Sign	NO	20	1,000	20,000.00
P	200x200mm Duct door Signage	NO	260	800	208,000.00
Q	450x200 mm Fire Exit Signage	NO	65	2,000	130,000.00
R	1000x300 mm Block Identification Signage	NO	1	6,000	6,000.00
	Carried to Collection				3,181,630.00
	COLLECTION				
	Total brought forward from page no: 10				4,353,150.00
	Total brought forward from page no: 11				3,894,550.00
	Total brought forward from above				3,181,630.00
	ELEMENT NO. 5	Carried to			
	DOORS	Main summary			11,429,330.00

Item	Description	Unit	QTY	Rate	Block A G+9 Amount
PROPOSED SOCIAL + AFFORDABLE UNITS BLOCK (TYPE A G+9) BILL NO.4 -BLOCK A ELEMENT NO 6 - EXTERNAL FINISHES EXTERNAL WALL FINISHES External Render <u>Cement and sand (1:3) render:wood floated: on concrete or blockwork:</u> <u>to</u>					
A	15mm thick to beams, Columns, Slab Moulds and walling externally	SM	5555	350	1,944,250.00
					-
B	Extra over horizontal and vertical pointing in 10mm thick rod in cement and sand mix (1:3) mortar including one coat Bituminous paint	SM	0	350	-
					-
	External Painting				-
					-
	<u>Prepare and apply one coat Alkali Resistant primer followed by two coats of silicon exterior Emulsion paint in accordance with the manufacturers written instructions and to the satisfaction of the architect to</u>				-
C	Concrete/masonry surfaces externally-Beam, Column and Slab Moulds	SM	5555	350	1,944,250.00
					-
	ROOF FINISHES				-
	Cement and sand (1:3) screeded beds: on concrete:				-
	complete with coloured pigmentation additives and hardener to:				-
					-
D	50mm average waterproofed lightweight screed laid to falls and crossfalls to roof slabs -upper roof including gutter bases	SM	617	400	246,800.00
					-
	<u>Prepare and apply to vertical/horizontal surfaces 4mm thick APP/EPDM water proofing or other equal and approved membrane with surface finish weighing 4kg/sm; laid on primer with torch-on process ;by an approved specialist all in accordance with the manufacturers instructions including provision of a written ten (10) year anti leak auarantee.</u>				-
E	4mm thick APP membrane applied to roof slabs	SM	617	1,400	863,800.00
					-
F	Ditto to skirting 200mm high	LM	358	280	100,240.00
					-
G	Dress membrane around 100mm rainwater outlet	No.	12	200	2,400.00
					-
	<u>The Following Flat roof concrete tiles fixed with approved adhesive, laid and jointed with waterproofing bituminous compound</u>				-
H	20mm thick interlocking Concrete tiles of size 225 x 225mm with and including cement sand (1:4) backing to falls	SM	582	1,500	873,000.00
					-
	ELEMENT NO. 6	Carried to			
	EXTERNAL FINISHES	Main summary			5,974,740.00

Item	Description	Unit	QTY	Rate	Block A G+9 Amount
	<u>PROPOSED SOCIAL + AFFORDABLE UNITS BLOCK</u> <u>(TYPE A G+9)</u>				
	BILL NO.4 -BLOCK A				
	ELEMENT NO 7 - INTERNAL FINISHES				
	<u>Internal Wall Finishes</u>				
	<u>Cement and sand (1:3) backings</u>				
A	15mm thick to receive Ceramic Wall tiles	SM	2757	350	964,950.00
B	To receive porcelain wall tiles (m.s.) (Lift Lobby)	SM	172	350	60,200.00
	<u>Ceramic wall tiles</u>				-
	<u>Allow a Prime Cost supply/ Purchase only rate of Ksh. 1000 per SM (In addition to the cost of purchase above, the contractor's rate shall include cost of transportation, offloading, storage, fixing including all necessary adhesives, grouting and accessories</u>				-
C	Supply and Fix ceramic wall tiles on prepared backings(m.s) with proprietary adhesive; jointed and pointed in matching coloured proprietary grouting; including pvc spacers and expansion joint as necessary: all to Architect's approval. - Wall Surfaces	SM	2757	1,650	4,549,050.00
	<u>Porcelain wall tiles</u>				-
	<u>Allow a Prime Cost supply/ Purchase only rate of Ksh. 1500 per SM (In addition to the cost of purchase above, the contractor's rate shall include cost of transportation, offloading, storage, fixing including all necessary adhesives, grouting and accessories</u>				-
D	Supply and Fix porcelain wall tiles on prepared backings(m.s) with proprietary adhesive; jointed and pointed in matching coloured proprietary grouting; including pvc spacers and expansion joint as necessary: all to Architect's approval. - Wall Surfaces	SM	172	2,150	369,800.00
	<u>15mm (minimum) two coat cement, sand (1:3) plaster complete with wire gauze anti-crack mechanism at the intersection of masonry walling and concrete beams as described to:-</u>				-
E	Concrete/masonry surfaces	SM	12110	350	4,238,500.00
	<u>Painting and Decoration</u>				-
	<u>Prepare, Skim and apply Emulsion or universal undercoat followed by 2 finishing coats of soft satin Emulsion paint in accordance with the manufacturers written instructions and to the satisfaction of the architect to</u>				-
F	Plastered concrete/masonry surfaces internally	SM	12110	300	3,633,000.00
	Carried to Collection				13,815,500.00

Item	Description	Unit	QTY	Rate	Block A G+9 Amount
	<u>Floor Finishes</u>				
	<u>Cement and sand (1:3) screeds, backings, beds etc</u>				
A	32mm bed finished to receive floor Tiles (m.s)	SM	3487	350	1,220,450.00
					-
B	32mm Thick coloured cement sand screed mix 1:3 finished with red oxide to approval	SM	1036	350	362,600.00
					-
	<u>Cement and sand (1:3) backings etc</u>				-
					-
C	32mm bed finished to receive ceramic tiles to surfaces of Landings (m.s)	SM	103	350	36,050.00
					-
D	25 x 300 mm wide treads to receive ceramic tiles (m.s)	LM	432	105	45,360.00
					-
E	20 x 150mm risers to receive ceramic tiles (m.s)	LM	480	53	25,200.00
					-
	<u>Ceramic Floor tiles</u>				-
					-
	<u>Allow a Prime Cost supply/ Purchase only rate of Ksh. 1000 per SM (In addition to the cost of purchase above, the contractor's rate shall include cost of transportation, offloading, storage, fixing including all necessary adhesives, grouting and accessorie</u>				-
					-
F	Supply and Fix Ceramic tiles; on prepared bed(m.s) with proprietary adhesive; jointed and pointed in matching coloured proprietary grouting: including pvc spacers and expansion joint as necessary: all to Architect's approval.	SM	2365	1,650	3,902,250.00
					-
G	Ditto Non Slip Ceramic Tiles	SM	1122	1,650	1,851,300.00
					-
H	Ditto 100mm wide Wall Skirtings	LM	3996	165	659,340.00
					-
I	Non Slip Ceramic Tiles to surfaces of Landings	SM	103	1,650	169,950.00
					-
J	Non Slip Ceramic Tiles to 300 mm wide treads	LM	432	495	213,840.00
					-
K	Non Slip Ceramic Tiles to 150mm risers	LM	480	248	118,800.00
					-
	<u>Wet areas waterproofing - Kitchens, Toilets etc.</u>				-
					-
					-
L	Liquid membrane - High build bitumen/ rubber latex emulsion with excellent adhesion, liquid applied to form a seamless flexible water proof/ vapour proof membrane as as per the manufacturer's printed instructions to; solid content =65%, Rubber Content=10% in dried film)				-
					-
M	To floors and walls	m ²	1567	1,500	2,350,500.00
					-
	Staircase soffit finishes				-
					-
	<u>15mm (minimum) two coat cement, sand (1:3) plaster complete with wire gauze anti-crack mechanism at the intersection of masonry walling and concrete beams as described to:-</u>				-
					-
N	Soffits of staircase landing	SM	103	350	36,050.00
					-
O	Ditto to sloping soffites exceeding 15° from horizontal	SM	148	350	51,800.00
					-
P	Staircase string 300mm extreme girth and cut to profile of steps	LM	247	105	25,935.00
					-
	Carried to Collection				11,069,425.00

Item	Description	Unit	QTY	Rate	Bid A.G. Amount
	<u>Paint works</u>				
	<u>Prepare, skim and apply Emulsion or universal undercoat followed by 3 coats of soft satin Emulsion paint in accordance with the manufacturers written instructions and to the satisfaction of the architect to</u>				
A	Soffits of staircase landing	SM	103	300	30,900.00
B	Ditto to sloping soffites exceeding 15° from horizontal	SM	148	300	- 44,400.00
C	Staircase string 300mm extreme girth and cut to profile of steps	LM	247	90	- 22,230.00
	<u>Ceiling finishes</u>				- -
	<u>15mm (minimum) two coat cement, sand (1:3) plaster complete with wire gauze anti-crack mechanism at the intersection of masonry walling and concrete beams as described to:-</u>				- -
D	Soffits of Concrete surfaces	SM	5044	350	1,765,400.00 - -
	<u>Painting and Decoration</u>				- -
	<u>Prepare, skim and apply Emulsion or universal undercoat followed by 3 coats of soft satin Emulsion paint in accordance with the manufacturers written instructions and to the satisfaction of the architect to</u>				- -
E	Plastered ceilings	SM	5044	300	1,513,200.00
	Carried to Collection				3,376,130.00

Item	Description	Unit	QTY	Rate	Block A G+9 Amount
	COLLECTION Total brought forward from page no: 12 Total brought forward from page no: 13 Total brought forward from page no: 14				13,815,500.00 11,069,425.00 3,376,130.00
	<u>ELEMENT NO. 7</u> Carried to INTERNAL FINISHES Main summary				28,261,055.00

Item	Description	Unit	QTY	Rate	Block A G+9 Amount
	PROPOSED SOCIAL + AFFORDABLE UNITS BLOCK (TYPE A G+9) BILL NO.4 -BLOCK A ELEMENT NO 8- BALUSTRADING AND RAILING <u>Balustrades and staircase railings</u>				
A	900mm high mild Steel balustrade; comprising 60 x 10mm mild Steel balusters at 900mm centres; bolted to base plate and tread (m.s), with 7No. 25mm diameter horinzontal bars, and 75x4mm diameter CHS mild Steel handrail part welded into 60x10mm balustrades; to Architects drawings	LM	159	5,500	874,500.00
B	50mm diameter x 3mm CHS handrail supported by 50 x 50 x 3mm thick SHS balusters anchored to slab with and including w/steel bolts and plates at 1800mm centres smooth welded 100mm high (Staircase)	LM	90	800	72,000.00
	<u>Prepare, prime and apply one undercoat and two finishing coats first quality gloss oil paint on.</u>				- - -
C	General metal surfaces of balustrading (both sides measured overall)	SM	400	300	120,000.00
	<u>ELEMENT NO. 8</u> Carried to the <u>BALUSTRADE AND RAILING</u> Main summary				1,066,500.00

Item	Description	Unit	QTY	Rate	Block A G+9 Amount
	PROPOSED SOCIAL + AFFORDABLE UNITS BLOCK (TYPE A G+9) BILL NO.4 -BLOCK A ELEMENT NO 9 - JOINERY FITTINGS <u>Allow for providing materials, labour and construct fixtures and fittings as per Architects drawings of the following JOINERY FITTINGS AND FIXTURES complete with associated iron mongery;</u> <u>NOTE: All blockboard, MDF boards,etc in joinery works shall be lipped with hardwood lipping all round before fixing.</u> <u>High level cupboards</u> <u>400mm Wide high level kitchen cupboards in 18mm laminated mdf sides & shelves complete with doors, top, bottom & divisions with and including all necessary ironmongery; to Architect's details</u>				
A	High level storage cupboard units 600mm high x 400mm deep <u>Low level kitchen cupboards</u> <u>Low level kitchen worktops with 600x600x10mm porcelain tiles top on 100mm thick reinforced concrete slab with A142 BRC mesh, formwork to soffits and slab edges, plater to soffits of slab, screed to top slab, 100mm thick plastered steeper walls, including 300x300x6mm thick ceramic wall tiles on both sides of the wall: 100mm plastered mass concrete plinths in concrete class 15MPa: 18mm laminated mdf sides & shelves complete with doors, shelves, drawers, cutting tiles for kitchen sink (m.s) & all necessary ironmongery; to Architect's details and approval</u>	LM	230.00	5,000	1,150,000.00 - -
B	Low level kitchen cupboards 850mm high x 600mm deep <u>Bedroom Wardrobes</u> <u>600mm Wide x 2200mm high wardrobes in 18mm laminated mdf sides, divisions, drawers, shelves ; ; complete with sliding/ side hung doors, shelves, divisions, shoe racks, hanging rails, drawers, & all necessary ironmongery; 100mm plastered mass concrete plinths in concrete class 15MPa ; to Architect's details</u>	LM	230	9,500	- 2,185,000.00 - -
C	Bedroom wardrobes size 2200mm high x 600mm deep in bedrooms Duct doors <u>50mm thick laminated MDF duct doors; complete with frames, ironmongery and all necessary paintwork</u>	LM	217	14,000	- 3,038,000.00 - -
D	Duct doors size 300mm wide x 2100mm high	NO.	10	5,000	50,000.00
E	Duct doors size 350mm wide x 2100mm high	NO.	10	5,000	50,000.00
F	Duct doors size 600mm wide x 2100mm high	NO.	50	7,000	350,000.00
G	Duct doors size 500mm wide x 2100mm high	NO.	20	7,000	140,000.00
H	Duct doors size 700mm wide x 2100mm high	NO.	20	7,000	140,000.00
I	Duct doors size 800mm wide x 2100mm high	NO.	20	7,000	140,000.00
	ELEMENT NO. 9 JOINERY & FITTINGS	Carried to the Main summary			7,243,000.00

Item	Description	Unit	QTY	Rate	Block A G+9 Amount
A	PROPOSED SOCIAL + AFFORDABLE UNITS BLOCK (TYPE A G+9) ELEMENT NO 10 - BUILDERS WORKS IN CONNECTION WITH SERVICES (BWICWS) <u>Plumbing, Drainage and Mechanical Installations</u> Inspect all drawings and Bills of Quantities for Plumbing, Drainage and Mechanical Installations and allow for all Builder's work associated with the installations; including cutting away and making good after installing a concealed drainage system; including cutting or leaving all holes notches, mortices, sinkings and chases in both the structure and its coverings; including but not limited to forming recess in masonry	Item		100,000	100,000.00
	<u>Electrical Services Installations</u> Inspect all drawings and Bills of Quantities for Electrical Installations and allow for all Builder's work associated with the Electrical installations, including cutting away for and making good after Electrician installing a concealed conduit system including cutting or leaving all holes notches, mortices, sinkings and chases in both the structure and its coverings; including but not limited to forming recess in masonry	Item		100,000	100,000.00
	ELEMENT NO. 10				
	Carried to the BUILDERS WORKS IN CONNECTION WITH SERVICES (BWICWS) Main summary				200,000.00

Item	Description	Unit	QTY	Rate	Block A G+9 Amount
<u>PROPOSED SOCIAL + AFFORDABLE UNITS BLOCK</u> <u>(TYPE A G+9)</u> BILL NO.4 -BLOCK A MAIN SUMMARY					
			Page No.		Amount (Kshs.)
1	Substructures		A/4		16,975,980.00
2	Reinforced Concrete Frame		A/5		49,967,547.00
3	Walling		A/6		13,260,900.00
4	Windows		A/8		6,969,180.00
5	Doors		A/10		11,429,330.00
6	External Finishes		A/11		5,974,740.00
7	Internal Finishes		A/15		28,261,055.00
8	Balustrade and Railing		A/16		1,066,500.00
9	Joinery and Fittings		A/17		7,243,000.00
10	Builders works in connection with services		A/18		200,000.00
<u>TOTAL FOR 1NO. TYPE A (G+9) BLOCK</u>					141,348,232.00
NO. OF BLOCKS					
MULTIPLY BY 3.NO OF BLOCK					3.00
<u>TOTAL FOR TYPE A (G+9) BLOCK(S) CARRIED TO GRAND SUMMARY</u>					424,044,696.00

BLOCK TYPE B

Item	Description	Unit	QTY	Rate	Amount
	<u>Disposal of excavated material</u>				
A	Load, wheel and cart away surplus excavated material to a Local Authority designated dumping site or fill soil heaps as away from site instructed by the Project Engineer.	CM	1484	200	296,800.00
					-
B	Return, fill and ram selected excavated material around sides of foundations.	CM	862	100	86,200.00
					-
	Fillings				-
					-
C	Make up levels using approved imported materials: compacted in layers not exceeding 300mm thick with a 10 tonne roller: to the satisfaction of the Structural Engineer.	CM	1407	1,450	2,040,150.00
					-
D	300mm thick hardcore bed: hand packed : compacted in layers not exceeding 150mm thick: to the satisfaction of the Structural Engineer	SM	938	500	469,000.00
					-
E	50 mm Stone dust/ Murrum blinding to surfaces of hardcore	SM	938	120	112,560.00
					-
	Anti - termite to treatment				-
F	Approved anti-termite treatment, with ten-year guarantee, sprayed to surfaces of hardcore strictly in accordance with manufacturer's instructions.	SM	938	150	140,700.00
					-
	Damp-proof membrane				-
G	1000 gauge polythene or other equal and approved damp-proof membrane, laid over blinded hardcore (measured separately) with 300mm side and end laps (measured nett-allow for laps); 1 No. layer: bedded in and including cement and sand (1:3) mortar	SM	938	100	93,800.00
					-
	Concrete Blinding				-
					-
	In situ concrete Class 15MPa: vibrated:				-
					-
H	50 mm thick blinding under raft	SM	938	400	375,200.00
					-
H	50 mm thick blinding under column bases	SM	0	400	-
					-
I	Ditto; under ground beams	SM	94	400	37,600.00
					-
	In- situ vibrated reinforced concrete Class 25MPa: in:				-
					-
I	Column bases	CM	0	13,200	-
					-
J	Ground Beams	CM	19	13,200	250,800.00
					-
N	100mm thick surface bed	SM	672	1,320	887,040.00
					-
	In- situ vibrated reinforced concrete Class 30MPa: in:				-
					-
J	Raft	CM	704	14,200	9,996,800.00
					-
K	Columns	CM	18	14,200	255,600.00
					-
L	200mm thick Lift pit shaft wall	SM	24	2,840	68,160.00
	Carried to collection				15,110,410.00

Item	Description	Unit	QTY	Rate	Amount
	Ribbed reinforcement steel bars to KS 573:2014 : Grade 500 high tensile strength including bends, hooks, tying wire and distance blocks; to S.E's detail (Provisional)				
A	Assorted reinforcement	KGS	31,160.00	165	5,141,400.00
					-
	Mesh fabric reinforcement to KS/EAS 412;2 (2019) BRC A142;200 x 200mm, weighing 2.22kg/m² (measured net - no allowance) for laps; in two layers - top & bottom; including bends, tying wire and spacer blocks)				-
					-
B	In ground floor slab	SM	672	400	268,800.00
					-
	<u>Modular steel frame with steel plates covering formwork and/or marine board formwork: to:</u>				-
					-
C	Sides of the raft	SM	103	500	51,500.00
					-
D	Sides of ground beam	SM	188	500	94,000.00
					-
E	Vertical sides to columns	SM	219	500	109,500.00
					-
F	Vertical sides to lift shaft walls	SM	48	500	24,000.00
					-
G	Edge of slab not exceeding 150mm girth	LM	162	75	12,150.00
					-
	Foundation Walling				-
					-
	Natural quarry stone walling with a minimum of 7.0 N/mm² bedded and jointed in cement and sand (1:4) mortar, reinforced with and including 25 x 3 mm thick hoop iron strips at every alternate course as described in;				-
					-
H	200mm thick walls in foundations	SM	266	1,450	385,700.00
					-
	<u>Pavings</u>				-
					-
I	Supply and lay 600 x 600 x 50mm reinforced concrete precast paving slabs around the building including laying, spreading and compacting 100mm thick approved sand bed blinding, on and including 150mm thick compacted hardcore to Engineer's approval.	SM	97	1,400	135,800.00
					-
	<u>Plinth</u>				-
					-
	<u>25mm Thick cement and sand (1:3) rendering on concrete or masonry ; wood float finished; to</u>				-
					-
J	Plinths externally	SM	49	350	17,150.00
					-
	Two coats black bitumastic paint on:				-
					-
K	Rendered surfaces	SM	49	300	14,700.00
					-
	Carried to collection				6,254,700.00

Item	Description	Unit	QTY	Rate	Amount
	<u>Cement/sand (1:3) screed with approved waterproofing admixture, steel trowelled hard and smooth to receive waterproofing (m/s)</u>				
A	20mm thick water proof cement/sand (1:3) screed to lift pit floor prepared to receive masterseal water proofing	m ²	9	350	3,150.00
B	12mm thick water proof cement/sand (1:3) render to wall prepared to receive water proofing	m ²	26	350	9,100.00
	<u>Waterproofing</u>				-
	<u>CRYSTALLINE WATERPROOFING</u>				-
	<u>All areas indicated shall be waterproofed by approved cementious system , provide 10 year guarantee, all to manufacturer's specifications and instructions as described:</u>				-
	<u>Two coat slurry application: waterproofing powder: 1kg per m2 per coat, minimum 2 coats to seal all expansion joints, holes, repaired areas and angle fillet or as per manufacturers instructions</u>				-
	<u>Application of render coat: waterproofing powder: 1kg per m2 at 4.5mm thick on slabs. Rate shall allow for hacking and preparing all concrete surfaces or as per manufacturers instructions</u>				-
C	Horizontal surfaces of lift base	m ²	9	1,300	11,700.00
D	Vertical surfaces of Lift shaft walls	m ²	26	1,300	33,800.00
	Carried to collection				57,750.00
	COLLECTION				
	Total brought forward from page no:				1,306,240.00
	Total brought forward from page no:				15,110,410.00
	Total brought forward from page no:				6,254,700.00
	Total brought forward from above				57,750.00
	<u>ELEMENT NO. 1</u>	Carried to			
	<u>SUBSTRUCTURES</u>	Main summary			22,729,100.00

Item	Description	Unit	QTY	Rate	Amount
	PROPOSED APARTMENTS FOR AFFORDABLE HOUSING PROGRAM (TYPE B G+9)				
	BILL NO.1-BUILDERS WORKS				
	ELEMENT No 2 - R.C FRAME				
	<u>In situ concrete class 30MPa: vibrated: reinforced</u>				
A	Columns	CM	273	14,200	3,876,600.00
					-
B	Lift shaft wall	SM	415	2,840	1,178,600.00
					-
	<u>In situ concrete class 25MPa: vibrated: reinforced</u>				-
					-
C	Beams	CM	312	13,200	4,118,400.00
					-
D	Upper Roof Beams	CM	14	13,200	184,800.00
					-
E	130mm thick suspended slabs	SM	4641	1,716	7,963,956.00
					-
F	Ditto but 150mm thick	SM	2094	1,980	4,146,120.00
					-
G	150mm thick Tank Slab over stairwell	SM	48	1,980	95,040.00
					-
H	150 mm thick landing	SM	101	1,980	199,980.00
					-
I	Staircases	CM	40	13,200	528,000.00
					-
	Ribbed reinforcement steel bars to KS 573:2014 : Grade 500 high tensile strength including bends, hooks, tying wire and distance blocks; to S.E's detail (Provisional)				-
					-
J	Assorted reinforcement	Kg	165793	165	27,355,845.00
					-
	<u>Modular steel frame with steel plates covering formwork and/or marine board formwork: to</u>				-
					-
K	Sides of columns	Sm	3240	500	1,620,000.00
					-
L	Sides and soffits of beams	Sm	3086	500	1,543,000.00
					-
M	Soffits of suspended slabs	Sm	6782	500	3,391,000.00
					-
N	Edges of slab over 150mm but not exceeding 225mm girth	Lm	2236	113	251,550.00
					-
O	To sloping soffits of staircases	SM	154	500	77,000.00
					-
P	Soffits of landings	SM	101	500	50,500.00
					-
Q	Riser of steps over 150 mm but not exceeding 225 mm girth	LM	480	113	54,000.00
					-
R	Staircase string 300mm extreme girth and cut to profile of steps	LM	268	150	40,200.00
					-
S	Edges of landing over 150 but ot exceeding 225mm high	Lm	96	113	10,800.00
					-
T	Edges of lift shaft openings 150-225mm high	LM	114	113	12,825.00
					-
U	Sides of lift walls	SM	829	500	414,500.00
					-
	<u>ELEMENT NO. 2</u>	Carried to			
	<u>R.C FRAME</u>	Main summary			57,112,716.00

Item	Description	Unit	QTY	Rate	Amount
	PROPOSED APARTMENTS FOR AFFORDABLE HOUSING PROGRAM (TYPE B G+9)				
	BILL NO.1-BUILDERS WORKS				
	ELEMENT No 3-WALLING				
	<u>WALLING</u>				
	<u>External Walling</u>				
	<i>Machine cut quarry stone walling with a minimum of 7.0 N/mm2 average compressive strength ;bedded and jointed in cement and sand (1:4) mortar, reinforced with and including 25 x 3 mm thick hoop iron strips at every alternate course as described in;</i>				
A	200mm thick walling Externally	Sm	3440	1,450	4,988,000.00
					-
B	200mm thick parapet walling	Sm	193	1,450	279,850.00
					-
	<u>Internal Walling</u>				-
	<i>Machine cut quarry stone walling with a minimum of 7.0 N/mm2 average compressive strength ;bedded and jointed in cement and sand (1:4) mortar, reinforced with and including 25 x 3 mm thick hoop iron strips at every alternate course as described in;</i>				-
					-
C	200mm thick walling Internally	Sm	2377	1,450	3,446,650.00
					-
D	150mm thick walling Internally	Sm	2973	1,300	3,864,900.00
					-
E	Approved hessian based damp proof course to 200mm thick walling in cement/sand mortar	Lm	282	100	28,200.00
					-
F	Approved hessian based damp proof course to 150mm thick walling in cement/sand mortar	Lm	179	75	13,425.00
					-
	<u>Precast Concrete Breeze Ventilation Blocks</u>				-
					-
G	150 x 150mm wide x 50mm thick Pre Cast Concrete flower 3D breeze ventilation blocks bedded and jointed in cement and sand (1:4) mortar	Sm	405	1,800	729,000.00
					-
	COPING				-
					-
	600 x 350 wide x 100mm thick insitu reinforced concrete class 20Mpa coping, throated and weathered and jointing to columns with cement sand 1:4 mortar	No	84	700	58,800.00
					-
H	Ditto 300 x 100mm concrete coping to walls twice weathered and throated;	LM	209	600	125,400.00
					-
	Lintols				-
					-
I	200mm x 200mm Deep lintols in reinforced concrete class 20MPa with and including 4No T10 and T8 stirups at 200mm centres; complete with formwork	LM	798.00	1,200	957,600.00
	<u>ELEMENT NO. 3</u>	Carried to			
	<u>WALLING</u>	Main summary			14,491,825.00

Item	Description	Unit	QTY	Rate	Amount
	<u>Precast concrete window cill finishing fair on all exposed surfaces and hoisting and placing in position, bedding, jointing and pointing in pigmented cement and sand (1:3) mortar</u>				
A	150 x 25mm thick Precast concrete window sill	LM	808	550	444,400.00
	<u>Curtain rods;</u>				-
B	1.5mm thick, 20mm thick diameter twin powdercoated mild steel rod complete accessories to approval	LM	624	1,300	811,200.00
	Carried to collection				1,255,600.00
	COLLECTION				
	Total brought forward from page no:				6,337,100.00
	Total brought forward from page no:				1,255,600.00
	<u>ELEMENT NO. 4</u>				
	<u>WINDOWS</u>				
	Carried to the Main summary				7,592,700.00

Item	Description	Unit	QTY	Rate	Amount
	PROPOSED APARTMENTS FOR AFFORDABLE HOUSING PROGRAM (TYPE B G+9)				
	BILL NO.1-BUILDERS WORKS				
	ELEMENT NO 5-DOORS				
	<u>External Doors</u>				
	<u>Hardwood Panelled Doors</u>				
	50mm thick approved hardwood PANEL doors with 12.5mm thick Meru oak hardwood lipping to Architect's details, specifications and approval				
	A prime cost sum at the rate indicated below has been allowed for fabrication of the above specified Hardwood panelled door leaves by AHP juakali artisans as approved by the Project Manager/Architect.				
	The contractor's unit rate shall include the cost of transport, storage, fixing and all associated accesories in addition to the PC Rate.				
A	Double leaf door size 1500 x 2400mm high (PC Rate Kshs 41,000)	NO	3	43,200	129,600.00
B	Single leaf Size 900 x 2400mm high (- PC Rate Kshs 25,000)	NO	80	27,000	2,160,000.00
	<u>Frames and frame finishes in mahogany timber:</u>				
C	25 x 25mm quadrant beading (PC Rate Kshs 75)	LM	483	100	48,300.00
D	25 x 50mm architrave with two labours, plugged (PC Rate Kshs 150)	LM	483	200	96,600.00
E	150mm x 50mm frame with three labours; chamfered edges; plugged (PC Rate Kshs 1400)	LM	483	1,550	748,650.00
	<u>Internal Doors</u>				
	<u>Flush timber doors</u>				
	50 mm thick Semi Solid cored flush doors with plywood facing to receive painting (m.s) all to Architects details, specifications and approval				
	A prime cost sum at the rate indicated below for fabrication of the above specified Timber Flush door leaves by AHP juakali artisans as approved by the Project Manager/Architect.				
	The contractor's unit rate shall include the cost of transport, storage, fixing and all associated accesories in addition to the PC Rate.				
F	Door size 900mm x 2400mm High comprising of 1 No Opennable leaf size 800 x 2100mm high including fixed fanlight size 900 x 300mm high in 4mm clear glass (measured separately) (PC Rate Kshs 4,000)	NO	260	5,200	1,352,000.00
G	Ditto 800 x 2100mm high comprising of 1No. Opennable leaf size 700 x 2100mm high (PC Rate Kshs 3,500)	NO	200	4,700	940,000.00
	4mm Thick clear sheet glass fixing with matching timber glazing beads to timber frames				
H	In panes exceeding 0.1 sqm but not exceeding 0.5 square metres.	SM	70	1,100	77,000.00
	<u>Frames and frame finishes in cypress Timber</u>				
I	25 x 25mm quadrant (PC Rate Kshs 75)	LM	2482	100	248,200.00
J	25 x 50mm architrave with two labours, plugged (PC Rate Kshs 150)	LM	2482	200	496,400.00
K	150mm x 50mm transome with three labours; chamfered edges; plugged (PC Rate Kshs 800)	LM	234	950	222,300.00
L	150mm x 50mm frame with three labours; chamfered edges; plugged (PC Rate Kshs 800)	LM	2482	950	2,357,900.00
	Carried to collection				8,876,950.00

Item	Description	Unit	QTY	Rate	Amount
	<u>Painting and decorating</u>				
	<u>Priming back of frame with an aluminium or equivalent and approved wood primer</u>				
A	Surfaces not exceeding 100mm girth	LM	5930	30	177,900.00
B	Surfaces over 100mm but not exceeding 200mm girth	LM	3199	60	191,940.00
	<u>Prepare Knot, prime, stop and apply one undercoat and two finishing coats first grade timber quality paint to wood surfaces as per the manufacturer's printed instructions</u>				-
C	General timber surfaces	SM	1867	300	560,100.00
D	Surfaces over 200mm but not exceeding 300mm girth	LM	3199	90	287,910.00
E	Architraves: not exceeding 100 mm girth	LM	2965	30	88,950.00
F	Quadrant beading : not exceeding 100 mm girth	LM	2965	30	88,950.00
	Ironmongery				-
	Supply and fix the following ironmongery to timber complete with matching screws and keys to the approval of the Architect				-
G	100mm pressed steel Butt Hinges	PRS	815	350	285,250.00
H	Stainless steel 3 Lever Mortice Door Lock with handle furniture set;(keyhole escutcheons, cylinder and latch)	NO	343	3,500	1,200,500.00
I	Ditto: but 2 Lever Door Lock with handle	NO	200	2,800	560,000.00
J	Door fixing cramps	NO	543	120	65,160.00
K	200 x 75 x 3mm perspex door signage with door numbers as per Architect detail	NO	0	500	-
L	Block sign with block type, name and number lin exterior quality paint as per Architect detail	NO	0	2,000	-
	<u>Supply and fix the following purpose-made signage in printed vinyl panels in approved background and text colours in accordance to the architects later signage design fixed to finished surfaces in the approved adhesive and matching screws as applicable to:</u>				-
M	200 x 50mm Main door signage	NO	130	400	52,000.00
N	150x450mm Floor Level Indicator Sign	NO	20	1,000	20,000.00
O	500x100 Floor Wayfinding/Directional Sign	NO	20	1,000	20,000.00
P	200x200mm Duct door Signage	NO	260	800	208,000.00
Q	450x200 mm Fire Exit Signage	NO	65	2,000	130,000.00
R	1000x300 mm Block Identification Signage	NO	1	6,000	6,000.00
	Carried to Collection				3,942,660.00
	COLLECTION				
	Total brought forward from page no:				8,876,950.00
	Total brought forward from page no:				3,942,660.00
	<u>ELEMENT NO. 5</u>	Carried to			
	<u>DOORS</u>	Main summary			12,819,610.00

Item	Description	Unit	QTY	Rate	Amount
	PROPOSED APARTMENTS FOR AFFORDABLE HOUSING PROGRAM (TYPE B G+9)				
	BILL NO.1-BUILDERS WORKS				
	ELEMENT NO 6 - EXTERNAL FINISHES				
	EXTERNAL WALL FINISHES				
	External Render				
	<u>Cement and sand (1:3) render:wood floated: on concrete or blockwork: to</u>				
A	15mm thick to beams, Columns, Slab Moulds and walling externally	SM	5,332.00	350	1,866,200.00
					-
B	Extra over horizontal and vertical pointing in 10mm thick rod in cement and sand mix (1:3) mortar including one coat Bituminous paint	SM	0	350	-
					-
	External Painting				-
	<u>Prepare and apply one coat Alkali Resistant primer followed by two finishing coats of silicon exterior Emulsion paint in accordance with the manufacturers written instructions and to the satisfaction of the architect to</u>				-
					-
C	Concrete/masonry surfaces externally-Beam, Column and Slab Moulds	SM	5332	350	1,866,200.00
					-
	ROOF FINISHES				-
	Cement and sand (1:3) screeded beds: on concrete: complete with coloured pigmentation additives and hardener to:				-
					-
D	50mm average waterproofed lightweight screed laid to falls and crossfalls to roof slabs -upper roof including gutter bases	SM	720	400	288,000.00
					-
	<u>Prepare and apply to vertical/horizontal surfaces 4mm thick APP/EPDM water proofing or other equal and approved membrane with surface finish weighing 4kg/sm; laid on primer with torch-on process ;by an approved specialist all in accordance with the manufacturers instructions including provision of a written ten (10) year anti leak guarantee.</u>				-
					-
E	4mm thick APP membrane applied to roof slabs	SM	720	1,400	1,008,000.00
					-
F	Ditto to skirting 200mm high	LM	186	280	52,080.00
					-
G	Dress membrane around 100mm rainwater outlet	No.	12	200	2,400.00
					-
	<u>The Following Flat roof concrete tiles fixed with approved adhesive, laid and jointed with waterproofing bituminous compound</u>				-
H	20mm thick interlocking Concrete tiles of size 225 x 225mm with and including cement sand (1:4) backing to falls	SM	720	1,500	1,080,000.00
	ELEMENT NO. 6	Carried to			
	EXTERNAL FINISHES	Main summary			6,162,880.00

Item	Description	Unit	QTY	Rate	Amount
	PROPOSED APARTMENTS FOR AFFORDABLE HOUSING PROGRAM (TYPE B G+9)				
	BILL NO.1-BUILDERS WORKS				
	ELEMENT NO 7 - INTERNAL FINISHES				
	<u>Internal Wall Finishes</u>				
	<u>Cement and sand (1:3) backings</u>				
A	15mm thick to receive ceramic Wall tiles	SM	2,647.00	350	926,450.00
					-
B	To receive ceramic wall tiles (m.s.) (Lift Lobby)	SM	141.00	350	49,350.00
					-
	<u>Ceramic wall tiles</u>				-
					-
	<u>Allow a Prime Cost supply/Purchase only rate of Ksh. 1000 per SM (In addition to the cost of purchase above, the contractor's rate shall include cost of transportation, offloading, storage, fixing including all necessary adhesives, grouting and accessories</u>				-
C	Supply and Fix approved ceramic wall tiles on prepared backings(m.s) with proprietary adhesive; jointed and pointed in matching coloured proprietary grouting: including pvc spacers and expansion joint as necessary: all to Architect's approval. - Wall Surfaces	SM	2647	1,650	4,367,550.00
					-
	<u>Porcelain wall tiles</u>				-
	<u>Allow a Prime Cost supply/Purchase only rate of Ksh. 1500 per SM (In addition to the cost of purchase above, the contractor's rate shall include cost of transportation, offloading, storage, fixing including all necessary adhesives, grouting and accessories</u>				-
D	Supply and Fix porcelain wall tiles on prepared backings(m.s) with proprietary adhesive; jointed and pointed in matching coloured proprietary grouting: including pvc spacers and expansion joint as necessary: all to Architect's approval. - Wall Surfaces	SM	141.00	2,150	303,150.00
					-
	<u>15mm (minimum) two coat cement, sand (1:3) plaster complete with wire gauze anti-crack mechanism at the intersection of masonry walling and concrete beams as described to:-</u>				-
					-
F	Concrete/masonry surfaces	SM	12882	350	4,508,700.00
					-
	<u>Painting and Decoration</u>				-
	<u>Prepare, Skim and apply Emulsion or universal undercoat followed by two finishing coats of soft satin Emulsion paint in accordance with the manufacturers written instructions and to the satisfaction of the architect to</u>				-
					-
G	Plastered concrete/masonry surfaces internally	SM	12882	300	3,864,600.00
					-
	Carried to Collection				14,019,800.00

Item	Description	Unit	QTY	Rate	Amount
	<u>Floor Finishes</u>				
	<u>Cement and sand (1:3) screeds, backings, beds etc</u>				
A	32mm bed finished to receive Floor Tiles (m.s)	SM	5074	350	1,775,900.00
					-
	<u>Cement and sand (1:3) backings etc</u>				-
					-
B	32mm bed finished to receive ceramic tiles to surfaces of Landings (m.s)	SM	101	350	35,350.00
					-
C	25 x 300 mm wide treads to receive ceramic tiles (m.s)	LM	437	105	45,885.00
					-
D	20 x 150mm risers to receive ceramic tiles (m.s)	LM	460	53	24,150.00
					-
	<u>Ceramic Floor tiles</u>				-
					-
	<u>Allow a Prime Cost supply/Purchase only rate of Ksh. 1000 per SM (In addition to the cost of purchase above, the contractor's rate shall include cost of transportation, offloading, storage, fixing including all necessary adhesives, grouting and accessories</u>				-
					-
E	Supply and Fix Ceramic tiles, on prepared bed(m.s) with proprietary adhesive; jointed and pointed in matching coloured proprietary grouting: including pvc spacers and expansion joint as necessary: all to Architect's approval.	SM	2689	1,650	4,436,850.00
					-
F	Ditto Non Slip Ceramic Tiles	SM	2385	1,650	3,935,250.00
					-
G	Ditto 100mm wide Wall Skirtings	LM	7535	165	1,243,275.00
					-
H	Non Slip Ceramic Tiles to surfaces of Landings	SM	101	1,650	166,650.00
					-
J	Non Slip Ceramic Tiles to 300 mm wide treads	LM	437	495	216,315.00
					-
K	Non Slip Ceramic Tiles to 150mm risers	LM	460	248	113,850.00
					-
	<u>Wet areas waterproofing - Kitchens, Toilets etc.</u>				-
					-
	Liquid membrane - High build bitumen/ rubber latex emulsion with excellent adhesion, liquid applied to form a seamless flexible water proof/ vapour proof membrane as as per the manufacturer's printed instructions to: solid content =65%, Rubber Content=10% in dried film)				-
					-
L	To floors and walls	SM	2385	1,500	3,577,500.00
					-
	<u>Staircase soffit finishes</u>				-
					-
	<u>15mm (minimum) two coat cement, sand (1:3) plaster complete with wire gauze anti-crack mechanism at the intersection of masonry walling and concrete beams as described to:-</u>				-
					-
M	Soffits of staircase landing	SM	101	350	35,350.00
					-
N	Ditto to sloping soffites exceeding 15° from horizontal	SM	154	350	53,900.00
					-
P	Staircase string 300mm extreme girth and cut to profile of steps	LM	268	105	28,140.00
					-
	<u>Paint works</u>				-
					-
	<u>Prepare, skim and apply Emulsion or universal undercoat followed by two finishing coats of soft satin Emulsion paint in accordance with the manufacturers written instructions and to the satisfaction of the architect to</u>				-
					-
Q	Soffits of staircase landing	SM	101	300	30,300.00
					-
R	Ditto to sloping soffites exceeding 15° from horizontal	SM	154	300	46,200.00
					-
S	Staircase string 300mm extreme girth and cut to profile of steps	LM	268	90	24,120.00
					-
	Carried to Collection				15,788,985.00

Item	Description	Unit	QTY	Rate	Amount
	PROPOSED APARTMENTS FOR AFFORDABLE HOUSING PROGRAM (TYPE B G+9) BILL NO.1-BUILDERS WORKS ELEMENT NO 8- BALUSTRADING AND RAILING <u>Balustrades and staircase railings</u> A 900mm high mild Steel balustrade; comprising 60 x 10mm mild Steel balusters at 900mm centres; bolted to base plate and tread (m.s), with 7No. 25mm diameter horizontal bars, and 75x4mm diameter CHS mild Steel handrail part welded into 60x10mm balustrades; to Architects drawings LM 82 5,500 451,000.00 - B 50mm diameter x 3mm CHS handrail supported by 50 x 50 x 3mm thick SHS balusters anchored to slab with and including w/steel bolts and plates at 1800mm centres smooth welded 100mm high (Staircase) LM 92 800 73,600.00 - C <u>Prepare, prime and apply one undercoat and two finishing coats first quality gloss oil paint on</u> General metal surfaces of balustrading (both sides measured overall) SM 117 300 35,100.00 -				
	ELEMENT NO. 8 BALUSTRADE AND RAILING	Carried to the Main summary			559,700.00

Item	Description	Unit	QTY	Rate	Amount
	PROPOSED APARTMENTS FOR AFFORDABLE HOUSING PROGRAM (TYPE B G+9)				
	BILL NO.1-BUILDERS WORKS				
	ELEMENT NO 9 - JOINERY FITTINGS				
	Allow for providing materials, labour and construct fixtures and fittings as per Architects drawings of the following JOINERY FITTINGS AND FIXTURES complete with associated iron mongery;				
	NOTE: All blockboard, MDF boards,etc in joinery works shall be lipped with hardwood lipping all round before fixing.				
	<u>High level cupboards</u>				
	<u>400mm Wide high level kitchen cupboards in 18mm laminated mdf sides & shelves complete with doors, top, bottom & divisions with and including all necessary ironmongery; to Architect's details</u>				
A	High level storage cupboard units 600mm high x 300mm deep	LM	140	5,000	700,000.00
					-
B	Ditto 600mm high x 300mm deep	LM	14	5,000	70,000.00
					-
	<u>Low level kitchen cupboards</u>				-
					-
	<u>Low level kitchen worktops with 600x600x10mm porcelain tiles top on 100mm thick reinforced concrete slab with A142 BRC mesh, formwork to soffits and slab edges, plater to soffits of slab, screed to top slab, 100mm thick plastered steeper walls, including 300x300x6mm thick ceramic wall tiles on both sides of the wall: 100mm plastered mass concrete plinths in concrete class 15MPa: 18mm laminated mdf sides & shelves complete with doors, shelves, drawers, cutting tiles for kitchen sink (m.s) & all necessary ironmongery; to Architect's details and approval</u>				-
					-
C	Low level kitchen cupboards below concrete worktop total girth grouped together 850mm high x 550mm deep	LM	168	9,500	1,596,000.00
					-
D	Ditto 1850mm long x 850mm high x 550mm deep	LM	19	9,500	180,500.00
					-
	<u>Bedroom Wardrobes</u>				-
	<u>600mm Wide x 2200mm high wardrobes in 18mm laminated mdf sides, divisions, drawers, shelves ; : complete with sliding/ side hung doors, shelves, divisions, shoe racks, hanging rails, drawers, & all necessary ironmongery; 100mm plastered mass concrete plinths in concrete class 15MPa ; to Architect's details</u>				-
E	Bedroom wardrobes size 2200mm high x 600mm deep in bedrooms	LM	135	14,000	1,890,000.00
					-
F	Ditto size 2200mm high x 600mm deep in bedrooms	LM	132	14,000	1,848,000.00
					-
	Duct doors				-
	50mm thick laminated MDF duct doors; complete with frames, ironmongery and all necessary paintwork to:				-
					-
H	Duct doors size 650mm wide x 2200mm high	NO.	44	7,000	308,000.00
					-
J	Duct doors size 400mm wide x 2200mm high	NO.	88	5,000	440,000.00
	ELEMENT NO. 9				
	JOINERY & FITTINGS				
	Carried to the Main summary				
					7,032,500.00

Item	Description	Unit	QTY	Rate	Amount
	<u>PROPOSED APARTMENTS FOR AFFORDABLE HOUSING PROGRAM (TYPE B G+9)</u>				
	BILL NO.1-BUILDERS WORKS MAIN SUMMARY				
1	Substructures				22,729,100.00
2	Reinforced Concrete Frame				57,112,716.00
3	Walling				14,491,825.00
4	Windows				7,592,700.00
5	Doors				12,819,610.00
6	External Finishes				6,162,880.00
7	Internal Finishes				34,217,085.00
8	Balustrade and Railing				559,700.00
9	Joinery and Fittings				7,032,500.00
10	Builders Works				200,000.00
	<u>TOTAL FOR 1NO. TYPE B (G+9) BLOCK</u>				162,918,116.00
	MULTIPLY BY 1.NO OF BLOCKS	X 1			1.00
	<u>TOTAL FOR TYPE B (G+9) BLOCK(S) CARRIED TO GRAND SUMMARY</u>				162,918,116.00

SOCIAL HALL

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ITEM	DESCRIPTION	UNIT	QTY	RATE	AMOUNT
	<u>Disposal of excavated material</u>				
A	Return, fill and ram selected excavated	CM	80	100	8,000.00
					-
B	Load,wheel and cart away surplus	CM	74	200	14,800.00
					-
	Fillings				-
					-
C	Make up levels using approved imported materials:	CM	230	1,450	333,500.00
					-
D	300mm thick hardcore bed: hand packed : compacted in lavers not	SM	328	500	164,000.00
					-
E	50 mm Thick Murram Blinding to	SM	328	120	39,360.00
	Anti - termite to treatment				-
					-
F	Approved anti-termite treatment, with ten-year	SM	328	150	49,200.00
					-
	Damp-proof membrane				-
					-
G	1000 gauge polythene or other equal and approved damp-proof	SM	328	100	32,800.00
					-
	Concrete Blinding				-
					-
	Class 15MPa:				-
					-
H	50 mm Thick under	SM	30	400	12,000.00
					-
J	Ditto strip	SM	83	400	33,200.00
					-
	In- situ vibrated				-
					-
K	Column bases	CM	6	13,200	79,200.00
					-
L	Columns	CM	2	13,200	26,400.00
					-
M	Strip foundation	CM	17	13,200	224,400.00
					-
N	100mm thick surface	SM	328	1,320	432,960.00
					-
	Carried to collection				1,449,820.00

ITEM	DESCRIPTION	UNIT	QTY	RATE	AMOUNT
A	Ribbed reinforcement steel bars to KS reinforcement	Kg	2,046	165	337,590.00
					-
	Mesh fabric reinforcement to KS/EAS 412;2 (2019) BRC				-
					-
B	Mesh reference A142 weighing 2.22	SM	328	400	131,200.00
					-
	<u>Modular steel frame with steel plates</u>				-
					-
C	Sides of column base	SM	115	500	57,500.00
					-
D	Edge of slab, over	LM	93	113	10,462.50
					-
	Foundation Walling				-
					-
	stone walling with a minimum of 7.0 N/mm² bedded and				-
					-
E	200mm thick walls in	SM	166	1,450	240,700.00
					-
	<u>Pavings</u>				-
					-
F	Supply and lay 600 x 600 x 50mm reinforced concrete precast paving slabs around the building	SM	56	1,400	78,400.00
					-
	<u>Plinth</u>				-
					-
	<u>20mm thick cement and sand</u>				-
					-
G	Plinths externally	SM	56	350	19,600.00
					-
	bitumastic paint				-
					-
H	Rendered surfaces	SM	56	300	16,800.00
					-
Carried to collection					892,252.50

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ITEM	DESCRIPTION	UNIT	QTY	RATE	AMOUNT
	<u>SOCIAL HALL</u> BILL NO.1-BUILDERS WORKS ELEMENT NO 4-WINDOWS <u>METAL WORK</u> <u>PURPOSE - MADE</u> Supply, fabricate and fix the following purpose made small pane mild steel casement windows to be fabricated from approved mild steel sections (atleast 14g 2mm thick) comprising of frame and casement incorporating permanent hooded high level ventilation panels infilled with mosquito gauze : window supplied complete with and including 12mm solid square burglar proofing bars fixed at 200mm centres both ways and metal fixing lugs including building into wall and making good, and all necessary iron mongery viz hinges, fasteners, and hasp including shop priming window with red oxide primer before delivery to site:- A prime cost sum of Kshs 6,000 per Sqm has been allowed for fabrication of the above specified Steel casement Windows by				
A	Window, overall size 2400 X 1500mm	NO	13	25,200	327,600.00
B	Ditto Size 1200 x	NO	6	12,600	75,600.00
	<u>Glazing</u>				-
D	4mm Thick clear sheet glass panes	SM	46	1,100	50,600.00
E	Ditto; obscure	SM	11	1,200	13,200.00
	<u>Painting and</u>				-
	<u>On Metal work</u>				-
	<u>prepare and apply aerosol spray</u>				-
					-
F	General window and	SM	114	300	34,200.00
	Carried to Collection				501,200.00

ITEM	DESCRIPTION	UNIT	QTY	RATE	AMOUNT
	<u>Precast concrete window cill finishing fair on all exposed surfaces</u>				
A	150 x 25mm thick	LM	27	550	14,850.00
					-
	<u>Curtain rods;</u>				-
					-
B	1.5mm thick, 20mm thick diameter twin	LM	32	1,300	41,600.00
	Carried to collection				56,450.00
	COLLECTION				
	Total brought forward from page no:		SH/7		501,200.00
	Total brought forward from page no:		SH/8		56,450.00
	<u>ELEMENT NO. 4 WINDOWS</u>	Carried to the Main summary			557,650.00

ITEM	DESCRIPTION	UNIT	QTY	RATE	AMOUNT
	PROPOSED SOCIAL HALL BILL NO. 1- BUILDING WORKS ELEMENT NO 5- <u>External Doors</u> <u>Mild steel</u> <u>casement doors</u> Heavy gauge purpose made mild steel door comprising 40 x 25 x3mm stiles, top and bottom stiles, 4 No Intermediate rails, 1.5mm steel sheet				
A	Door overall size 900 x 2400mm	NO	5	19,440	97,200.00
B	Ditto Size 1500 x 2400mm	NO	1	32,400	32,400.00
	<u>Internal Doors</u>				-
	Flush timber doors				-
	50 mm thick Semi Solid cored flush doors with plywood facing to A prime cost sum at the rate indicated below has been allowed for fabrication of the above specified Hardwood panelled door leaves by AHP				-
C	Door size 900mm x 2060mm high (	NO	5	5,200	26,000.00
	<u>finishes in</u>				-
E	25 x 25mm	LM	30	100	3,000.00
F	25 x 50mm architrave with two	LM	30	200	6,000.00
G	frame with three labours; chamfered	LM	30	950	28,500.00
Carried to collection					193,100.00

ITEM	DESCRIPTION	UNIT	QTY	RATE	AMOUNT
	<u>Painting and</u>				
	<u>Prepare and apply</u>				
A	Surfaces not	LM	30	30	900.00
B	Surfaces over	LM	30	60	1,800.00
	<u>Prepare Knot, prime,</u>				-
	<u>stop and apply one</u>				-
	<u>undercoat and two</u>				-
C	General timber	SM	18	300	5,400.00
D	Surfaces not	LM	90	60	5,400.00
	Ironmongery				-
	following				-
	matching screws				-
	the Architect				-
E	100mm pressed	Pairs	8	350	2,800.00
F	2 Lever Door Lock	NO	5	2,800	14,000.00
G	Door fixing cramps	NO	50	120	6,000.00
	Carried to Collection				36,300.00
	COLLECTION				
	Total brought forward from page no:		SH/9		193,100.00
	Total brought forward from page no:		SH/10		36,300.00
	ELEMENT NO. 5	Carried to			
	DOORS	Main summary			229,400.00

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ITEM	DESCRIPTION	UNIT	QTY	RATE	AMOUNT
	<u>PROPOSED</u> <u>SOCIAL HALL</u> <u>BILL NO.1-</u> <u>REPAIRS WORKS</u> ELEMENT NO 7 - <u>Finishes</u> <u>15mm thick to</u>				
A	Internal wall	SM	22	350	7,560.00
	<u>Ceramic wall tiles</u>				-
	<u>Allow a Prime Cost</u> <u>supply/ Purchase</u> <u>only rate of Ksh.</u> <u>1000 per SM (In</u> <u>addition to the cost</u>				-
B	Supply and Fix approved ceramic wall tiles on prepared backings(m.s) with	SM	22	1,650	35,640.00
	<u>15mm (minimum)</u> <u>two coat cement,</u> <u>sand (1:3) plaster</u>				-
C	surfaces Internally	SM	414	350	144,900.00
D	Ditto to window cills, door Jambs	LM	125	70	8,750.00
	<u>Decoration</u> <u>Prepare, Skim and</u> <u>apply Emulsion or</u> <u>universal undercoat</u> <u>followed by two</u>				-
E	Plastered concrete / masonry	SM	414	300	124,200.00
F	Ditto to window cills, door Jambs	LM	125	60	7,500.00
Carried to Collection					328,550.00

ITEM	DESCRIPTION	UNIT	QTY	RATE	AMOUNT
	<u>Floor Finishes</u>				
	<u>Cement and sand</u> <u>(1:3) backing on</u>				
A	Floor surfaces	SM	303	350	106,050.00
	Ceramic Floor tiles				-
	<u>Allow a Prime Cost</u> <u>supply/Purchase</u> <u>only rate of Ksh.</u> <u>1000 per SM (In</u> <u>addition to the cost</u> Supply and Fix approved ceramic wall tiles on prepared backings(m.s) with				-
B	Floor Surfaces	SM	252	1,650	415,800.00
C	Ditto Non Slip	SM	51	1,650	84,150.00
D	Ditto 100mm wide	LM	183	165	30,195.00
	<u>Step finishes</u>				-
	<u>Cement and sand</u> <u>(1:3) backing on</u>				-
E	300 mm wide treads	LM	58	105	6,090.00
F	150mm high risers	LM	58	53	3,045.00
	<u>Supply and Fix non-</u> <u>slip Ceramic floor</u> <u>tiles or equal and</u> <u>approved; as</u> <u>manufactured by</u> <u>Saj ceramic or equal</u>				-
G	300 mm wide	LM	58	495	28,710.00
H	150mm high risers	LM	58	248	14,355.00
Carried to Collection					688,395.00

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ITEM	DESCRIPTION	UNIT	QTY	RATE	AMOUNT
	<u>SOCIAL HALL</u>				
	MAIN SUMMARY		<u>PAGE</u>		
1	Substructures		SH/4		2,482,512.50
2	Reinforced Concrete Frame		SH/5		2,096,070.00
3	Walling		SH/6		642,517.50
4	Windows		SH/8		557,650.00
5	Doors		SH/10		229,400.00
6	External Finishes		SH/11		121,852.50
7	Internal Finishes		SH/14		1,016,945.00
8	Roofing		SH/15		1,674,125.00
	<u>TOTAL FOR</u>				
	<u>SOCIAL HALL</u>				
	CARRIED TO				8,821,072.50

POWER HOUSE

Item	DESCRIPTION	Unit	QTY	Rate	Amount
	<u>PROPOSED POWER HOUSE</u>				
	BILL NO.6-BUILDERS WORKS				
	ELEMENT NO 1 - SUBSTRUCTURES (ALL PROVISIONAL)				
	<u>Site Clearance</u>				
A	Clear site of all grass, hedges, shrubs, bushes grub up roots, load and remove from site and dispose at designated local authority areas.	SM	24	30	720.00
					-
B	Excavate average 200mm deep to remove top vegetable soil, load, remove from site and dump in designated local authority dump site.	SM	24	150	3,600.00
					-
C	Excavate to reduced levels in varying depths not exceeding 1.5m deep from existing ground levels.	Cm	36	300	10,800.00
					-
D	Ditto exceeding 1.5m deep but not exceeding 3.0m from existing reduced levels.	Cm	12	400	4,800.00
E	Excavate for Strip foundations depth starting from Reduced ground levels.	Cm	3	300	900.00
					-
F	Excavate for duct bank depth starting from Reduced ground levels.	Cm	4	300	1,200.00
					-
G	Extra over all type of excavation for excavating in soft rock	Cm	3	1,000	3,000.00
					-
H	Ditto excavation in hard rock class I	Cm	3	1,400	4,200.00
					-
	Disposal of water				-
I	Allow for keeping the whole of the excavation free from all spring and running water by pumping or any other such means as may be necessary	Item		5,000	5,000.00
					-
	Planking and strutting				-
					-
J	Allow for maintaining and upholding the sides of excavations and keeping excavations clear of all fallen materials, rubbish etc	Item		5,000	5,000.00
					-
	Carried to collection				39,220.00

Item	DESCRIPTION	Unit	QTY	Rate	Amount
	<u>Disposal of excavated material</u>				
A	Return, fill and ram selected excavated material around foundations.	CM	65	100	6,500.00
					-
B	Load, wheel and cart away surplus excavated material to a Local Authority designated dumping site or fill soil heaps as away from site instructed by the Project Engineer.	CM	15	200	3,000.00
					-
	Fillings				-
					-
C	Make up levels using approved imported materials: compacted in layers not exceeding 300mm thick with a roller: to the satisfaction of the Structural Engineer.	CM	49	1,450	71,050.00
					-
D	300mm thick hardcore bed: hand packed : compacted in layers not exceeding 150mm thick: to the satisfaction of the Structural Engineer	SM	36	500	18,000.00
					-
F	50 mm Thick Murram Blinding to surfaces of hardcore	SM	36	120	4,320.00
					-
	Anti - termite to treatment				-
					-
G	Approved anti-termite treatment, with ten-year guarantee, sprayed to surfaces of hardcore strictly in accordance with manufacturer's instructions.	SM	36	150	5,400.00
					-
	Damp-proof membrane				-
					-
H	1000 gauge polythene or other equal and approved damp-proof membrane, laid over blinded hardcore (measured separately) with 300mm side and end laps (measured nett-allow for laps); 1 No. layer: bedded in and including cement and sand (1:3) mortar	SM	36	100	3,600.00
					-
	Concrete Blinding				-
					-
	In situ concrete Class 15MPa: vibrated:				-
					-
I	50 mm Thick under strip foundation	SM	14	400	5,600.00
					-
J	50 mm Thick under duct bank	SM	5	400	2,000.00
					-
	In- situ vibrated reinforced concrete Class 25MPa: in:				-
					-
L	Columns	CM	1	13,200	13,200.00
					-
N	Strip foundation	CM	3	13,200	39,600.00
					-
O	100mm thick duct bank sides	SM	13	1,320	17,160.00
					-
P	100mm thick duct bank bottom	SM	4	1,320	5,280.00
					-
Q	150 mm thick horizontal ground floor slab	SM	20	1,980	39,600.00
					-
	Carried to collection				234,310.00

Item	DESCRIPTION	Unit	QTY	Rate	Amount
	Ribbed reinforcement steel bars to KS 573:2014 : Grade 500 high tensile strength including bends, hooks, tying wire and distance blocks; to S.E's detail (Provisional)				
A	Assorted reinforcement	KG	303	165	49,995.00
					-
	Mesh fabric reinforcement to KS/EAS 412;2 (2019) BRC A142;200 x 200mm, weighing 2.22kg/m² (measured net - no allowance) for laps; in two layers - top & bottom; including bends, tying wire and spacer blocks)				-
					-
B	In floor beds	SM	20	400	8,000.00
					-
	<u>Modular steel frame with steel plates covering formwork and/or marine board formwork: to:</u>				-
					-
C	Vertical sides to columns	SM	10	500	5,000.00
					-
E	Vertical sides to strip footing	SM	9	500	4,500.00
					-
F	Vertical sides to duck bank sides	SM	27	500	13,500.00
					-
G	Vertical sides to duck bank bottom	LM	49	75	3,675.00
					-
H	Edge of slab not exceeding 150mm girth	LM	48	75	3,600.00
					-
	Foundation Walling				-
					-
	Natural quarry stone walling with a minimum of 7.0 N/mm² bedded and jointed in cement and sand (1:4) mortar, reinforced with and including 25 x 3 mm thick hoop iron strips at every alternate course as described in;				-
					-
I	200mm thick walls in foundations	SM	46	1,450	66,700.00
					-
	<u>Pavings</u>				-
					-
J	Supply and lay 600 x 600 x 50mm reinforced concrete precast paving slabs around the building including laying, spreading and compacting 100mm thick approved sand bed blinding, on and including 150mm thick compacted hardcore to Engineer's approval.	SM	27	1,400	37,800.00
					-
	<u>Plinth</u>				-
					-
	<u>25mm Thick cement and sand (1:3) rendering on concrete or masonry ; wood float finished; to</u>				-
					-
K	Plinths externally	SM	10	350	3,500.00
					-
	Two coats black bitumastic paint on:				-
					-
L	Rendered surfaces	SM	10	300	3,000.00
					-
	Carried to collection				199,270.00

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Item	DESCRIPTION	Unit	QTY	Rate	Amount
	<u>PROPOSED POWER HOUSE</u> BILL NO.1-BUILDERS WORKS ELEMENT No 3-WALLING <u>WALLING</u> <u>External Walling</u> <i>Machine cut quarry stone walling with a minimum of 7.0 N/mm² average compressive strength ;bedded and jointed in cement and sand (1:4) mortar, reinforced with and including 25 x 3 mm thick hoop iron strips at every alternate course as described in;</i>				
A	200mm thick walling Externally	Sm	40	1,450	58,000.00 -
B	200mm thick parapet walling	Sm	8	1,450	11,600.00 - -
	<u>Precast Concrete Breeze Ventilation Blocks</u>				- -
C	305 x 305 mm wide x 150 mm thick Pre Cast Concrete louvered breeze ventilation blocks bedded and jointed in cement and sand (1:4) mortar including vermin proof gauge wire	Sm	1	1,800	1,800.00 -
	<u>Damp proof courses</u>				- -
D	Approved hessian based damp proof course to 200mm thick walling in cement/sand mortar	Lm	17	100	1,700.00 -
	<u>Window Cill</u>				- -
	<i>Precast concrete window cill finishing fair on all exposed surfaces and hoisting and placing in position, bedding, jointing and pointing in pigmented cement and sand (1:3) mortar</i>				- -
E	150 x 25mm thick Precast concrete window sill	Lm	7	550	3,850.00
	<u>ELEMENT NO. 3</u> Carried to <u>WALLING</u> Main summary				76,950.00

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Item	DESCRIPTION	Unit	QTY	Rate	Amount
A	<u>PROPOSED POWER HOUSE</u> BILL NO.1-BUILDERS WORKS ELEMENT NO 5 - EXTERNAL FINISHES EXTERNAL WALL FINISHES External Finish <u>Cement and sand (1:3) render:wood floated: on concrete or blockwork:</u> <u>to</u> 15mm thick to columns/beams surfaces externally Keying Extra over horizontal and vertical pointing in 10mm thick rod in cement and sand mix (1:3) mortar including one coat Bituminous paint External Painting <u>Prepare and apply one coat Alkali Resistant primer followed by two finishing coats of silicon exterior Emulsion paint in accordance with the manufacturers written instructions and to the satisfaction of the architect to</u>	SM	17	350	5,950.00 - -
B		SM	56	350	19,600.00 - - -
C	Columns surfaces externally	SM	56	350	- 19,600.00
	<u>ELEMENT NO. 5</u> Carried to EXTERNAL FINISHES Main summary				45,150.00

Item	DESCRIPTION	Unit	QTY	Rate	Amount
	<u>PROPOSED POWER HOUSE</u>				
	BILL NO.1-BUILDERS WORKS				
	ELEMENT NO 6 - INTERNAL FINISHES				
	<u>Internal Wall Finishes</u>				
	<u>15mm (minimum) two coat cement, sand (1:3) plaster complete with wire gauze anti-crack mechanism at the intersection of masonry walling and concrete beams as described to:-</u>				
A	Concrete/masonry surfaces	SM	42	350	14,700.00
B	Ditto to Jambs and reveals not exceeding 200mm girth	LM	30	70	2,100.00
	<u>Painting and Decoration</u>				
	<u>Prepare, Skim and apply Emulsion or universal undercoat followed by two finishing coats of soft satin Emulsion paint in accordance with the manufacturers written instructions and to the satisfaction of the architect to</u>				-
C	Plastered concrete/masonry surfaces internally	SM	42	300	12,600.00
D	Ditto to Jambs and reveals not exceeding 200mm girth	LM	30	60	1,800.00
	<u>Floor Finishes</u>				
	<u>Plain in-situ concrete (1:3:6/20 mm agg.)</u>				-
E	40 mm Thick polished plain concrete on existing Reinforced concrete base; level or to falls; mixed with hardener ; power floated to smooth finish to approval	SM	20	350	7,000.00
	<u>Duct bank cover plate</u>				-
F	6 mm thick chequered plate in duct bank covers including Forming 50 mm wide x 50 mm deep rebate to both sides of channel at the top; with and including 50 mm x 50 mm x 5 mm thick mild steel angle framing cast in concrete with mild steel anchor bars and grouted to both rebated sides of channel; all welding ground to a smooth finish; including priming metal surfaces with red oxide primer before delivery to site and three coats gloss oil automotive spray paint finish; all to Architect's and Structural Engineer's approval	SM	24	14,200	340,800.00
G	Allow a provisional sum for foundation holes for 100 mm LG x 12.5 diamter bolts	Item		10,000	10,000.00
H	Allow a provisional sum for plates, bolts, nuts, washers e.t.c	Item		10,000	10,000.00
	<u>Ceiling finishes</u>				
	<u>15mm (minimum) two coat cement, sand (1:3) plaster complete with wire gauze anti-crack mechanism at the intersection of masonry walling and concrete beams as described to:-</u>				-
J	Soffits of Concrete surfaces	SM	24	350	8,400.00
	<u>Painting and Decoration</u>				
	<u>Prepare, Skim and apply Emulsion or universal undercoat followed by two finishing coats of soft satin Emulsion paint in accordance with the manufacturers written instructions and to the satisfaction of the architect to</u>				-
K	Plastered ceilings	SM	24	300	7,200.00
	<u>ELEMENT NO. 6 Carried to</u>				
	INTERNAL FINISHES				414,600.00

Item	DESCRIPTION	Unit	QTY	Rate	Amount
	PROPOSED POWER HOUSE BILL NO.1-BUILDERS WORKS ELEMENT NO 7 - ROOFING ROOF FINISHES Cement and sand (1:4) screeded beds: on concrete: complete with coloured pigmentation additives and hardener to:				
A	50mm average waterproofed lightweight screed laid to falls and crossfalls to roof slabs -upper roof including gutter bases <u>Prepare and apply to vertical/horizontal surfaces 4mm thick APP/EPDM water proofing or other equal and approved membrane with surface finish weighing 4kg/sm; laid on primer with torch-on process ;by an approved specialist all in accordance with the manufacturers instructions including provision of a written ten (10) year anti leak guarantee.</u>	SM	31	400	12,400.00 - -
B	4mm thick APP membrane applied to roof slabs	SM	31	1,400	43,400.00
C	Ditto to skirting 200mm high	LM	20	280	5,600.00
D	Dress membrane around 100mm rainwater outlet <u>The Following Flat roof concrete tiles fixed with approved adhesive, laid and jointed with waterproofing bituminous compound</u>	No.	2	200	400.00 - - -
E	20mm thick interlocking Concrete tiles of size 225 x 225mm	SM	31	1,500	46,500.00
	ELEMENT NO. 7 ROOFING Carried to the Main summary				108,300.00

Item	DESCRIPTION	Unit	QTY	Rate	Amount
	<u>PROPOSED POWER HOUSE</u>				
	BILL NO.1-BUILDERS WORKS				
	MAIN SUMMARY		<u>PAGE</u>		<u>AMOUNT (KSHS)</u>
1	Substructures		C/4		472,800.00
2	Reinforced Concrete Frame		C/5		244,960.00
3	Walling		C/6		76,950.00
5	Doors		C/7		67,500.00
6	External Finishes		C/8		45,150.00
7	Internal Finishes		C/9		414,600.00
8	Roofing		C/10		108,300.00
	<u>TOTAL FOR POWER HOUSE TO GRAND SUMMARY</u>				1,430,260.00

GUARD HOUSE

Item	Description	Unit	QTY	Rate	Amount
	<u>PROPOSED GUARD HOUSE</u>				
	BILL NO.1-BUILDERS WORKS				
	ELEMENT NO 1 - SUBSTRUCTURES (ALL PROVISIONAL)				
	<u>Site Clearance</u>				
A	Clear site of all grass, hedges, shrubs, bushes grub up roots, load and remove from site and dispose at designated local authority areas.	SM	15	30	450.00
					-
B	Excavate average 300mm deep to remove top vegetable soil, load, remove from site and dump in designated local authority dump site.	SM	15	150	2,250.00
					-
C	Excavate for Strip foundations depth not exceeding 1.50 metres starting from Reduced ground levels.	CM	10	300	3,000.00
					-
D	Ditto to column bases	CM	2	300	600.00
					-
E	Extra over all type of excavation for excavating in soft rock	CM	1	1,000	1,000.00
					-
F	Ditto excavation in hard rock class I	CM	1	1,400	1,400.00
					-
	Disposal of water				-
G	Allow for keeping the whole of the excavation free from all spring and running water by pumping or any other such means as may be necessary	Item		5,000	5,000.00
					-
	Planking and strutting				-
					-
H	Allow for maintaining and upholding the sides of excavations and keeping excavations clear of all fallen materials, rubbish etc	Item		5,000	5,000.00
					-
	Carried to collection				18,700.00

Item	Description	Unit	QTY	Rate	Amount
	<u>Disposal of excavated material</u>				
A	Load, wheel and cart away surplus excavated material to a Local Authority designated dumping site or fill soil heaps as away from site instructed by the Project Engineer.	CM	1.00	200	200.00
B	Return, fill and ram selected excavated material around sides of foundations.	CM	12	100	1,200.00
	Fillings				-
C	Make up levels using approved imported materials: compacted in layers not exceeding 300mm thick with a roller: to the satisfaction of the Structural Engineer.	CM	0	1,450	-
D	300mm thick hardcore bed: hand packed : compacted in layers not exceeding 150mm thick: to the satisfaction of the Structural Engineer	SM	12	500	6,000.00
E	50 mm Stone dust/ Murrum blinding to surfaces of hardcore	SM	15	120	1,800.00
	Anti - termite to treatment				-
F	Approved anti-termite treatment, with ten-year guarantee, sprayed to surfaces of hardcore strictly in accordance with manufacturer's instructions.	SM	15	150	2,250.00
	Damp-proof membrane				-
G	1000 gauge polythene or other equal and approved damp-proof membrane, laid over blinded hardcore (measured separately) with 300mm side and end laps (measured nett-allow for laps); 1 No. layer: bedded in and including cement and sand (1:3) mortar	SM	15	100	1,500.00
	Concrete Blinding				-
	In situ concrete Class 15MPa: vibrated:				-
H	50 mm thick blinding under column bases	SM	6	400	2,400.00
I	50 mm thick blinding under strip foundations	SM	8	400	3,200.00
	In- situ vibrated reinforced concrete Class 25 MPa: in:				-
J	Column bases	CM	2	13,200	26,400.00
K	Strip foundations	CM	2	13,200	26,400.00
L	100mm thick surface bed	SM	15	1,320	19,800.00
M	Steps	CM	0	13,200	-
	In- situ vibrated reinforced concrete Class 25 MPa: in:				-
N	Columns	CM	1	13,200	13,200.00
	Carried to collection				104,350.00

Item	Description	Unit	QTY	Rate	Amount
A	Ribbed reinforcement steel bars to KS 573:2014 : Grade 500 high tensile strengthincluding bends, hooks, tying wire and distance blocks; to S.E's detail (Provisional)	KGS	264	165	43,560.00
B	Assorted reinforcement	KGS	264	165	-
C	Mesh fabric reinforcement to KS/EAS 412;2 (2019) BRC A142;200 x 200mm, weighing 2.22kg/m² (measured net - no allowance) for laps; in two layers - top & bottom; including bends, tying wire and spacer blocks)	KGS	264	165	-
D	In ground floor slab	SM	15	400	6,000.00
E	<u>Modular steel frame with steel plates covering formwork and/or marine board formwork: to:</u>	SM	15	400	-
F	Sides of column bases	SM	7	500	3,500.00
G	Vertical sides to columns	SM	8	500	4,000.00
H	Edge of slab not exceeding 150mm girth	LM	15	75	1,125.00
I	Edges of risers 75 - 150mm high	LM	14	75	1,050.00
J	Foundation Walling				-
K	Natural quarry stone walling with a minimum of 7.0 N/mm² bedded and jointed in cement and sand (1:4) mortar, reinforced with and including 25 x 3 mm thick hoop iron strips at every alternate course as described in;				-
L	200mm thick walls in foundations	SM	18	1,450	26,100.00
M	Pavings				-
N	Supply and lay 600 x 600 x 50mm reinforced concrete precast paving slabs around the building including laying, spreading and compacting 100mm thick approved sand bed blinding, on and including 150mm thick compacted hardcore to Engineer's approval.	SM	7	1,400	9,800.00
O	Plinth				-
P	25mm Thick cement and sand (1:4) rendering on concrete or masonry ; wood float finished; to				-
Q	Plinths externally	SM	5	350	1,750.00
R	Two coats black bitumastic paint on:				-
S	Rendered surfaces	SM	5	300	1,500.00
T					
	Carried to collection				98,385.00

Item	Description	Unit	QTY	Rate	Amount
	COLLECTION				
	Total brought forward from page no:				18,700.00
	Total brought forward from page no:				104,350.00
	Total brought forward from page no:				98,385.00
	<u>ELEMENT NO. 1</u>				
	<u>SUBSTRUCTURES</u>				
	Carried to Main summary				221,435.00

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Item	Description	Unit	QTY	Rate	Amount
	<u>PROPOSED GUARD HOUSE</u>				
	BILL NO.1-BUILDERS WORKS				
	ELEMENT No 3-WALLING				
	<u>WALLING</u>				
	<u>External Walling</u>				
	<i>Machine cut quarry stone walling with a minimum of 7.0 N/mm² average compressive strength; bedded and jointed in cement and sand (1:4) mortar, reinforced with and including 25 x 3 mm thick hoop iron strips at every alternate course as described in;</i>				
A	200mm thick walling Externally	SM	24	1,450	34,800.00
					-
B	200mm thick parapet walling	SM	6	1,450	8,700.00
					-
	<u>Internal Walling</u>				
	<i>Machine cut quarry stone walling with a minimum of 7.0 N/mm² average compressive strength; bedded and jointed in cement and sand (1:4) mortar, reinforced with and including 25 x 3 mm thick hoop iron strips at every alternate course as described in;</i>				
					-
C	200mm thick walling Internally	SM	5	1,450	7,250.00
					-
D	150mm thick walling Internally	SM	0	1,300	-
					-
E	Approved hessian based damp proof course to 200mm thick walling in cement/sand mortar	LM	11	100	1,100.00
					-
F	Approved hessian based damp proof course to 150mm thick walling in cement/sand mortar	LM	0	75	-
					-
	<u>Precast Concrete Breeze Ventilation Blocks</u>				
					-
G	150 x 150mm wide x 50mm thick Pre Cast Concrete flower 3D breeze ventilation blocks bedded and jointed in cement and sand (1:4) mortar	SM	0	1,800	-
					-
	COPING				
H	300 x 100mm insitu reinforced concrete class 20Mpa coping, throated and weathered and jointing to columns with cement sand 1:4 mortar	LM	11	700	7,700.00
					-
	Lintols				
					-
					-
I	200mm x 200mm Deep lintols in reinforced concrete class 20MPa with and including 4No T10 and T8 stirups at 200mm centres; complete with formwork	LM	2	1,200	2,400.00
	<u>ELEMENT NO. 3</u>	Carried to			
	<u>WALLING</u>	Main summary			61,950.00

Item	Description	Unit	QTY	Rate	Amount
	<u>PROPOSED GUARD HOUSE</u>				
	BILL NO.1-BUILDERS WORKS				
	ELEMENT NO 4-WINDOWS				
	<u>MILD STEEL WINDOWS</u>				
	Supply, fabricate and fix the following purpose made small pane mild steel casement windows to be fabricated from approved mild steel sections (atleast 14g 2mm thick) comprising of frame and casement incorporating permanent hooded high level ventilation panels infilled with mosquito gauze : window supplied complete with and including 12mm solid square burglar proofing bars fixed at 200mm centres both ways and metal fixing lugs including building into wall and making good, and all necessary iron mongery viz hinges, fasteners, and hasp including shop priming window with red oxide primer before delivery to site:-				
	A prime cost sum of Kshs 6,000 per Sqm has been allowed for fabrication of the above specified Steel casement Windows by AHP juakali artisans as approved by the Project Manager/Architect.				
	The contractor's unit rate shall include the cost of transport, storage, fixing and all associated accesories in addition to the PC Rate.				
A	Window, overall size 1900 X 1500mm high to Architects Details	NO	1	19,950	19,950.00
B	Ditto Size 650 x 1500mm high (bedroom)	NO	1	6,300	6,300.00
C	Ditto Size 1000 x 1350mm high (Kitchen)	NO	0	9,450	-
D	Ditto Size 600 x 900mm high (WC/SH)	NO	1	3,780	-
	<u>Glazing</u>				-
E	4mm Thick clear sheet glass panes over 0.1 but not exceeding 0.5 square meters; fixing with premium putty	SM	4	1,100	4,400.00
F	Ditto; obscure	SM	1	1,200	-
	<u>Painting and Decorations</u>				-
	<u>On Metal work</u>				-
	<u>Prepare and apply aerosol spray painting in one finishing coats of approved first grade to: -</u>				-
G	General window and grille surfaces; over 300mm girth internal	SM	4	300	1,200.00
	Carried to Collection				36,830.00

Item	Description	Unit	QTY	Rate	Amount
A	<u>Precast concrete window cill finishing fair on all exposed surfaces and hoisting and placing in position, bedding, jointing and pointing in pigmented cement and sand (1:3) mortar</u>	LM	3	550	
	150 x 25mm thick Precast concrete window sill				1,650.00
	Carried to collection				1,650.00
	COLLECTION				
	Total brought forward from page no:				36,830.00
	Total brought forward from page no:				1,650.00
	<u>ELEMENT NO. 4</u>	Carried to the			
	<u>WINDOWS</u>	Main summary			38,480.00

Item	Description	Unit	QTY	Rate	Amount
	<u>PROPOSED GUARD HOUSE</u>				
	BILL NO.1-BUILDERS WORKS				
	ELEMENT NO 5-DOORS				
	<u>External Doors</u>				
	<u>Flush timber doors</u>				
	50 mm thick Semi Solid cored flush doors with plywood facing to receive painting (m.s) all to Architects details, specifications and approval				
	A prime cost sum at the rate indicated below for fabrication of the above specified Timber Flush door leaves by AHP juakali artisans as approved by the Project Manager/Architect.				
	The contractor's unit rate shall include the cost of transport, storage, fixing and all associated accesories in addition to the PC Rate.				
A	Door size 900mm x 2400mm High comprising of 1 No Opennable leaf size 800 x 2100mm high including fixed fanlight size 900 x 300mm high in 4mm clear glass (measured separetely) (PC Rate Kshs 4,000)	NO	2	5,200	10,400.00
B	Ditto 800 x 2100mm high comprising of 1No. Opennable leaf size 700 x 2100mm high (PC Rate Kshs 3,500)	NO	0	4,700	-
	4mm Thick clear sheet glass fixing with matching timber glazing beads to timber frames				-
C	In panes exceeding 0.1 sqm but not exceeding 0.5 square metres.	SM	1	1,100	1,100.00
	<u>Frames and frame finishes in cypress Timber</u>				-
D	25 x 25mm quadrant (PC Rate Kshs 75)	LM	5	100	500.00
E	25 x 50mm architrave with two labours, plugged (PC Rate Kshs 150)	LM	8	200	1,600.00
F	150mm x 50mm transome with three labours; chamfered edges; plugged (PC Rate Kshs 800)	LM	8	950	7,600.00
G	150mm x 50mm frame with three labours; chamfered edges; plugged (PC Rate Kshs 800)	LM	8	950	7,600.00
	Carried to collection				28,800.00

Item	Description	Unit	QTY	Rate	Amount
	<u>Painting and decorating</u>				
	<u>Priming back of frame with an aluminium or equivalent and approved wood primer</u>				
A	Surfaces not exceeding 100mm girth	LM	13	30	390.00
					-
B	Surfaces over 100mm but not exceeding 200mm girth	LM	15	60	900.00
					-
	<u>Prepare Knot, prime, stop and apply one undercoat and two finishing coats first grade timber quality paint to wood surfaces as per the manufacturer's printed instructions</u>				-
C	General timber surfaces	SM	9	300	2,700.00
					-
D	Surfaces over 200mm but not exceeding 300mm girth	LM	15	90	1,350.00
					-
E	Architraves: not exceeding 100 mm girth	LM	8	30	240.00
					-
F	Quadrant beading : not exceeding 100 mm girth	LM	5	30	150.00
					-
	Ironmongery				-
					-
	Supply and fix the following ironmongery to timber complete with matching screws and keys to the approval of the Architect				-
					-
G	100mm pressed steel Butt Hinges	PRS	3	350	1,050.00
					-
H	Stainless steel 3 Lever Mortice Door Lock with handle furniture set;(keyhole escutcheons, cylinder and latch)	NO	0	3,500	-
					-
I	Ditto: but 2 Lever Door Lock with handle	NO	2	2,800	5,600.00
					-
J	Door fixing cramps	NO	0	120	-
					-
K	200 x 75 x 3mm perspex door signage with door numbers as per Architect detail	NO	2	500	1,000.00
					-
L	Block sign with block type, name and number in exterior quality paint as per Architect detail	NO	0	2,000	-
					-
	Carried to Collection				13,380.00
	COLLECTION				
	Total brought forward from page no:				28,800.00
	Total brought forward from page no:				13,380.00
	ELEMENT NO. 5	Carried to			
	DOORS	Main summary			42,180.00

Item	Description	Unit	QTY	Rate	Amount
	<u>PROPOSED GUARD HOUSE</u>				
	BILL NO.1-BUILDERS WORKS				
	ELEMENT NO 6 - EXTERNAL FINISHES				
	EXTERNAL WALL FINISHES				
	External Render				
	<u>Cement and sand (1:3) render:wood floated: on concrete or blockwork: to</u>				
A	15mm thick to beams, Columns, Slab Moulds and walling externally	SM	7	350	2,450.00
					-
B	Extra over horizontal and vertical pointing in 10mm thick rod in cement and sand mix (1:3) mortar including one coat Bituminous paint	LM	29	350	10,150.00
					-
	External Painting				-
	<u>Prepare and apply one coat Alkali Resistant primer followed by two finishing coats of silicon exterior Emulsion paint in accordance with the manufacturers written instructions and to the satisfaction of the architect to</u>				-
					-
C	Concrete/masonry surfaces externally-Beam, Column and Slab Moulds	SM	7	350	2,450.00
					-
	ROOF FINISHES				-
	Cement and sand (1:4) screeded beds: on concrete: complete with coloured pigmentation additives and hardener to:				-
					-
					-
D	50mm average waterproofed lightweight screed laid to falls and crossfalls to roof slabs -upper roof including gutter bases	SM	12	400	4,800.00
					-
	<u>Prepare and apply to vertical/horizontal surfaces 4mm thick APP/EPDM water proofing or other equal and approved membrane with surface finish weighing 4kg/sm; laid on primer with torch-on process ;by an approved specialist all in accordance with the manufacturers instructions including provision of a written ten (10) year anti leak guarantee.</u>				-
					-
E	4mm thick APP membrane applied to roof slabs	SM	12	1,400	16,800.00
					-
F	Ditto to skirting 200mm high	LM	11	280	3,080.00
					-
G	Dress membrane around 100mm rainwater outlet	No.	2	200	400.00
					-
	<u>The Following Flat roof concrete tiles fixed with approved adhesive, laid and jointed with waterproofing bituminous compound</u>				-
					-
H	20mm thick interlocking Concrete tiles of size 225 x 225mm	SM	12	1,500	18,000.00
					-
	ELEMENT NO. 6	Carried to			
	EXTERNAL FINISHES	Main summary			58,130.00

Item	Description	Unit	QTY	Rate	Amount
	<u>PROPOSED GUARD HOUSE</u>				
	BILL NO.1-BUILDERS WORKS				
	ELEMENT NO 7 - INTERNAL FINISHES				
	<u>Internal Wall Finishes</u>				
	<u>Cement and sand (1:3) backings</u>				
A	15mm thick to receive ceramic Wall tiles	SM	12	350	4,200.00
					-
B	To receive ceramic wall tiles (m.s.)	SM	24	350	8,400.00
					-
	<u>Ceramic wall tiles</u>				-
					-
	<u>Allow a Prime Cost supply/ Purchase only rate of Ksh. 1000 per SM (In addition to the cost of purchase above, the contractor's rate shall include cost of transportation, offloading, storage, fixing including all necessary adhesives, grouting and accessories</u>				-
C	Supply and Fix approved ceramic wall tiles on prepared backings(m.s) with proprietary adhesive; jointed and pointed in matching coloured proprietary grouting; including pvc spacers and expansion joint as necessary: all to Architect's approval. - Wall Surfaces	SM	12	1,650	19,800.00
					-
	<u>Ceramic wall tiles</u>				-
					-
	<u>Allow a Prime Cost supply/ Purchase only rate of Ksh. 1000 per SM (In addition to the cost of purchase above, the contractor's rate shall include cost of transportation, offloading, storage, fixing including all necessary adhesives, grouting and accessories</u>				-
D	Supply and Fix approved ceramic wall tiles on prepared backings(m.s) with proprietary adhesive; jointed and pointed in matching coloured proprietary grouting; including pvc spacers and expansion joint as necessary: all to Architect's approval. - Wall Surfaces	SM	24	1,650	39,600.00
					-
	<u>15mm (minimum) two coat cement, sand (1:3) plaster complete with wire gauze anti-crack mechanism at the intersection of masonry walling and concrete beams as described to:-</u>				-
					-
E	Concrete/masonry surfaces	SM	8	350	2,800.00
					-
	<u>Painting and Decoration</u>				-
					-
	<u>Prepare, Skim and apply Emulsion or universal undercoat followed by two finishing coats of soft satin Emulsion paint in accordance with the manufacturers written instructions and to the satisfaction of the architect to</u>				-
					-
F	Plastered concrete/masonry surfaces internally	SM	8	300	2,400.00
					-
	Carried to Collection				77,200.00

Item	Description	Unit	QTY	Rate	Amount
	COLLECTION Total brought forward from page no: 77,200.00 Total brought forward from page no: 7,800.00				
	ELEMENT NO. 7 Carried to INTERNAL FINISHES Main summary				85,000.00

Item	Description	Unit	QTY	Rate	Amount
	<u>PROPOSED GUARD HOUSE</u>				
	BILL NO.1-BUILDERS WORKS MAIN SUMMARY				
1	Substructures				221,435.00
2	Reinforced Concrete Frame				238,422.50
3	Walling				61,950.00
4	Windows				38,480.00
5	Doors				42,180.00
6	External Finishes				58,130.00
7	Internal Finishes				85,000.00
	<u>TOTAL FOR 1NO. GUARD HOUSE</u>				745,597.50
	NO. OF BLOCKS				
	MULTIPLY BY 1.NO OF GUARD HOUSE	X 1			1.00
	<u>TOTAL FOR GUARD HOUSE CARRIED TO GRAND SUMMARY</u>				745,597.50

GARBAGE RECEPTACLE

[illegible]

Item	Description	Unit	QTY	Rate	Amount
	<u>Disposal of excavated material</u>				
A	Load, wheel and cart away surplus excavated material to a Local Authority designated dumping site or fill soil heaps as away from site instructed by the Project Engineer.	CM	2	200	400.00
					-
B	Return, fill and ram selected excavated material around sides of foundations.	CM	70	100	7,000.00
					-
	Fillings				-
					-
C	Make up levels using approved imported materials: compacted in layers not exceeding 300mm thick with a roller: to the satisfaction of the Structural Engineer.	CM	0	1,450	-
					-
D	300mm thick hardcore bed: hand packed : compacted in layers not exceeding 150mm thick: to the satisfaction of the Structural Engineer	SM	112	500	56,000.00
					-
E	50 mm Stone dust/ Murrum blinding to surfaces of hardcore	SM	112	120	13,440.00
					-
	Anti - termite to treatment				-
F	Approved anti-termite treatment, with ten-year guarantee, sprayed to surfaces of hardcore strictly in accordance with manufacturer's instructions.	SM	112	150	16,800.00
					-
	Damp-proof membrane				-
G	1000 gauge polythene or other equal and approved damp-proof membrane, laid over blinded hardcore (measured separately) with 300mm side and end laps (measured nett-allow for laps); 1 No. layer: bedded in and including cement and sand (1:3) mortar	SM	112	100	11,200.00
					-
	Concrete Blinding				-
					-
	In situ concrete Class 15MPa: vibrated:				-
					-
H	50 mm thick blinding under column bases	SM	14	400	5,600.00
					-
I	50 mm thick blinding under strip foundations	SM	45	400	18,000.00
					-
	In- situ vibrated reinforced concrete Class 25 MPa: in:				-
					-
J	Column bases	CM	4	13,200	52,800.00
					-
K	Strip foundations	CM	9	13,200	118,800.00
					-
L	100mm thick surface bed	SM	112	1,320	147,840.00
					-
M	Columns	CM	1	13,200	13,200.00
					-
	Carried to collection				461,080.00

Item	Description	Unit	QTY	Rate	Amount
A	Ribbed reinforcement steel bars to KS 573:2014 : Grade 500 high tensile strength including bends, hooks, tying wire and distance blocks; to S.E's detail (Provisional)				
	Assorted reinforcement	KGS	812	165	133,980.00
	Mesh fabric reinforcement to KS/EAS 412;2 (2019) BRC A142;200 x 200mm, weighing 2.22kg/ m² (measured net - no allowance) for laps; in two layers - top & bottom; including bends, tying wire and spacer blocks)				-
B	In ground floor slab	SM	112	400	44,800.00
	<u>Modular steel frame with steel plates covering formwork and/or marine board formwork: to:</u>				-
					-
C	Sides of column bases	SM	17	500	8,500.00
D	Sides of Strip foundations	SM	30	500	15,000.00
E	Vertical sides to columns	SM	17	500	8,500.00
F	Edge of slab not exceeding 150mm girth	LM	43	75	3,225.00
	Foundation Walling				-
	Natural quarry stone walling with a minimum of 7.0 N/mm² bedded and jointed in cement and sand (1:4) mortar, reinforced with and including 25 x 3 mm thick hoop iron strips at every alternate course as described in;				-
G	200mm thick walls in foundations	SM	98	1,450	142,100.00
	Pavings				-
					-
H	Supply and lay 600 x 600 x 50mm reinforced concrete precast paving slabs around the building including laying, spreading and compacting 100mm thick approved sand bed blinding, on and including 150mm thick compacted hardcore to Engineer's approval.	SM	45	1,400	63,000.00
	Plinth				-
					-
	25mm Thick cement and sand (1:4) rendering on concrete or masonry ; wood float finished; to				-
J	Plinths externally	SM	30	350	10,500.00
	Two coats black bitumastic paint on:				-
					-
K	Rendered surfaces	SM	30	300	9,000.00
	Carried to collection				438,605.00
	COLLECTION				
	Total brought forward from page no:				58,760.00
	Total brought forward from page no:				461,080.00
	Total brought forward from page no:				438,605.00
	ELEMENT NO. 1	Carried to			
	SUBSTRUCTURES	Main summary			958,445.00

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Item	Description	Unit	QTY	Rate	Amount
A	<u>PROPOSED GARBAGE RECEPTACLE</u> BILL NO.1-BUILDERS WORKS ELEMENT NO 4-ROOF IT5 roofing sheets on steel trusses (m/s) with approved galvanised hook bolts, nuts and washers including side and end laps fixed to and including 100x50x4mm rafters spaced at 900mm c/c with 50x50x3mm SHS purlins at 600x600mm c/c with and including all welded and bolted connections : delivery to site and erection with and including one shop coat red oxide, zinc chromate or similar approved primer: complete to manufacturer's specifications	SM	101	4,000	404,000.00
	<u>ELEMENT NO. 4</u> Carried to the ROOF Main summary				404,000.00

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Item	Description	Unit	QTY	Rate	Amount
	BILL NO.1-BUILDERS WORKS MAIN SUMMARY				
1	Substructures				958,445.00
2	Reinforced Concrete Frame				244,395.00
3	Walling				249,650.00
4	Roof				404,000.00
5	Doors				164,160.00
6	External Finishes				50,050.00
7	Internal Finishes				151,550.00
	<u>TOTAL FOR 1NO. GARBAGE RECEPTACLES</u>				2,222,250.00
	NO. OF BLOCKS				
	MULTIPLY BY 1.NO OF RECEPTACLES	X 1			1.00
	<u>TOTAL FOR GARBAGE RECEPTACLES CARRIED TO GRAND SUMMARY</u>				2,222,250.00

BOUNDARY WALL

ITEM	DESCRIPTION	QTY	UNIT	RATE	AMOUNT
	<u>BOUNDARY WALL</u>				
	<u>ELEMENT NO.1</u>				
	<u>SUBSTRUCTURES</u>				
	<u>(ALL PROVISIONAL)</u>				
	<u>Siteworks and Excavations</u>				
A	Clear site of all grass, hedges, shrubs, bushes including grubbing up of roots, cart away arising debris and burn them.	SM	0	30	-
					-
	<u>Excavations</u>				-
					-
B	Excavate for Strip foundations depth not exceeding 1.50 metres starting groundlevel	CM	0	300	-
					-
C	Ditto to column base	CM	4	300	1,200.00
					-
E	Extra over excavation for excavating in soft rock	CM	0	1,000	-
					-
F	Extra over excavation for excavating in Hard rock class type 1	CM	0	1,400	-
					-
	<u>Disposal of water</u>				-
G	Allow an item for keeping all excavations free from all spring and running water by pumping or any other such means.	ITEM	1	10,000	10,000.00
					-
	<u>Planking and strutting</u>				-
					-
H	Allow an item for maintaining and upholding the sides of excavations and keeping excavations clear of all fallen materials.	ITEM	1	10,000	10,000.00
					-
					-
					-
	Carried to Collection				21,200.00

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ITEM	DESCRIPTION	QTY	UNIT	RATE	AMOUNT
	<u>SUBSTRUCTURES-(CONTINUED)</u>				
	Modular steel frame with steel plates covering formwork and/or marine board formwork: to:				
A	Sides of strip footing	SM	0	500	-
B	Ditto colum bases	SM	5	500	2,500.00
	Ribbed reinforcement steel bars to KS 573:2014 : Grade 500 high tensile strength including bends, hooks, tying wire and distance blocks; to S.E's detail (Provisional)				-
C	Assorted Reinforcement (8 - 16mm bars)	KG	114	165	18,810.00
	Foundation Wall				-
	Natural quarry stones rough dressed with a minimum compressive strength of 7.0N/mm² ;bedded and jointed in cement and sand(1:4) mortar;reinforcced with 25 x 3mm thick iron strips at alternate courses.				-
D	200mm Thick walling	SM	0	1,450	-
					-
	Carried to Collection				21,310.00

ITEM	DESCRIPTION	QTY	UNIT	RATE	AMOUNT
	<u>ELEMENT NO. 2</u>				
	<u>REINFORCED CONCRETE SUPERSTRUCTURE</u>				
	In- situ vibrated reinforced concrete Class 20 MPa: in:				
A	Columns	CM	1	13,000	13,000.00
B	Ring Beam	CM	0	13,000	-
	Modular steel frame with steel plates covering formwork and/or marine board formwork: to:				-
C	Vertical sides of columns	SM	11	500	5,500.00
	<u>Ribbed reinforcement steel bars to KS 573:2014 : Grade 500 high tensile strengthincluding bends, hooks, tying wire and distance blocks; to S.E's detail (Provisional)</u>				-
D	Assorted Reinforcement (8 - 16mm bars)	KG	108	165	17,820.00
					-
	TOTAL FOR ELEMENT NO. 2 CARRIED TO				36,320.00
	(REINFORCED CONCRETE) SUMMARY				

ITEM	DESCRIPTION	QTY	UNIT	RATE	AMOUNT
	ELEMENT NO. 3 WALLING <u>Machine cut quarry stone walling with a minimum of 7.0 N/mm2 average compressive strength to B.S 5390;bedded and jointed in cement and sand (1:4) mortar, reinforced with and including 25 x 3 mm thick hoop iron strips at every alternate course as described in;</u>				
A	200mm Thick walling	SM	0	1,450	- - -
B	Precast concrete class 20/20 coping as described in; 600 x 350 wide x 100mm thick insitu cast reinforced concrete coping, throated and weathered and jointing to columns with cement sand 1:4 mortar	NO	4	700	- 2,800.00
C	Ditto 300 x 100mm concrete coping to walls twice weathered and throated;	LM	0	600	- -
	TOTAL FOR ELEMENT NO. 3 CARRIED TO SUMMARY (WALLING)				2,800.00

ITEM	DESCRIPTION	QTY	UNIT	RATE	AMOUNT
	<u>ELEMENT NO. 4</u>				
	<u>EXTERNAL FINISHES</u>				
A	Extra over walling for smooth chisel dressing with flush pointed vertical joints and recessed horizontal joints 10 mm rod in cement and sand mix (1:3) mortar including one coat of Bituminous paint	SM	0	350	-
	<u>15mm (minimum) two coat lime plaster complete with wire gauze anti-crack mechanism at the intersection of masonry walling and concrete beams as described to:-</u>				-
B	Columns	SM	7	350	2,450.00
C	Ring beam	SM	0	350	-
	GATES				-
	Mild steel sections as described in;				-
D	6000mm wide x 2400mm high double gate comprising of 50x50x3mm RHS framing and middle rail 50x25x3mm RHS Vertical infill members at 225mm centres; priming with red oxide primer; purpose made ironmongery; all necessary lugs and grouting as per details (all with roller at the ground)	NO	1	144,000	144,000.00
E	Ditto pedestrian gate size 900x1800 ditto	NO	2	16,200	32,400.00
	Green Fencing with Chainlink				-
F	2400mm High chainlink fencing : comprising 16 gauge chainlink fixed to and including precast concrete posts :holes for and including 8No strands 16 gauge galvanised barbed wire : 16 gauge high tensile galvanised straining vertical wire at 250 centres : 700 x 600 x 650mm deep plain concrete (class 20) base at 3000mm centres : strutting posts : including planting and maintenace to maturity of bouganvillea shrubs along the fenceline; all necessary excavations and disposal	LM	400	2,800	1,120,000.00
	TOTAL FOR ELEMENT NO. 4 CARRIED TO (EXTERNAL FINISHES) SUMMARY				1,298,850.00

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CIVIL WORKS (ROADS)

NYALI AHP SITE - CIVIL WORKS

BILL NO. 15 - CIVIL WORKS

Element No. 1: Preliminary and General Items

ITEM	DESCRIPTION	UNIT		RATE (KSH)	AMOUNT (KSH)
A	Allow Provisional sum of Kenya shillings (KSh 100,000/=) for materials testing as instructed by the Engineer.	Prov. Sum			100,000.00
B	Extra Over on Item 1.01 for Contractors overheads and profits	Percentage		5%	5,000.00
Element No. 1 TOTAL CARRIED TO SUMMARY PAGE					105,000.00

Element No. 2: Site Clearance and Topsoil Stripping

ITEM	DESCRIPTION	UNIT		RATE (KShs)	AMOUNT (KShs)
A	Clear site including removal of trees (girth less than 300 mm), hedges, bushes and other vegetation and other deleterious materials, grub up roots and backfilling of holes left by removal of stumps and roots in accordance with the Specifications, as shown on the drawings and as instructed by Engineer.	Sm	1710.00	50.00	85,500.00
B	Cutting of trees of all girth above 300 mm including cutting of trunks, branches, uprooting and removal of all materials and stacking within the Right of Way and complete with filling of depressions/pits by earth including liaison with concerned authorities for obtaining permissions.				-
	(i) Girth from 300 mm to 600 mm	Nº	40.00	1,100.00	44,000.00
	(ii) Girth above 600 mm but up to 900 mm	Nº	40.00	1,800.00	72,000.00
	(iii) Girth above 900 mm but up to 1800 mm	Nº	20.00	3,000.00	60,000.00
C	Transpotation of the existing trees of girth above 600 mm girth as instructed by Engineer, including shifting of the tree and storing at locations as instructed by the Engineer.	Nº	100.00	3,000.00	300,000.00
Total of Element No. 2 (Carried Forward to Summary					561,500.00

Element No. 3: Earthworks					
ITEM	DESCRIPTION	UNIT		RATE (KShs)	AMOUNT (KShs)
	No separate payments shall be made for the overhaul of the material and the cost of such haulage shall be included in the rates and or prices.				
A	Cut to spoil in soft material	m3	855.00	300.00	256,500.00
					-
B	As Item 5.01 but in hard material	m3	855.00	700.00	598,500.00
					-
C	Provide, spread, water, process and compact 300 mm improved subgrade to 100% MDD (AASHTO T99) in two layers of 150 mm thickness	m2	1710.00	500.00	855,000.00
					-
D	Provide and compact soft material as fill material as shown in the drawing and as directed by the Engineer	m3	855.00	1,100.00	940,500.00
					-
E	Provide and fill in hard material as shown in the drawing and as directed by the Engineer.	m3	855.00	1,500.00	1,282,500.00
	Total of Element No. 3 (Carried Forward to Summary)				3,933,000.00

Element No. 4: Culvert and Drainage Works					
ITEM	DESCRIPTION	UNIT		RATE (KShs)	AMOUNT (KShs)
	No separate payment shall be made for the haulage of surplus or unsuitable excavated material and the cost of such haulage shall be included in the rates and/or prices				
A	Excavate in soft material for pipe culverts, subsoil drains, headwalls, wing walls, aprons, toe walls, drop inlets, mitre drains, catch water drains and median drains including support of trench sides, backfilling and compacting as specified or as instructed by the Engineer.	m3	72.00	300.00	21,600.00
B	As Item 8.01 but in hard material (any method)	m3	7.00	700.00	4,900.00
C	Allow for hacking in existing concrete drain for junction connections	m3	10.00	1,000.00	10,000.00
D	Allow for perforation and connecting to the existing drain including stoppage of inflowing water (hole approximately 600 widex800 high x 250 thick)	no.	10.00	10,000.00	100,000.00
E	Excavate/ desilt, grade to shape inlets outfalls, side drains to free flow conditions including cart to spoil any excess grass debris and soils as and where directed by the Engineer.	m3	900.00	300.00	270,000.00
F	Provide, lay and joint 450 mm Internal Diameter (I. D.) Reinforced Cement Concrete pipes. The rate to include backfilling and compaction to drain formation level	m	40.00	4,500.00	180,000.00
G	Ditto item 8.06 above but 600mm I.D. Reinforced Cement Concrete pipes	m	2.00	5,500.00	11,000.00
H	Ditto item 8.06 above but 900mm I.D. Reinforced Cement Concrete pipes	m	2.00	7,000.00	14,000.00
I	Provide place and compact class 25/20 concrete to headwalls, wingwalls, aprons and toe walls to pipe culverts.	m3	10.00	13,500.00	135,000.00
J	Provide place and compact 150mm class 15/20 concrete to beds and surround to 450mm diameter pipes (0.4059m3/m)	m3	16.00	6,500.00	104,000.00
K	Ditto item 8.11 above but 600mm I.D. Reinforced Cement Concrete pipes (0.5259m3/m)	m3	1.00	7,000.00	7,000.00
L	Ditto item 8.11 above but 900mm I.D. Reinforced Cement Concrete pipes (0.8118m3/m)	m3	1.00	7,000.00	7,000.00
M	Allow for in-situ lining with concrete Class 20/20 on outfall drains through built-up areas and limit of works areas and access roads	m3	2.00	13,000.00	26,000.00
N	Provide and joint 450mm diameter precast concrete invert block drain (IBD) channels with two double side precast side slabs of 600x225x75mm as lining for side drain including bedding and backfilling with selected material as directed by the Engineer.	m	600.00	2,500.00	1,500,000.00
O	Extra Over for precast side slabs of 600x225x75mm.	m	600.00	500.00	300,000.00
P	Provide all materials lay and joint shallow IBD as directed by the Engineer to form mitre drains. Rate to include provision of 100mm well compacted bed and jointed by 1:3 cement mortar	m	10.00	2,500.00	25,000.00
Q	Provide and lay 150mm thick grouted stone pitching with ratio 1:4 cement to Mortar, on culvert inlets and outlets and where directed by the Engineer.	m2	18.00	450.00	8,100.00
R	Concrete class 20/20 for side closed drains , including formwork and steel reinforcements as directed by the Engineer	m	5.00	13,000.00	65,000.00
S	Provide all materials and construct standard untrapped gully pot in concrete class 20/20 reinforced with BRC A142 and with 350x500mm polyresin frames and covers conforming with standard specifications	no.	5.00	5,500.00	27,500.00
T	Provide and place A142 fabric Mesh reinforcement or equivalent for wing walls, head walls, aprons, toe, inlets and outlets as directed by the Engineer	m2	9.00	400.00	3,600.00
U	Excavate in soft material for service ducts including support of trench sides, backfilling and compacting as specified or as instructed by the Engineer.	m3	10.00	300.00	3,000.00
V					-
W	As Item 8.20 but in hard material (any method)	m3	5.00	700.00	3,500.00
					-
X	Provide and lay 450 Dia service ducts of length 10 m each as per the drawings and as instructed by the Engineer	No.	2.00	6,400.00	12,800.00
Total of Element No. 4 (Carried Forward to Summary)					2,839,000.00

Element No. 5: Concrete Works					
ITEM	DESCRIPTION	UNIT		RATE (KSH)	AMOUNT (KSH)
	CONCRETE				
A	Provide and fix on the carriageway and parking interlocking concrete paved unshaped blocks (monolithic single layer precast concrete blocks) of any specified colour/size & shape, with approved pattern of 60 mm thick having average crushing strength of 50 N/mm ² on average thickness of 50 mm complete with uniformly graded river sand cushioning properly compacted with a mechanical compactor to required level, grade and camber as instructed by Engineer. Rate to include bedding sand and that to fill the joints, ties and edge restraints	m ²	720.00	1,100.00	792,000.00
					-
B	Extra over item 17.01 for laying blocks at speed bumps	m ²	300.00	500.00	150,000.00
					-
C	Ditto item 17.01 above but for 60mm heavy duty blocks at the walkway	m ²	300.00	500.00	150,000.00
					-
D	Provide, lay in place and joint 600x600x50mm well cured paving slabs on 50mm well compacted sand/quarry dust bed to footpaths/islands, Cloth drying areas and around the blocks as stipulated in the special Specifications.- TWO ROWS OF PCC slab	m ²	720.00	1,050.00	756,000.00
Element No. 5 TOTAL CARRIED TO SUMMARY PAGE					1,848,000.00
Element No. 6: Road Furniture					
ITEM	DESCRIPTION	UNIT		RATE (KShs)	AMOUNT (KShs)
	Road Marking and Road Signs				
	Road Marking				
A	Provide and lay hot applied thermoplastic road marking compound in approved colour and shade (ASTM 9) for road marking on bituminous surface on centerline, 100 mm, edge line 150 mm wide 3.0 mm thick, using fully automatic extrusion machine and using pre-melter for melting thermoplastic material including cleaning the surface of all dirt, dust, and other foreign matter, complete with demarcation at site/pre-marking, finishing and managing the traffic movements. Marking to be done as per the specifications, detailed drawings and as instructed by the Engineer.				
	(i) For lane marking (broken lines) with white paint, 100 mm wide	m ²	300.00	400.00	120,000.00
	(ii) For lane marking with yellow paint, 100mm wide	m ²	450.00	400.00	180,000.00
	(iii) For raised kerb lines with black paint, 150 mm wide	m ²	600.00	400.00	240,000.00
					-
B	Provide and lay hot applied thermoplastic road marking compound in approved colour and shade (ASTM 9) for road marking on bituminous road surface on pedestrian crossings, chevrons, directional arrows, give way and stop lines mm thick using fully automatic extrusion machine and using pre-melter for melting thermoplastic. Material, including dispensing drop on glass beads of approved make and as per BS 6088 at the rate of 250g/m ² including cleaning the surface of all dirt, dust, and other foreign matter, complete with demarcation at site/ pre-marking, finishing and managing the traffic control. Marking to be done as per the specifications, detailed drawings and as instructed by the Engineer.	m ²	100.00	400.00	40,000.00
					-
	Other Road Furniture				-
					-
					-
C	Provide, lay and Joint complete with hauncing as shown on the drawings and as instructed by the Engineer				-
	(i) Raised Kerbs	m	600.00	1,100.00	660,000.00
	(ii) Flush Kerbs	m	600.00	750.00	450,000.00
					-
D	Provide, lay and Joint complete with hauncing as shown on the drawings and as instructed by the Engineer 100 x 125 mm channels for the walkways and shallow drains	m	1200.00	400.00	480,000.00
Total of Element No. 6 (Carried Forward to Summary					2,170,000.00

Element	DESCRIPTION	
1	Preliminary and General Items	105,000.00
2	Site Clearance and Topsoil Stripping	561,500.00
3	Earthworks	3,933,000.00
4	Culverts and Drainage Works	2,839,000.00
5	Concrete Works	1,848,000.00
6	Road Furniture	2,170,000.00
A	Sub-total A	11,456,500.00

UNDERGROUND WATER TANK

ITEM NO	DESCRIPTION	UNIT	QTY	RATE	AMOUNT
	<u>EXTERNAL WORKS</u>				
	<u>UNDERGROUND WATER TANK</u>				
	<u>SUBSTRUCTURE (ALL PROVISIONAL)</u>				
	<u>Excavations</u>				
A	Excavate over site to remove top soil average 200mm deep and cart away.	m ²	480	150	72,000.00
					-
B	Excavate for pits commencing at reduced excavation level and not exceeding 1.50m deep.	m ³	720	300	216,000.00
					-
C	Excavate for pits commencing at reduced excavation level and not exceeding 1.50-3m deep	m ³	720	400	288,000.00
					-
D	Excavate for pits commencing at reduced excavation level and not exceeding 3-4.5m deep	m ³	2,472	500	1,236,000.00
					-
E	Extra excavation for excavating in all types of rock and cart away.	m ³	1,174	1,000	1,174,000.00
					-
F	Remove surplus excavated material from site.	m ³	3,912	200	782,400.00
					-
	<u>Disposal</u>				-
					-
G	Allow for keeping the whole of the excavation free from all spring and running water by pumping or any other such means as may be necessary	Item	1	10,000	10,000.00
					-
H	Allow for planking and strutting to sides of already excavated surfaces and new excavated surfaces to keep excavations free from all fallen materials.	Item	1	10,000	10,000.00
					-
	<u>Fillings</u>				-
					-
J	Approved 300mm wide granular filling layers not exceeding 150mm thick	m ²	480	500	240,000.00
					-
	<u>50mm Plain Concrete Blinding (Class 15) to:</u>				-
					-
K	Floor slab	m ²	480	400	192,000.00
	Carried to Collection				4,220,400.00

ITEM NO	DESCRIPTION	UNIT	QTY	RATE	AMOUNT
	<u>REINFORCED CONCRETE FRAME</u>				
	<u>Vibrated reinforced concrete (class 30); Fiber mesh to be dosed in the concrete at minimum dosage of 0.9kg/m³</u>				
A	Beams	m ²	13	14,200	184,600.00
B	250mm RC wall	m ²	138	3,550	489,900.00
C	300mm RC wall	m ²	616	4,260	2,624,160.00
D	300mm bottom slab	m ²	480	4,260	2,044,800.00
E	200mm top slab	m ²	480	2,840	1,363,200.00
F	Extra over for boxing to form opening size 600 x 450mm through 200mm thick reinforced concrete slab; rebated edges	No	4	400	1,600.00
G	Sump	No	4	12,000	48,000.00
	<u>Deformed high yield steel ribbed bars reinforcement as before described (all provisional).</u>				-
H	Assorted reinforcement	kg	62,344	165	10,286,760.00
	<u>Sawn formwork to:-</u>				-
J	Sides of beams	m ²	85	500	42,500.00
K	RC wall	m ²	1,508	500	754,000.00
L	Soffits of suspended slab	m ²	480	500	240,000.00
M	Edge of floor slab 150 - 225mm high.	m	88	113	9,900.00
N	Edge of floor slab 225 - 300mm high.	m	88	150	13,200.00
	<u>Water Bar</u>				-
P	250mm wide PVC water bar	m	107	900	96,660.00
	Carried to Collection				18,199,280.00
	<i>Underground Water Tank</i>				

ITEM NO	DESCRIPTION	UNIT	QTY	RATE	AMOUNT
	<u>FINISHES</u>				
	<u>WALL FINISHES</u>				
	<u>15mm cement and sand rendering on concrete surfaces finished with a wood float to:-</u>				
A	12mm Two coat external lime plaster on walls finished smooth with a steel trowel.	m²	1,508	350	527,800.00
	<u>Waterproofing</u>				-
	<u>25mm thick 2 coats lime waterproofed plaster in "penetron Plus" as described;</u>				-
B	300mm thick floor slab	m²	480	800	384,000.00
C	Ditto to walls	m²	1,508	800	1,206,400.00
	<u>Plastic manhole</u>				-
E	Heavy duty plastic manhole cover and frame size 600x450mm wide with cover set in frame with grease	No	4	6,500	26,000.00
F	Supply and fix approved 450mm, 10mm dia solid rungs wide stainless steel step ladder at 300mm centres fixed with SS bolts and nuts to structural details	No	4	12,000	48,000.00
	Carried to Collection				2,192,200.00
	<u>UNDERGROUND WATER TANK</u>				
	<u>COLLECTION</u>				
	Total brought forward from Page No.				4,220,400.00
	Total brought forward from Page No.				18,199,280.00
	Total brought forward from Page No.				2,192,200.00
	Carried Forward to Summary				24,611,880.00

MECHANICAL INSTALLATION WORKS

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
1.0	<u>Internal Plumbing Installations</u>				
	Supply, deliver install, Test and Commission:				
	PP-R (Polypropylene Random Co-polymer) pipes PN 20 and fittings and fusion joints to (KS ISO 15874 Part 1, 2, 3 & 5) of approved manufacturer. Rates must allow for all Metal/plastic threaded adaptors where required for the connection of sanitary fixtures, valves, sockets, sliding and fixed joints etc. as required in the running lengths of pipework. The pipes shall run in floors and wall chase. The pipes will be pressure tested before the plastering of wall commences and as per the manufacturers recommended testing pressures.				
(i)	<u>3 BR AHP</u>				
	Pipe Works				
A	32mm Ø PPR Pipe	2	LM	300	600.00
B	Ditto 25mm Ø	20	LM	150	3,000.00
	Extra Over Pipe Work				
	Elbows/ Bends				
C	32mm Ø Elbows/ Bends	2	No.	100	200.00
D	Ditto 25mm Ø	14	No.	90	1,260.00
	Equal/Unequal Tees				
E	32 x 32 x 32mm	1	No.	100	100.00
F	25 x 25 x 25mm	7	No.	90	630.00
	Reducers				
G	32 x 25mm reducer	1	No.	100	100.00
	Male/Female brass threaded adaptor				
J	25 x 20mm male/female threaded adaptor	2	No.	130	260.00
K	25 x 15mm ditto	4	LM	130	520.00
L	25 x 15mm male threaded bend	14	LM	130	1,820.00
Total Carried to Next Page					8,490.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Total from Previous Page				8,490.00
	Isolating valves				
M	25mm Ø Gate Valve as "Pegler" or Equivalent	1	No.	2,200	2,200.00
	Shut off Angle Valve				
N	Brass plated 1/2" angle valve	7	No.	600	4,200.00
	Check Meter				
O	25mm diameter water check meter	1	No.	8,500	8,500.00
	Testing and Commissioning				
P	Allow for pressure testing of the Plumbing installation to the satisfaction of the Engineer including provision of necessary pipe plugs.	1	Item	1,500	1,500.00
	Sub Total for 1 Units				24,890.00
	TOTAL FOR 1 No. 3BR AHP (Per Floor)	1		x 1	24,890.00
Total Cost of Plumbing Installation for 3BR AHP per Floor					24,890.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
(ii)	3 Room Social				
	Pipe Works				
A	32mm Ø PPR Pipe	2	LM	200	400.00
B	Ditto 25mm Ø	15	LM	150	2,250.00
	Extra Over Pipe Work				
	Elbows/ Bends				
C	32mm Ø Elbows/ Bends	2	No.	100	200.00
D	Ditto 25mm Ø	8	No.	90	720.00
	Equal/Unequal Tees				
E	32 x 32 x 32mm	1	No.	100	100.00
F	25 x 25 x 25mm	4	No.	90	360.00
	Reducers				
G	32 x 25mm reducer	1	No.	100	100.00
	Male/Female brass threaded adaptor				
J	25 x 20mm male/female threaded adaptor	2	No.	130	260.00
K	25 x 15mm ditto	4	LM	130	520.00
L	25 x 15mm male threaded bend	8	LM	130	1,040.00
	Isolating valves				
M	25mm Ø Gate Valve as "Pegler" or Equivalent	1	No.	2,200	2,200.00
	Shut off Angle Valve				
N	Brass plated 1/2" angle valve	4	No.	600	2,400.00
	Check Meter				
O	25mm diameter water check meter	1	No.	8,500	8,500.00
	Testing and Commissioning				
P	Allow for pressure testing of the Plumbing installation to the satisfaction of the Engineer including provision of necessary pipe plugs.	1	Item	1,500	1,500.00
	Sub Total for 1 Units				20,550.00
	TOTAL FOR 1 No. 3Room Social (Per Floor)	1		x 1	20,550.00
Total Cost of Plumbing Installation for 3Room Social per Floor					20,550.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
(iii)	2 BR AHP				
	Pipe Works				
A	32mm Ø PPR Pipe	2	LM	200	400.00
B	Ditto 25mm Ø	15	LM	150	2,250.00
	Extra Over Pipe Work				
	Elbows/ Bends				
C	32mm Ø Elbows/ Bends	2	No.	100	200.00
D	Ditto 25mm Ø	8	No.	90	720.00
	Equal/Unequal Tees				
E	32 x 32 x 32mm	1	No.	100	100.00
F	25 x 25 x 25mm	4	No.	90	360.00
	Reducers				
G	32 x 25mm reducer	1	No.	100	100.00
	Male/Female brass threaded adaptor				
J	25 x 20mm male/female threaded adaptor	2	No.	130	260.00
K	25 x 15mm ditto	4	LM	130	520.00
L	25 x 15mm male threaded bend	8	LM	130	1,040.00
	Isolating valves				
M	25mm Ø Gate Valve as "Pegler" or Equivalent	1	No.	2,200	2,200.00
	Shut off Angle Valve				
N	Brass plated 1/2" angle valve	4	No.	600	2,400.00
	Check Meter				
O	25mm diameter water check meter	1	No.	8,500	8,500.00
	Testing and Commissioning				
P	Allow for pressure testing of the Plumbing installation to the satisfaction of the Engineer including provision of necessary pipe plugs.	1	Item	1,500	1,500.00
	Sub Total for 1 Units				20,550.00
	TOTAL FOR 4 No. 2BR AHP (Per Floor)	4		x 4	82,200.00
Total Cost of Plumbing Installation for 2BR AHP per Floor					82,200.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
(iv)	2 Room Social				
	Pipe Works				
A	32mm Ø PPR Pipe	2	LM	200	400.00
B	Ditto 25mm Ø	15	LM	150	2,250.00
	Extra Over Pipe Work				
	Elbows/ Bends				
C	32mm Ø Elbows/ Bends	2	No.	100	200.00
D	Ditto 25mm Ø	8	No.	90	720.00
	Equal/Unequal Tees				
E	32 x 32 x 32mm	1	No.	100	100.00
F	25 x 25 x 25mm	4	No.	90	360.00
	Reducers				
G	32 x 25mm reducer	1	No.	100	100.00
	Male/Female brass threaded adaptor				
J	25 x 20mm male/female threaded adaptor	2	No.	130	260.00
K	25 x 15mm ditto	4	LM	130	520.00
L	25 x 15mm male threaded bend	8	LM	130	1,040.00
	Isolating valves				
M	25mm Ø Gate Valve as "Pegler" or Equivalent	1	No.	2,200	2,200.00
	Shut off Angle Valve				
N	Brass plated 1/2" angle valve	4	No.	600	2,400.00
	Check Meter				
O	25mm diameter water check meter	1	No.	8,500	8,500.00
	Testing and Commissioning				
P	Allow for pressure testing of the Plumbing installation to the satisfaction of the Engineer including provision of necessary pipe plugs.	1	Item	1,500	1,500.00
	Sub Total for 1 Units				20,550.00
	TOTAL FOR 2 No. 2 Room Social (Per Floor)	2		x 2	41,100.00
Total Cost of Plumbing Installation for 2Room Social per Floor					41,100.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
(v)	1 Room Social				
	Pipe Works				
A	32mm Ø PPR Pipe	2	LM	200	400.00
B	Ditto 25mm Ø	15	LM	150	2,250.00
	Extra Over Pipe Work				
	Elbows/ Bends				
C	32mm Ø Elbows/ Bends	2	No.	100	200.00
D	Ditto 25mm Ø	8	No.	90	720.00
	Equal/Unequal Tees				
E	32 x 32 x 32mm	1	No.	100	100.00
F	25 x 25 x 25mm	4	No.	90	360.00
	Reducers				
G	32 x 25mm reducer	1	No.	100	100.00
	Male/Female brass threaded adaptor				
J	25 x 20mm male/female threaded adaptor	2	No.	130	260.00
K	25 x 15mm ditto	4	LM	130	520.00
L	25 x 15mm male threaded bend	8	LM	130	1,040.00
	Isolating valves				
M	25mm Ø Gate Valve as "Pegler" or Equivalent	1	No.	2,200	2,200.00
	Shut off Angle Valve				
N	Brass plated 1/2" angle valve	4	No.	600	2,400.00
	Check Meter				
O	25mm diameter water check meter	1	No.	8,500	8,500.00
	Testing and Commissioning				
P	Allow for pressure testing of the Plumbing installation to the satisfaction of the Engineer including provision of necessary pipe plugs.	1	Item	1,500	1,500.00
	Sub Total for 1 Units				20,550.00
	TOTAL FOR 1 No. 1Room Social (Per Floor)	1		x 1	20,550.00
Total Cost of Plumbing Installation for 1Room Social per Floor					20,550.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
(vi)	<u>Studio Apartment</u>				
	Pipe Works				
A	32mm Ø PPR Pipe	2	LM	200	400.00
B	Ditto 25mm Ø	15	LM	150	2,250.00
	Extra Over Pipe Work				
	Elbows/ Bends				
C	32mm Ø Elbows/ Bends	2	No.	100	200.00
D	Ditto 25mm Ø	8	No.	90	720.00
	Equal/Unequal Tees				
E	32 x 32 x 32mm	1	No.	100	100.00
F	25 x 25 x 25mm	4	No.	90	360.00
	Reducers				
G	32 x 25mm reducer	1	No.	100	100.00
	Male/Female brass threaded adaptor				
J	25 x 20mm male/female threaded adaptor	2	No.	130	260.00
K	25 x 15mm ditto	4	LM	130	520.00
L	25 x 15mm male threaded bend	8	LM	130	1,040.00
	Isolating valves				
M	25mm Ø Gate Valve as "Pegler" or Equivalent	1	No.	2,200	2,200.00
	Shut off Angle Valve				
N	Brass plated 1/2" angle valve	4	No.	600	2,400.00
	Check Meter				
O	25mm diameter water check meter	1	No.	8,500	8,500.00
	Testing and Commissioning				
P	Allow for pressure testing of the Plumbing installation to the satisfaction of the Engineer including provision of necessary pipe plugs.	1	Item	1,500	1,500.00
	Sub Total for 1 Units				20,550.00
	TOTAL FOR 2 No. Studios (Per Floor)	2		x 2	41,100.00
Total Cost of Plumbing Installation for 2Room Social per Floor					41,100.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
(vii)	<u>1 No. Extra Apartment on Ground Floor</u>				
	Pipe Works				
A	32mm Ø PPR Pipe	2	LM	200	400.00
B	Ditto 25mm Ø	15	LM	150	2,250.00
	Extra Over Pipe Work				
	Elbows/ Bends				
C	32mm Ø Elbows/ Bends	2	No.	100	200.00
D	Ditto 25mm Ø	8	No.	90	720.00
	Equal/Unequal Tees				
E	32 x 32 x 32mm	1	No.	100	100.00
F	25 x 25 x 25mm	4	No.	90	360.00
	Reducers				
G	32 x 25mm reducer	1	No.	100	100.00
	Male/Female brass threaded adaptor				
J	25 x 20mm male/female threaded adaptor	2	No.	130	260.00
K	25 x 15mm ditto	4	LM	130	520.00
L	25 x 15mm male threaded bend	8	LM	130	1,040.00
	Isolating valves				
M	25mm Ø Gate Valve as "Pegler" or Equivalent	1	No.	2,200	2,200.00
	Shut off Angle Valve				
N	Brass plated 1/2" angle valve	4	No.	600	2,400.00
	Check Meter				
O	25mm diameter water check meter	1	No.	8,500	8,500.00
	Testing and Commissioning				
P	Allow for pressure testing of the Plumbing installation to the satisfaction of the Engineer including provision of necessary pipe plugs.	1	Item	1,500	1,500.00
	Sub Total for 1 Units				20,550.00
	TOTAL FOR 1 No. Studio (Extra)	1		x1	20,550.00
Total Cost of Plumbing Installation for the Extra Studio					20,550.00

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ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
2	Risers and Roof Levels				
	Pipe Works				
A	75mm Ø PPR pipe	15	LM	700	10,500.00
B	Ditto 63mm Ø	15	LM	600	9,000.00
B	Ditto 50mm Ø	15	LM	600	9,000.00
C	Ditto 40mm Ø	12	LM	450	5,400.00
	Extra Over Pipe Work				
	Elbows/ Bends				
D	75mm Ø Elbows/ Bends	12	No.	550	6,600.00
E	Ditto 63mm Ø	10	No.	450	4,500.00
F	Ditto 50mm Ø	8	No.	250	2,000.00
G	Ditto 40mm Ø	8	No.	200	1,600.00
H	Ditto 32mm Ø	8	No.	100	800.00
I	Ditto 25mm Ø	8	No.	90	720.00
	Equal/Unequal Tees				
J	63 x 63 x 63mm	10	No.	450	4,500.00
K	40 x 40 x 40mm	8	No.	200	1,600.00
L	40 x 40 x 32mm	8	No.	200	1,600.00
M	32 x 32 x 32mm	4	No.	100	400.00
N	32 x 32 x 25mm	4	No.	100	400.00
O	25 x 25 x 25mm	12	No.	90	1,080.00
	Reducers				
P	63 x 50mm reducer	6	No.	450	2,700.00
Q	50 x 40mm reducer	6	No.	250	1,500.00
R	40 x 32mm reducer	6	No.	200	1,200.00
S	40 x 32mm reducer	6	No.	200	1,200.00
T	32 x 25mm reducer	8	No.	100	800.00
	Male/Female brass threaded adaptor				
U	40 x 32mm male/female threaded adaptor	5	No.	800	4,000.00
V	32 x 25mm male/female threaded adaptor	7	No.	600	4,200.00
Total Carried to the Next Page					75,300.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Total from Previous Page				75,300.00
	Isolating valves				
W	65mm Ø Gate Valve as "Pegler" or Equivalent	1	No.	15,500	15,500.00
X	40mm Ø Ditto	1	No.	4,400	4,400.00
	Testing and Commissioning				
Y	Allow for pressure testing of the Plumbing installation to the satisfaction of the Engineer including provision of necessary pipe plugs.	1	Item	1,500	1,500.00
	<i>Sub Total for 1 Riser</i>				96,700.00
	TOTAL FOR 13 No. Risers	13		x 13	1,257,100.00
Total Cost of Riser & Roof Level Plumbing					1,257,100.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
3	SANITARY FITTINGS AND ACCESSORIES INSTALL ONLY				
A	Water Closet (WC) Pan Close Coupled dual flush Floor Standing Close Coupled WC Bowl complete with seat cover and cistern, WC connector relevant fittings & accessories	121	No.	1,000	121,000.00
B	Wash Basin WHB basin Full pedestal Wash Basin 450x485x230mm White Complete with bottle traps, flexible connection hoses and other accessories with cws only tap	121	No.	1,000	121,000.00
C	Shower Fittings Shower fitting c/w 15mm diameter chrome plated stop cork, Instant shower fitting and 1/2" Cobra 211- 15 Star Pillar tap as Pegler or equal and equivalent	121	No.	800	96,800.00
D	Bathroom Accessories Medium Washroom bathroom set Consisting of Toilet roll holder, soap dish holder, brush holder and Coat Hooke	121	Set	200	24,200.00
E	kitchen sink Stainless steel kitchen sink single drain, single bowl complete overflow and 40mm diameter plastic tubular p-trap PVC Bottle Trap and waste 1.5in x 40, Long Neck Wall type Bib Tap	111	No.	700	77,700.00
F	WHB Mirror 6mm thick polished plate glass silver backed mirror with bevelled edges, screwed to wall with 4No. Chrome plated dome screws complete with 5mm thick foam layer back rest and of sizes: 600mm by 450mm	121	Set	300	36,300.00
G	Fire Blanket Fire Blanket In Soft PVC Bag, Plastic box packing (Blanket Size 1m x 1m)	111	No.	800.00	88,800.00
Total Cost of Sanitary Fittings & Accessories					565,800.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	FIRE PROTECTION				
A	Hose Reel and Associated Pipework Supply and Install automatic 30 meters long, 25mm diameter hosereel and nozzle installed to KS 2003:2006	10	No.	20,000	200,000.00
	Associated Pipework GI Class 'B' tubing to KS06-259 with screwed and socketed joints to KS ISO 7-1:1994 including all range piping, fittings, hanagers, supports, brackets, and supports				
B	50mm diameter	65	LM	1,300	84,500.00
C	25mm ditto	13	LM	950	12,350.00
	Extra Over Piping For Fittings:-				
	Elbows/Various Bends				
D	50mm bend/elbow	7	No.	200	1,400.00
E	25mm ditto	20	No.	70	1,400.00
	Equal/Unequal tees				
F	50 x 50 x 50mm tee	10	No.	500	5,000.00
G	50 x 50 x 25mm ditto	19	No.	350	6,650.00
	Reducers				
H	50 x 25mm reducer	10	No.	200	2,000.00
	Unions				
I	50mm diameter union	2	No.	550	1,100.00
J	25mm ditto	20	No.	250	5,000.00
	Valves				
K	25mm diameter quarter Turn hose reel isolation valve to be as PEGLER or approved equivalent.	20	No.	3,000	60,000.00
	Painting				
L	Allow for Wire brushing , cleaning and painting of the complete fire fighting pipework installation with one coat of red oxide primer, undercoat, and gloss coat to specifications	1	Item	10,000	10,000.00
Total Cost Carried For hose reel and associated pipework Installations					389,400.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	<u>Portable Fire Protection Equipment + Hose Reel Pump</u>				
	Supply, deliver, install, test and commission portable fire protection equipment with initial fill complete with all the necessary mounting accessories.				
A	9 litres, carbon dioxide gas extinguisher	10	No.	9,500	95,000.00
B	4.5kg ABC dry powder extinguisher	10	No.	9,500	95,000.00
C	Supply and fix signs indicating the words "FIRE POINT" in 80mm high letters	10	No.	2,500	25,000.00
	Testing and Commissioning				
D	Allow for testing and commissioning of the fire pump installation to the satisfaction of the Engineer.	1	Item	5,000	5,000.00
Total Cost For Portable fire Extinguishers + Hose Reel Pump Installation					220,000.00

ITEM	DESCRIPTION	UNIT	QTY	RATE	AMOUNT
	<u>DRY RISER</u>				
	Supply , deliver and install Galvernised Iron (GI) Pipe KS 06-259 Class B, Groove fittings including fixing and jointing. Tenderers must allow in their pipework prices for Galvanised Support Brackets, Clamping screw, Threaded Rod, Bolts & all the couplings, connectors, joints etc. required in the running length of pipework and also where necessary, for pipe fixing clips, holderbats plugged and screwed, brackets and pipe sleeves through structural members.				
1	100mm diameter Black Pipe	LM	48	5,000	240,000.00
	<u>Extra over Galvernised Iron (GI) Pipe tubing for the following:</u>				
	<u>Bends</u>				
2	100mm diameter bend	No.	8	3,000	24,000.00
	<u>Tees</u>				
3	100mm ditto	No.	10	3,000	30,000.00
	<u>Reducing Bushes</u>				
4	100x65mm Black Pipe reducing bush	No.	10	2,000	20,000.00
	<u>Couplers</u>				
5	100mm diameter Black Pipe union	No.	9	2,000	18,000.00
	<u>Flanges</u>				
6	65mm dia Black Pipe flanges(pair, including bolts, nuts and gasket)	No.	10	4,000	40,000.00
	<u>Landing Valve</u>				
7	65mm diameter landing valve as Merry -Weather "Equery" constant pressure outlet with flanged inlet and 1No. 65mm dia female instantaneous outlet with blank cap and chain.	No.	10	16,500	165,000.00
	<u>Fire Hose</u>				
8	65mmdia x30m long canvas hose as specified	No.	1	10,000	10,000.00
9	Hose cradle for above canvas hose	No.	1	9,500	9,500.00
10	65mm dia branch pipe complete with nozzle/spray	No.	10	9,500	95,000.00
	<u>Gate Valves</u>				
11	100mm dia approved high pressure screw- down full way non-rising stem solid wedge disc gate valve to BS 5154 PN 16 for Series B Rating with wheel head and joints to steel tubing. As "Crane Model 156 " or equal and approved	No.	1	10,000	10,000.00
12	100mm 2-way horizontal mounted pumping in breeching inlet to BS 5041 Part 3 and horizontal inlet box with door fitted with Georgian wired glass panel to BS5041 Part 5. Inclusive of Breeching inlet cabinet as specified complete with access break glass and painted	Item	1	20,000	20,000.00
	TOTAL TO NEXT PAGE				681,500.00

ITEM	DESCRIPTION	UNIT	QTY	RATE	AMOUNT
	<i>Balance brought forward</i>				681,500.00
14	25mm dia Air Relief Valve as specified	No.	1	7,500	7,500.00
15	Wire brush , clean, and paint complete installation with one coat of red oxide primer, undercoat, and gloss coat to Architects colour including banding and colour coding to British Standards	Sum	1	5,000	5,000.00
	<u>Pressure testing and Painting</u>				
16	Allow for pressure testing of the entire Dry riser installation and obtain relevant test certificates endorsed by the Engineer or his representative.	Sum	1	5,000	5,000.00
	Sub Total For 1No. Block				699,000.00
	TOTAL COST FOR DRY RISER TO MAIN SUMMARY PAGE				699,000.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	<u>Foul Drainage (Ground Floor)</u>				
	Supply, deliver and install the following uPVC (Unplasticized Polyvinyl chloride) to (KS ISO 1452 Part 1, 2, 3, 4 & 5) Heavy gauge soil and waste system. Allow for all the various sizes of adaptor, connectors, sockets, holderbats, clips etc not measured but required for the satisfactory functioning of the system.				
	<u>Piping</u>				
A	32mm diameter heavy duty grey uPVC pipe	60	LM	180	10,800.00
B	40mm ditto	60	LM	250	15,000.00
C	50mm ditto	120	LM	350	42,000.00
D	75mm ditto	80	LM	350	28,000.00
E	100mm ditto	80	LM	1,000	80,000.00
F	100mm diameter heavy duty golden brown pipe	155	LM	750	116,250.00
G	150mm diameter heavy duty golden brown pipe	300	LM	1,000	300,000.00
	<u>Extra over uPVC Pipeworks</u>				
H	32mm sweep bend	15	No.	100	1,500.00
I	40mm sweep bend	60	No.	100	6,000.00
J	50mm sweep bend	15	No.	150	2,250.00
K	75mm sweep bend	8	No.	250	2,000.00
L	100mm sweep bend	22	No.	350	7,700.00
M	150mm sweep bend	16	No.	350	5,600.00
N	40mm 450 bend	22	No.	110	2,420.00
O	50mm 450 bend	15	No.	110	1,650.00
P	40mm sweep tee	25	No.	110	2,750.00
Q	40 x 32mm reducer	15	No.	80	1,200.00
R	75 x 40mm ditto	15	No.	180	2,700.00
S	150 x 40mm ditto	8	No.	350	2,800.00
T	40mm access plug	22	No.	150	3,300.00
U	150MM vent cowl	15	No.	750	11,250.00
	Total Carried to Next Page				645,170.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Total From Previous Page				645,170.00
V	100 x 50mm floor trap set complete with plastic grating	17	No.	800	13,600.00
	Gulley Trap				
AA	Gully trap comprising of 100mm diameter golden brown uPVC gully piece, 100mm diameter uPVC trap spigot outlet with screws and washers, and 300 x 300mm masonry gully trap chamber with mild steel plate and a heavy duty iron cover.	25	No.	4,500	112,500.00
	Manholes				
BB	Construct manhole/ inspection chamber size 450 x 600 x 750mm deep internally in 200mm stone walls, 150mm concrete bed, water proof plaster, forming drain channels, medium duty cover frame in cast iron with recessed cover with concrete infill and all necessary formwork, excavation and soil disposal.	32	No.	15,000	480,000.00
	Excavations				
CC	Excavate trench for pipe not exceeding 100mm diameter and not exceeding 1.5m deep (average 600mm deep) and make good as before.	120	LM	300	36,000.00
DD	Allow for hydrostatic pressure testing of drainage installation including provision of pipe plugs and other required fittings.	1	Item	5,000	5,000.00
	Total Cost For Ground Floor Foul Drainage Installation				1,292,270.00

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ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	<u>Foul Drainage (1St Floor to 9th Floor)</u>				
	Supply, deliver and install the following uPVC (Unplasticized Polyvinyl chloride) to (KS ISO 1452 Part 1, 2, 3, 4 & 5) Heavy gauge soil and waste system. Allow for all the various sizes of adaptor, connectors, sockets, holderbats, clips etc not measured but required for the satisfactory functioning of the system.				
	Piping				
A	32mm diameter heavy duty grey uPVC pipe	20	LM	180	3,600.00
B	40mm ditto	70	LM	250	17,500.00
C	50mm ditto	40	LM	350	14,000.00
D	100mm ditto	40	LM	750	30,000.00
E	150mm ditto	120	LM	1,000	120,000.00
	Extra over uPVC Pippings				
F	32mm sweep bend	15	No.	150	2,250.00
G	40mm sweep bend	82	No.	200	16,400.00
H	50mm sweep bend	15	No.	250	3,750.00
I	40mm 450sweep bend	25	No.	200	5,000.00
J	50mm 450sweep bend	15	No.	250	3,750.00
K	50mm y-connector	8	No.	250	2,000.00
L	50mm sweep tee	4	No.	250	1,000.00
M	40mm sweep tee	32	No.	200	6,400.00
N	40 x 32mm reducer	14	No.	200	2,800.00
O	150 x 40mm reducer	8	No.	600	4,800.00
P	150 x 50mm reducer	8	No.	600	4,800.00
Q	150 x 100mm reducer	22	No.	600	13,200.00
R	40mm access plug	25	No.	150	3,750.00
S	100mm access plug	15	No.	320	4,800.00
T	100mm single branch	2	No.	500	1,000.00
U	150mm single branch	17	No.	900	15,300.00
	Total Carried to Next Page				276,100.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Total From Previous Page				276,100.00
X	100 x 50mm floor trap set complete with polycarbonate flush grating	17	No.	3,500	59,500.00
Y	Allow for hydrostatic pressure testing of drainage installation including provision of pipe plugs and other required fittings.	1	Item	5,000	5,000.00
	Sub Total for 1 Floor				340,600.00
	TOTAL FOR 9 Floors	9		x 9	3,065,400.00
	Total Cost For 1st to 9th floor Foul Drainage Installation				3,065,400.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Rain Water Drainage				
	Supply, deliver and install UPVC (Unplasticized Polyvinyl chloride) to (KS ISO 1452 Part 1, 2, 3, 4 & 5) rainwater pipework. Allow for all flanges, couplings, nipples, connector joints, fixing clips holder bats etc as required in running length of pipework but not measured.				
	Piping				
A	100mm ditto	400	LM	300	120,000.00
	Extra over piping for fittings:-				
B	100mm ditto	16	No.	350	5,600.00
C	100mm sweep bend	2	No.	600	1,200.00
D	100mm single branch	13	No.	450	5,850.00
E	100mm double branch	7	No.	750	5,250.00
F	100mm flat roof rain water outlets	10	No.	1,200	12,000.00
	Testing and Commissioning				
G	Allow for testing and commissioning of the rain water drainage installation to the satisfaction of the engineer.	1	Item	5,000	5,000.00
					-
					-
	Total Cost For Rain Water Drainage Installation				154,900.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
1.0	Internal Plumbing Supply, deliver install, Test and Commission: PP-R (Polypropylene Random Co-polymer) pipes PN 20 and fittings and fusion joints to (KS ISO 15874 Part 1, 2, 3 & 5) of approved				
(i)	3 BR AHP				
A	Pipe Works 32mm Ø PP-R	2	LM	300	600.00
B	Ditto 25mm Ø	20	LM	150	3,000.00
	Extra Over Pipe Work				
C	Elbows/ Bends 32mm Ø	2	No.	100	200.00
D	Ditto 25mm Ø	14	No.	90	1,260.00
	Equal/Unequal Tees				
E	32 x 32 x 32mm	1	No.	100	100.00
F	25 x 25 x 25mm	7	No.	90	630.00
	Reducers 32 x 25mm				
G	reducers	1	No.	100	100.00
	Male/Female brass threaded adaptor				
J	25 x 20mm male/female	2	No.	130	260.00
K	25 x 15mm ditto	4	LM	130	520.00
L	25 x 15mm male threaded bend	14	LM	130	1,820.00
Total Carried to Next Page					8,490.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Total from Previous Page				8,490.00
M	Isolating valves 25mm O Gate Valves	1	No.	2,200	2,200.00
N	Shut off Angle Valves plated 1/2" angle valve	7	No.	600	4,200.00
O	Check Meter 25mm diameter water check	1	No.	8,500	8,500.00
P	Testing and Commissioning Allow for pressure testing of the Plumbing installation to	1	Item	1,500	1,500.00
	SUB TOTAL for 1 Unit				24,890.00
	TOTAL FOR 1 No. 3RD AHP	1		x 1	24,890.00
Total Cost of Plumbing Installation for 3BR AHP per Floor					24,890.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
(ii)	3 BR Market				
	Pipe Works				
A	52mm Ø Pipe	2	LM	300	600.00
B	Ditto 25mm Ø	20	LM	150	3,000.00
	Extra Over Pipe				
	Elbows/ Bends				
C	52mm Ø	2	No.	100	200.00
D	Ditto 25mm Ø	14	No.	90	1,260.00
	Equal/Unequal				
E	32 x 32 x 32mm	1	No.	100	100.00
F	25 x 25 x 25mm	7	No.	90	630.00
	Reducers				
G	52 x 25mm	1	No.	100	100.00
	Male/Female brass threaded adaptor				
J	25 x 25mm	2	No.	130	260.00
K	25 x 15mm ditto	4	LM	130	520.00
L	25 x 15mm male threaded head	14	LM	130	1,820.00
	Isolating valves				
M	25mm Ø Gate Valve	1	No.	2,200	2,200.00
	90° or Angle				
N	Valve plate	7	No.	600	4,200.00
	Check Meter				
O	25mm diameter water check	1	No.	8,500	8,500.00
	Testing and Commissioning				
P	Allow for pressure testing of the Plumbing installation to	1	Item	1,500	1,500.00
	SUB TOTAL for 1 Fluite				24,890.00
	TOTAL FOR 3 No. 2BR Market	3		x 3	74,670.00
Total Cost of Plumbing Installation for 3BR Market per Floor					74,670.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
(iii)	2 BR AHP				
	Pipe Works				
A	32mm Ø Pipe	2	LM	200	400.00
B	Ditto 25mm Ø	15	LM	150	2,250.00
	Extra Over Pipe				
	Elbows/ Bends				
C	32mm Ø	2	No.	100	200.00
D	Ditto 25mm Ø	8	No.	90	720.00
	Equal/Unequal				
E	32 x 32 x 32mm	1	No.	100	100.00
F	25 x 25 x 25mm	4	No.	90	360.00
	Reducers				
G	32 x 25mm	1	No.	100	100.00
	Male/Female brass threaded adaptor				
J	25 x 20mm	2	No.	130	260.00
K	25 x 15mm ditto	4	LM	130	520.00
L	25 x 15mm male threaded head	8	LM	130	1,040.00
	Isolating valves				
M	25mm Ø Gate Valve	1	No.	2,200	2,200.00
	90° or Angle				
N	Valve plate 1/2" angle valve	4	No.	600	2,400.00
	Check Meter				
O	25mm diameter water check	1	No.	8,500	8,500.00
	Testing and Commissioning				
P	Allow for pressure testing of the Plumbing installation to	1	Item	1,500	1,500.00
	SUB TOTAL for 1 Unit				20,550.00
	TOTAL FOR 1 No. 2BR AHP	1		x 1	20,550.00
Total Cost of Plumbing Installation for 2BR AHP per Floor					20,550.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
(iv)	2 BR Market				
	Pipe Works				
A	25mm Ø Pipe	2	LM	200	400.00
B	Ditto 25mm Ø	20	LM	150	3,000.00
	Extra Over Pipe				
	Elbows/ Bends				
C	25mm Ø	2	No.	100	200.00
D	Ditto 25mm Ø	14	No.	90	1,260.00
	Equal/Unequal				
E	32 x 32 x 32mm	1	No.	100	100.00
F	25 x 25 x 25mm	7	No.	90	630.00
	Reducers				
G	32 x 25mm	1	No.	100	100.00
	Male/Female brass threaded adaptor				
J	25 x 20mm	2	No.	130	260.00
K	25 x 15mm ditto	4	LM	130	520.00
L	25 x 15mm male threaded head	14	LM	130	1,820.00
	Isolating valves				
M	25mm Ø Gate Valve	1	No.	2,200	2,200.00
	90° or Angle				
N	Valve plate 1/2" angle valve	7	No.	600	4,200.00
	Check Meter				
O	25mm diameter water check	1	No.	8,500	8,500.00
	Testing and Commissioning				
P	Allow for pressure testing of the Plumbing installation to	1	Item	1,500	1,500.00
	SUB TOTAL for 1 Unit				24,690.00
	TOTAL FOR 3 No. 2BR Market	3		x 3	74,070.00
Total Cost of Plumbing Installation for 3BR Market per Floor					74,070.00

[illegible]

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
2	Wisers and Roof Levels				
	Pipe Works				
A	75mm Ø FRP pipe	15	LM	700	10,500.00
B	Ditto 63mm Ø	15	LM	600	9,000.00
B	Ditto 50mm Ø	15	LM	600	9,000.00
C	Ditto 40mm Ø	12	LM	450	5,400.00
	Extra Over Pipe Work				
	Elbows/ Bends				
D	75mm Ø Elbow / Bend	12	No.	550	6,600.00
E	Ditto 63mm Ø	10	No.	450	4,500.00
F	Ditto 50mm Ø	8	No.	250	2,000.00
G	Ditto 40mm Ø	8	No.	200	1,600.00
H	Ditto 32mm Ø	8	No.	100	800.00
I	Ditto 25mm Ø	8	No.	90	720.00
	Equal/Unequal Tees				
J	63 x 63 x 63mm	10	No.	450	4,500.00
K	40 x 40 x 40mm	8	No.	200	1,600.00
L	40 x 40 x 32mm	16	No.	200	3,200.00
M	32 x 32 x 32mm	4	No.	100	400.00
N	32 x 32 x 25mm	4	No.	100	400.00
O	25 x 25 x 25mm	12	No.	90	1,080.00
	Reducers				
P	63 x 50mm	6	No.	450	2,700.00
Q	50 x 40mm	6	No.	250	1,500.00
R	40 x 32mm	6	No.	200	1,200.00
S	40 x 32mm	6	No.	200	1,200.00
T	32 x 25mm	8	No.	100	800.00
	Male/Female				
U	40 x 32mm	5	No.	800	4,000.00
V	32 x 25mm	7	No.	600	4,200.00
Total Carried to the Next Page					76,900.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Total from Previous Page				76,900.00
W	Isolating valves 60mm Ø Gate	1	No.	15,500	15,500.00
X	Valves as 40mm Ø Ditto	1	No.	4,400	4,400.00
Y	Testing and Commissioning Allow for pressure testing of the Plumbing installation to	1	Item	1,500	1,500.00
	Sub Total for 1 Piece				98,300.00
	TOTAL FOR 8 No. Pieces	8		x 8	786,400.00
Total Cost of Riser & Roof Level Plumbing					786,400.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
3	SANITARY FITTINGS AND ACCESSORIES				
A	Water Closet (WC) Pan Close Coupled dual flush Floor Standing Close Coupled WC	15	No.	1,000	15,000.00
B	Wash Basin WHB basin Full pedestal Wash Basin 450x485x230mm	15	No.	1,000	15,000.00
C	Shower Fittings Shower fitting c/w 15mm diameter chrome plated	15	No.	800	12,000.00
D	Bathroom Accessories Medium Washroom bathroom set	15	Set	200	3,000.00
E	kitchen sink Stainless steel kitchen sink single drain, single bowl complete	8	No.	700	5,600.00
F	WHB Mirror 6mm thick polished plate glass silver backed mirror with bevelled	15	Set	300	4,500.00
G	Fire Blanket Fire blanket in Soft PVC Bag, Plastic box	8	No.	800	6,400.00
H	Bip Tap Splash Long Neck wall type Bib Tap complete with	4	No.	200	800.00
	SUB TOTAL for 1 Floor				62,300.00
	TOTAL FOR 10 Floors	10		x 10	623,000.00
Total Cost of Sanitary Fittings & Accessories					623,000.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	FIRE PROTECTION				
A	Hose Reel and Associated Pipework Supply and Install automatic 30 meters long	10	No.	20,000	200,000.00
B	Associated Pipework GI Class 'B' tubing to KS06- 259 with screwed and 50mm diameter	65	LM	1,300	84,500.00
C	25mm ditto	13	LM	950	12,350.00
	Extra Over Elbows/Various				
D	50mm 90° elbow	7	No.	200	1,400.00
E	25mm ditto	20	No.	70	1,400.00
	Equal/Unequal				
F	50 x 50 x 50mm	10	No.	500	5,000.00
G	50 x 50 x 25mm ditto	19	No.	350	6,650.00
	Reducers				
H	50 x 25mm reducers	10	No.	200	2,000.00
	Unions				
I	50mm diameter union	2	No.	550	1,100.00
J	25mm ditto	20	No.	250	5,000.00
	Valves				
K	25mm diameter quarter Turn hose reel	20	No.	3,000	60,000.00
	Painting				
L	Allow for Wire brushing , cleaning and painting of the complete fire	1	Item	10,000	10,000.00
Total Cost Carried For hose reel and associated pipework Installations					389,400.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Portable Fire				
	Supply, deliver, install, test and commission				
A	9 litres, carbon dioxide ABC dry powder	10	No.	9,500	95,000.00
B	4.5kg ABC dry powder	10	No.	9,500	95,000.00
C	Supply and fix signs indicating	10	No.	2,500	25,000.00
D	Testing and Allow for testing and	1	Item	5,000	5,000.00
Total Cost For Portable fire Extinguishers + Hose Reel Pump Installation					220,000.00

ITEM	DESCRIPTION	UNIT	QTY	RATE	AMOUNT
1	<u>DRY RISER</u> Supply , deliver and install Galvernised Iron (GI) Pipe KS 06-259 Class B, Groove fittings including fixing and jointing. Tenderers must allow in their 100mm diameter	LM	48	5,000	240,000.00
<u>Extra over Galvernised Iron (GI) Pipe tubing for the following:</u>					
2	<u>Bends</u> 100mm diameter	No.	8	3,000	24,000.00
3	<u>Tees</u> 100mm ditto	No.	10	3,000	30,000.00
4	<u>Reducing Bushes</u> 100x65mm Black	No.	10	2,000	20,000.00
5	<u>Couplers</u> 100mm diameter	No.	9	2,000	18,000.00
6	<u>Flanges</u> 65mm dia Black I	No.	10	4,000	40,000.00
7	<u>Landing Valve</u> 65mm diameter landing valve as Merry -Weather "Emergency"	No.	10	16,500	165,000.00
8	<u>Fire Hose</u> 65mmdia x30m l	No.	1	10,000	10,000.00
9	Hose cradle for a	No.	1	9,500	9,500.00
10	65mm dia branch	No.	10	9,500	95,000.00
11	<u>Gate Valves</u> 100mm dia approved high pressure screw-down full way non-rising stem	No.	1	10,000	10,000.00
12	100mm 2-way horizontal mounted pumping in breeching inlet to BS 5041 Part 3	Item	1	20,000	20,000.00
TOTAL TO NEXT PAGE					681,500.00

ITEM	DESCRIPTION	UNIT	QTY	RATE	AMOUNT
	<i>Balance brought forward</i>				681,500.00
14	25mm dia Air Re	No.	1	7,500	7,500.00
15	Wire brush , clean, and paint complete installation with one coat of red oxide primer	Sum	1	5,000	5,000.00
	<u>Pressure testing and Painting</u>				
16	Allow for pressure testing of the entire Dry riser installation	Sum	1	5,000	5,000.00
	Sub Total For 1No. Block				699,000.00
TOTAL COST FOR DRY RISER TO MAIN SUMMARY PAGE					699,000.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Room Drainage (Ground Floor)				
	Supply, deliver and install the following uPVC (Unplasticized Polyvinyl chloride) to (KS ISO 1452				
	Piping				
A	52mm diameter	24	LM	180	4,320.00
B	40mm ditto	36	LM	250	9,000.00
C	50mm ditto	72	LM	350	25,200.00
D	75mm ditto	90	LM	350	31,500.00
E	100mm ditto	90	LM	1,000	90,000.00
F	100mm diameter	120	LM	750	90,000.00
G	150mm diameter	300	LM	1,000	300,000.00
	Extra over				
H	52mm sweep	13	No.	100	1,300.00
I	40mm sweep	60	No.	100	6,000.00
J	50mm sweep	13	No.	150	1,950.00
K	75mm sweep	6	No.	250	1,500.00
L	100mm sweep	15	No.	350	5,250.00
M	150mm sweep	6	No.	350	2,100.00
N	40mm 45° bend	20	No.	110	2,200.00
O	50mm Ditto	15	No.	110	1,650.00
P	75mm Ditto	65	No.	120	7,800.00
Q	150mm Ditto	60	No.	150	9,000.00
R	40mm sweep tee	20	No.	110	2,200.00
S	50mm sweep tee	18	No.	120	2,160.00
T	75mm ditto	15	No.	120	1,800.00
U	40 x 52mm	60	No.	80	4,800.00
V	75 x 40mm ditto	20	No.	180	3,600.00
W	150 x 40mm	6	No.	350	2,100.00
X	40mm access	20	No.	150	3,000.00
Y	50mm access	13	No.	160	2,080.00
Z	75mm access	13	No.	200	2,600.00
	Total Carried to Next Page				613,110.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Total From Previous Page				613,110.00
AA	100 x 50mm floor trap set complete with 100mm vent pipe	25	No.	800	20,000.00
BB		15	No.	750	11,250.00
CC	Gulley Trap Gully trap comprising of 100mm diameter golden brown uPVC gully piece. 100mm	25	No.	4,500	112,500.00
DD	Manholes Construct manhole/ inspection chamber size 450 x 600 x 750mm deep internally in	32	No.	15,000	480,000.00
EE	Excavations Excavate trench for pipe not exceeding 100mm diameter	120	LM	300	36,000.00
II	Allow for hydrostatic pressure testing of drainage	1	Item	5,000	5,000.00
	Total Cost For Ground Floor Foul Drainage Installation				1,277,860.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Four Drainage (1st Floor to 8th)				
	Supply, deliver and install the following uPVC (Unplasticized Polyvinyl chloride) to (KS ISO 1452				
A	Piping 52mm diameter	18	LM	180	3,240.00
B	40mm ditto	60	LM	250	15,000.00
C	50mm ditto	36	LM	350	12,600.00
D	100mm ditto	30	LM	750	22,500.00
E	150mm ditto	120	LM	1,000	120,000.00
	Extra over				
F	52mm sweep	13	No.	150	1,950.00
G	40mm sweep	80	No.	200	16,000.00
H	50mm sweep	13	No.	250	3,250.00
I	40mm 450 sweep	24	No.	200	4,800.00
J	50mm 450 sweep	13	No.	250	3,250.00
K	50mm y-connector	6	No.	250	1,500.00
L	50mm sweep tee	2	No.	250	500.00
M	40mm sweep tee	30	No.	200	6,000.00
N	40 x 52mm	13	No.	200	2,600.00
O	100 x 40mm	6	No.	600	3,600.00
P	100 x 50mm	6	No.	600	3,600.00
Q	100 x 100mm	20	No.	600	12,000.00
R	40mm access	23	No.	150	3,450.00
S	100mm access	13	No.	320	4,160.00
T	100mm single	1	No.	500	500.00
U	150mm single	15	No.	900	13,500.00
V	100mm vvc	13	No.	650	8,450.00
W	52mm/100mm p tee	15	No.	450	6,750.00
	Total Carried to Next Page				269,200.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Total From Previous Page				269,200.00
X	100 x 50mm floor trap set	15	No.	3,500	52,500.00
Y	Allow for hydrostatic pressure testing of drainage	1	Item	5,000	5,000.00
	Sub Total for 1 Floor				326,700.00
	TOTAL FOR 9 Floors	9		x 9	2,940,300.00
	Total Cost For 1st to 10th floor Foul Drainage Installation				2,940,300.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Rain water Drainage				
	Supply, deliver and install UPVC (Unplasticized Polyvinyl chloride) to (KS ISO 1452				
A	Piping 100mm ditto	400	LM	300	120,000.00
B	extra over piping for 100mm ditto	16	No.	350	5,600.00
C	100mm sweep	2	No.	600	1,200.00
D	100mm single branch	13	No.	450	5,850.00
E	100mm double branch	7	No.	750	5,250.00
F	100mm flat roof rain water	10	No.	1,200	12,000.00
G	Testing and Commissioning Allow for testing and commissioning	1	Item	5,000	5,000.00
Total Cost For Rain Water Drainage Installation					154,900.00

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ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	<u>EXTERNAL WATER RETICULATION</u>				
	Supply, install below HDPE PN 16 to (KS ISO 4427 Part 1, 2, 3 & 5) ground pipe providing, fixing, jointing, testing in position & commissioning, including excavation & backfilling as required				
	Note				
	Pipe connection, tees must be electrofusion/butt fusion fittings and be included in the pipe lengths rate. The pipes will be pressure tested before the plastering of wall commences and as per the manufacturers recommended testing pressures.				
	Underground Piping complete with Electro-fusion joining of the pipes works				
A	100Ø HDPE	750	m	1,800	1,350,000.00
B	Ditto 63Ø	100	m	1,500	150,000.00
C	Ditto 50Ø	200	m	1,000	200,000.00
	Isolation Valves & Valve Chamber				
	Supply and Install Isolation valves to (KS ISO 4427 Part 1, 2, 3 & 5) Standards. Standard precast concrete valve chamber made of concrete (1:3:6) base, including formwork, excavations backfilling and disposal.				
D	Ditto 100Ø	15	No.	35,000	525,000.00
E	Ditto 75Ø		No.	30,000	
F	Ditto 63Ø		No.	28,000	
G	Ditto 50Ø	20	No.	25,000	500,000.00
	Total Carried to Next Page				2,725,000.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	From previous page				2,725,000.00
	Check/ Non- return Valve Valve Chamber & Valve Chamber				
	Supply and Install brass check valves to (KS ISO 4427 Part 1, 2, 3 & 5) Standards. Standard precast concrete valve chamber made of concrete (1:3:6) base, including formwork, excavations backfilling and disposal.				
H	Ditto 100Ø	15	No.	40,000	600,000.00
I	Ditto 75Ø		No.		
J	Ditto 50Ø	20	No.	25,000	500,000.00
	Water Meter				
	Supply and Install brass Water Meter to the engineer's approval as kent or equal and approved. Meters to include meter chambers in the rates				
K	Water meter 100Ø	1	No.	75,000	75,000.00
L	Water meter 75Ø		No.		
M	Water meter 63Ø		No.		
N	Water meter 50Ø	5	No.	30,000	150,000.00
	Garden Stand Pipe				
O	Stand pipe 15mm 2Metres long GMS stand pipe each complete with 15mm lockabler bib tap	20	No.	2,000	40,000.00
	Total Carried to Next Page				4,090,000.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	From previous page				4,090,000.00
	Supply, deliver install, Test and Commission: PP-R (Polypropylene Random Co-polymer) pipes PN 20 and fittings and fusion joints to (KS ISO 15874 Part 1, 2, 3 & 5) of approved manufacturer. Rates must allow for all Metal/plastic threaded adaptors where required for the connection of sanitary fixtures, valves, sockets, sliding and fixed joints etc. as required in the running lengths of pipework. The pipes shall run in floors and wall chase. The pipes will be				
	Underground PPR Piping				
P	PPR 25Ø	250	m	200	50,000.00
Q	Ditto 32Ø	250	m	250	62,500.00
	Water Connection				
	Making necessary connection from the Local municipal line (water supply) which is available near the project site, the scope of work include cutting the corporation road and making the same after connection. The contractor responsibility shall include submitting the necessary papers.				
R		1	Lot	50,000	50,000.00
	Pipe Sleeves 100mm diameter heavy duty PVC Class 41 pipe sleeves for crossing over pathways and driveways. The sleeves will be encased in 150mm concrete surround.				
S		60	m	200	12,000.00
	Total Carried to Next Page				4,264,500.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
1.0	<u>Internal Plumbing Installations</u>				
	Supply, deliver install, Test and Commission: PP-R (Polypropylene Random Co-polymer) pipes PN 20 and fittings and fusion joints to (KS ISO 15874 Part 1, 2, 3 & 5) of approved manufacturer. Rates must allow for all Metal/plastic threaded adaptors where required for the connection of sanitary fixtures, valves, sockets, sliding and fixed joints etc. as required in the running lengths of pipework. The pipes shall run in floors and wall chase. The pipes will be pressure tested before the plastering of wall commences and as per the manufacturers recommended testing pressures.				
	<u>Community Centre</u>				
	Pipe Works				
A	32mm Ø PPR Pipe	40	LM	300	12,000.00
B	Ditto 25mm Ø	33	LM	150	4,950.00
	Extra Over Pipe Work				
	Elbows/ Bends				
C	32mm Ø Elbows/ Bends	20	No.	100	2,000.00
D	Ditto 25mm Ø	18	No.	90	1,620.00
	Equal/Unequal Tees				
E	32 x 32 x 32mm	21	No.	100	2,100.00
F	25 x 25 x 25mm	14	No.	90	1,260.00
	Reducers				
G	32 x 25mm reducer	1	No.	100	100.00
	Male/Female brass threaded adaptor				
J	25 x 20mm male/female threaded adaptor	12	No.	130	1,560.00
K	25 x 15mm ditto	10	LM	130	1,300.00
L	25 x 15mm male threaded bend	13	LM	130	1,690.00
Total Carried to Next Page					28,580.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Total from Previous Page				28,580.00
M	Isolating valves 32mm Ø Gate Valve as "Pegler" or Equivalent	2	No.	3,600	7,200.00
N	Shut off Angle Valve Brass plated 1/2" angle valve	16	No.	600	9,600.00
O	Check Meter 50 mm diameter water check meter	1	No.	18,000	18,000.00
P	Testing and Commissioning Allow for pressure testing of the Plumbing installation to the satisfaction of the Engineer including provision of necessary pipe plugs.	1	Item	2,000	2,000.00
	TOTAL Internal Plumbing				65,380.00
Total Cost of Plumbing Installation					65,380.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
2	Risers and Roof Levels				
	Pipe Works				
A	50 mm Ø PPR pipe	28	LM	600	16,800.00
B	Ditto 40mm Ø	12	LM	450	5,400.00
C	Ditto 32mm Ø	13	LM	300	3,900.00
D	Ditto 25mm Ø	6	LM	160	960.00
	Extra Over Pipe Work				
	Elbows/ Bends				
E	50mm Ø Elbows/ Bends	4	No.	250	1,000.00
F	Ditto 40mm Ø	3	No.	200	600.00
G	Ditto 32mm Ø	3	No.	100	300.00
H	Ditto 25mm Ø	2	No.	90	180.00
	Equal/Unequal Tees				
I	50 x 40 x 50mm	4	No.	250	1,000.00
J	40 x 40 x 32mm	3	No.	200	600.00
K	32 x 32 x 32mm	4	No.	100	400.00
L	32 x 32 x 25mm	4	No.	100	400.00
M	25 x 25 x 25mm	2	No.	90	180.00
	Reducers				
O	50 x 40mm reducer	2	No.	250	500.00
P	40 x 32mm reducer	2	No.	200	400.00
Q	40 x 32mm reducer	2	No.	200	400.00
R	32 x 25mm reducer	1	No.	100	100.00
	Male/Female brass threaded adaptor				
S	40 x 32mm male/female threaded adaptor	2	No.	800	1,600.00
T	32 x 25mm male/female threaded adaptor	2	No.	600	1,200.00
Total Carried to the Next Page					35,920.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Total from Previous Page				35,920.00
U	Isolating valves 50 mm Ø Gate Valve as "Pegler" or Equivalent	2	No.	6,500	13,000.00
V	Testing and Commissioning Allow for pressure testing of the Plumbing installation to the satisfaction of the Engineer including provision of necessary pipe plugs.	1	Item	2,000	2,000.00
Total Cost of Riser & Roof Level Plumbing					50,920.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
3	SANITARY FITTINGS AND ACCESSORIES				
A	Water Closet (WC) Pan Floor Standing Close Coupled WC Bowl complete with seat cover and cistern, flexible connection hoses, WC connector relevant fittings & accessories <i>(Quote Install Only)</i>	7	No.	1,000	7,000.00
B	Wash Basin Wall Hang Wash Basin 450x485x230mm White Complete with bottle traps, flexible connection hoses and other accessories with cws only tap <i>(Quote Install Only)</i>	9	No.	1,000	9,000.00
C	WHB Mirror 6mm thick polished plate glass silver backed mirror with bevelled edges, screwed to wall with 4No. Chrome plated dome screws complete with 5mm thick foam layer back rest and of sizes: 600mm by 450mm	9	Set	300	2,700.00
D	Disabled Water Closet (WC) Pan Physically challenges set as Armitage Shanks Doc M Contour 21+ close coupled right hand corner pack, WC pan, Wash basin, water saving delay fill cistern with spatula lever, grab rails, hinged support rail with toilet roll holder, seat no cover with retaining buffers, copper tails on TMV3 mixer tap or equal & approved <i>(Quote Install Only)</i>	2	No.	10,000	20,000.00
E	Urinal Urinal bowl in white colour of size 450 x 685mm with built in spreader and concealed waste trap complete with stainless steel fixing bolts and caps. Complete with exposed Flash Valve <i>(Quote Install Only)</i>	3	No.	3,500	10,500.00
F	Flushing and Sterilization Allow for flushing and sterilization of the entire system to the satisfaction of the Engineer.	1	Item	2,000	2,000.00
Total Carried to Next Page					51,200.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	FIRE PROTECTION				
	Hose Reel and Associated Pipework				
A	Supply and Install automatic 30 meters long, 25mm diameter hosereel and nozzle installed to KS 2003:2006	2	No.	20,000	40,000.00
	Associated Pipework				
	GI Class 'B' tubing to KS06-259 with screwed and socketed joints to KS ISO 7-1:1994 including all range piping, fittings, hanagers, supports, brackets, and supports				
B	50mm diameter	25	LM	1,300	32,500.00
C	25mm ditto	4	LM	950	3,800.00
	Extra Over Piping For Fittings:-				
	Elbows/Various Bends				
D	50mm bend/elbow	6	No.	200	1,200.00
E	25mm ditto	5	No.	70	350.00
	Equal/Unequal tees				
F	50 x 50 x 50mm tee	3	No.	500	1,500.00
G	50 x 50 x 25mm ditto	3	No.	500	1,500.00
	Reducers				
H	50 x 25mm reducer	3	No.	200	600.00
	Unions				
I	50mm diameter union	2	No.	550	1,100.00
J	25mm ditto	8	No.	250	2,000.00
	Valves				
K	25mm diameter quarter Turn hose reel isolation valve to be as PEGLER or approved equivalent.	2	No.	3,000	6,000.00
	Painting				
L	Allow for Wire brushing , cleaning and painting of the complete fire fighting pipework installation with one coat of red oxide primer, undercoat, and gloss coat to specifications	1	Item	5,000	5,000.00
Total Cost Carried For hose reel and associated pipework Installations					95,550.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	<u>Portable Fire Protection Equipment</u> Supply, deliver, install, test and commission portable fire protection equipment with initial fill complete with all the necessary mounting accessories.				
A	9 litres, carbon dioxide gas extinguisher	4	No.	9,500	38,000.00
B	4.5kg ABC dry powder extinguisher	4	No.	9,500	38,000.00
C	Supply and fix signs indicating the words "FIRE POINT" in 80mm high letters	4	No.	2,500	10,000.00
	<u>Testing and Commissioning</u> Allow for testing and commissioning of the fire pump installation to the satisfaction of the Engineer.	1	Item	5,000	5,000.00
Total Cost For Portable fire Extinguishers					91,000.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	<u>Foul Drainage</u>				
	Supply, deliver and install the following uPVC(Unplasticized Polyvinyl chloride) to(KS ISO 1452 Part 1, 2, 3, 4 & 5) Heavy gauge soil and waste system. Allow for all the various sizes of adaptor, connectors, sockets, holderbats, clips etc not measured but required for the satisfactory functioning of the system.				
	<u>Piping</u>				
A	32mm diameter heavy duty grey uPVC pipe	32	LM	180	5,760.00
B	40mm ditto	50	LM	250	12,500.00
C	50mm ditto	56	LM	350	19,600.00
D	75mm ditto	12	LM	350	4,200.00
E	100mm ditto	32	LM	1,000	32,000.00
F	100mm diameter heavy duty golden brown pipe	80	LM	750	60,000.00
G	150mm diameter heavy duty golden brown pipe	44	LM	1,000	44,000.00
	<u>Extra over uPVC Pipeworks</u>				
H	32mm sweep bend	15	No.	100	1,500.00
I	40mm sweep bend	60	No.	100	6,000.00
J	50mm sweep bend	15	No.	150	2,250.00
K	75mm sweep bend	8	No.	250	2,000.00
L	100mm sweep bend	22	No.	350	7,700.00
M	150mm sweep bend	16	No.	350	5,600.00
N	40mm 450 bend	22	No.	110	2,420.00
O	50mm 450 bend	15	No.	110	1,650.00
P	40mm sweep tee	25	No.	110	2,750.00
Q	40 x 32mm reducer	15	No.	80	1,200.00
R	75 x 40mm ditto	15	No.	180	2,700.00
S	150 x 40mm ditto	8	No.	350	2,800.00
T	40mm access plug	22	No.	150	3,300.00
U	100mm bend WC connector	13	No.	1,200	15,600.00
V	40mm bottle 'p' trap	15	No.	150	2,250.00
	Total Carried to Next Page				237,780.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Total From Previous Page				237,780.00
Z	100 x 50mm floor trap set complete with polycarbonate flush grating	17	No.	3,500	59,500.00
	Gulley Trap				
AA	Gully trap comprising of 100mm diameter golden brown uPVC gully piece, 100mm diameter uPVC trap spigot outlet with screws and washers, and 300 x 300mm masonry gully trap chamber with mild steel plate and a heavy duty iron cover.	4	No.	4,500	18,000.00
	Manholes				
BB	Construct manhole/ inspection chamber size 450 x 600 x 750mm deep internally in 200mm stone walls, 150mm concrete bed, water proof plaster, forming drain channels, medium duty cover frame in cast iron with recessed cover with concrete infill and all necessary formwork, excavation and soil disposal.	10	No.	15,000	150,000.00
	Excavations				
CC	Excavate trench for pipe not exceeding 100mm diameter and not exceeding 1.5m deep (average 600mm deep) and make good as before.	120	LM	300	36,000.00
DD	Allow for hydrostatic pressure testing of drainage installation including provision of pipe plugs and other required fittings.	1	Item	20,000	20,000.00
	Total Cost For Ground Floor Foul Drainage Installation				521,280.00

[illegible]

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
1.0	Internal Plumbing Supply, deliver install, Test and Commission:				
	PP-R (Polypropylene Random Copolymer) pipes PN 20 and fittings and fusion joints to (KS ISO 15874 Part 1, 2, 3 & 5) of approved				
(i)	<u>3 BR AHP</u>				
A	Pipe Works 20mm Ø PP-R	10	LM	200	2,000.00
B	Ditto 20mm Ø	2	LM	150	300.00
	Extra Over Pipe Works				
C	Elbows/ Bends 20mm Ø	3	No.	100	300.00
D	Ditto 20mm Ø	2	No.	90	180.00
7	Equal/Unequal Tees				
E	25 x 20 x 25mm	1	No.	100	100.00
F	Reducers 25 x 20mm	2	No.	100	200.00
	Male/Female brass threaded adaptor				
9.00	25 x 20mm	2	No.	130	260.00
H	25 x 20mm	1	LM	130	130.00
I	25 x 20mm	1	LM	130	130.00
Total Carried to Next Page					3,600.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Total from Previous Page				3,600.00
J	Isolating valves 20mm Ø Gate Valves	1	No.	2,000	2,000.00
K	Shut off Angle Brass plate 1/2" angle	3	No.	600	1,800.00
L	Check Meter 20mm diameter water check	1	No.	8,500	8,500.00
M	Testing and Commissioning Allow for pressure testing of the Plumbing	1	Item	1,500	1,500.00
	Sub Total for 1				17,400.00
	TOTAL	1		x 1	17,400.00
Total Cost of Plumbing Installation Guard House					17,400.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
2.0	SANITARY FITTINGS				
A	Water Closet (WC) Pan Close Coupled dual flush Floor Standing Close Coupled WC	1	No.	1,000	1,000.00
B	Wash Basin WHB basin Full pedestal Wash Basin 450x485x730mm	1	No.	1,000	1,000.00
C	WHB Mirror 6mm thick polished plate glass silver backed mirror with bevelled	1	Set	300	300.00
D	Shower Fitting Shower fitting c/w 15mm diameter chrome plated	1	No.	600	600.00
	<i>Sub Total for 1</i>				2,900.00
	TOTAL	1		x 1	2,900.00
Total Cost of Sanitary Fittings & Accessories					2,900.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	<u>Foul Drainage</u>				
	Supply, deliver and install the following uPVC (Unplasticized Polyvinyl chloride) to				
	<u>Piping</u>				
A	52mm diameter heavy duty	2	LM	180	360.00
B	40mm ditto	3	LM	250	750.00
C	50mm ditto	6	LM	350	2,100.00
D	100mm ditto	4	LM	1,000	4,000.00
E	100mm diameter heavy	8	LM	750	6,000.00
	<u>Extra over</u>				
F	52mm sweep	1	No.	100	100.00
G	40mm sweep	1	No.	100	100.00
H	50mm sweep	2	No.	150	300.00
I	100mm sweep head	1	No.	450	450.00
J	40mm 45° bend	1	No.	110	110.00
K	50mm Ditto	1	No.	110	110.00
L	50mm sweep	2	No.	200	400.00
M	40° x 52mm	1	No.	80	80.00
N	40mm access plate	1	No.	150	150.00
	Total Carried to Next Page				15,010.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Total From Previous Page				15,010.00
AA	100 x 50mm floor trap set complete with	2	No.	800	1,600.00
BB	Gulley Trap Gully trap comprising of 100mm diameter golden brown uPVC gully piece, 100mm	1	No.	4,500	4,500.00
CC	Manholes Construct manhole/ inspection chamber size 450 x 600 x 750mm deep internally in 200mm stone	1	No.	14,000	14,000.00
DD	Excavations Excavate trench for pipe not exceeding 100mm	8	LM	300	2,400.00
EE	Allow for hydrostatic pressure testing of drainage	1	Item	5,000	5,000.00
Total Cost For Ground Floor Foul Drainage Installation					42,510.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Rain water Drainage				
	Supply, deliver and install UPVC (Unplasticized Polyvinyl chloride) to (KS ISO 1452				
A	Piping 100mm ditto	6	LM	300	1,800.00
B	Extra over piping for 100mm ditto	2	No.	350	700.00
C	100mm sweep	2	No.	600	1,200.00
D	100mm single	2	No.	450	900.00
E	100mm double	2	No.	750	1,500.00
F	100mm flat roof rain water	1	No.	1,200	1,200.00
Total Cost For Rain Water Drainage Installation					7,300.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
A	ROOF TANKS TOP TANK Deluxe heavy duty rotationally moulded polyethylene cylindrical vertical close end tank of nominal capacity of 1000 litres and Tank Capacity : 1000 litres	1	No	12,000	12,000.00
Total Cost For Roof Tanks					12,000.00

ITEM	DESCRIPTION	AMOUNT (KES)
	<u>SUMMARY PAGE</u>	
1.0	Total Cost for Internal Plumbing	17,400.00
3.0	Total Cost for Sanitary Fittings	2,900.00
6.0	Total Cost For Ground Floor Foul Drainage Installation	42,510.00
8.0	Total Cost For Rain Water Drainage Installation	7,300.00
9.0	Total Cost For Roof Tanks	12,000.00
	Total carried to Mechanical Works Main Summary Page	82,110.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
1.0	<u>Internal Plumbing Installations</u>				
	Supply, deliver install, Test and Commission:				
	PP-R (Polypropylene Random Co-polymer) pipes PN 20 and fittings and fusion joints to (KS ISO 15874 Part 1, 2, 3 & 5) of approved manufacturer. Rates must allow for all Metal/plastic threaded adaptors where required for the connection of sanitary fixtures, valves, sockets, sliding and fixed joints etc. as required in the running lengths of pipework. The pipes shall run in floors and wall chase. The pipes will be pressure tested before the plastering of wall commences and as per the manufacturers recommended testing pressures.				
	Pipe Works				
A	25mm Ø PPR Pipe	6	LM	200	1,200.00
B	Ditto 20mm Ø	2	LM	150	300.00
	Extra Over Pipe Work				
	Elbows/ Bends				
C	25mm Ø Elbows/ Bends	3	No.	100	300.00
D	Ditto 20mm Ø	2	No.	90	180.00
	Equal/Unequal Tees				
E	25 x 20 x 25mm	1	No.	100	100.00
	Reducers				
F	25 x 20mm reducer	2	No.	100	200.00
	Male/Female brass threaded adaptor				
G	25 x 20mm male/female threaded adaptor	2	No.	130	260.00
9.00	25 x 15mm ditto	1	LM	130	130.00
I	25 x 15mm male threaded bend	1	LM	130	130.00
Total Carried to Next Page					2,800.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Total from Previous Page				2,800.00
	Isolating valves				
J	25mm Ø Gate Valve as "Pegler" or Equivalent	1	No.	2,000	2,000.00
	Testing and Commissioning				
L	Allow for pressure testing of the Plumbing installation to the satisfaction of the Engineer including provision of necessary pipe plugs.	1	Item	1,500	1,500.00
	Sub Total				6,300.00
	TOTAL	1		x 1	6,300.00
Total Cost of Plumbing Installation Guard House					6,300.00

ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	Total From Previous Page				9,850.00
EE	Gulley Trap Gully trap comprising of 100mm diameter golden brown uPVC gully piece, 100mm diameter uPVC trap spigot outlet with screws and washers, and 300 x 300mm masonry gully trap chamber with mild steel plate and a heavy duty iron cover.	1	No.	4,500	4,500.00
GG	Manholes Construct manhole/ inspection chamber size 450 x 600 x 750mm deep internally in 200mm stone walls, 150mm concrete bed, water proof plaster, forming drain channels, medium duty cover frame in cast iron with recessed cover with concrete infill and all necessary formwork, excavation and soil disposal.	1	No.	14,000	14,000.00
HH	Excavations Excavate trench for pipe not exceeding 100mm diameter and not exceeding 1.5m deep (average 600mm deep) and make good as before.	4	LM	300	1,200.00
Total Cost For Ground Floor Foul Drainage Installation					29,550.00

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ITEM	DESCRIPTION	QTY.	UNIT	RATE (KES)	AMOUNT (KES)
	ROOF WATER STORAGE TANK Heavy duty rotationally moulded polyethylene cylindrical vertical close end tank of nominal capacity of 1000 Litres and complete with lid and ball valve, float switch. Tanks to be complete with inlet, outlet and overflow connections and tank connectors for the same as described below: - - 1 No. 32mm diameter inlet pipe connection - 1 No. 50mm diameter outlet pipe connection - 1 No. 40 mm diameter overflow pipe connection				
A	Tank Capacity : 500 litres	1	No	8,500	8,500.00
	Total Cost For Roof Tanks				8,500.00

PROPOSED CONSTRUCTION OF EMURUA DIKIRR AHP G+9 GRAND SUMMARY PAGE

ITEM	DESCRIPTION	Unit	Qty	RATE (KSHS)	AMOUNT (KSHS)
1	SUMMARY FOR MECHANICAL SERVICES FOR AFFORDABLE HOUSING BLOCK TYPE A	No	3	9,968,320	29,904,960.00
2	SUMMARY FOR MECHANICAL SERVICES FOR AFFORDABLE HOUSING BLOCK TYPE B	No	1	9,032,660	9,032,660.00
3	SUMMARY FOR COMMUNITY CENTER	No	1	969,630	969,630.00
4	SUMMARY FOR GATE HOUSE	No	1	82,110	82,110.00
5	SUMMARY FOR GARBAGE RECEPTACLE	No	1	52,050	52,050.00
6	SUMMARY FOR EXTERNAL RETICULATION	No	1	4,289,500	4,289,500.00
	FOR PROPOSED CONSTRUCTION OF EMURUA DIKIRR G+9 AFFORDABLE HOUSING PROJECT				44,330,910.00

8.00	OMMISSION SUM				
ITEM	DESCRIPTION	Unit	Qty	RATE (KSHS)	AMOUNT (KSHS)
8.10	Sanitary Supply Only	Item	1	35,000,000	35,000,000
8.20	Booster Pumps Supply & Installation (50m ³ /hr @ 7.5Bar)	Set	1	2,400,000	2,400,000
8.30	32,000 Liters GRP Tanks Roof Level	No	4	1,000,000	4,000,000
8.40	12,000 Liters GRP Tank, 9m Above Ground	No	1	1,000,000	1,000,000
8.50	Sump Pumps 10m ³ /hr @ 1bar	No	4	150,000.00	600,000
8.60	Hose Reel Pumps (100 l/min@ 3Bar)	Set	5	200,000	1,000,000
8.70	Borehole Drilling & Equipping Estimate	No.	1	4,500,000	4,500,000
8.80	External Reticulation Borehole Water	Item	1	101,800	101,800
8.90	External Reticulation Council Water	Item	1	131,800	131,800
9.00	2,050 PE Waste Water Treatment Plant	Item	1	20,602,500	20,602,500
	Sub Total				69,336,100

Amount in Words: Kenya Shillings **Forty Four Million, Three Hundred and Thirty Thousand, Nine Hundred and Ten Shillings Only**

Official Stamp & Address:

50797-00200 Nairobi



Tenderer's Signature.

Date: 03/02/2026

Witness' Name: **Damiana Kimeu**

Witness' Signature:

Address

50797-00200 Nairobi

Date: 03/02/2026

ELECTRICAL INSTALLATION WORKS

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	<u>BILL NO. 1</u>				
	<u>PROPOSED TYPICAL SOCIAL & AFFORDABLE UNITS BLOCK TYPE A</u>				
(i)	<u>GROUND FLOOR</u>				
A.	<u>2 BEDROOM AHP UNITS</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
1.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	6	1,100.00	6,600.00
	ii) Ditto as in item No. 1.01 but for two way switching	No.	4	1,400.00	5,600.00
1.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang one way switch	No.	3	250.00	750.00
	ii) 10A one gang two way switch	No.	5	300.00	1,500.00
	iii) 10A two gang two way switch	No.	2	350.00	700.00
1.03	LIGHTING FITTINGS				
	a) Ceiling rose complete with 3 core flex cable, lamp holder and 11W led lamp as crabtree or approved equivalent Type 'CR'	No.	6	300.00	1,800.00
	b) Spherical screwneck luminaire with opal glass and 16W compact fluorescent lamp as Astra, Micromark, Thorn or approved equivalent. "Type BF"	No.	2	300.00	600.00
Total Carried Forward to the Next Pg.					17,550.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				17,550.00
	POWER AND SOCKET OUTLET POINTS				
1.04	13A SP socket point completely wired in 3 x 2.5mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories but excluding the 13A SP switched socket outlet.	No.	7	1,600.00	11,200.00
1.05	Socket Outlet plates i) 13A SP twin switched socket outlet as MK 2536 or approved equivalent.	No.	7	500.00	3,500.00
1.06	Instant shower water heater power points completely wired in 3 x 4mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories, but excluding the 40A DP connection unit	No.	1	2,500.00	2,500.00
1.07	40A DP switched connection unit with neon indicator as MK or approved equivalent	No.	1	1,000.00	1,000.00
1.08	Cooker outlet power points completely wired in 3 x 6.0mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories, but excluding the 45A Cooker connection unit	No.	1	2,700.00	2,700.00
1.09	45A Moulded plate Cooker Control Unit complete with 13A fused Socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
1.10	45A moulded plate cooker connection unit as MK/Crabtree or approved equivalent:	No.	1	1,500.00	1,500.00
Total Carried Forward to the Next Pg.					41,450.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				41,450.00
1.11	Domestic door bell point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories but excluding the bell.	No.	1	1,500.00	1,500.00
1.12	240V/12V Ding dong domestic door bell	No.	1	2,000.00	2,000.00
	TELEPHONE, DATA AND TV POINTS POINTS				
1.13	Data point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking cover but excluding the Data socket outlet.	No.	1	1,000.00	1,000.00
1.14	DTV Socket point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking cover but excluding the DTV socket outlet.	No.	1	1,000.00	1,000.00
1.15	300x 300x 100mm deep 16G galvanized adaptable	No.	2	1,500.00	3,000.00
1.16	25mm diameter PVC HG conduit for ICT services and linking the Draw Boxes to the ICT ducts	LM	45	90.00	4,050.00
Total Carried Forward to the Next Pg.					54,000.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				54,000.00
	POWER RETICULATION AND DISTRIBUTION BOARDS				
1.17	6 ways 100A SPN Consumer Unit 'A' complete with 100A SPN integral isolator, but excluding the MCBs ,and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
1.18	MCBs for item above as Schneider Electric Acti 9				
	i)10 A SP MCB	No.	1	600.00	600.00
	ii)32 A SP MCB	No.	2	600.00	1,200.00
	iii) 45 A SP MCB	No.	1	600.00	600.00
	iv) Blanking plates	No.	2	100.00	200.00
1.19	Sub-mains circuits for CU 'A ' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the LV switchboard below the staircase.	LM	45	700.00	31,500.00
Total Carried Forward to Ground Floor Collection Pg.					94,100.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
B.	<u>3 BEDROOM AHP UNIT</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
1.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	4	1,250.00	5,000.00
	ii) Ditto as in item No. 1.01 but for two way switching	No.	6	1,400.00	8,400.00
1.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang one way switch	No.	4	250.00	1,000.00
	ii) 10A one gang two way switch	No.	8	300.00	2,400.00
	iii) 10A two gang two way switch	No.	1	350.00	350.00
1.03	LIGHTING FITTINGS				
	a) Ceiling rose complete with 3 core flex cable, lamp holder and 11W led lamp as crabtree or approved equivalent Type 'CR'	No.	7	300.00	2,100.00
	b) Spherical screwneck luminaire with opal glass and 16W compact fluorescent lamp as Astra, Micromark, Thorn or approved equivalent. "Type BF"	No.	3	300.00	900.00
Total Carried Forward to the Next Pg.					20,150.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				20,150.00
	POWER AND SOCKET OUTLET POINTS				
1.04	13A SP socket point completely wired in 3 x 2.5mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories but excluding the 13A SP switched socket outlet.	No.	8	1,600.00	12,800.00
1.05	Socket Outlet plates i) 13A SP twin switched socket outlet as MK 2536 or equal & approved equivalent.	No.	8	500.00	4,000.00
1.06	Instant shower water heater power points completely wired in 3 x 4mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories, but excluding the 40A DP connection unit	No.	2	2,500.00	5,000.00
1.07	40A DP switched connection unit with neon indicator as MK or approved equivalent	No.	2	1,000.00	2,000.00
1.08	Cooker outlet power points completely wired in 3 x 6.0mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories, but excluding the 45A DP connection unit	No.	1	2,500.00	2,500.00
1.09	45A Moulded plate Cooker Control Unit complete with 13A fused socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
1.10	45A Moulded plate Cooker Control Unit complete with 13A fused Socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
Total Carried Forward to the Next Pg.					49,450.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				49,450.00
1.11	Domestic door bell point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories but excluding the bell.	No.	1	1,500.00	1,500.00
1.12	240V/12V Ding dong domestic door bell	No.	1	2,000.00	2,000.00
	TELEPHONE, DATA AND TV POINTS POINTS				
1.13	Data point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking cover but excluding the telephone socket outlet.	No.	1	1,000.00	1,000.00
1.14	DTV Socket point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking cover but excluding the DTV socket outlet.	No.	1	1,000.00	1,000.00
1.15	300x 300x 100mm deep 16G galvanized adaptable box.	No.	2	1,500.00	3,000.00
1.16	25mm diameter PVC HG conduit for ICT services and linking the Draw Boxes to the ICT ducts	LM	45	90.00	4,050.00
Total Carried Forward to the Next Pg.					62,000.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				62,000.00
	POWER RETICULATION AND DISTRIBUTION BOARDS				
1.17	6 ways 100A SPN Consumer Unit 'A' complete with 100A SPN integral isolator ,but excluding the MCBs, and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
1.18	MCBs for item above as Schneider Electric Acti 9				
	i)10 A SP MCB	No.	1	600.00	600.00
	ii)32 A SP MCB	No.	3	600.00	1,800.00
	iii) 45 A SP MCB	No.	1	600.00	600.00
	iv) Blanking plates	No.	1	100.00	100.00
1.19	Sub-mains circuits for CU 'A ' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the Lv switchboard below the staircase.	LM	45	700.00	31,500.00
Total Carried Forward to the Ground Floor Collection Pg.					102,600.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
C.	<u>STUDIO/1 ROOM UNITS</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
1.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	1	1,250.00	1,250.00
	ii) Ditto as in item No. 1.01 but for two way switching	No.	2	1,400.00	2,800.00
1.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang one way switch	No.	1	250.00	250.00
	ii) 10A one gang two way switch	No.	2	300.00	600.00
	iii) 10A two gang two way switch	No.	1	350.00	350.00
1.03	LIGHTING FITTINGS				
	a) Ceiling rose complete with 3 core flex cable, lamp holder and 11W led lamp as crabtree or approved equivalent Type 'CR'	No.	2	300.00	600.00
	b) Spherical screwneck luminaire with opal glass and 16W compact fluorescent lamp as Astra, Micromark, Thorn or approved equivalent. "Type BF"	No.	1	300.00	300.00
Total Carried Forward to the Next Pg.					6,150.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				6,150.00
	POWER AND SOCKET OUTLET POINTS				
1.04	13A SP socket point completely wired in 3 x 2.5mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories but excluding the 13A SP switched socket outlet.	No.	4	1,600.00	6,400.00
1.05	Socket Outlet plates i) 13A SP twin switched socket outlet as MK 2536 or equal & approved equivalent.	No.	4	500.00	2,000.00
1.06	Instant shower water heater power points completely wired in 3 x 4mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories ,but excluding the 40A DP connection unit	No.	1	2,500.00	2,500.00
1.07	40A DP switched connection unit with neon indicator as MK or approved equivalent	No.	1	1,000.00	1,000.00
1.08	Cooker outlet power points completely wired in 3 x 6.0mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories ,but excluding the 45A DP connection unit	No.	1	2,500.00	2,500.00
1.09	45A Moulded plate Cooker Control Unit complete with 13A fused Socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
1.10	45A moulded plate cooker connection unit as MK/Crabtree or approved equivalent:	No.	1	1,500.00	1,500.00
Total Carried Forward to the Next Pg.					23,550.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				23,550.00
1.11	Domestic door bell point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories but excluding the bell.	No.	1	1,500.00	1,500.00
1.12	240V/12V Ding dong domestic door bell	No.	1	2,000.00	2,000.00
	TELEPHONE, DATA AND TV POINTS POINTS				
1.13	Data point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking plate but excluding the telephone socket outlet.	No.	1	1,000.00	1,000.00
1.14	DTV Socket point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories but excluding the DTV socket outlet.	No.	1	1,000.00	1,000.00
1.15	300x 300x 100mm deep 16G galvanized adaptable box.	No.	2	1,500.00	3,000.00
1.16	25mm diameter PVC HG conduit for ICT services and linking the Draw Boxes to the ICT ducts	LM	30	90.00	2,700.00
Total Carried Forward to the Next Pg.					34,750.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				34,750.00
	POWER RETICULATION AND DISTRIBUTION BOARDS				
1.17	6 ways 100A SPN Consumer Unit 'A' complete with 100A SPN integral isolator ,but excluding the MCBs ,and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
1.18	MCBs for item above as Schneider Electric Acti 9 or approved equivalent				
	i)10 A SP MCB	No.	1	600.00	600.00
	ii)32 A SP MCB	No.	3	600.00	1,800.00
	iii) 45 A SP MCB	No.	1	600.00	600.00
	iv) Blanking plates	No.	1	100.00	100.00
1.19	Sub-mains circuits for CU 'A ' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the LV switchboard below the staircase.	LM	30	700.00	21,000.00
Total Carried Forward to the Next Pg.					64,850.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
C.	<u>2 ROOM SOCIAL UNITS</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
1.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	2	1,250.00	2,500.00
	ii) Ditto as in item No. 1.01 but for two way switching	No.	3	1,400.00	4,200.00
1.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang one way switch	No.	1	250.00	250.00
	ii) 10A one gang two way switch	No.	3	300.00	900.00
	iii) 10A two gang two way switch	No.	1	350.00	350.00
1.03	LIGHTING FITTINGS				
	a) Ceiling rose complete with 3 core flex cable, lamp holder and 11W led lamp as crabtree or approved equivalent Type 'CR'	No.	4	300.00	1,200.00
	b) Spherical screwneck luminaire with opal glass and 16W compact fluorescent lamp as Astra, Micromark, Thorn or approved equivalent. "Type BF"	No.	1	300.00	300.00
Total Carried Forward to the Next Pg.					9,700.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				9,700.00
	POWER AND SOCKET OUTLET POINTS				
1.04	13A SP socket point completely wired in 3 x 2.5mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories but excluding the 13A SP switched socket outlet.	No.	6	1,600.00	9,600.00
1.05	Socket Outlet plates i) 13A SP twin switched socket outlet as MK 2536 or equal & approved equivalent.	No.	6	500.00	3,000.00
1.06	Instant shower water heater power points completely wired in 3 x 4mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories ,but excluding the 40A DP connection unit	No.	1	2,500.00	2,500.00
1.07	40A DP switched connection unit with neon indicator as MK or approved equivalent	No.	1	1,000.00	1,000.00
1.08	Cooker outlet power points completely wired in 3 x 6.0mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories ,but excluding the 45A DP connection unit	No.	1	2,500.00	2,500.00
1.09	45A Moulded plate Cooker Control Unit complete with 13A fused Socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
1.10	45A moulded plate cooker connection unit as MK/Crabtree or approved equivalent:	No.	1	1,500.00	1,500.00
Total Carried Forward to the Next Pg.					31,300.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				31,300.00
1.11	Domestic door bell point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories but excluding the bell.	No.	1	1,500.00	1,500.00
1.12	240V/12V Ding dong domestic door bell	No.	1	2,000.00	2,000.00
	TELEPHONE, DATA AND TV POINTS POINTS				
1.13	Data point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking plate but excluding the telephone socket outlet.	No.	1	1,000.00	1,000.00
1.14	DTV Socket point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories but excluding the DTV socket outlet.	No.	1	1,000.00	1,000.00
1.15	300x 300x 100mm deep 16G galvanized adaptable box.	No.	2	1,500.00	3,000.00
1.16	25mm diameter PVC HG conduit for ICT services and linking the Draw Boxes to the ICT ducts	LM	30	90.00	2,700.00
Total Carried Forward to the Next Pg.					42,500.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				42,500.00
	POWER RETICULATION AND DISTRIBUTION BOARDS				
1.17	6 ways 100A SPN Consumer Unit 'A' complete with 100A SPN integral isolator ,but excluding the MCBs ,and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
1.18	MCBs for item above as Schneider Electric Acti 9 or approved equivalent				
	i)10 A SP MCB	No.	1	600.00	600.00
	ii)32 A SP MCB	No.	3	600.00	1,800.00
	iii) 45 A SP MCB	No.	1	600.00	600.00
	iv) Blanking plates	No.	1	100.00	100.00
1.19	Sub-mains circuits for CU 'A ' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the LV switchboard below the staircase.	LM	30	700.00	21,000.00
Total Carried Forward to the Collection Pg.					72,600.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
E.	<u>3 ROOM SOCIAL UNIT</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
1.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	2	1,250.00	2,500.00
	ii) Ditto as in item No. 1.01 but for two way switching	No.	5	1,400.00	7,000.00
1.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang one way switch	No.	1	250.00	250.00
	ii) 10A one gang two way switch	No.	4	300.00	1,200.00
	iii) 10A two gang two way switch	No.	3	350.00	1,050.00
1.03	LIGHTING FITTINGS				
	a) Ceiling rose complete with 3 core flex cable, lamp holder and 11W led lamp as crabtree or approved equivalent Type 'CR'	No.	6	300.00	1,800.00
	b) Spherical screwneck luminaire with opal glass	No.	1	300.00	300.00
Total Carried Forward to the Next Pg.					14,100.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				14,100.00
	POWER AND SOCKET OUTLET POINTS				
1.04	13A SP socket point completely wired in 3 x 2.5mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories but excluding the 13A SP switched socket outlet.	No.	7	1,600.00	11,200.00
1.05	Socket Outlet plates i) 13A SP twin switched socket outlet as MK 2536 or equal & approved equivalent.	No.	7	500.00	3,500.00
1.06	Instant shower water heater power points completely wired in 3 x 4mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories, but excluding the 40A DP connection unit	No.	1	2,500.00	2,500.00
1.07	40A DP switched connection unit with neon indicator as MK or approved equivalent	No.	1	1,000.00	1,000.00
1.08	Cooker outlet power points completely wired in 3 x 6.0mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories, but excluding the 45A DP connection unit	No.	1	2,500.00	2,500.00
1.09	45A Moulded plate Cooker Control Unit complete with 13A fused socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
1.10	45A Moulded plate Cooker Control Unit complete with 13A fused Socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
Total Carried Forward to the Next Pg.					37,800.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				37,800.00
1.11	Domestic door bell point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories but excluding the bell.	No.	1	1,500.00	1,500.00
1.12	240V/12V Ding dong domestic door bell	No.	1	2,000.00	2,000.00
	TELEPHONE, DATA AND TV POINTS POINTS				
1.13	Data point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking cover but excluding the telephone socket outlet.	No.	1	1,000.00	1,000.00
1.14	DTV Socket point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking cover but excluding the DTV socket outlet.	No.	1	1,000.00	1,000.00
1.15	300x 300x 100mm deep 16G galvanized adaptable box.	No.	2	2,500.00	5,000.00
1.16	25mm diameter PVC HG conduit for ICT services and linking the Draw Boxes to the ICT ducts	Lm.	30	90.00	2,700.00
Total Carried Forward to the Next Pg.					<u>51,000.00</u>

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				51,000.00
	POWER RETICULATION AND DISTRIBUTION BOARDS				
1.17	6 ways 100A SPN Consumer Unit `A' complete with 100A SPN integral isolator ,but excluding the MCBs, and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
1.18	MCBs for item above as Schneider Electric Acti 9				
	i)10 A SP MCB	No.	1	600.00	600.00
	ii)32 A SP MCB	No.	3	600.00	1,800.00
	iii) 45 A SP MCB	No.	1	600.00	600.00
	iv) Blanking plates	No.	1	100.00	100.00
1.19	Sub-mains circuits for CU 'A ' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the Lv switchboard below the staircase.	LM	35	700.00	24,500.00
Total Carried Forward to Ground Floor Collection Page					84,600.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
F.	<u>CORRIDOR/ LIFTS LOBBY/ROOF AREA</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
1.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for two way switching, but excluding the switch and luminaire.	No.	50	1,250.00	62,500.00
1.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang two way switch	No.	4	300.00	1,200.00
	ii) 10A two gang two way switch	No.	4	350.00	1,400.00
1.03	LIGHTING FITTINGS				
	a) Circular shaped surface mounted led luminaire	No.	14	3,850.00	53,900.00
	b) Emergency Maintained Circular shaped surface	No.	3	4,200.00	12,600.00
	c) 8W bulkhead luminaire ,polycarbonate base, and clear diffuser and installed inside the lift shaft As Thorn EWB/ETB bulkhead. "Type G"	No.	2	2,500.00	5,000.00
	d) Bulkhead fitting with moulded glass diffuser	No.	36	3,500.00	126,000.00
Total Carried Forward to the Next Pg.					262,600.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				262,600.00
1.04	CCTV points in designaed locations for surveillance, comprising of 25mm diameter HG PVC conduits complete with draw wire and the blanking plate. Allow for draw boxes where all the conduits from each security point converge.	No.	6	1,200.00	7,200.00
1.05	300x 300x 100mm deep 16 G galvanized adaptable box.	No.	2	2,500.00	5,000.00
	LIGHTING CONTROL				
1.06	Presence sensor complete with assocaited wiring (light and motion sensor) as Osram Duo with the following characteristics: Operation voltage 220 - 240V 50Hz, detection area-12m dia minimum, settable light value, switch-off delay (if no motion detected), installation height - 5m minimum for staircase lights	No.	4	6,500.00	26,000.00
1.07	24A, 240V, 2P DB/CU mounted silent operation contactor for Switching External lights complete with all accessories and as CRABTREE CIK24 or an approved equivalent	No.	2	2,500.00	5,000.00
1.08	Photocell control unit and wired to energize the contactors complete with a D.P override switch as THORN QPK or approved equivalent	No.	2	2,500.00	5,000.00
1.09	Water booster power points completely wired in 5 x 6mm ² single core PVC insulated copper cables drawn in 38mm diameter heavy gauge PVC conduits including all accessories ,but excluding the the 40A TPN isolator	No.	1	5,000.00	5,000.00
1.10	40A TPN isolator as MK or approved equivalent	No.	1	3,000.00	3,000.00
Total Carried Forward to the Next Pg.					318,800.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				318,800.00
1.11	Lifts isolators power points completely wired in 5 x 6mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories ,but excluding the the 63A TPN isolator	No.	2	8,500.00	17,000.00
1.12	63A TPN isolator as MK or approved equivalent	No.	2	6,500.00	13,000.00
1.13	8 ways 100A TPN distribution boards 'T' complete with 100A TPN integral isolator ,but excluding the MCBs ,and as Schneider Electric Acti 9 or approved equivalent.	No.	2	35,000.00	70,000.00
1.14	4 ways 100A SPN Consumer Unit 'C' complete with 100A SPN integral isolator ,but excluding the MCBs ,and as Schneider Electric Acti 9 or approved equivalent for common services.	No.	2	6,000.00	12,000.00
1.15	MCBs for item above as Schneider Electric Acti 9				
	i) 10 A SP MCB	No.	6	600.00	3,600.00
	ii) 32 A SP MCB	No.	4	600.00	2,400.00
	iii) 40 A SP MCB	No.	5	600.00	3,000.00
	iv) 32 A TP MCB	No.	3	1,500.00	4,500.00
	v) Blanking plates	No.	8	100.00	800.00
1.16	Sub-mains circuits for DB 'T' above comprising of 4 core 25mm ² PVC/SWA/PVC cable drawn in cable tray/ 50mm diameter heavy gauge conduits and running from the Lv switchboard	LM	50	2,500.00	125,000.00
Total Carried Forward to the Next Pg.					570,100.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				570,100.00
1.17	Sub-mains circuits for CU 'C ' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from common distribution board.	LM	30	700.00	21,000.00
1.18	200mm x 50mm galvanised metallic cable tray (Telecom, DTV, Internet Cables) gauge 14 (riser duct) c/w all mounting accessories bends, rawl bolts, threaded bolts, brackets,	LM.	60	2,000.00	120,000.00
1.19	200mm x 50mm galvanised metallic cable tray (Power Cables) gauge 14 (riser duct) c/w all mounting accessories bends, rawl bolts, threaded bolts, brackets,	LM.	60	2,500.00	150,000.00
Total Carried Forward to Ground Floor Collection Pg.					861,100.00

ITEM	DESCRIPTION	QTY	RATE (KSH)	AMOUNT (KSH)
	GROUND FLOOR SUMMARY PAGE			
1	Total Amount for 2 Bedroom AHP Unit	4	94,100.00	376,400.00
2	Total Amount for 3 Bedroom AHP Unit	1	102,600.00	102,600.00
3	Total Amount for Studio/1 Room AHP Unit	4	64,850.00	259,400.00
4	Total Amount for 2 Room Social Unit	2	72,600.00	145,200.00
5	Total Amount for 3 Room Social Unit	1	84,600.00	84,600.00
6	Total Amount for Common Area	1	861,100.00	861,100.00
Total Carried Forward to Ground Floor Collection Pg.				1,829,300.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
(ii)	<u>TYPICAL 1ST - 9TH FLOOR</u>				
A.	<u>2 BEDROOM AHP UNITS</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
1.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	6	1,250.00	7,500.00
	ii) Ditto as in item No. 1.01 but for two way switching	No.	4	1,400.00	5,600.00
1.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang one way switch	No.	3	250.00	750.00
	ii) 10A one gang two way switch	No.	5	300.00	1,500.00
	iii) 10A two gang two way switch	No.	2	350.00	700.00
1.03	LIGHTING FITTINGS				
	a) Ceiling rose complete with 3 core flex cable, lamp holder and 11W led lamp as crabtree or approved equivalent Type 'CR'	No.	6	300.00	1,800.00
	b) Spherical screwneck luminaire with opal glass and 16W compact fluorescent lamp as Astra, Micromark, Thorn or approved equivalent. "Type BF"	No.	2	300.00	600.00
Total Carried Forward to the Next Pg.					18,450.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				18,450.00
	POWER AND SOCKET OUTLET POINTS				
1.04	13A SP socket point completely wired in 3 x 2.5mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories but excluding the 13A SP switched socket outlet.	No.	7	1,600.00	11,200.00
1.05	Socket Outlet plates i) 13A SP twin switched socket outlet as MK 2536 or approved equivalent.	No.	7	500.00	3,500.00
1.06	Instant shower water heater power points completely wired in 3 x 4mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories, but excluding the 40A DP connection unit	No.	1	2,500.00	2,500.00
1.07	40A DP switched connection unit with neon indicator as MK or approved equivalent	No.	1	1,000.00	1,000.00
1.08	Cooker outlet power points completely wired in 3 x 6.0mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories, but excluding the 45A Cooker connection unit	No.	1	2,500.00	2,500.00
1.09	45A Moulded plate Cooker Control Unit complete with 13A fused Socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
1.10	45A moulded plate cooker connection unit as MK/Crabtree or approved equivalent:	No.	1	1,500.00	1,500.00
Total Carried Forward to the Next Pg.					42,150.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				42,150.00
1.11	Domestic door bell point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories but excluding the bell.	No.	1	1,500.00	1,500.00
1.12	240V/12V Ding dong domestic door bell	No.	1	2,000.00	2,000.00
	TELEPHONE, DATA AND TV POINTS POINTS				
1.13	Data point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking cover but excluding the Data socket outlet.	No.	1	1,000.00	1,000.00
1.14	DTV Socket point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking cover but excluding the DTV socket outlet.	No.	1	1,000.00	1,000.00
1.15	300x 300x 100mm deep 16G galvanized adaptable box.	No.	2	1,500.00	3,000.00
1.16	25mm diameter PVC HG conduit for ICT services and linking the Draw Boxes to the ICT ducts	LM	45	90.00	4,050.00
Total Carried Forward to the Next Pg.					54,700.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				54,700.00
	POWER RETICULATION AND DISTRIBUTION BOARDS				
1.17	6 ways 100A SPN Consumer Unit 'A' complete with 100A SPN integral isolator, but excluding the MCBs ,and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
1.18	MCBs for item above as Schneider Electric Acti 9				
	i)10 A SP MCB	No.	1	600.00	600.00
	ii)32 A SP MCB	No.	2	600.00	1,200.00
	iii) 45 A SP MCB	No.	1	600.00	600.00
	iv) Blanking plates	No.	2	100.00	200.00
1.19	Sub-mains circuits for CU 'A ' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the LV switchboard below the staircase.	LM	65	700.00	45,500.00
Total Carried Forward to Typical Floor Collection Pg.					108,800.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
B.	<u>3 BEDROOM AHP UNIT</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
1.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	4	1,250.00	5,000.00
	ii) Ditto as in item No. 1.01 but for two way switching	No.	6	1,400.00	8,400.00
1.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang one way switch	No.	4	250.00	1,000.00
	ii) 10A one gang two way switch	No.	8	300.00	2,400.00
	iii) 10A two gang two way switch	No.	1	350.00	350.00
1.03	LIGHTING FITTINGS				
	a) Ceiling rose complete with 3 core flex cable, lamp holder and 11W led lamp as crabtree or approved equivalent Type 'CR'	No.	7	300.00	2,100.00
	b) Spherical screwneck luminaire with opal glass and 16W compact fluorescent lamp as Astra, Micromark, Thorn or approved equivalent. "Type BF"	No.	3	300.00	900.00
Total Carried Forward to the Next Pg.					20,150.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				20,150.00
	POWER AND SOCKET OUTLET POINTS				
1.04	13A SP socket point completely wired in 3 x 2.5mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories but excluding the 13A SP switched socket outlet.	No.	8	1,600.00	12,800.00
1.05	Socket Outlet plates i) 13A SP twin switched socket outlet as MK 2536 or equal & approved equivalent.	No.	8	500.00	4,000.00
1.06	Instant shower water heater power points completely wired in 3 x 4mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories, but excluding the 40A DP connection unit	No.	2	2,500.00	5,000.00
1.07	40A DP switched connection unit with neon indicator as MK or approved equivalent	No.	2	1,000.00	2,000.00
1.08	Cooker outlet power points completely wired in 3 x 6.0mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories, but excluding the 45A DP connection unit	No.	1	2,500.00	2,500.00
1.09	45A Moulded plate Cooker Control Unit complete with 13A fused socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
1.10	45A Moulded plate Cooker Control Unit complete with 13A fused Socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
Total Carried Forward to the Next Pg.					49,450.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				49,450.00
1.11	Domestic door bell point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories but excluding the bell.	No.	1	1,500.00	1,500.00
1.12	240V/12V Ding dong domestic door bell	No.	1	2,000.00	2,000.00
	TELEPHONE, DATA AND TV POINTS POINTS				
1.13	Data point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking cover but excluding the telephone socket outlet.	No.	1	1,000.00	1,000.00
1.14	DTV Socket point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking cover but excluding the DTV socket outlet.	No.	1	1,000.00	1,000.00
1.15	300x 300x 100mm deep 16G galvanized adaptable box.	No.	2	1,500.00	3,000.00
1.16	25 mm diameter PVC HG conduit for ICT services and linking the Draw Boxes to the ICT ducts	LM	45	90.00	4,050.00
Total Carried Forward to the Next Pg.					62,000.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				62,000.00
	POWER RETICULATION AND DISTRIBUTION BOARDS				
1.17	6 ways 100A SPN Consumer Unit 'A' complete with 100A SPN integral isolator ,but excluding the MCBs, and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
1.18	MCBs for item above as Schneider Electric Acti 9				
	i)10 A SP MCB	No.	1	600.00	600.00
	ii)32 A SP MCB	No.	3	600.00	1,800.00
	iii) 45 A SP MCB	No.	1	600.00	600.00
	iv) Blanking plates	No.	1	100.00	100.00
1.19	Sub-mains circuits for CU 'A ' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the Lv switchboard below the staircase.	LM	65	700.00	45,500.00
Total Carried Forward to Typical Floor Collection Pg.					116,600.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
C.	<u>STUDIO/1 ROOM UNITS</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
1.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	1	1,250.00	1,250.00
	ii) Ditto as in item No. 1.01 but for two way switching	No.	2	1,400.00	2,800.00
1.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang one way switch	No.	1	250.00	250.00
	ii) 10A one gang two way switch	No.	2	300.00	600.00
	iii) 10A two gang two way switch	No.	1	350.00	350.00
1.03	LIGHTING FITTINGS				
	a) Ceiling rose complete with 3 core flex cable, lamp holder and 11W led lamp as crabtree or approved equivalent Type 'CR'	No.	2	300.00	600.00
	b) Spherical screwneck luminaire with opal glass and 16W compact fluorescent lamp as Astra, Micromark, Thorn or approved equivalent. "Type BF"	No.	1	300.00	300.00
Total Carried Forward to the Next Pg.					6,150.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				6,150.00
	POWER AND SOCKET OUTLET POINTS				
1.04	13A SP socket point completely wired in 3 x 2.5mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories but excluding the 13A SP switched socket outlet.	No.	4	1,600.00	6,400.00
1.05	Socket Outlet plates i) 13A SP twin switched socket outlet as MK 2536 or equal & approved equivalent.	No.	4	500.00	2,000.00
1.06	Instant shower water heater power points completely wired in 3 x 4mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories ,but excluding the 40A DP connection unit	No.	1	2,500.00	2,500.00
1.07	40A DP switched connection unit with neon indicator as MK or approved equivalent	No.	1	1,000.00	1,000.00
1.08	Cooker outlet power points completely wired in 3 x 6.0mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories ,but excluding the 45A DP connection unit	No.	1	2,500.00	2,500.00
1.09	45A Moulded plate Cooker Control Unit complete with 13A fused Socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
1.10	45A moulded plate cooker connection unit as MK/Crabtree or approved equivalent:	No.	1	1,500.00	1,500.00
Total Carried Forward to the Next Pg.					23,550.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				23,550.00
1.11	Domestic door bell point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories but excluding the bell.	No.	1	1,500.00	1,500.00
1.12	240V/12V Ding dong domestic door bell	No.	1	2,000.00	2,000.00
	TELEPHONE, DATA AND TV POINTS POINTS				
1.13	Data point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking plate but excluding the telephone socket outlet.	No.	1	1,000.00	1,000.00
1.14	DTV Socket point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories but excluding the DTV socket outlet.	No.	1	1,000.00	1,000.00
1.15	300x 300x 100mm deep 16G galvanized adaptable box.	No.	2	1,500.00	3,000.00
1.16	25mm diameter PVC HG conduit for ICT services and linking the Draw Boxes to the ICT ducts	LM	30	90.00	2,700.00
Total Carried Forward to the Next Pg.					34,750.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				34,750.00
	POWER RETICULATION AND DISTRIBUTION BOARDS				
1.17	6 ways 100A SPN Consumer Unit 'A' complete with 100A SPN integral isolator ,but excluding the MCBs ,and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
1.18	MCBs for item above as Schneider Electric Acti 9				
	i)10 A SP MCB	No.	1	600.00	600.00
	ii)32 A SP MCB	No.	3	600.00	1,800.00
	iii) 45 A SP MCB	No.	1	600.00	600.00
	iv) Blanking plates	No.	1	100.00	100.00
1.19	Sub-mains circuits for CU 'A ' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the LV switchboard below the staircase.	LM	35	700.00	24,500.00
Total Carried Forward to Typical Floor Collection Pg.					68,350.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
D.	<u>2 ROOM SOCIAL UNITS</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
1.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	2	1,250.00	2,500.00
	ii) Ditto as in item No. 1.01 but for two way switching	No.	3	1,400.00	4,200.00
1.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A two gang one way switch	No.	1	250.00	250.00
	ii) 10A one gang two way switch	No.	3	300.00	900.00
	iii) 10A two gang two way switch	No.	1	350.00	350.00
1.03	LIGHTING FITTINGS				
	a) Ceiling rose complete with 3 core flex cable, lamp holder and 11W led lamp as crabtree or approved equivalent Type 'CR'	No.	4	300.00	1,200.00
	b) Spherical screwneck luminaire with opal glass and 16W compact fluorescent lamp as Astra, Micromark, Thorn or approved equivalent. "Type BF"	No.	1	300.00	300.00
Total Carried Forward to the Next Pg.					9,700.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				9,700.00
	POWER AND SOCKET OUTLET POINTS				
1.04	13A SP socket point completely wired in 3 x 2.5mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories but excluding the 13A SP switched socket outlet.	No.	6	1,600.00	9,600.00
1.05	Socket Outlet plates i) 13A SP twin switched socket outlet as MK 2536 or equal & approved equivalent.	No.	6	500.00	3,000.00
1.06	Instant shower water heater power points completely wired in 3 x 4mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories, but excluding the 40A DP connection unit	No.	1	2,500.00	2,500.00
1.07	40A DP switched connection unit with neon indicator as MK or approved equivalent	No.	1	1,000.00	1,000.00
1.08	Cooker outlet power points completely wired in 3 x 6.0mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories ,but excluding the 45A DP connection unit	No.	1	2,500.00	2,500.00
1.09	45A Moulded plate Cooker Control Unit complete with 13A fused socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
1.10	45A moulded plate cooker connection unit as MK/Crabtree or approved equivalent:	No.	1	1,500.00	1,500.00
Total Carried Forward to the Next Pg.					31,300.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				31,300.00
1.11	Domestic door bell point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories but excluding the bell.	No.	1	1,500.00	1,500.00
1.12	240V/12V Ding dong domestic door bell	No.	1	2,000.00	2,000.00
	TELEPHONE, DATA AND TV POINTS POINTS				
1.13	Data point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking plate but excluding the data socket outlet.	No.	1	1,000.00	1,000.00
1.14	DTV Socket point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories but excluding the DTV socket outlet.	No.	1	1,000.00	1,000.00
1.15	300x 300x 100mm deep 16G galvanized adaptable box.	No.	2	1,500.00	3,000.00
1.16	25mm diameter PVC HG conduit for ICT services and linking the Draw Boxes to the ICT ducts	LM	30	90.00	2,700.00
Total Carried Forward to the Next Pg.					42,500.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				42,500.00
	POWER RETICULATION AND DISTRIBUTION BOARDS				
1.17	6 ways 100A SPN Consumer Unit 'A' complete with 100A SPN integral isolator, but excluding the MCBs ,and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
1.18	MCBs for item above as Schneider Electric Acti 9				
	i)10 A SP MCB	No.	1	600.00	600.00
	ii)32 A SP MCB	No.	3	600.00	1,800.00
	iii) 45 A SP MCB	No.	1	600.00	600.00
	iv) Blanking plates	No.	1	100.00	100.00
1.19	Sub-mains circuits for CU 'A ' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the LVswitchboard below the staircase.	LM	30	700.00	21,000.00
Total Carried Forward to Typical Floor Collection Pg.					72,600.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
E.	<u>3 ROOM SOCIAL UNIT</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
1.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	2	1,250.00	2,500.00
	ii) Ditto as in item No. 1.01 but for two way switching	No.	5	1,400.00	7,000.00
1.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang one way switch	No.	1	250.00	250.00
	ii) 10A one gang two way switch	No.	4	300.00	1,200.00
	iii) 10A two gang two way switch	No.	3	350.00	1,050.00
1.03	LIGHTING FITTINGS				
	a) Ceiling rose complete with 3 core flex cable, lamp holder and 11W led lamp as crabtree or approved equivalent Type 'CR'	No.	6	300.00	1,800.00
	b) Spherical screwneck luminaire with opal glass and 16W compact fluorescent lamp as Astra, Micromark, Thorn or approved equivalent. "Type BF"	No.	1	300.00	300.00
Total Carried Forward to the Next Pg.					14,100.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				14,100.00
	POWER AND SOCKET OUTLET POINTS				
1.04	13A SP socket point completely wired in 3 x 2.5mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories but excluding the 13A SP switched socket outlet.	No.	7	1,600.00	11,200.00
1.05	Socket Outlet plates i) 13A SP twin switched socket outlet as MK 2536 or equal & approved equivalent.	No.	7	500.00	3,500.00
1.06	Instant shower water heater power points completely wired in 3 x 4mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories, but excluding the 40A DP connection unit	No.	1	2,500.00	2,500.00
1.07	40A DP switched connection unit with neon indicator as MK or approved equivalent	No.	1	1,000.00	1,000.00
1.08	Cooker outlet power points completely wired in 3 x 6.0mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories, but excluding the 45A DP connection unit	No.	1	2,500.00	2,500.00
1.09	45A Moulded plate Cooker Control Unit complete with 13A fused socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
1.10	45A Moulded plate Cooker Control Unit complete with 13A fused Socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
Total Carried Forward to the Next Pg.					37,800.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				37,800.00
1.11	Domestic door bell point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories but excluding the bell.	No.	1	1,500.00	1,500.00
1.12	240V/12V Ding dong domestic door bell	No.	1	2,000.00	2,000.00
	TELEPHONE, DATA AND TV POINTS POINTS				
1.13	Data point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking cover but excluding the telephone socket outlet.	No.	1	1,000.00	1,000.00
1.14	DTV Socket point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking cover but excluding the DTV socket outlet.	No.	1	1,000.00	1,000.00
1.15	300x 300x 100mm deep 16G galvanized adaptable box.	No.	2	1,500.00	3,000.00
1.16	25mm diameter PVC HG conduit for ICT services and linking the Draw Boxes to the ICT ducts	Lm.	30	90.00	2,700.00
Total Carried Forward to the Next Pg.					49,000.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				49,000.00
	POWER RETICULATION AND DISTRIBUTION BOARDS				
1.17	6 ways 100A SPN Consumer Unit 'A' complete with 100A SPN integral isolator ,but excluding the MCBs, and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
1.18	MCBs for item above as Schneider Electric Acti 9				
	i)10 A SP MCB	No.	1	600.00	600.00
	ii)32 A SP MCB	No.	3	600.00	1,800.00
	iii) 45 A SP MCB	No.	1	600.00	600.00
	iv) Blanking plates	No.	1	100.00	100.00
1.19	Sub-mains circuits for CU 'A ' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the Lv switchboard below the staircase.	LM	45	700.00	31,500.00
Total Carried Forward to Typical Floor Collection Pg.					89,600.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
F.	<u>CORRIDOR/ LIFTS LOBBY AREA</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
1.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for two way switching, but excluding the switch and luminaire.	No.	17	1,250.00	21,250.00
1.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang two way switch	No.	4	300.00	1,200.00
	ii) 10A two gang two way switch	No.	4	350.00	1,400.00
1.03	LIGHTING FITTINGS				
	a) Circular shaped surface mounted led luminaire	No.	14	3,850.00	53,900.00
	b) Emergency Maintained Circular shaped surface	No.	3	4,200.00	12,600.00
	c) 8W bulkhead luminaire ,polycarbonate base, and clear diffuser and installed inside the lift shaft As Thorn EWB/ETB bulkhead. "Type G"	No.	2	2,500.00	5,000.00
Total Carried Forward to Next Pg.					95,350.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				95,350.00
1.04	CCTV points in designated locations for surveillance, comprising of 25mm diameter HG PVC conduits complete with draw wire and the blanking plate. Allow for draw boxes where all the conduits from each security point converge.	No.	6	1,200.00	7,200.00
1.05	300x 300x 100mm deep 16G galvanized adaptable box.	No.	2	2,500.00	5,000.00
	LIGHTING CONTROL				
1.06	Presence sensor complete with associated wiring (light and motion sensor) as Osram Duo with the following characteristics: Operation voltage 220 - 240V 50Hz, detection area-12m dia minimum, settable light value, switch-off delay (if no motion detected), installation height - 5m minimum	No.	3	6,500.00	19,500.00
1.07	4 ways 100A SPN Consumer Unit 'A' complete with 100A SPN integral isolator, including the MCBs, and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
1.08	Sub-mains circuits for CU 'A' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the LV switchboard below the staircase.	LM	40	700.00	28,000.00
Total Carried Forward to Next Pg.					161,050.00

ITEM	DESCRIPTION	QTY	RATE (KSH)	AMOUNT (KSH)
	TYPICAL FLOOR SUMMARY			
1	Total Amount for 2 Bedroom AHP Unit	36	108,800.00	3,916,800.00
2	Total Amount for 3 Bedroom AHP Unit	9	116,600.00	1,049,400.00
3	Total Amount for Studio AHP Unit	18	68,350.00	1,230,300.00
4	Total Amount for 1 Room Social Unit	9	68,350.00	615,150.00
5	Total Amount for 2 Room Social Unit	18	72,600.00	1,306,800.00
6	Total Amount for 3 Room Social Unit	9	89,600.00	806,400.00
7	Total Amount for Common Area	9	161,050.00	1,449,450.00
Total for 9No. Typical Floor Carried Forward to Main Summary Page				10,374,300.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
(iii)	<u>EARTHING AND LIGHTNING PROTECTION</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	Air Termination 2000mm x15mmØ multiple point pure copper AirRods/ Termination with spikes as Furse Part No. RA240 or approved equivalent	No.	3	1,250.00	3,750.00
1.01					
1.02	Copper Air Rod Base as Furse Part No. SD105-H or approved equivalent	No.	3	1,400.00	4,200.00
1.03	Copper Junction Clamps for copper tape as Furse Part No. CN105-H or approved equivalent	No.	3	250.00	750.00
1.04	25mm x 3mm Tinned Copper Tape as Furse TC230 or approved equivalent	Lm.	300	1,200.00	360,000.00
1.05	Down Conductor 25mm x 3mm Tinned Copper Tape as Furse TC230 or approved equivalent	Lm.	200	900.00	180,000.00
1.06	Screwdown copper test clamp as Furse CT305 or approved equivalent	No.	3	1,400.00	4,200.00
1.07	38mm Ø HG PVC conduits for drawing the down conductor above.	Lm.	200	120.00	24,000.00
Total Carried Forward to Next Pg.					576,900.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				576,900.00
	Earth Termination				
1.08	15mm Ø, 1500mm long solid copper earth rod c/w driving stud, coupling, and spike as Furse RC011 or approved equivalent	No	3	3,000.00	9,000.00
1.09	Earth rod to tape clamp type A as Furse CR108 or approved equivalent	No.	3	2,500.00	7,500.00
1.10	Concrete inspection earth pit Cat. No. PT 005 with 5 hole earth bar as Furse Cat. No. PT 006.	No.	2	2,500.00	5,000.00
1.11	600mm x 600mm copper earth mat made from 25mm x 3mm copper tape at 300mm spacing, buried to permanent moisture level and complete with all clamps and 6m long 25mm x 3mm copper tape clamped to the down conductor, soil conditioning agents (marconite or bentonite) as necessary to achieve earthing resistance value below 10-Ohms	Lot	2	6,000.00	12,000.00
1.12	Allow for earthing tests for the above and submission of the report to the engineer to BS7671 & BS62305 standards	No.	2	5,000.00	10,000.00
1.13	Bonding Bonding and clamping to all metal work including water pipes, gas pipes, hand-rails, smatv system, window frames, cladding, metal roof etc. and the main earth for the building.	Item	2	20,000.00	40,000.00
Total Carried Forward to Main Summary Pg.					660,400.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
(iv)	<u>LV SWITCHBOARD-GROUND FLOOR</u>				
	TAKE DELIVERY, STORE, INSTALL AND COMMISSION THE FOLLOWING:-				
1.01	Install purpose made Free-standing, fully wired front access metal clad main switchboard cubicle suitable for floor / wall mounting and manufactured in 14SWG galvanised mild steel sheet, to be finished in cream (or appropriate colour) powder coating. Wiring as shown on the schematic (the other details as per Particular Specification) and in compliance with IEC 60439 and Local KS standards, complete with the following:-	No.	1	50,000.00	50,000.00
(i)	200Amps TP/N copper busbars - KPLC supply (Sealable)				
(ii)	200Amps MCCB as mains incomer adjustable				
(iii)	Space for 1No. 3-phase KPLC Common Services meter (kW, kWh, kVA, V, I etc), 100A TP MCB				
(iv)	24No. 63A DP MCBs for the units				
(v)	2No. 100A TP MCCBs for common area				
(vi)	100A TP sub changeover for common area				
(vii)	Space for KPLC cut-outs and 12No. 1-phase meters including studs for mounting KPLC seals				
(viii)	Sealable studs for all cover plate screws and all necessary accessories				
(ix)	6mm perspex viewing window for each section				
(x)	Heavy duty rubber lining for all the perspex viewing windows				
(xi)	10 No. Spare ways				
(xii)	Complete earthing system to KPLC requirements				
Total Carried Forward to Next Pg.					50,000.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				50,000.00
(v)	<u>LV SWITCHBOARD-TYPICAL FLOOR</u>				
	TAKE DELIVERY, STORE, INSTALL AND COMMISSION THE FOLLOWING:-				
1.01	Install purpose made Free-standing, fully wired front access metal clad main switchboard cubicle suitable for floor / wall mounting and manufactured in 14SWG galvanised mild steel sheet, to be finished in cream (or appropriate colour) powder coating. Wiring as shown on the schematic (the other details as per Particular Specification) and in compliance with IEC 60439 and Local KS standards, complete with the following:-	No.	9	20,000.00	180,000.00
(i)	100Amps TP/N copper busbars - KPLC supply (Sealable)				
(ii)	100Amps MCCB as mains incomer adjustable				
(iii)	22No. 63A DP MCBs for the units				
(iv)	Space for KPLC cut-outs and 11No. 1-phase meters including studs for mounting KPLC seals				
(v)	Sealable studs for all cover plate screws and all necessary accessories				
(vi)	6mm perspex viewing window for each section				
(vii)	Heavy duty rubber lining for all the perspex viewing windows				
(viii)	10 No. Spare ways				
(ix)	Complete earthing system to KPLC requirements				
Total Carried Forward to Next Pg.					230,000.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				230,000.00
1.02	Earthing for the switch-board under this section comprising 25x3mm copper tape lead, 1800mm long x 15mm diameter copper earth electrode as Furse or approved equivalent complete with driving stud and tape to rod clamp, 300mm x 300mm x 300mm deep concrete inspection earth pit with removable waterproof cover, 600mm x 600mm copper earth mat, soil conditioning agents (marconite and bentonite etc) necessary to achieve earthing value below 5-Ohms and all other necessary accessories	No.	1	20,000.00	20,000.00
Total Carried Forward to Main Summary Pg.					250,000.00

ITEM	DESCRIPTION	AMOUNT (KSH)
	BLOCK TYPE A SUMMARY PAGE	
1	Total Amount for Ground Floor B/F Page D-25	1,829,300.00
2	Total Amount for Typical 1st - 9th Floor B/F Page D-48	10,374,300.00
3	Total Amount for Lightning Protection B/F Page D-50	660,400.00
4	Total Amount for LV Switchboard B/F Page D-52	250,000.00
TOTAL AMOUNT FOR BILL No. 1: TYPICAL 1NO. BLOCK TYPE A CARRIED FORWARD TO PRICE COLLECTION PAGE		13,114,000.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	<u>BILL NO. 2</u>				
	<u>PROPOSED TYPICAL AFFORDABLE & MARKET UNITS BLOCK TYPE B</u>				
(i)	<u>GROUND FLOOR</u>				
A.	<u>2 BEDROOM AHP UNITS</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
2.01	i) Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	3	1,100.00	3,300.00
	ii) Ditto as in item No. 2.01 but for two way switching	No.	6	1,400.00	8,400.00
2.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang one way switch	No.	3	250.00	750.00
	ii) 10A one gang two way switch	No.	6	300.00	1,800.00
	iii) 10A two gang two way switch	No.	2	350.00	700.00
2.03	LIGHTING FITTINGS				
	a) Ceiling rose complete with 3 core flex cable, lamp holder and 11W led lamp as crabtree or approved equivalent Type 'CR'	No.	6	300.00	1,800.00
	b) Spherical screwneck luminaire with opal glass and 16W compact lamp as Astra, Micromark, Thorn or approved equivalent. "Type BF"	No.	2	300.00	600.00
Total Carried Forward to the Next Pg.					17,350.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				17,350.00
	POWER AND SOCKET OUTLET POINTS				
2.04	13A SP socket point completely wired in 3 x 2.5mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories but excluding the 13A SP switched socket outlet.	No.	8	1,600.00	12,800.00
2.05	Socket Outlet plates i) 13A SP twin switched socket outlet as MK 2536 or approved equivalent.	No.	8	500.00	4,000.00
2.06	Instant shower water heater power points completely wired in 3 x 4mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories, but excluding the 40A DP connection unit	No.	1	2,500.00	2,500.00
2.07	40A DP switched connection unit with neon indicator as MK or approved equivalent	No.	1	1,000.00	1,000.00
2.08	Cooker outlet power points completely wired in 3 x 6.0mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories, but excluding the 45A Cooker connection unit	No.	1	2,500.00	2,500.00
2.09	45A Moulded plate Cooker Control Unit complete with 13A fused Socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
2.10	45A moulded plate cooker connection unit as MK/Crabtree or approved equivalent:	No.	1	1,500.00	1,500.00
Total Carried Forward to the Next Pg.					43,150.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				43,150.00
2.11	Domestic door bell point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories but excluding the bell.	No.	1	1,500.00	1,500.00
2.12	240V/12V Ding dong domestic door bell as Oxford or approved equivalent	No.	1	2,000.00	2,000.00
	TELEPHONE, DATA AND TV POINTS POINTS				
2.13	Data point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking cover but excluding the Data socket outlet.	No.	1	1,000.00	1,000.00
2.14	DTV Socket point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking cover but excluding the DTV socket outlet.	No.	1	1,000.00	1,000.00
2.15	300x 300x 100mm deep 16G galvanized adaptable box.	No.	2	1,500.00	3,000.00
2.16	25mm diameter PVC HG conduit for ICT services and linking the Draw Boxes to the ICT ducts	Lm.	30	90.00	2,700.00
Total Carried Forward to Next Pg.					54,350.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				54,350.00
	POWER RETICULATION AND DISTRIBUTION BOARDS				
2.17	6 ways 100A SPN Consumer Unit 'A' complete with 100A SPN integral isolator, but excluding the MCBs ,and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
2.18	MCBs for item above as Schneider Electric Acti 9				
	i)10 A SP MCB	No.	1	600.00	600.00
	ii)32 A SP MCB	No.	2	600.00	1,200.00
	iii) 45 A SP MCB	No.	1	600.00	600.00
	iv) Blanking plates	No.	2	100.00	200.00
2.19	Sub-mains circuits for CU 'A ' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the LV switchboard below the staircase.	LM	40	700.00	28,000.00
Total Carried Forward to Ground Floor Collection Page					90,950.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
B.	<u>3 BEDROOM AHP UNITS</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
2.01	i) Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	4	1,250.00	5,000.00
	ii) Ditto as in item No. 2.01 but for two way switching	No.	8	1,400.00	11,200.00
2.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang one way switch	No.	5	250.00	1,250.00
	ii) 10A one gang two way switch	No.	8	300.00	2,400.00
	iii) 10A two gang two way switch	No.	2	350.00	700.00
2.03	LIGHTING FITTINGS				
	a) Ceiling rose complete with 3 core flex cable, lamp holder and 11W led lamp as crabtree or approved equivalent Type 'CR'	No.	9	300.00	2,700.00
	b) Spherical screwneck luminaire with opal glass and 16W compact lamp as Astra, Micromark, Thorn or approved equivalent. "Type BF"	No.	3	300.00	900.00
Total Carried Forward to the Next Pg.					24,150.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				24,150.00
	POWER AND SOCKET OUTLET POINTS				
2.04	13A SP socket point completely wired in 3 x 2.5mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories but excluding the 13A SP switched socket outlet.	No.	11	1,600.00	17,600.00
2.05	Socket Outlet plates i) 13A SP twin switched socket outlet as MK 2536 or equal & approved equivalent.	No.	11	500.00	5,500.00
2.06	Instant shower water heater power points completely wired in 3 x 4mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories, but excluding the 40A DP connection unit	No.	2	2,500.00	5,000.00
2.07	40A DP switched connection unit with neon indicator as MK or approved equivalent	No.	2	1,000.00	2,000.00
2.08	Cooker outlet power points completely wired in 3 x 6.0mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories, but excluding the 45A DP connection unit	No.	1	2,500.00	2,500.00
2.09	45A Moulded plate Cooker Control Unit complete with 13A fused socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
2.10	45A Moulded plate Cooker Control Unit complete with 13A fused Socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
Total Carried Forward to the Next Pg.					59,750.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				59,750.00
2.11	Domestic door bell point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories but excluding the bell.	No.	1	1,500.00	1,500.00
2.12	240V/12V Ding dong domestic door bell as Oxfrud or approved equivalent	No.	1	2,000.00	2,000.00
	TELEPHONE, DATA AND TV POINTS POINTS				
2.13	Data point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking cover but excluding the telephone socket outlet.	No.	1	1,000.00	1,000.00
2.14	DTV Socket point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking cover but excluding the DTV socket outlet.	No.	1	1,000.00	1,000.00
2.15	300x 300x 100mm deep 16G galvanized adaptable box.	No.	2	1,500.00	3,000.00
2.16	25mm diameter PVC HG conduit for ICT services and linking the Draw Boxes to the ICT ducts	Lm.	20	90.00	1,800.00
Total Carried Forward to the Next Pg.					70,050.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				70,050.00
	POWER RETICULATION AND DISTRIBUTION BOARDS				
2.17	6 ways 100A SPN Consumer Unit 'A' complete with 100A SPN integral isolator ,but excluding the MCBs, and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
2.18	MCBs for item above as Schneider Electric Acti 9				
	i)10 A SP MCB	No.	1	600.00	600.00
	ii)32 A SP MCB	No.	3	600.00	1,800.00
	iii) 45 A SP MCB	No.	1	600.00	600.00
	iv) Blanking plates	No.	1	100.00	100.00
2.19	Sub-mains circuits for CU 'A ' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the Lv switchboard below the staircase.	LM	45	700.00	31,500.00
Total Carried Forward to Ground Floor Collection Page					110,650.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
C.	<u>2 BEDROOM MARKET UNITS</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
2.01	i) Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	5	1,250.00	6,250.00
	ii) Ditto as in item No. 2.01 but for two way switching	No.	5	400.00	2,000.00
2.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang one way switch	No.	4	250.00	1,000.00
	ii) 10A two gang one way switch	No.	1	280.00	280.00
	iii) 10A one gang two way switch	No.	6	300.00	1,800.00
	iv) 10A two gang two way switch	No.	2	350.00	700.00
2.03	LIGHTING FITTINGS				
	a) Ceiling rose complete with 3 core flex cable, lamp holder and 11W led lamp as crabtree or approved equivalent Type 'CR'	No.	7	300.00	2,100.00
	b) Spherical screwneck luminaire with opal glass and 16W compact lamp as Astra, Micromark, Thorn or approved equivalent. "Type BF"	No.	2	300.00	600.00
Total Carried Forward to the Next Pg.					14,730.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				14,730.00
	POWER AND SOCKET OUTLET POINTS				
2.04	13A SP socket point completely wired in 3 x 2.5mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories but excluding the 13A SP switched socket outlet.	No.	9	1,600.00	14,400.00
2.05	Socket Outlet plates i) 13A SP twin switched socket outlet as MK 2536 or equal & approved equivalent.	No.	9	500.00	4,500.00
2.06	Instant shower water heater power points completely wired in 3 x 4mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories ,but excluding the 40A DP connection unit	No.	2	2,500.00	5,000.00
2.07	40A DP switched connection unit with neon indicator as MK or approved equivalent	No.	2	1,000.00	2,000.00
2.08	Cooker outlet power points completely wired in 3 x 6.0mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories ,but excluding the 45A DP connection unit	No.	1	2,500.00	2,500.00
2.09	45A Moulded plate Cooker Control Unit complete with 13A fused Socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
2.10	45A moulded plate cooker connection unit as MK/Crabtree or approved equivalent:	No.	1	1,500.00	1,500.00
Total Carried Forward to the Next Pg.					46,130.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				46,130.00
2.11	Domestic door bell point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories but excluding the bell.	No.	1	1,500.00	1,500.00
2.12	240V/12V Ding dong domestic door bell	No.	1	2,000.00	2,000.00
	TELEPHONE, DATA AND TV POINTS				
2.13	Data point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking plate but excluding the telephone socket outlet.	No.	1	1,000.00	1,000.00
2.14	DTV Socket point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories but excluding the DTV socket outlet.	No.	1	1,000.00	1,000.00
2.15	300x 300x 100mm deep 16G galvanized adaptable box.	No.	2	1,500.00	3,000.00
2.16	25mm diameter PVC HG conduit for ICT services and linking the Draw Boxes to the ICT ducts	Lm.	40	90.00	3,600.00
Total Carried Forward to the Next Pg.					58,230.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				58,230.00
	POWER RETICULATION AND DISTRIBUTION BOARDS				
2.17	6 ways 100A SPN Consumer Unit 'A' complete with 100A SPN integral isolator ,but excluding the MCBs ,and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
2.18	MCBs for item above as Schneider Electric Acti 9				
	i)10 A SP MCB	No.	1	600.00	600.00
	ii)32 A SP MCB	No.	3	600.00	1,800.00
	iii) 45 A SP MCB	No.	1	600.00	600.00
	iv) Blanking plates	No.	1	100.00	100.00
2.19	Sub-mains circuits for CU 'A ' above comprising of 3 x 10mm2 single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the LV switchboard below the staircase.	LM	40	700.00	28,000.00
Total Carried Forward to Ground Floor Collection Page					95,330.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
D.	<u>3 BEDROOM MARKET UNITS</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
2.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	5	1,250.00	6,250.00
	ii) Ditto as in item No. 2.01 but for two way switching	No.	7	1,400.00	9,800.00
2.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang one way switch	No.	5	250.00	1,250.00
	ii) 10A one gang two way switch	No.	6	300.00	1,800.00
	iii) 10A two gang two way switch	No.	4	350.00	1,400.00
2.03	LIGHTING FITTINGS				
	a) Ceiling rose complete with 3 core flex cable, lamp holder and 11W led lamp as crabtree or approved equivalent Type 'CR'	No.	8	300.00	2,400.00
	b) Spherical screwneck luminaire with opal glass and 16W compact fluorescent lamp as Astra, Micromark, Thorn or approved equivalent. "Type BF"	No.	3	300.00	900.00
Total Carried Forward to the Next Pg.					23,800.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				23,800.00
	POWER AND SOCKET OUTLET POINTS				
2.04	13A SP socket point completely wired in 3 x 2.5mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories but excluding the 13A SP switched socket outlet.	No.	12	1,600.00	19,200.00
2.05	Socket Outlet plates i) 13A SP twin switched socket outlet as MK 2536 or equal & approved equivalent.	No.	12	500.00	6,000.00
2.06	Instant shower water heater power points completely wired in 3 x 4mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories, but excluding the 40A DP connection unit	No.	2	2,500.00	5,000.00
2.07	40A DP switched connection unit with neon indicator as MK or approved equivalent	No.	2	1,000.00	2,000.00
2.08	Cooker outlet power points completely wired in 3 x 6.0mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories ,but excluding the 45A DP connection unit	No.	1	2,500.00	2,500.00
2.09	45A Moulded plate Cooker Control Unit complete with 13A fused socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
2.10	45A moulded plate cooker connection unit as MK/Crabtree or approved equivalent:	No.	1	1,500.00	1,500.00
Total Carried Forward to the Next Pg.					61,500.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				61,500.00
2.11	Domestic door bell point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories but excluding the bell.	No.	1	1,500.00	1,500.00
2.12	240V/12V Ding dong domestic door bell as Oxford or approved equivalent	No.	1	2,000.00	2,000.00
	TELEPHONE, DATA AND TV POINTS POINTS				
2.13	Data point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking plate but excluding the data socket outlet.	No.	1	1,000.00	1,000.00
2.14	DTV Socket point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories but excluding the DTV socket outlet.	No.	1	1,000.00	1,000.00
2.15	300x 300x 100mm deep 16G galvanized adaptable box.	No.	2	2,500.00	5,000.00
2.16	32mm diameter PVC HG conduit for ICT services and linking the Draw Boxes to the ICT ducts	Lm.	40	90.00	3,600.00
Total Carried Forward to the Next Pg.					<u>75,600.00</u>

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				75,600.00
	POWER RETICULATION AND DISTRIBUTION BOARDS				
2.17	6 ways 100A SPN Consumer Unit 'A' complete with 100A SPN integral isolator, but excluding the MCBs ,and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
2.18	MCBs for item above as Schneider Electric Acti 9				
	i)10 A SP MCB	No.	1	600.00	600.00
	ii)32 A SP MCB	No.	3	600.00	1,800.00
	iii) 45 A SP MCB	No.	1	600.00	600.00
	iv) Blanking plates	No.	1	100.00	100.00
2.19	Sub-mains circuits for CU 'A ' above comprising of 3 x 10mm2 single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the LVswitchboard below the staircase.	LM	40	700.00	28,000.00
Total Carried Forward to Ground Floor Collection Page					112,700.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
E.	<u>CORRIDOR/ LIFTS LOBBY/ROOF AREA</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
2.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for two way switching, but excluding the switch and luminaire.	No.	40	1,250.00	50,000.00
2.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang two way switch	No.	4	300.00	1,200.00
	ii) 10A two gang two way switch	No.	4	300.00	1,200.00
2.03	LIGHTING FITTINGS				
	a) Circular shaped surface mounted led luminaire as Phillips Coreline Cat No. WL131V LED12S/840 PSED EL3 WH "Type N"	No.	15	3,850.00	57,750.00
	b) Emergency Maintained Circular shaped surface mounted led luminaire as Phillips Coreline Cat No. WL131V LED12S/840 PSED EL3 WH "Type Ne"	No.	3	4,200.00	12,600.00
	c) 8W bulkhead luminaire ,polycarbonate base, and clear diffuser and installed inside the lift shaft As Thorn EWB/ETB bulkhead. "Type G"	No.	2	2,500.00	5,000.00
	d) Bulkhead fitting with moulded glass diffuser & Die Cast Aluminium base c/w 16W PL Lamp as Massive Sunderland Cat. No. 71410/01/31 for security lights. "Type B"	No.	28	3,500.00	98,000.00
Total Carried Forward to Next Pg.					225,750.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				225,750.00
2.04	CCTV points in designated locations for surveillance, comprising of 25mm diameter HG PVC conduits complete with draw wire and the blanking plate. Allow for draw boxes where all the conduits from each security point converge.	No.	6	1,200.00	7,200.00
2.05	300x 300x 100mm deep 16 G galvanized adaptable box.	No.	2	2,500.00	5,000.00
	LIGHTING CONTROL				
2.06	Presence sensor complete with associated wiring (light and motion sensor) as Osram Duo with the following characteristics: Operation voltage 220 - 240V 50Hz, detection area-12m dia minimum, settable light value, switch-off delay (if no motion detected), installation height - 5m minimum for staircase	No.	4	6,500.00	26,000.00
2.07	24A, 240V, 2P DB/CU mounted silent operation contactor for Switching External lights complete with all accessories and as CRABTREE CIK24 or an approved equivalent	No.	2	2,500.00	5,000.00
2.08	Photocell control unit and wired to energize the contactors complete with a D.P override switch as THORN QPK or approved equivalent	No.	2	2,500.00	5,000.00
2.09	Water booster power points completely wired in 5 x 6mm ² single core PVC insulated copper cables drawn in 38mm diameter heavy gauge PVC conduits including all accessories ,but excluding the the 40A TPN isolator	No.	1	5,000.00	5,000.00
Total Carried Forward to Next Pg.					<u>278,950.00</u>

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				278,950.00
2.10	40A TPN isolator as MK or approved equivalent	No.	1	3,000.00	3,000.00
2.11	Lifts isolators power points completely wired in 5 x 6mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories ,but excluding the the 63A TPN isolator	No.	2	8,500.00	17,000.00
2.12	63A TPN isolator as MK or approved equivalent	No.	2	6,500.00	13,000.00
2.13	8 ways 100A TPN distribution boards 'T' complete with 100A TPN integral isolator ,but excluding the MCBs ,and as Schneider Electric Acti 9 or approved equivalent.	No.	2	35,000.00	70,000.00
2.14	4 ways 100A SPN Consumer Unit 'C' complete with 100A SPN integral isolator ,but excluding the MCBs ,and as Schneider Electric Acti 9 or approved equivalent for common services.	No.	2	6,000.00	12,000.00
2.15	MCBs for item above as Schneider Electric Acti 9				
	i) 10 A SP MCB	No.	6	600.00	3,600.00
	ii) 32 A SP MCB	No.	4	600.00	2,400.00
	iii) 40 A SP MCB	No.	5	600.00	3,000.00
	iv) 32 A TP MCB	No.	3	1,500.00	4,500.00
	v) Blanking plates	No.	8	100.00	800.00
2.16	Sub-mains circuits for DB 'T' above comprising of 4 core 25mm ² PVC/SWA/PVC cable drawn in cable tray/ 50mm diameter heavy gauge conduits and running from the Lv switchboard	LM	50	2,700.00	135,000.00
Total Carried Forward to Next Pg.					543,250.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				543,250.00
2.17	Sub-mains circuits for CU 'C' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from common distribution board.	LM	30	700.00	21,000.00
2.18	200mm x 50mm galvanised metallic cable tray (Telecom, DTV, Internet Cables) gauge 14 (riser duct) c/w all mounting accessories bends, rawl bolts, threaded bolts, brackets,	LM.	60	2,000.00	120,000.00
2.19	300mm x 50mm galvanised metallic cable tray (Power Cables) gauge 14 (riser duct) c/w all mounting accessories bends, rawl bolts, threaded bolts, brackets,	LM.	60	2,500.00	150,000.00
Total Carried Forward to Ground Floor Collection Page					834,250.00

ITEM	DESCRIPTION	QTY	RATE (KSH)	AMOUNT (KSH)
	GROUND FLOOR SUMMARY			
1	Total Amount for 2 Bedroom AHP Unit	1	90,950.00	90,950.00
2	Total Amount for 3 Bedroom AHP Unit	1	110,650.00	110,650.00
3	Total Amount for 2 Bedroom Market Unit	3	95,330.00	285,990.00
4	Total Amount for 3 Bedroom Market Unit	3	112,700.00	338,100.00
5	Total Amount for Ground Floor Lobby	1	834,250.00	834,250.00
Total Carried Forward to Main Summary Page				1,659,940.00

ITEM NO.	DESCRIPTION	UNIT	QTY	RATE KShs	AMOUNT KShs
	<u>BILL NO. 2</u>				
(ii)	<u>TYPICAL 1ST - 9TH FLOOR</u>				
A.	<u>2 BEDROOM AHP UNITS</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
2.01	i) Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	3	1,250.00	3,750.00
	ii) Ditto as in item No. 2.01 but for two way switching	No.	6	1,400.00	8,400.00
2.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang one way switch	No.	3	250.00	750.00
	ii) 10A one gang two way switch	No.	6	300.00	1,800.00
	iii) 10A two gang two way switch	No.	2	350.00	700.00
2.03	LIGHTING FITTINGS				
	a) Ceiling rose complete with 3 core flex cable, lamp holder and 11W led lamp as crabtree or approved equivalent Type 'CR'	No.	6	300.00	1,800.00
	b) Spherical screwneck luminaire with opal glass and 16W compact lamp as Astra, Micromark, Thornor equal & approved equivalent. "Type BF"	No.	2	300.00	600.00
Total Carried Forward to Next Pg.					17,800.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				17,800.00
	POWER AND SOCKET OUTLET POINTS				
2.04	13A SP socket point completely wired in 3 x 2.5mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories but excluding the 13A SP switched socket outlet.	No.	8	1,600.00	12,800.00
2.05	Socket Outlet plates i) 13A SP twin switched socket outlet as MK 2536 or equal & approved equivalent.	No.	8	500.00	4,000.00
2.06	Instant shower water heater power points completely wired in 3 x 4mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories ,but excluding the 40A DP connection unit	No.	1	2,500.00	2,500.00
2.07	40A DP switched connection unit with neon indicator as MK or approved equivalent	No.	1	1,000.00	1,000.00
2.08	Cooker outlet power points completely wired in 3 x 6.0mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories ,but excluding the 45A DP connection unit	No.	1	2,500.00	2,500.00
2.09	45A Moulded plate Cooker Control Unit complete with 13A fused socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
2.10	45A moulded plate cooker connection unit as MK/Crabtree or approved equivalent:	No.	1	1,500.00	1,500.00
Total Carried Forward to Next Pg.					43,600.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				43,600.00
2.11	Domestic door bell point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories but excluding the bell.	No.	1	1,500.00	1,500.00
2.12	240V/12V Ding dong domestic door bell	No.	1	2,000.00	2,000.00
	TELEPHONE, DATA AND TV POINTS				
2.13	Data point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking plate but excluding the data socket outlet.	No.	1	1,000.00	1,000.00
2.14	DTV Socket point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories but excluding the DTV socket outlet.	No.	1	1,000.00	1,000.00
2.15	300x 300x 100mm deep 16G galvanized adaptable box.	No.	2	1,500.00	3,000.00
2.16	25mm diameter PVC HG conduit for ICT services and linking the Draw Boxes to the ICT ducts	Lm.	30	90.00	2,700.00
Total Carried Forward to Next Pg.					54,800.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				54,800.00
	POWER RETICULATION AND DISTRIBUTION BOARDS				
2.17	6 ways 100A SPN Consumer Unit 'A' complete with 100A SPN integral isolator ,but excluding the MCBs ,and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
2.18	MCBs for item above as Schneider Electric Acti 9				
	i)10 A SP MCB	No.	1	600.00	600.00
	ii)32 A SP MCB	No.	2	600.00	1,200.00
	iii) 45 A SP MCB	No.	1	600.00	600.00
	iv) Blanking plates	No.	2	100.00	200.00
2.19	Sub-mains circuits for CU 'A ' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the Lv switchboard below the staircase.	LM	40	700.00	28,000.00
Total Carried Forward to First Floor Collection Page					91,400.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
B.	<u>3 BEDROOM AHP UNITS</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
2.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	4	1,250.00	5,000.00
	ii) Ditto as in item No. 2.01 but for two way switching	No.	8	1,400.00	11,200.00
2.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang one way switch	No.	5	250.00	1,250.00
	ii) 10A one gang two way switch	No.	8	300.00	2,400.00
	iii) 10A two gang two way switch	No.	2	350.00	700.00
2.03	LIGHTING FITTINGS				
	a) Ceiling rose complete with 3 core flex cable, lamp holder and 11W led lamp as crabtree or approved equivalent Type 'CR'	No.	9	300.00	2,700.00
	b) Spherical screwneck luminaire with opal glass and 16W compact fluorescent lamp as Astra, Micromark, Thornor equal & approved equivalent. "Type BF"	No.	3	300.00	900.00
Total Carried Forward to Next Pg.					24,150.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				24,150.00
	POWER AND SOCKET OUTLET POINTS				
2.04	13A SP socket point completely wired in 3 x 2.5mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories but excluding the 13A SP switched socket outlet.	No.	11	1,600.00	17,600.00
2.05	Socket Outlet plates i) 13A SP twin switched socket outlet as MK 2536 or equal & approved equivalent.	No.	11	500.00	5,500.00
2.06	Instant shower water heater power points completely wired in 3 x 4mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories ,but excluding the 40A DP connection unit	No.	2	2,500.00	5,000.00
2.07	40A DP switched connection unit with neon indicator as MK or approved equivalent	No.	2	1,000.00	2,000.00
2.08	Cooker outlet power points completely wired in 3 x 6.0mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories ,but excluding the 45A DP connection unit	No.	1	2,500.00	2,500.00
2.09	45A Moulded plate Cooker Control Unit complete with 13A fused socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
2.10	45A moulded plate cooker connection unit as MK/Crabtree or approved equivalent:	No.	1	1,500.00	1,500.00
Total Carried Forward to Next Pg.					59,750.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				59,750.00
2.11	Domestic door bell point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories but excluding the bell.	No.	1	1,500.00	1,500.00
2.12	240V/12V Ding dong domestic door bell as Oxford or approved equivalent	No.	1	2,000.00	2,000.00
	TELEPHONE, DATA AND TV POINTS POINTS				
2.13	Data point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking plate but excluding the data socket outlet.	No.	1	1,000.00	1,000.00
2.14	DTV Socket point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories but excluding the DTV socket outlet.	No.	1	1,000.00	1,000.00
2.15	300x 300x 100mm deep 16G galvanized adaptable box.	No.	2	1,500.00	3,000.00
2.16	25mm diameter PVC HG conduit for ICT services and linking the Draw Boxes to the ICT ducts	Lm.	20	90.00	1,800.00
Total Carried Forward to Next Pg.					70,050.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				70,050.00
	POWER RETICULATION AND DISTRIBUTION BOARDS				
2.17	6 ways 100A SPN Consumer Unit 'A' complete with 100A SPN integral isolator ,but excluding the MCBs ,and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
2.18	MCBs for item above as Schneider Electric Acti 9				
	i)10 A SP MCB	No.	1	600.00	600.00
	ii)32 A SP MCB	No.	3	600.00	1,800.00
	iii) 45 A SP MCB	No.	1	600.00	600.00
	iv) Blanking plates	No.	1	100.00	100.00
2.19	Sub-mains circuits for CU 'A ' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the Lv switchboard below the staircase.	LM	65	700.00	45,500.00
Total Carried Forward to First Floor Collection Page					124,650.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
C.	<u>2 BEDROOM MARKET UNITS</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
2.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	5	1,250.00	6,250.00
	ii) Ditto as in item No. 2.01 but for two way switching	No.	5	1,400.00	7,000.00
2.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang one way switch	No.	4	250.00	1,000.00
	ii) 10A two gang one way switch	No.	1	280.00	280.00
	iii) 10A one gang two way switch	No.	6	300.00	1,800.00
	iv) 10A two gang two way switch	No.	2	350.00	700.00
2.03	LIGHTING FITTINGS				
	a) Ceiling rose complete with 3 core flex cable, lamp holder and 11W led lamp as crabtree or approved equivalent Type 'CR'	No.	7	300.00	2,100.00
	b) Spherical screwneck luminaire with opal glass and 16W compact lamp as Astra, Micromark, Thornor equal & approved equivalent. "Type BF"	No.	2	300.00	600.00
Total Carried Forward to Next Page					19,730.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				19,730.00
	POWER AND SOCKET OUTLET POINTS				
2.04	13A SP socket point completely wired in 3 x 2.5mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories but excluding the 13A SP switched socket outlet.	No.	9	1,600.00	14,400.00
2.05	Socket Outlet plates i) 13A SP twin switched socket outlet as MK 2536 or equal & approved equivalent.	No.	9	500.00	4,500.00
2.06	Instant shower water heater power points completely wired in 3 x 4mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories ,but excluding the 40A DP connection unit	No.	2	2,500.00	5,000.00
2.07	40A DP switched connection unit with neon indicator as MK or approved equivalent	No.	2	1,000.00	2,000.00
2.08	Cooker outlet power points completely wired in 3 x 6.0mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories ,but excluding the 45A DP connection unit	No.	1	2,500.00	2,500.00
2.09	45A Moulded plate Cooker Control Unit complete with 13A fused socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
2.10	45A moulded plate cooker connection unit as MK/Crabtree or approved equivalent:	No.	1	1,500.00	1,500.00
Total Carried Forward to Next Page					51,130.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				51,130.00
2.11	Domestic door bell point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories but excluding the bell.	No.	1	1,500.00	1,500.00
2.12	240V/12V Ding dong domestic door bell as Oxford or approved equivalent	No.	1	2,000.00	2,000.00
	TELEPHONE, DATA AND TV POINTS POINTS				
2.13	Data point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking plate but excluding the data socket outlet.	No.	1	1,000.00	1,000.00
2.14	DTV Socket point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories but excluding the DTV socket outlet.	No.	1	1,000.00	1,000.00
2.15	300x 300x 100mm deep 16G galvanized adaptable box.	No.	2	1,500.00	3,000.00
2.16	25mm diameter PVC HG conduit for ICT services and linking the Draw Boxes to the ICT ducts	Lm.	40	90.00	3,600.00
Total Carried Forward to Next Page					63,230.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				63,230.00
	POWER RETICULATION AND DISTRIBUTION BOARDS				
2.17	6 ways 100A SPN Consumer Unit 'A' complete with 100A SPN integral isolator ,but excluding the MCBs ,and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
2.18	MCBs for item above as Schneider Electric Acti 9 or approved equivalent				
	i)10 A SP MCB	No.	1	600.00	600.00
	ii)32 A SP MCB	No.	3	600.00	1,800.00
	iii) 45 A SP MCB	No.	1	600.00	600.00
	iv) Blanking plates	No.	1	100.00	100.00
2.19	Sub-mains circuits for CU 'A ' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the Lv switchboard below the staircase.	LM	60	700.00	42,000.00
Total Carried Forward to First Floor Collection Page					114,330.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
D.	<u>3 BEDROOM MARKET UNITS</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
2.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	5	1,250.00	6,250.00
	ii) Ditto as in item No. 2.01 but for two way switching	No.	7	1,400.00	9,800.00
2.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang one way switch	No.	5	250.00	1,250.00
	ii) 10A one gang two way switch	No.	6	300.00	1,800.00
	iii) 10A two gang two way switch	No.	4	350.00	1,400.00
2.03	LIGHTING FITTINGS				
	a) Ceiling rose complete with 3 core flex cable, lamp holder and 11W led lamp as crabtree or approved equivalent Type 'CR'	No.	8	300.00	2,400.00
	b) Spherical screwneck luminaire with opal glass and 16W compact fluorescent lamp as Astra, Micromark, Thornor equal & approved equivalent. "Type BF"	No.	3	300.00	900.00
Total Carried Forward to Next Page					23,800.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				23,800.00
	POWER AND SOCKET OUTLET POINTS				
2.04	13A SP socket point completely wired in 3 x 2.5mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories but excluding the 13A SP switched socket outlet.	No.	12	1,600.00	19,200.00
2.05	Socket Outlet plates i) 13A SP twin switched socket outlet as MK 2536 or equal & approved equivalent.	No.	12	500.00	6,000.00
2.06	Instant shower water heater power points completely wired in 3 x 4mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories ,but excluding the 40A DP connection unit	No.	2	2,500.00	5,000.00
2.07	40A DP switched connection unit with neon indicator as MK or approved equivalent	No.	2	1,000.00	2,000.00
2.08	Cooker outlet power points completely wired in 3 x 6.0mm ² single core PVC insulated copper cables drawn in 32mm diameter heavy gauge PVC conduits including all accessories ,but excluding the 45A DP connection unit	No.	1	2,500.00	2,500.00
2.09	45A Moulded plate Cooker Control Unit complete with 13A fused socket outlet as MK/Crabtree or approved equivalent.	No.	1	1,500.00	1,500.00
2.10	45A moulded plate cooker connection unit as MK/Crabtree or approved equivalent:	No.	1	1,500.00	1,500.00
Total Carried Forward to Next Page					61,500.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				61,500.00
2.11	Domestic door bell point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories but excluding the bell.	No.	1	1,500.00	1,500.00
2.12	240V/12V Ding dong domestic door bell as Oxford or approved equivalent	No.	1	2,000.00	2,000.00
	TELEPHONE, DATA AND TV POINTS POINTS				
2.13	Data point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking plate but excluding the data socket outlet.	No.	1	1,000.00	1,000.00
2.14	DTV Socket point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories but excluding the DTV socket outlet.	No.	1	1,000.00	1,000.00
2.15	300x 300x 100mm deep 16G galvanized adaptable box.	No.	2	1,500.00	3,000.00
2.16	25mm diameter PVC HG conduit for ICT services and linking the Draw Boxes to the ICT ducts	Lm.	40	90.00	3,600.00
Total Carried Forward to Next Page					73,600.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				73,600.00
	POWER RETICULATION AND DISTRIBUTION BOARDS				
2.17	6 ways 100A SPN Consumer Unit 'A' complete with 100A SPN integral isolator ,but excluding the MCBs ,and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
2.18	MCBs for item above as Schneider Electric Acti 9 or approved equivalent				
	i)10 A SP MCB	No.	1	600.00	600.00
	ii)32 A SP MCB	No.	3	600.00	1,800.00
	iii) 45 A SP MCB	No.	1	600.00	600.00
	iv) Blanking plates	No.	1	100.00	100.00
2.19	Sub-mains circuits for CU 'A ' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the Lv switchboard below the staircase.	LM	60	700.00	42,000.00
Total Carried Forward to First Floor Collection Page					124,700.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
E.	<u>CORRIDOR/LIFTS LOBBY AREA</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
2.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for two way switching, but excluding the switch and luminaire.	No.	18	1,250.00	22,500.00
2.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang two way switch	No.	4	300.00	1,200.00
	ii) 10A two gang two way switch	No.	4	350.00	1,400.00
2.03	LIGHTING FITTINGS				
	a) Circular shaped surface mounted led luminaire as Phillips Coreline Cat No. WL131V LED12S/840 PSED EL3 WH "Type N"	No.	15	3,850.00	57,750.00
	b) Emergency Maintained Circular shaped surface mounted led luminaire as Phillips Coreline Cat No. WL131V LED12S/840 PSED EL3 WH "Type Ne"	No.	3	4,200.00	12,600.00
Total Carried Forward to Next Page					95,450.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				95,450.00
2.04	CCTV points in designated locations for surveillance, comprising of 25mm diameter HG PVC conduits complete with draw wire and the blanking plate. Allow for draw boxes where all the conduits from each security point converge.	No.	6	1,200.00	7,200.00
2.05	300x 300x 100mm deep 16G galvanized adaptable box.	No.	2	2,500.00	5,000.00
	LIGHTING CONTROL				
2.06	Presence sensor complete with associated wiring (light and motion sensor) as Osram Duo with the following characteristics: Operation voltage 220 - 240V 50Hz, detection area-12m dia minimum, settable light value, switch-off delay (if no motion detected), installation height - 5m minimum - Staircase lights	No.	3	6,500.00	19,500.00
2.07	4 ways 100A SPN Consumer Unit 'A' complete with 100A SPN integral isolator, including the MCBs, and as Schneider Electric Acti 9 or approved equivalent.	No.	1	6,000.00	6,000.00
2.08	Sub-mains circuits for CU 'A' above comprising of 3 x 10mm ² single core PVC insulated copper cables drawn in 32mm heavy gauge conduits and running from the Lv switchboard below the staircase.	LM	30	700.00	21,000.00
Total Carried Forward to First Floor Collection Page					154,150.00

ITEM	DESCRIPTION	QTY	RATE (KSH)	AMOUNT (KSH)
	TYPICAL FLOOR SUMMARY			
1	Total Amount for 2 Bedroom AHP Unit	9	91,400.00	822,600.00
2	Total Amount for 3 Bedroom AHP Unit	9	124,650.00	1,121,850.00
3	Total Amount for 2 Bedroom Market Unit	27	114,330.00	3,086,910.00
4	Total Amount for 3 Bedroom Market Unit	27	124,700.00	3,366,900.00
5	Total Amount for Lift Lobby Summary	9	154,150.00	1,387,350.00
Total for 9No. Typical Floors Carried Forward to Main Summary Page				9,785,610.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
(iii)	<u>EARTHING AND LIGHTNING PROTECTION</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	Air Termination				
2.01	2000mm x15mmØ multiple point pure copper AirRods/ Termination with spikes as Furse Part No. RA240 or approved equivalent	No.	3	1,250.00	3,750.00
2.02	Copper Air Rod Base as Furse Part No. SD105-H or approved equivalent	No.	3	1,400.00	4,200.00
2.03	Copper Junction Clamps for copper tape as Furse Part No. CN105-H or approved equivalent	No.	3	250.00	750.00
2.04	25mm x 3mm Tinned Copper Tape as Furse TC230 or approved equivalent	Lm.	300	1,500.00	450,000.00
	Down Conductor				
2.05	25mm x 3mm Tinned Copper Tape as Furse TC230 or approved equivalent	Lm.	300	1,500.00	450,000.00
2.06	Screwdown copper test clamp as Furse CT305 or approved equivalent	No.	3	1,400.00	4,200.00
2.07	38mm Ø HG PVC conduits for drawing the down conductor above.	Lm.	200	120.00	24,000.00
Total Carried Forward to Next Page					936,900.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				936,900.00
2.08	Earth Termination 15mm Ø, 1500mm long solid copper earth rod c/w driving stud, coupling, and spike as Furse RC011 or approved equivalent	No	3	3,000.00	9,000.00
2.09	Earth rod to tape clamp type A as Furse CR108 or approved equivalent	No.	3	2,500.00	7,500.00
2.10	Concrete inspection earth pit Cat. No. PT 005 with 5 hole earth bar as Furse Cat. No. PT 006.	No.	2	2,500.00	5,000.00
2.11	600mm x 600mm copper earth mat made from 25mm x 3mm copper tape at 300mm spacing, buried to permanent moisture level and complete with all clamps and 6m long 25mm x 3mm copper tape clamped to the down conductor, soil conditioning agents (marconite or bentonite) as necessary to achieve earthing resistance value below 10-Ohms	Lot	2	6,000.00	12,000.00
2.12	Allow for earthing tests for the above and submission of the report to the engineer to BS7671 & BS62305 standards	No.	2	5,000.00	10,000.00
2.13	Bonding Bonding and clamping to all metal work including water pipes, gas pipes, hand-rails, smatv system, window frames, cladding, metal roof etc. and the main earth for the building.	Item	2	20,000.00	40,000.00
Total Carried Forward to Main Summary Page					1,020,400.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
(iv)	<u>LV SWITCHBOARD-GROUND FLOOR</u>				
2.01	TAKE DELIVERY, STORE, INSTALL AND COMMISSION THE FOLLOWING:- Install purpose made Free-standing, fully wired front access metal clad main switchboard cubicle suitable for floor / wall mounting and manufactured in 14SWG galvanised mild steel sheet, to be finished in cream (or appropriate colour) powder coating. Wiring as shown on the schematic (the other details as per Particular Specification) and in compliance with IEC 60439 and Local KS standards, complete with the following:-	No.	1	50,000.00	50,000.00
(i)	200Amps TP/N copper busbars - KPLC supply (Sealable)				
(ii)	200Amps MCCB as mains incomer adjustable				
(iii)	Space for 1No. 3-phase KPLC Common Services meter (kW, kWh, kVA, V, I etc), 100A TP MCB				
(iv)	16No. 63A DP MCBs for the units				
(v)	2No. 100A TP MCCBs for common area				
(vi)	100A TP sub changeover for common area				
(vii)	Space for KPLC cut-outs and 8No. 1-phase meters including studs for mounting KPLC seals				
(viii)	Sealable studs for all cover plate screws and all necessary accessories				
(ix)	6mm perspex viewing window for each section				
(x)	Heavy duty rubber lining for all the perspex viewing windows				
(xi)	10 No. Spare ways				
(xii)	Complete earthing system to KPLC requirements				
Total Carried Forward to Next Page					50,000.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				50,000.00
(v)	<u>LV SWITCHBOARD-TYPICAL FLOORS</u>				
	TAKE DELIVERY, STORE, INSTALL AND COMMISSION THE FOLLOWING:-				
2.01	Install purpose made Free-standing, fully wired front access metal clad main switchboard cubicle suitable for floor / wall mounting and manufactured in 14SWG galvanised mild steel sheet, to be finished in cream (or appropriate colour) powder coating. Wiring as shown on the schematic (the other details as per Particular Specification) and in compliance with IEC 60439 and Local KS standards, complete with the following:-	No.	9	20,000.00	180,000.00
(i)	100Amps TP/N copper busbars - KPLC supply (Sealable)				
(ii)	100Amps MCCB as mains incomer adjustable				
(iii)	16No. 63A DP MCBs for the units				
(iv)	Space for KPLC cut-outs and 8No. 1-phase meters including studs for mounting KPLC seals				
(v)	Sealable studs for all cover plate screws and all necessary accessories				
(vi)	6mm perspex viewing window for each section				
(vii)	Heavy duty rubber lining for all the perspex viewing windows				
(viii)	10 No. Spare ways				
(ix)	Complete earthing system to KPLC requirements				
Total Carried Forward to Next Page					230,000.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				230,000.00
2.02	Earthing for the switch-board under this section comprising 25x3mm copper tape lead, 1800mm long x 15mm diameter copper earth electrode as Furse or approved equivalent complete with driving stud and tape to rod clamp, 300mm x 300mm x 300mm deep concrete inspection earth pit with removable waterproof cover, 600mm x 600mm copper earth mat, soil conditioning agents (marconite and bentonite etc) necessary to achieve earthing value below 5-Ohms and all other necessary accessories	No.	1	20,000.00	20,000.00
Total Carried Forward to Main Summary Page					250,000.00

ITEM	DESCRIPTION	AMOUNT
BLOCK TYPE B SUMMARY PAGE		
1	Total Amount for Ground Floor B/F Page D-21	1,659,940.00
2	Total Amount for Typical 1st - 9th Floor B/F Page D-40	9,785,610.00
3	Total Amount for Lightning Protection B/F Page D-42	1,020,400.00
4	Total Amount for LV Switchboard B/F Page D-45	250,000.00
TOTAL AMOUNT FOR BILL No. 2: TYPICAL 1NO. BLOCK TYPE B CARRIED FORWARD TO PRICE COLLECTION PAGE		12,715,950.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	BILL NO. 11				
1.00	<u>SOCIAL HALL</u>				
	<u>LIGHTING</u>				
	<u>POINTS</u>				
	<u>DETAILS &</u>				
	<u>Supply, install,</u>				
	<u>Connect, Test</u>				
	<u>and Set to work</u>				
1.01	Lighting points wired in 3x1.5mm ² PVC insulated single core (SC) copper cables drawn in 20mm diameter	No.	124	1,100.00	136,400.00
1.02	DLRO Bulb for Two way	No.	8	1,400.00	11,200.00
1.03	DLRO Bulb for Emergency	No.	12	1,400.00	16,800.00
	<u>Supply, install,</u>				
	<u>Connect, Test</u>				
	<u>and Set to work</u>				
	<u>10A White</u>				
	<u>moulded side</u>				
1.04	10A 1 gang 1 way switch	No.	12	250.00	3,000.00
1.05	10A 2 gang 2 way switch	No.	4	300.00	1,200.00
	<u>Light fittings,</u>				
	<u>complete with</u>				
	<u>lamps of specified</u>				
1.06	1 type P 1200 x 300MM 220/240V x 30 watt Warm	No.	64	6,000.00	384,000.00
1.07	1 type D - Round Light with LED	No.	60	2,500.00	150,000.00
Total Carried Forward to Next Page					702,600.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/L from Previous Page				702,600.00
1.08	Spherical screwneck luminaire with opal glass and	No.	20	3,500.00	70,000.00
1.09	Type Exit	No.	20	3,500.00	70,000.00
1.10	d) Bulkhead fitting with moulded glass diffuser & Die	No.	20	1,500.00	30,000.00
1.11	Presence sensor complete with associated	No.	6	6,500.00	39,000.00
	Operation voltage 220 - 240V 50Hz, detection area-				
1.12	24A, 240V, 2P DB/CU mounted silent operation	No.	2	2,500.00	5,000.00
1.13	Photocell control unit and wired to	No.	2	1,600.00	3,200.00
	POWER				
	Supply, install, connect and set				
1.14	13A SP socket point completely wired in 3 x 2.5mm ² single	No.	60	1,600.00	96,000.00
	Socket Outlet				
	i) 13A SP twin switched socket	No.	60	500.00	30,000.00
Total Carried Forward to Next Page					1,045,800.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/L from Previous Page				1,045,800.00
1.15	Cooker outlet power points completely wired in 3 x 6.0mm ² single phase PVC	No.	1	2,500.00	2,500.00
1.16	45A Moulded plate Cooker Control Unit	No.	1	1,500.00	1,500.00
1.17	45A moulded plate cooker	No.	1	1,500.00	1,500.00
1.18	300x 300x 100mm deep 16	No.	4	2,500.00	10,000.00
1.19	Water booster power points completely wired in 5 x 6mm ² single phase PVC	No.	1	2,500.00	2,500.00
1.20	40A TPN isolator as MK	No.	1	1,500.00	1,500.00
	<u>ELV CABLE</u>				
1.21	ICT Points Supply, install and connect data outlet point consisting of average 20 meters of 25 mm diameter	No.	20	1,500.00	30,000.00
1.22	CCTV Points Supply, install and connect CCTV outlet point consisting of average 20 meters of 25 mm diameter	No.	10	1,500.00	15,000.00
Total Carried Forward to Next Page					1,110,300.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/L from Previous Page				1,110,300.00
1.23	fire Alarm Points Outlets for fire Alarm points drawn in 25mm diameter heavy gauge PVC conduits	No.	10	1,500.00	15,000.00
1.24	POWER Supply, install, connect and set to work the 8 ways 100A TPN distribution boards 'T' boards	No.	1	30,000.00	30,000.00
	i) 10 A SP MCB	No.	10	600.00	6,000.00
	ii) 32 A SP MCB	No.	10	600.00	6,000.00
	iii) 40 A SP MCB	No.	1	600.00	600.00
	iv) 32 A TP	No.	1	1,500.00	1,500.00
	v) Blanking	No.	4	100.00	400.00
	vi) earthing of the Distribution	Item.	4	2,500.00	10,000.00
1.25	Sub-mains circuits for DB 'T' above comprising of 4	LM	150	600.00	90,000.00
	200mm x 50mm galvanised metallic cable tray (Telecom, Data)	LM.	200	2,000.00	400,000.00
	200mm x 50mm galvanised metallic cable tray (Power)	LM.	200	2,000.00	400,000.00
TOTAL AMOUNT FOR BILL No. 11 SOCIAL HALL CARRIED FORWARD TO GRAND SUMMARY PAGE					2,069,800.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
i)	EXTENAL SERVICES				
	Supply, Install, test and Commission the following;				
	STREET LIGHTING				
A	Cylindrical-conical pole 8 meters with single arm and tube, in hot-dipped galvanized steel painted with anchor bolts complete with photovoltaic solar panel minimum 305Wp LED light 1x2B - 80W nominal - 24V - 4000K - >190lm/W NiMH battery 24V 4P, on top of pole,	No	35	62,000.00	2,170,000.00
B	Type ML- 20m Highmast c/w 200 W flood light complete with with inbuilt charge controller ,automatic light sensor and a solar panel complete with all necessary wiring,fixing bracket and all other neccessary accessories	No.	1	1,400,000.00	1,400,000.00
ii)	POWER HOUSE ELECTRICS (4No.)				
C	All lighting points wired in 1.5mm ² PVC cu cables in 20mm diameterHG PVC conduit embedded in wall fabrics, roof structure and floor slabs complete with switches as shown on the contract drawings	No	30	1,500.00	45,000.00
	POWER				
D	4way 63A SPN CU as SCHNEIDER or Equivalent and Approved complete with MCBs or equal and approved.	No	3	6,000.00	18,000.00
E	All socket outlet points wired in 2.5mm ² PVC cable in conduit embedded in floor slabs and in metal trunking complete with 13A twin socket outlet as shown on the drawing	No	12	2,100.00	25,200.00
c	Provisions for points including laying of conduits and provision of outlet boxes for the following:				
	i) Surveillance Cameras	No	6	1,200.00	7,200.00
Total Carried Forward to Next Page					3,665,400.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				3,665,400.00
	DATA				
H	600 x 600 x 600 concrete manhole complete with Manhole covers.	No	30	15,000.00	450,000.00
I	100mm diameter HG PVC ducts for KPLC Ltd, laid at a depth of 600mm.	Lm.	600	600.00	360,000.00
J	Trenching to a depth of 700mm, and backfilling after laying of ducts itemised above.	Lm.	600	550.00	330,000.00
iii)	PUMP ROOM ELECTRICS				
K	All lighting points wired in 1.5mm ² PVC cu cables in 20mm diameterHG PVC conduit embedded in wall fabrics, roof structure and floor slabs complete with switches as shown on the contract drawings	No	20	1,400.00	28,000.00
	POWER				
L	Supply, install and connect 6 way TP/N distribution board for power supply in riser duct complete with 250 Amp integral isolator and MCBs as specified.	No	5	35,000.00	175,000.00
M	All socket outlet points wired in 2.5mm ² PVC cable in conduit embedded in floor slabs and in metal trunking complete with 13A twin socket outlet as shown on the drawing	No	15	2,000.00	30,000.00
N	400A TPN isolator as KATCO or approved equivalent	No	10	28,000.00	280,000.00
M	Submains circuit from the External Common Loads LV Board to Pump Room DB in electrical ducts comprising of 95mm ² XLPE/PVC/SWA 4c + 35mm ² sc ECC Copper cables laid in PVC Duct and Trays	Lm.	200	10,500.00	2,100,000.00
Total Carried Forward to Next Page					7,418,400.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				7,418,400.00
	<u>EXTERNAL LOADS SWITCHBOARD</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
N	Supply and install purpose made Free-standing, fully wired front access metal clad main switchboard cubicle suitable for floor / wall mounting and manufactured in 14SWG galvanised mild steel sheet, to be finished in cream (or appropriate colour) powder coating. Wiring as shown on the schematic (the other details as per Particular Specification), complete with the following:-	No.	1	20,000.00	20,000.00
(i)	320Amps TP/N copper busbars - KPLC supply (Sealable)				
(ii)	320Amps MCCB as mains incomer adjustable				
(iii)	Space for 1No. 3-phase KPLC Common Services meter (kW, kWh, kVA, V, I etc),				
(iv)	Outgoers: 1No. 250A TP MCCBs for the Pump Room DB				
(v)	5No. 63A TP MCCBs for the External Loads Feeder Pillars				
(vi)	Sealable studs for all cover plate screws and all necessary accessories				
(vii)	6mm perspex viewing window for each section				
(viii)	Heavy duty rubber lining for all the perspex viewing windows				
(ix)	4 No. Spare ways				
O	Feeder Pillars, 125A 6-WAY TPN Feeder Pillar with 125A TPN incomer and 10x20A SP MCBs outgoers and 3No. spareways for External power distribution to to Engineer's approval	No.	6	35,500.00	213,000.00
Total Carried Forward to Next Page					7,651,400.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				7,651,400.00
P	Submains circuit from the External Common Loads LV Board to Feeder Pillars in electrical ducts comprising of 16mm ² XLPE/PVC/SWA 4c + 6mm ² sc ECC Copper cables laid in PVC Duct and Trays	Lm.	1000	2,050.00	2,050,000.00
Total Carried Forward to Main Summary Page					<u>9,701,400.00</u>

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	BILL NO. 4				
i)	<u>INCOMING POWER</u>				
	<u>KPLC RELATED WORKS</u>				
A	150mm diameter Heavy gauge duct for incoming underground H.V. cables with 1:3:6 concrete surround	M	3000	850.00	2,550,000.00
B	900 x 900 x 900 concrete manhole complete with Manhole covers and Hatari(Danger) Sign indelibly engraved at the top	No	150	16,000.00	2,400,000.00
C	Hatari(Danger) sign concrete slabs	No	150	500.00	75,000.00
D	Attendance on Kenya Power and Lighting Co. Ltd.	Sum	6	25,000.00	150,000.00
E	Complete Earthing system to KPLC requirements	Sum	6	10,000.00	60,000.00
F	Any other item to complete the installation in this section.	Sum	1	10,000.00	10,000.00
					5,245,000.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	<u>BILL NO. 8</u>				
	<u>GATE HOUSE</u>				
A.	<u>GUARD HOUSE</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
7.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	4.00	1,100.00	4,400.00
7.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:- i) 10A one gang one way switch	No.	2.00	300.00	600.00
7.03	LIGHTING FITTINGS				
	a) Circular shaped surface mounted led luminaire as Phillips Coreline Cat No. WL131V LED12S/840 PSED EL3 WH "Type N"	No.	1.00	2,000.00	2,000.00
	b) Bulkhead fitting with moulded glass diffuser & Die Cast Aluminium base c/w 16W PL Lamp as Massive Sunderland Cat. No. 71410/01/31 for security lights. "Type B"	No.	3.00	1,500.00	4,500.00
7.04	CCTV points in designaed locations for surveillance, comprising of 25mm diameter HG PVC conduits complete with draw wire and the blanking plate. Allow for draw boxes where all the conduits from each security point converge.	No.	2.00	1,200.00	2,400.00
Total Carried Forward to Next Page					13,900.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				13,900.00
7.05	300x 300x 100mm deep 16 G galvanized adaptable box.	No.	1.00	2,500.00	2,500.00
7.06	24A, 240V, 2P DB/CU mounted silent operation contactor for Switching External lights complete with all accessories and as CRABTREE CIK24 or an approved equivalent	No.	1.00	2,500.00	2,500.00
7.07	Photocell control unit and wired to energize the contactors complete with a D.P override switch as THORN QPK or approved equivalent	No.	1.00	2,500.00	2,500.00
	POWER AND SOCKET OUTLET POINTS				
7.08	13A SP socket point completely wired in 3 x 2.5mm ² single core PVC insulated copper cables drawn in 25mm diameter heavy gauge PVC conduits including all accessories but excluding the 13A SP switched socket outlet.	No.	3.00	1,600.00	4,800.00
7.09	Socket Outlet plates i) 13A SP twin switched socket outlet as MK 2536 or equal & approved equivalent.	No.	3.00	500.00	1,500.00
	TELEPHONE, DATA AND TV POINTS POINTS				
7.10	Data point comprising of draw wire in 25mm diameter Heavy Gauge PVC conduits including all accessories and blanking plate but excluding the data socket outlet.	No.	1.00	1,000.00	1,000.00
7.11	6 ways 100A SPN Consumer Unit 'G' complete with 100A SPN integral isolator, but excluding the MCBs ,and as Schneider Electric Acti 9 or approved equivalent.	No.	1.00	6,000.00	6,000.00
Total Carried Forward to Next Page					<u>34,700.00</u>

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				34,700.00
7.12	MCBs for item above as Schneider Electric Acti 9 or approved equivalent				
	i)10 A SP MCB	No.	1.00	600.00	600.00
	ii)32 A SP MCB	No.	1.00	600.00	600.00
	iii) Blanking plates	No.	4.00	100.00	400.00
7.13	Sub-mains circuits for CU 'G' above comprising of 2 core 6mm2 PVC/SWA/PVC cable drawn in 32mm diameter heavy gauge conduits for gate house	LM	35.00	600.00	21,000.00
TOTAL AMOUNT FOR BILL No. 7 GATE HOUSE CARRIED FORWARD TO GRAND SUMMARY PAGE					57,300.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	<u>BILL NO. 8</u>				
	<u>GARBAGE RECEPTACLE</u>				
A.	<u>GARBAGE RECEPTACLE</u>				
	SUPPLY, DELIVER , INSTALL ,SET TO WORK AND COMMISSION THE FOLLOWING:-				
	LIGHTING POINTS				
7.01	Lighting point completely wired in 3 x 1.5mm ² single core PVC insulated copper cables drawn in 20mm diameter heavy gauge PVC conduits including all accessories for one way switching, but excluding the switch and luminaire.	No.	4.00	1,100.00	4,400.00
	ii) Ditto as in item No. 5.01 but for two way switching	No.	4.00	1,400.00	5,600.00
7.02	10A white moulded switch plate as Crabtree, MK Logic, Clipsal E- Series or approved equivalent as follows:-				
	i) 10A one gang one way switch	No.	4.00	250.00	1,000.00
	ii) 10A one gang two way switch	No.	2.00	300.00	600.00
7.03	LIGHTING FITTINGS				
	a) Circular shaped surface mounted led luminaire as Phillips Coreline Cat No. WL131V LED12S/840 PSED EL3 WH "Type N"	No.	4.00	2,000.00	8,000.00
	b) Bulkhead fitting with moulded glass diffuser & Die Cast Aluminium base c/w 16W PL Lamp as Massive Sunderland Cat. No. 71410/01/31 for security lights. "Type B"	No.	4.00	1,500.00	6,000.00
7.04	CCTV points in designaed locations for surveillance, comprising of 25mm diameter HG PVC conduits complete with draw wire and the blanking plate. Allow for draw boxes where all the conduits from each security point converge.	No.	2.00	1,200.00	2,400.00
Total Carried Forward to Next Page					28,000.00

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				28,000.00
7.05	300x 300x 100mm deep 16 G galvanized adaptable box.	No.	1.00	2,500.00	2,500.00
7.06	24A, 240V, 2P DB/CU mounted silent operation contactor for Switching External lights complete with all accessories and as CRABTREE CIK24 or an approved equivalent	No.	1.00	2,500.00	2,500.00
7.07	Photocell control unit and wired to energize the contactors complete with a D.P override switch as THORN QPK or approved equivalent	No.	1.00	2,500.00	2,500.00
7.08	6 ways 100A SPN Consumer Unit 'G' complete with 100A SPN integral isolator, but excluding the MCBs ,and as Schneider Electric Acti 9 or approved equivalent.	No.	1.00	6,000.00	6,000.00
Total Carried Forward to Next Page					<u>41,500.00</u>

ITEM	DESCRIPTION	UNIT	QTY	RATE (KSH)	AMOUNT (KSH)
	Total B/F from Previous Pg.				41,500.00
7.09	MCBs for item above as Schneider Electric Acti 9 or approved equivalent				
	i)10 A SP MCB	No.	1.00	600.00	600.00
	ii)32 A SP MCB	No.	1.00	600.00	600.00
	iii) Blanking plates	No.	4.00	100.00	400.00
7.13	Sub-mains circuits for CU 'G' above comprising of 2 core 6mm ² PVC/SWA/PVC cable drawn in 32mm diameter heavy gauge conduits for gate house	LM	35.00	600.00	21,000.00
TOTAL AMOUNT FOR BILL No. 7 GARBAGE RECEPTACLE CARRIED FORWARD TO GRAND SUMMARY PAGE					64,100.00

PROPOSED EMURUA DIKIRR AFFORDABLE HOUSING PROJECT
ELECTRICAL INSTALLATIONS BILL OF QUANTITIES
GRAND SUMMARY PAGE

ITEM	DESCRIPTION	Unit	Qty	RATE (KSHS)	AMOUNT (KSHS)
1	SUMMARY FOR INCOMING POWER & KPLC RELATED COSTS	Lot	1	5,245,000.00	5,245,000.00
2	SUMMARY FOR ELECTRICAL SERVICES FOR AFFORDABLE HOUSING BLOCK TYPE A	No	3	13,114,000.00	39,342,000.00
3	SUMMARY FOR ELECTRICAL SERVICES FOR AFFORDABLE HOUSING BLOCK TYPE B	No	1	12,715,950.00	12,715,950.00
4	SUMMARY FOR ELECTRICAL SERVICES FOR AFFORDABLE HOUSING COMMUNITY CENTRE	No	1	2,069,800.00	2,069,800.00
5	SUMMARY FOR EXTERNAL ELECTRICS	No	1	9,701,400.00	9,701,400.00
6	SUMMARY FOR ELECTRICAL SERVICES FOR GUARD HOUSE	Lot	1	57,300.00	57,300.00
7	SUMMARY FOR ELECTRICAL SERVICES FOR GARBAGE RECEPTACLE	Lot	1	64,100.00	64,100.00
	TOTALS FOR ELECTRICAL INSTALLATION SERVICES FOR PROPOSED EMURUA DIKIRR AFFORDABLE HOUSING PROJECT				69,195,550.00

Omissions

1.00	No. Lifts Estimate at KES.22,000,000.00				
2.00	Generators Estimate at KES.8,000,000.00				
3.00	CCTV Installations at KES.2,250,000.00				
4.00	LV Switchboards at KES.7,875,000.00				
5.00	Capital Contribution to KPLC For Power Connection at KES.2,700,000.00				
6.00	MATV Installation at KES. 2,250,000.00				
7.00	Commercial Stalls Installation at KES. 800,000.00				

Amount in Words: Kenya Shillings...**Sixty Nine Million,One Hundred and Ninety Five Thousand, Five Hundred and Fifty Shillings Only**.....

Official Stamp & Address:

50797-00200 Nairobi

Tenderer's Signature:



Date: 03/02/2026



Witness' Name: Damiana Kimeu Witness' Signature:



Address: 50797 - 00200 Nairobi.....

Date: 03/02/2026

PROVISIONAL SUMS & PRIME COST SUMS

ITEM	DESCRIPTION	UNIT	QTY	RATE	AMOUNT
	<u>PRIME COST SUMS</u>				
	<u>Ground Breaking & Project Launch</u>				
A	Allow a prime cost of Five Hundred Thousand Only (KSHs. 500,000) for ground breaking, project launch, commissioning and project handover	SUM	1	500,000	500,000.00
B	Allow for Profits, Overheads and Attendance	Percentage		1.2%	6,000.00
	<u>Marketing and sales support</u>				
C	Allow a prime cost of One Million (KSHs. 1,000,000) for Marketing and sales support to Boma Yangu	SUM	1	1,000,000	1,000,000.00
D	Allow for Profits, Overheads and Attendance	Percentage		1.2%	12,000.00
	<u>Renders and Printing</u>				
E	Allow a prime cost of Five Hundred Thousand (KSHs. 500,000) for preparation of renders and printing	SUM	1	500,000	500,000.00
F	Allow for Profits, Overheads and Attendance	Percentage		1.2%	6,000.00
	<u>PROVISIONAL SUMS</u>				
G	Clothes Drying Areas	Item		400,000	400,000.00
I	Landscaping	Item		1,200,000	1,200,000.00
I	Retaining Walls and Existing Ground Improvements	Item		4,000,000	4,000,000.00
J	Play fields	Item		3,000,000	3,000,000.00
K	Signages	Item		1,500,000	1,500,000.00
L	Improvement of community facilities in the neighbourhood. Community project to be advised by the Project Engineer	Item		25,000,000	25,000,000.00
	TOTALS FOR PROVISIONAL & PRIME COST SUMS CARRIED TO GRAND SUMMARY			Kshs	37,124,000.00

	<u>ELECTRICAL INSTALLATIONS PROVISIONAL SUMS / EXCLUSIONS</u>				
A	Supply, Installation and Commisioning of lifts to Apartment Blocks	SUM	1	22,000,000	22,000,000.00
B	Allow for Profits, Overheads and Attendance	Percentage		1.0%	220,000.00
C	Supply, Installation and Commisioning of Backup Generators	SUM	1	8,000,000	8,000,000.00
D	Allow for Profits, Overheads and Attendance	Percentage		1.0%	80,000.00
E	Supply, Installation and Commisioning of CCTV Installations	SUM	1	2,250,000	2,250,000.00
F	Allow for Profits, Overheads and Attendance	Percentage		1.0%	22,500.00
G	Fabrication, Supply and commisioning of LV Switchboards Distribution Boards	SUM	1	7,875,000	7,875,000.00
H	Allow for Profits, Overheads and Attendance	Percentage		1.0%	78,750.00
I	Capital Contribution to KPLC For Power Connection	SUM	1	2,700,000	2,700,000.00
J	Allow for Profits, Overheads and Attendance	Percentage		1.0%	27,000.00
I	Commercial Stalls Installations	SUM	1	800,000	800,000.00
J	Allow for Profits, Overheads and Attendance	Percentage		1.0%	8,000.00
	<u>MECHANICAL INSTALLATIONS PROVISIONAL SUMS / EXCLUSIONS</u>				
K	Supply only of Sanitary Fittings	SUM	1	35,000,000	35,000,000.00
L	Allow for Profits, Overheads and Attendance	Percentage		1.0%	350,000.00
M	Supply & Installations of Water Booster Pumps	SUM	1	2,400,000	2,400,000.00
N	Allow for Profits, Overheads and Attendance	Percentage		1.0%	24,000.00
O	Supply of GRP Roof Level Water Tanks	SUM	1	5,000,000	5,000,000.00
P	Allow for Profits, Overheads and Attendance	Percentage		1%	50,000.00
Q	Sump Pump	SUM	1	600,000	600,000.00
R	Allow for Profits, Overheads and Attendance	Percentage		1.0%	6,000.00
Q	Hose Reel Pumps	SUM	1	1,000,000	1,000,000.00
R	Allow for Profits, Overheads and Attendance	Percentage		1.0%	10,000.00
S	Borehole Drilling & Equiping	SUM	1	4,500,000	4,500,000.00
T	Allow for Profits, Overheads and Attendance	Percentage		1.0%	45,000.00
U	External Reticulation Borehole Water	SUM	1	101,800	101,800.00
V	Allow for Profits, Overheads and Attendance	Percentage		1.0%	1,018.00
W	External Reticulation Council Water	SUM	1	131,800	131,800.00
X	Allow for Profits, Overheads and Attendance	Percentage		1.0%	1,318.00
Y	2,050 PE Waste Water Treatement Plant	SUM	1	20,602,500	20,602,500.00
Z	Allow for Profits, Overheads and Attendance	Percentage		1.0%	206,025.00
	TOTAL FOR SERVICES INSTALLATIONS PROVISIONAL SUM CARRIED TO GRAND SUMMARY				114,090,711.00

GRAND SUMMARY

AFFORDABLE HOUSING PROGRAMME
PROPOSED CONSTRUCTION OF AFFORDABLE HOUSING AND ASSOCIATED SOCIAL INFRASTRUCTURE

GRAND SUMMARY

ITEM	DESCRIPTION	PAGE	FOR TENDERER USE ONLY	FOR OFFICIAL USE
1.00	PARTICULAR PRELIMINARIES		8,000,000.00	
2.00	GENERAL PRELIMINARIES		11,550,000.00	
3.00	PROJECT PROVISIONS		28,785,000.00	
5.00	BUILDERS WORK BLOCK TYPE A (G+9)		424,044,696.00	
6.00	BUILDERS WORK BLOCK TYPE B (G+9)		162,918,116.00	
6.00	SOCIAL HALL		8,821,072.50	
5.00	GATE HOUSE		745,597.50	
5.00	GARBAGE RECEPTACLE		2,222,250.00	
6.00	POWER HOUSE		1,430,260.00	
14.00	CHAIN LINK FENCE		1,408,580.00	
14.00	BOUNDARY WALLING			
15.00	CIVIL WORKS - ROADS		11,456,500.00	
17.00	UNDERGROUND WATER TANK		24,611,880.00	
18.00	PROVISIONAL SUMS & P C SUMS		37,124,000.00	
19.00	SERVICES INSTALLATION PROVISIONAL SUMS		114,090,711.00	
20.00	MECHANICAL INSTALLATION WORKS		44,330,910.00	
21.00	ELECTRICAL INSTALLATION WORKS		69,195,550.00	
	SUB-TOTAL		950,735,123.00	
	ADD CONTINGENCY 2%		19,014,702.46	
	Less: CONTRACTORS TENDER DISCOUNT			
	GRAND TOTAL CARRIED TO FORM OF TENDER (VAT INCLUSIVE)	KSHS	969,749,825.46	
	AMOUNT IN WORDS : KENYA SHILLINGS NINE HUNDRED AND SIXTY NINE MILLION,SEVEN HUNDRED AND FORTY NINE THOUSANS,EIGHT HUNDRED AND TWENTY FIVE SHILLINGS AND FORTY SIX CENTS ONLY TENDERER'S NAME MULTIPLEX CONSTRUCTION CO. LTD ADDRESS 50797-00200 NAIROBI DATE 03.02.2026 TENDERER'S SIGNATURE RODGERS SITIENEI WITNESS'S NAME..... DAMIANA KIMEU ADDRESS 50797-00200 NAIROBI DATE 03.02.2026 WITNESS SIGNATURE.....			

PROPOSED CONSTRUCTION OF AFFORDABLE HOUSING AND ASSOCIATED SOCIAL INFRASTRUCTURE
COST PER SQM CALCULATION

<u>CONSTRUCTION COST</u>			
1.0 BUILT UP STRUCTURES			
	Built Up Area	No. of Blocks	Area
2.1 Block Type A (G+9Floors)	5,351.03	3	16,053
2.2 Block Type B (G+9Floors)	6,874.99	1	6,875
2.1 Social hall	328	1	328
2.2 Guard House	21	1	21
2.3 Garbage Receptacle	112	1	112
2.4 Power House	24	1	24
			Total Area
			23,413.08
2.0 COST PER SQM CALCULATION			
2.1 Total Project Cost			-
2.2 Gross Built up Area			23,413
2.3 Cost per Sqm		Kshs	- Per Sqm